

Perceptions of Effective Organizational Leadership in
American Fire and Emergency Services

Monika S. Byrd

A Dissertation Submitted to the Faculty of
The Chicago School of Professional Psychology
In Partial Fulfillment of the Requirements
For the Degree of Doctor of Philosophy in Organizational Leadership

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Abstract

This study explored the perceptions of fire officers on effective organizational leadership in the American fire service to discover what is shared among varied perspectives and what is distinct, and to analyze how the perspectives align with existing leadership theories and models. It employed Q methodology, which includes both qualitative and quantitative data-collection and analysis. Ten distinct perspectives emerged from the data that have varying degrees of fit with existing theories, including (listed alphabetically) adaptive leadership theory, authentic leadership theory, boundary-spanning leadership theory, complexity leadership theory, crew resource management, leader-member exchange theory, learning organization model, the managerial grid, Mintzberg's model of management, public administrative leadership theory, regulatory focus theory, servant leadership theory, and transformational leadership. The perspectives differ in the number and the combination of leadership-related concepts they embody. The leadership constructs expressed include individual consideration, systems-thinking, community engagement, task-orientation, change agency, future-orientation, team-orientation, relational leadership, reflection, and inclusivity. Individual consideration is the only construct included in all ten perspectives. The distinctions among the ten perspectives include the level of importance given to engagement in the community, attention to the external influences on a fire department, the salience of a future orientation and work as change agents, monitoring and evaluating individuals and teams, and building and maintaining individual relationships. The number of different perspectives and the differences among the perspectives on effective organizational leadership indicate the need for future studies that identify any correlations with organizational performance measures to learn if any of the perspectives represent best practice. There are specific leadership traits, knowledge, skills, abilities, and behaviors that are shared

across all the perspectives as important to department-wide, organizational leadership. Organizational leaders in the fire service should consistently manifest integrity, honesty, confidence, decisiveness, fairness, and professionalism. They should know the elements of effective public relations and communications theory as well as psychology, the principles for effective labor/management relations, the community governance structure and people, and the 16 National Fire Service Life Safety Initiatives. Skills and abilities that were shared across the perspectives were communications skills, organizational skills, the ability to run effective meetings, and public speaking. Behaviors important for all the perspectives include supporting rigorous, regular training for Operations, providing the example and role-modeling the core values and mission of the fire service, and professionalism. These components of organizational leadership could become a core for leadership education and development and professional development in the American fire service.

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Chapter 1: Introduction to the Study

Nature of the Study

Fire and emergency services departments are critical to public safety. They reduce community risk through public education, fire prevention, code enforcement, and health and accident risk reduction, and when those efforts to reduce risk are not enough, what the public commonly refers to as the *Fire Department* is now the *All-hazards Emergency Response Department* (Onieal, 2003). The expectations for Fire Department contributions to public safety include responding to structure fires, wildland fires that encroach on community boundaries, transportation accidents, hazardous materials mitigation, and medical and rescue operations in natural or man-made disasters. Understanding leadership in organizations like USFES Departments has significance because of this critical role in public safety and as a publicly funded good worth sustaining (Neyfakh, 2013).

The Congressional Select Bipartisan Committee that investigated the preparation for and response to Hurricane Katrina was unequivocal about the importance of leadership in public safety. Donald Kettl concluded that leadership was the critical and missing element in the poor Katrina response in his testimony before the Senate Committee on Homeland Security and Governmental Affairs in June 2006, stating that there were “failures at all levels of government that significantly undermined and detracted from the heroic efforts of first responders, private individuals and organizations, faith-based groups, and others” (U.S. House, 2006, p. 1). More recent public safety disaster response efforts also revealed inconsistencies and failures of leadership. In 2012, as Hurricane Sandy approached the east coast of the United States, responders staging in New Jersey noted the absence of planning and organization, resulting in four days’ delay in deployment (Bucci et al., 2013; Schmeltz et al., 2013). Lack of effectively

coordinated responses to a supermarket fire and for a toxic, vinyl chloride spill, both in 2012 and both resulting in injuries and death among the responders, highlight the grave importance of leadership (Brinker et al., 2015; Crowe, Allen, & Bowes, 2014). In a different way, the tragic Oakland warehouse fire in 2016 that killed 36 people of the artists' community living there makes it clear that effective organizational leadership is vital to protect the public through adequate staffing and ensuring the resources exist for administrative and inspection services. The investigation revealed that the fire marshal's office had been understaffed for years and the building had not been inspected by the city or the fire department in three decades, and the building likely caught fire because of a faulty, dangerous electrical system that any inspector would have required immediate intervention (Rocha, Willon, St. John, & Karlamangla, 2016). Academic investigations of foreign public safety operations have also indicated that leadership unequivocally impacts the outcomes (Andrews, Ashworth, & Meier, 2014; Pillai & Williams, 2004; Sjöberg, Wallenius, & Larsson, 2011).

Leadership is also important *away* from the scenes of incidents and disasters. USFES Departments are complex organizations with many contexts embedded within them, and all of the contexts have the potential to impact overall effectiveness. These multiple, interacting contexts apart from the scenes of emergency response that require *department-wide* leadership include:

- Firehouses
- Training academies
- Administrative offices
- Communities, where fire prevention and life safety education are important endeavors
- Local government that provides oversight and accountability

- State or federal associations and agencies that develop standards and policies (International Fire Service Training Association, 2011, 2013).

The Wingspread Conference on fire service administration, education and research that first convened in 1966 and has convened every ten years since then, develops a set of statements of national significance to the United States fire and emergency services during each conference to guide coordinated, national efforts to improve fire department contributions to the public safety. The 1996 statements of national significance included one related to organizational leadership for the first time: “To move successfully into the future, the fire service needs leaders capable of developing and managing their organizations in dramatically changed environments.” Ten years later, the 2006 set of statements of national significance includes a statement about the role of the Fire Chief specifically: “It is the responsibility of the Fire Chief to be the community’s chief advocate for fire safety. . . The executive skill set of the Fire Chief will to a large part define the success of the organization.” Most recently, the 2016 statements of national significance went further: “Leaders in the United States fire and emergency services must develop positive human relations skills, knowledge, and abilities to manage in current times. These skills, knowledge and abilities must be reflected in the organizations’ recruitment and hiring practices” (The Johnson Foundation, 2016).

Figure 1 presents an example organizational chart for a USFES Department and how the distinct divisions within the Department ultimately report to the Fire Chief. Research has shown the correlations that exist between leadership and performance measures for nations, social movements, and organizations (Day, 2014; Hickman, 2010; Yukl, 2010). Fire service professionals agree that leadership is important in USFES organizations. For practitioners, the evidence that they consider leadership of great importance exists in books, memoirs and personal

testimonials (Brunacini, 2002; Rubin, 2012; Russell, 2014), in textbooks and training curricula (Barr & Eversole, 2003; Brame, 2001; Cochran, 2013; Coleman, Revere, & Craig, 2010; Compton, 2010; IFSTA, 2011; IPSLEI, 2014; Salka, 2005; Smeby, 2006); and in trade journals and blogs or conference presentations (Auernhammer, 1994; Hayes, 2014; Kastros, 2013; King, 2011; Kupietz, 2010; Revere, 2014; Waite, 2007).

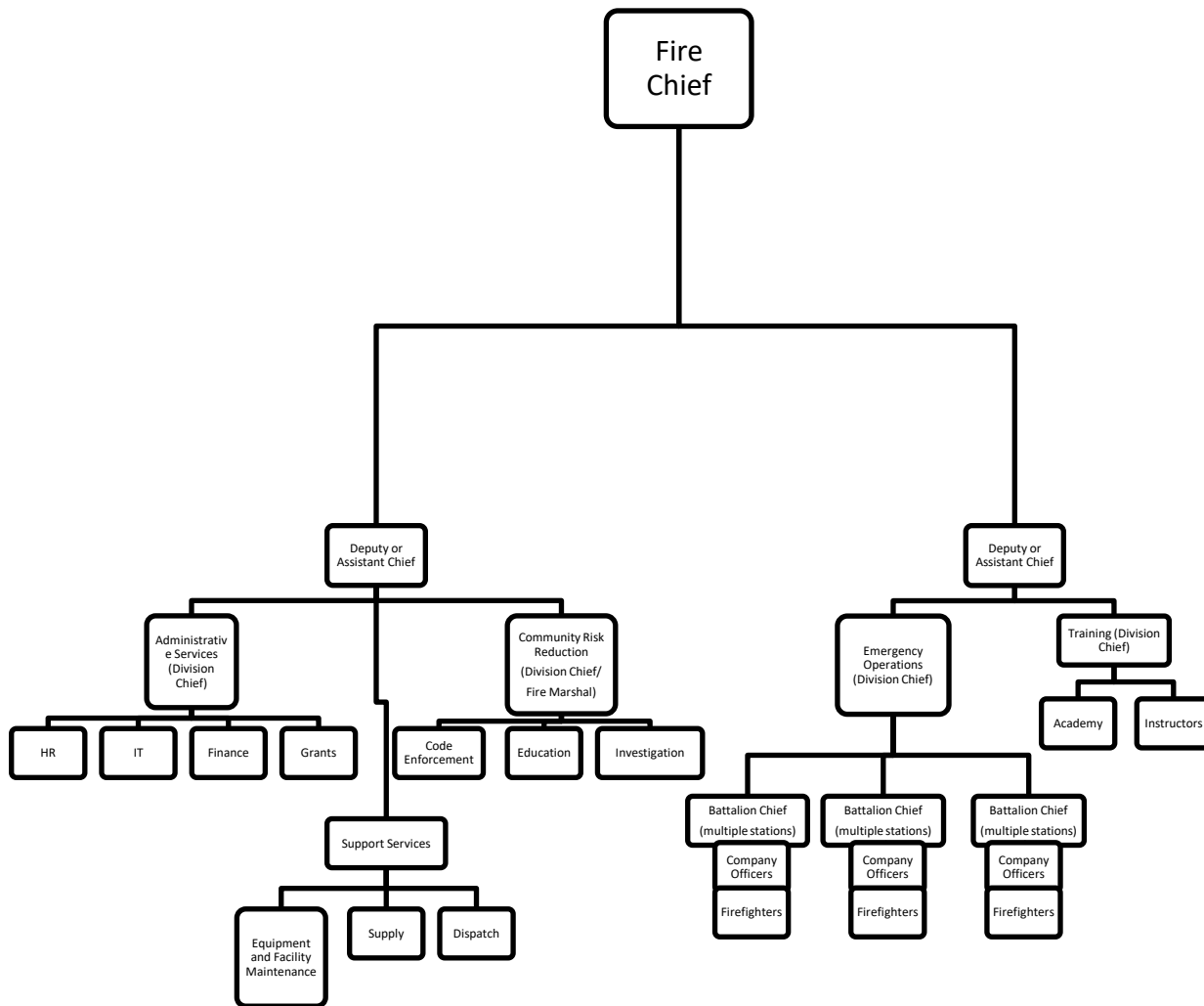


Figure 1. Example of an American municipal fire and emergency services department organizational chart (modified from IFSTA, 2011).

Researchers have examined leadership in USFES Departments with limitations to specific contexts and issues. Leadership research exists for the single contexts of emergency situations and the scenes of fire incidents or post-incident analysis (Dixon, 2014; Gasaway, 2008; Griffin, 2013; Jensen & Waugh, 2014; Klein, Calderwood, & Clinton-Cirocco, 2010).

Several studies consider leadership with a case study approach during specific events such as a merger of two Departments (Greco, 2013) or in a single or limited number of Departments (Griffin, 2013; Lowe & Barnes, 2000; Odom, 2011). Leadership research related to single, specific issues for the fire service includes the issues of:

- firefighter injuries (Hong, Chin, Phelps, Feld, & Vogel, 2012)
- health and wellness programs (Gist, Taylor, & Raak, 2011; Kuehl et al., 2013; Kuehl, Mabry, Elliot, Kuehl, & Favorite, 2013)
- volunteer retention (Carter, 2005; Smith, 2014)
- paid employee job satisfaction (Beaton, Johnson, Infield, Ollis, & Bond, 2001)
- education and leadership development (Ditch, 2012; Jones, 2007; Standridge, 2012)
- regional governance (Duncan, Matthew, & Scarpa, 2012).

There is also a single, industry-wide study of organizational leadership capacity, defined as the capacity of the organization's structure, processes, policies, and culture to respond to endogenous and exogenous stimuli, presented as challenges, opportunities, and threats to the organization (Kime, 2001). Finally, there are two theoretical studies (Carter, 2007; Woods & Branlat, 2011). These studies of single contexts and single issues have used a variety of theoretical frameworks. Outside of the academic literature, USFES leaders have presented additional theoretical frameworks based upon personal experience. Despite all this related research, the consideration of perspectives on department-wide USFES leadership *across* multiple contexts has not been discussed in the literature.

Recently, leadership theorists and researchers examining other industries are considering the varied contexts and levels of activity or types of events within the same organization to enhance our understanding of the complexity and social construction of leadership, which would

more closely relate to *department-wide* leadership. These studies present additional leadership theories and models that may relate to effective leadership in USFES Departments. Theorists are examining leadership in dangerous and risky organizational contexts and missions, but have not included fire and emergency services organizations (Griffin & Talati, 2014; Hannah & Parry, 2014; Hannah, Uhl-Bien, Avolio, & Cavarretta, 2009; Hoffman & Lord, 2013). Other leadership researchers have completed grounded studies in public administration organizations that also involve numerous interacting contexts (Trottier, Van Wart, & Wang, 2008; Van Wart, 2013). The study contributes to this trend to study leadership for complex organizations with multiple contexts, in an industry that these researchers have not yet examined. Chapter 2 discusses these studies further, and Appendix A includes Tables 37, 38, 39, and 40 summarizing the leadership research in the USFES.

Conclusions of the study may contribute to the industry standards for education and job performance requirements, which do not include mention of leadership. The National Fire Protection Association (NFPA) identifies job performance requirements (JPRs) for fire service officers and many operating standards for the fire service, but does not identify any *leadership* competencies for individuals or organizations (NFPA, 2014b). The absence of any leadership competencies in the NFPA JPRs is notable given the investigations of poorly executed emergency management and that practitioners in the fire service have pointed out the importance of leadership and the relationship between leadership and performance. USFES department-wide leadership is not a well-understood phenomenon, and that is perhaps why it is not included in the NFPA standards. This study of the self-referent perceptions of what is important for effective Department-wide leadership in USFES Departments contributes to the body of knowledge related to effective leadership in these complex organizations.

Individuals' perceptions of what they see, hear, and experience of a phenomenon such as leadership are personal, subjective, and affect their cognition, decisions, interactions and relationships. In short, individuals' perceptions of a phenomenon are an important primary, foundational source for knowledge of socially constructed phenomena such as leadership (Bryman, Stephens, & aCampo, 1996; Conger, 1998). It is important to learn about the perceptions of a phenomenon in order to more fully understand it. The study explores the main question about what perceptions exist within USFES Departments regarding effective department-wide leadership.

Background of the Problem

Since the 1973 National Commission on Fire Prevention and Control report called attention to the fact that the United States had more fatalities, injuries, and damages due to fire than any other industrialized country at the time, much more research than had ever been done before has contributed to improving the statistical picture. Technical and scientific research into building practices, building materials, consumer products (e.g., flame-retardant fabrics), fire behavior, and protective equipment for firefighters has contributed to reducing the fatalities, injuries, and property loss related to fire. Nevertheless, the goals set in 1973 have not been reached, and the United States remains in the worst 50% of industrialized countries for these statistics (Ahrens, 2013).

USFES Departments, defined as organizations and agencies accountable to a jurisdictional authority that is a city or county, provide emergency response services to the majority of the population of the United States (USFA, 2012). The remainder of the population is served by other kinds of organizations or agencies, such as fire protection districts or federal agencies. The 2012 United States Fire Administration census includes 26,482 organizations

fitting this definition of a USFES Department, estimated to be 88% of the total number of municipal departments in the United States (census reporting is voluntary), employing more than one million people. USFES Departments are significant because of these numbers and because they provide an important public good. Within this definition of USFES Departments, there are different classifications. Classifications include the size and type of jurisdictional community: urban, suburban, and rural. Classification details also specify if a USFES Department is comprised of volunteer professionals, or paid professionals, or a combination of both. Finally, there are USFES Departments that provide emergency medical response, and USFES Departments that do not, in jurisdictions with a separate emergency medical response agency. Practitioners point out that different classifications require different leadership, yet there has not been research conducted to discover what differences may exist in the leadership of these different classifications or understand the specific nature of leadership for the different classifications (Brown, A., personal communication, June 2014).

In USFES Departments of any classification, there are challenges and issues related to the many functions and contexts within them that create organizational complexity. For example, a USFES Department can have multiple stations to deliver fire suppression and emergency medical services, station administrative staff, central administrative staff, training staff, financial staff, fire prevention staff, fire investigation staff, community risk-reduction and life-safety education staff, and human resources staff. There is even complexity within a functional division like Operations, which includes the incident response areas of fire suppression and emergency medical services: structural fires may be industrial or residential; medical emergencies may involve a single victim or multiple victims, situations with hazardous materials, or rescue operations; incidents can be routine or non-routine, routine and complex or non-routine and

complex, or extreme incidents, in part depending upon the size of the responding team required and if the incident requires a multi-agency response and interoperability (Hannah & Parry, 2014). Organizational complexity resulting from the presence of multiple functional areas exists within every Department. Doubts exist related to whether the style or theory relevant to these different contexts within the same emergency response organization can be effective across the contexts (Canyon & Morrison, 2010; Hannah & Parry, 2014). The top Department leader or leaders, the Fire Chief, and any Deputy or Assistant Chiefs, must provide Department-wide leadership across the organizational complexity.

Fire and emergency services organizations also face a challenging paradox that makes an understanding of leadership important. Their mission involves risk, of necessity, so they cannot be risk-averse, yet must simultaneously espouse goals to minimize risk and prevent injuries and line-of-duty deaths, balancing high-risk operational obligations with the health and safety of personnel and the public. This paradox requires confronting disaster experiences from a broader perspective than the incident command frame by examining concepts and issues like heroism, community relations, post-traumatic stress, and survivor guilt (Eyre, 2014). It is important to do so because the United States has the second highest rate of firefighter injuries internationally (Burgess et al., 2014). It has been a long-standing goal in USFES to reduce the number of injuries and line-of-duty deaths (National Commission on Fire Prevention and Control, 1973). Nationally there has been progress in this area, but there are disparities among Departments related to these statistics (Ahrens, 2013).

The number of fires that USFES Departments respond to has declined, and USFES Departments have increased the number of firefighters over the same period. Does this mean that fire *prevention* has improved, (through new fire retardant and fire proof materials, life safety

education, building codes and enforcement), but not the quality of *firefighting*? Are the wrong solutions (more equipment and firefighters) implemented without evaluation? Why should the public pay for costly, ineffective or inefficient service? There are investigative journalists, economic researchers, and public policy analysts asking controversial questions like this (McChesney, 2002, 2015; Neyfakh, 2012; Tabarrok, 2013). The questions persist in spite of the weak correlation between number of fires and firefighter deaths ($r=0.464$, $R^2=.216$, $p=0.128$, at $\alpha=.05$). The number of fires alone cannot explain the variation between the number of fires and firefighter deaths (Griffith, Roberts, & Wakeham, 2015).

Additional questions relate to identifiable differences among international Departmental health and wellness approaches and risk management, and the rate of firefighter injuries. USFES Departments have higher rates of firefighter injuries that a comparative study indicates may be related to the difference in culture related to health and wellness, as well as risk management approaches (Burgess et al., 2014). These are organizational culture and leadership issues as much as policy issues (Pessemier, 2012). The comparative study is the first to correct for and moderate the effects of variables commonly cited to explain why the United States firefighter injury rate is higher, including differences among reporting systems and differing types of responses required. USFES Departments have responsibilities, such as responding first to medical calls and sometimes for transport to hospitals, but not all do, so the study included only lost-time injuries related to fire and hazardous materials responses to enable a comparison of equal factors. The study concluded that there is a difference in the firefighter-injury rates from country to country. USFES Departments have the highest rate of injuries among the countries of the study (United States, United Kingdom, Australia, and Japan). This study also illuminated significant differences in the rate of firefighter injuries from Department to Department *within* the U.S.,

again suggesting that studies to explain the differences, including the study of department-wide leadership, could contribute to reducing Department and industry firefighter injury rates.

Further background for the study includes professional standards for USFES Departments. Professional standards vary. The States all have their own education and training standards for all levels of fire service work. This includes job performance requirements and training necessary for each level of fire service rank. Standards usually are based upon those that the National Fire Protection Association and the International Fire Service Training Association develop, although the National Fallen Firefighter Foundation faults state fire agencies that regulate Departments and Departments for not keeping pace with the revision cycles and the research underlying the revisions in these standards (NFFF, 2007).

Fire officers receive technical, supervisory, managerial, and administrative training upon promotion, but it is important to note that no leadership education or training is included in standards followed by USFES Departments (Compton, 2010). National Fire Protection Association (NFPA) Standard 1021 that identifies the minimum job performance requirements for fire officer professional qualifications lists tasks, tools, equipment, materials, and evaluation parameters and/or performance outcomes, and prerequisite knowledge and skills to perform the job (NFPA, 2014); it does not include leadership competencies, models or theories, workshops, training, or education requirements. Another key organization providing technical, supervisory, managerial, and administrative training resources is the International Fire Service Training Association (IFSTA), but it does not have leadership curricula supported by empirical research in the USFES. An IFSTA textbook publication on “Progressive Leadership Principles, Concepts, and Tools: The Fire Officer’s Guide to Excellence in Leadership” (Compton, 2010) is a

testimonial from an experienced fire chief, and is not evidence-based, containing no citations or references.

There are optional education and training opportunities for USFES professionals that seek to close the gap created by the absence of leadership material in the NFPA standards and IFSTA materials, but the percentage of USFES professionals who took advantage of these opportunities is small (Compton, 2010). The most rigorous and in-depth leadership education available to USFES professionals is the United States Fire Administration (USFA) Executive Fire Officer Program (EFOP). It has less than 7,000 graduates and is the only leadership education offered at the National Fire Academy (NFA); it uses the adaptive leadership conceptual framework, but without empirical evidence that adaptive leadership is an effective framework for the United States Fire Service (NFA, 2014). Another training opportunity is the International Association of Fire Chiefs (IAFC) Chief Fire Officer Designation (CFO) program. It is a fee-based credentialing program that tracks and documents experience without research providing evidence that the experience tracked relates to Department effectiveness.

Economic constraints, technological advances, and political change have brought increased attention to USFES Departments (Barr & Eversole, 2003; Durso, 2015; Onieal, 2004, 2007; Revere, 2014; Rubin, 2014; Tabarrok, 2013). As the numbers of fires decrease and the other kinds of responses the USFES Departments execute have increased, it is no longer as straightforward to measure effectiveness (Tabarrok, 2013). In some respects, the irony of the situation seems missed or ignored; as USFES Departments have improved fire prevention and fire response, and have filled gaps in other areas such as medical responses, continuing efforts to improve community safety, the reason for their existence and expense has been questioned.

There is so much variance among USFES Departments that some are disbanded, and others are acclaimed (Rubin, 2014).

Problem Statements

Practical Problem

There are disparities among USFES Departments in effectiveness as measured by firefighter injury and line-of-duty death rates, and it is possible that there can be improvements in the national statistics related to civilian injuries and deaths and property loss related to fire and other community risk factors. In USFES Departments, a shared understanding of what characterizes effective department-wide leadership does not exist. The absence of a shared understanding of what characterizes department-wide leadership for Department effectiveness is a possible explanation for variations in USFES Department effectiveness. Further, the absence of a shared understanding of what characterizes department-wide leadership for Department effectiveness is a possible explanation for the lack of a standard or requirement related to leadership education or training in USFES. It is possible that a shared understanding of what characterizes effective department-wide leadership could operationally reduce injuries and fatalities among firefighters and civilians, and reduce community risk, through improved processes, coordination, and teamwork within Departments, and through influence of policy formation, revision, and implementation.

My personal interactions with professionals in the industry over the course of fifteen years in education and training settings have exposed discontinuities that also present practical problems. At the executive level, USFES Departments do not have standards or minimum required qualifications. At one end of the spectrum is a chief in a volunteer department who is illiterate and fire chiefs with high school diplomas as their highest level of education and 35

years of firefighting experience. At the other end of the spectrum are fire chiefs with M.B.A.s and Ph.D.s and 10 years of firefighting experience. There are fire chiefs with many other combinations of educational credentials and years of experience in between. There are individuals appointed to fire chief who skip ranks and others not considered because they have not held an assistant or deputy chief rank. There are fire chiefs who have experience in fire suppression and rescue operations, training, investigations, prevention and community education, and administration, and fire chiefs who have experience only in fire suppression and rescue operations (Brown, A., personal communication, June 11, 2014; Resurreccion, R., personal communication, July 16, 2016).

At non-executive levels, young fire professionals who may have years of experience and desire to move to new Departments – perhaps because a spouse has a great career opportunity in a new city – find that they have to start at the beginning as rookies in the academy, indicating that there is little, if any respect or credit given for knowledge, experience, wisdom, or leadership competencies gained in other Departments. I have met fire service professionals who sought out leadership and personal development opportunities and described situations when their colleagues treated them with disdain because they tried to bring new knowledge and ideas into their Departments (Schum, J., personal communication, October 10, 2015).

An additional practical problem arises from the culture of heroism that has called many to the profession. The heroism the public expects in extreme circumstances, and that fire suppression and rescue personnel expect of themselves, can become an excuse for a culture of complacency or an excuse to disregard standard operating procedures, or disrespect people in the industry who are not part of fire suppression and rescue operations. Fire service professionals have described examples to me ranging from disregard for precautions related to life safety

during emergencies, as basic as wearing seatbelts on the way to an incident, to disrespect for the people engaged with the “less heroic” aspects of the profession, like prevention and education work. It is another way that department-wide leadership is important so that such contradictions do not exist in a profession with a dangerous and vital mission to serve public safety with extreme personal risk for the people in the profession (Eyre, 2014; Franco, Blau, & Zimbardo, 2011; LeSage, Dyer, & Evans, 2010).

Ethical lapses occur with more regularity than they should among America’s heroes, ranging from the mundane, like unauthorized use of a government vehicle, to more extreme matters involving conflicts of interest, embezzlement, or sexual assault (International Fire Service Training Association, Fire Service Reputation Management Development Group, 2010). A shared understanding of what constitutes effective department-wide leadership can contribute to the development of solutions for these practical problems.

Research Problem

There is not an understanding of what characterizes effective USFES Department-wide leadership. It is important to know what characterizes effective Department-wide leadership because of the implications for the public safety and for firefighter safety. Understanding what perceptions of department-wide leadership exist will be foundational knowledge for future leadership research and recommendations for leadership education and training, and may contribute to improved USFES Department effectiveness as measured by reduced injuries and fatalities, and improved community safety. Research to explore the perceptions of department-wide leadership may also provide preliminary conclusions about differences due to department classification and geographic location. Within the industry, it is commonly noted and assumed that there are organizational culture and organizational leadership differences across

classifications of fire departments, such as all-volunteer, all-career, or combination departments, and across regions of the United States such as the East Coast and the West Coast (Beaton et al., 2001; Brown, 2014; Carter, 2005; Eyre, 2014; Holton & Roden, 2014; Smith, 2014).

Statement of Purpose

The purpose of the study is to explore the perceptions that exist among USFES Department officers regarding effective Department-wide leadership because foundational knowledge of USFES Department-wide leadership will be important to future USFES research, education, and training, and may contribute to reduced injuries and fatalities, and improved community safety. Literature does not provide the perspectives of USFES officers on effective *Department-wide* leadership in USFES Departments with rigorous data collection and analysis. There are two ways that the findings of this study to discover the self-referent perspectives of effective USFES Department-wide leadership have significance. First, practical significance is possible because deeper understanding of effective USFES Department-wide leadership may impact public safety, firefighter safety, education and training in the industry, and individual USFES employee well-being and career development. The findings also provide direction for future research of USFES Department leadership and for leadership education for fire service officers.

The study is limited in scope to municipal Departments – as distinguished from joint fire districts and state, federal, airport, or port agencies, and Departments at military installations that provide emergency response. Within the category of municipal Departments, there may be a variety of perspectives on effective Department-wide leadership because of the ways that municipal Departments vary. Municipal Departments vary regarding size, regarding the type of community (urban, suburban), regarding the types of industries, structures, risks, or hazards that

exist in the community, regarding the ratio of fire incidents to other kinds incidents, and regarding whether the professionals are paid or volunteer or a mix of both. There may also be a variety of perspectives because there are no industry standards regarding leadership education and training.

Research Questions

Three main questions guide the study:

1. What are the perspectives of officers in the USFES Departments regarding effective *Department-wide* leadership?
2. How do these perspectives differ and how are they similar?
3. How do these perspectives correspond to existing leadership constructs, concepts, theories, and models?

Conceptual Framework

The trade literature and the studies related to understanding USFES leadership include leader-centric, follower-centric, and dyadic or team-centric theories that will be discussed in greater depth in Chapter 2. The studies have not concerned department-wide, multi-contextual leadership, with one exception that examines organizational leadership capacities and not leaders (Kime, 1999, 2001). Kime (1999) examined fire service organizational leadership capacity and postulated that research in public administrative leadership theory is relevant to fire service organizational leadership capacity. Recent public administrative leadership research is informed by relational and networked leadership concepts and complex adaptive systems concepts that acknowledge the complexity within public organizations, complexity in the nature of the challenges public organizations exist to address, and complexity in the environment around them.

In the time since Kime's 1999 study, researchers have further refined the conceptual framework of public administrative leadership theory on the organizational level of analysis and the individual level of analysis (Van Wart, 2013). This study adds the individual level of analysis to Kime's study of USFES leadership and could confirm the relevance of public administrative leadership theory to the USFES for effective department-wide leadership. While this study was exploratory and abductive, and it did not frame the data collection and analysis within a single theory *a priori*, connections to multiple leadership theories or models that USFES studies have applied will be discussed in the analysis of the data (Creswell, 2013). Particular attention will be given to public administrative leadership theory because the only industry-wide study (Kime, 1999, 2001) provided a connection to public administrative leadership theory as relevant to USFES.

Public administrative leadership theory frames leadership as an action cycle with five segments important to public administrative settings (Van Wart, 2004, 2013). The first feature of the action cycle involves organizing and coordinating work with frequent, vigilant re-examination of bureaucratic processes to prevent over-formalization ("red tape") that can result in ineffectiveness and declining follower satisfaction. The second part of the action cycle involves external coordination to ensure that the needs and desires of the end users and stakeholders are being met and that there is alignment and adaptation to changes in the environment. The third segment of the action cycle involves resources and support services and consideration of the constraints that can be peculiar to the public sector, such as laws, regulations, and external budgetary control. The fourth area of the action cycle relates to the attention that public administrators give to the values and traits that studies of the public sector demonstrate are of critical importance to follower satisfaction, including honesty, integrity, and

fairness, a desire for excellence, a service motivation, and cultural sensitivity. The fifth area of the public administration leadership action cycle is concerned with power constructs and the use of power in administrative settings.

Methodological Approach

Q methodology provides a means to access and rigorously analyze multiple, subjective viewpoints. In the 1930s, William Stephenson first combined qualitative and quantitative methods to study “subjectivity” and called it Q methodology to distinguish it from the standard statistical studies, which he referred to as R methodology (Watts & Stenner, 2012). The rationale for using Q methodology in the study involves its unique methodological characteristics. It allows for the exploration and discovery of shared, common perceptions from complex scenarios differently than quantitative confirmatory factor analysis (Watts & Stenner, 2012). Q methodology can employ centroid factor analysis that presents all possible correlative solutions for examination, rather than presenting only the best mathematical correlations. It can also start with principal component mathematical factor analysis initially, and then the researcher manually rotates the factors to examine alternative solutions in addition to the mathematical correlations. This is important when the subject matter is not homogeneous – when there are different social constructions and contexts for the topic, and conflicting opinions, or multiple perspectives on the subject.

This non-homogeneity is relevant to USFES officer perspectives on leadership for effective USFES Departments. Consensus or a normative definition of what constitutes effective leadership in USFES Departments does not exist. There are indications in trade publications that perspectives may vary considerably related to the variations of Department type (Career, Volunteer, Combination), Department size, and geographic location. The amount and the flow of

communication about leadership in American fire and emergency services – the *concourse* – is large and complex. Q methodology is ideally suited for the purpose of exploring and discovering the different perspectives that exist in a large and complex concourse and to determine to what extent there are aspects of the perspectives that are shared and identify key distinguishing characteristics for the perspectives (Focht, 2002; Shinebourne, 2009; Watts & Stenner, 2012).

Q is a methodology with strong qualitative and quantitative elements that does not begin from a theoretical stand point. The methodology will not test any particular, *a priori* ideas about department-wide leadership. Q requires qualitative, self-referent interviews and reviewing other self-referent artifacts of the concourse such as trade literature and blogs to saturate, or complete, the concourse. USFES officers within the industry make their own points of view operant through a process called a Q sort. Q sorting involves arranging a collection of statements sampled from a complete communication concourse on a topic or phenomenon according to a subjective dimension such as “agreement/disagreement,” “most like my point of view/most unlike my point of view.” The resulting arrangement of the statements represents a participant’s subjective view of the subject matter in the study.

It is these arrangements that are analyzed with factor analysis to compare and contrast the different, subjective interpretations of the subject matter to discover the distinct or shared viewpoints that can then be further interpreted. This quantitative aspect of Q requires the researcher to complete statistical rotation across the sorts to maximize the exposure of difference and identify the number of significant factors. The statistical analysis also identifies aspects and factors that are held in common across a number of sorters, and illuminates the nature of the differences among the perspectives. For example, are the perspectives bi-polar with distinct

viewpoints on both ends of a single factor, or orthogonal with statistically significant loading on factors and holding nothing in common with other factors.

Brown (2006) stated, “In short, the mechanics of Q methodology make it difficult for any viewpoint to fall by the wayside unnoticed” (p. 373), which is an important feature for an exploratory study. Researchers do not hypothesize about significant factors or relationships among the viewpoints on a phenomenon in advance and test the hypotheses. The statistical analysis reveals the factors and relationships only after participants complete their sorts (McKeown & Thomas, 1988). Q methodology will reveal the self-referent perspectives that exist and are shaped by the experiences of officers in USFES Departments. The methodology and process for the study is discussed in greater detail in Chapter 3.

Definitions of Key Terms and Abbreviations

CPSE - Center for Public Safety Excellence, an organization providing accreditation of agencies and review and certification of education, training, experience, and other credentials of fire service professionals (publicsafetyexcellence.org, n.d.)

Department - Throughout this study the word *department*, as in *Fire Department* or *USFES Department*, shall refer to the organization or agency as a whole, comprised of subgroups for different functional areas; because these functional divisions in many organizations are often called *departments*, as in *accounting department*, to avoid confusion, if there is reference to a functional area within a USFES Department it will be called a division, sub-group or staff, as in the Fire Department’s *accounting division*, or the Department’s *paramedic staff*, or the Department’s *human resources staff*

DHS - The United States Department of Homeland Security, a United States federal agency overseeing 22 federal agencies responsible for preparation for and response to all hazards and disasters for the nation (IFSTA, 2011)

EFOP - Executive Fire Officer Program, an educational program of the United States Fire Administration's National Fire Academy consisting of four sequential courses focused on leadership development and knowledge deemed important for leaders in the fire service (USFA.FEMA.gov, 2015)

Firefighter - Active member of a fire department, a person trained in the techniques, equipment, processes, and incident command system for fire suppression and other emergency responses such as search and rescue (IFSTA, 2011)

Fire ground - The scene of an incident involving fire

Fire Officer - Active supervisor of a fire department

IAFC - International Association of Fire Chiefs, a professional membership association for providing representation, information, education, and services to enhance professional capacities (IFSTA, 2011)

IFSJLM - International Fire Service Journal of Leadership and Management – a peer reviewed journal published by Oklahoma State University

IFSTA - International Fire Service Training Association, a non-profit educational alliance for the development of training materials in fire and emergency services (IFSTA, 2011)

IPSLEI - International Public Safety Leadership & Ethics Institute, a non-profit organization founded in 2005 with a mission to provide leadership and ethics education and support the development of ethical leadership for people in service to the public safety

ISO - Insurance Services Office, Limited: provides data, analytics, and decision-support services for the property/casualty insurance industry, including ratings of USFES Departments

NFA - National Fire Academy, a division of the United States Fire Administration for providing training and education to USFES members

NFFF - National Fallen Firefighters Foundation, a non-profit organization with a mission to honor firefighters killed and injured in the line of duty, to provide resources to assist their survivors, and to provide research and support to reduce firefighter deaths and injuries (firehero.org, n.d.)

NFIRS - National Fire Incident Reporting System, fire departments report fire incident data to a state coordinator who compiles statewide fire incident data and forwards information to the United States Fire Administration (IFSTA, 2011)

NFPA - National Fire Protection Association, a non-profit educational and technical association that develops fire protection standards (IFSTA, 2011)

NIOSH - National Institute for Occupational Safety and Health, a United States Federal agency operating within the Center for Disease Control and Prevention of the Department of Health and Human Services that ensures workplace safety (IFSTA, 2011)

NIST - National Institute for Standards and Technology, a United States Federal agency within the Department of Commerce that provides technology research, measurements, and standards for United States industry (nist.gov, 2015)

PPE - Personal protective equipment, equipment worn to minimize exposure to a variety of hazards; in the fire service, PPE includes the protective clothing and self-contained breathing apparatus that protect a firefighter from extreme temperatures and smoke inhalation (IFSTA, 2011)

UL - Underwriters' Laboratories, independent fire research and testing laboratory that certifies equipment and materials (IFSTA, 2011)

USFA - United States Fire Administration, a Federal agency within the Federal Emergency Management Agency of the Department of Homeland Security charged with reducing American fire losses; activities include research, training, and education to improve fire prevention and control, and support for new state and local government programs (IFSTA, 2011)

USFES and USFEMS - United States fire and emergency services, United States fire and emergency medical services, the industry concerned with fire prevention and suppression and response to additional community hazards and disasters

Outline of Remaining Chapters

The literature review in Chapter 2 provides historical context regarding USFES research and a summary of the existing USFES leadership research. Chapter 2 also presents leadership research in related industries. Chapter 3 specifies the details of the study methodology, including the rationale for use, the procedures and process, sampling methodology for the concourse and for the participants, the participant experience, and an explanation of the minimal risk of ethical issues. The initial steps in the methodology—the development of the concourse and sampling the concourse to prepare the Q-sort materials for the data collection are also described in Chapter 3. Chapter 4 presents the data collected as Q-sorts and the statistical and interpretive analysis of the Q-sort data. The discussion of the statistical and interpretive analysis and presentation of the study findings makes up the first part of Chapter 5, and the recommendations based upon the findings, limitations, recommendations for future study, and implications make up the final parts of Chapter 5.

Chapter 2: Review of the Literature

Overview

The general purpose of this study is to contribute to the body of knowledge related to USFES organizational leadership. This chapter is a review of the USFES antecedent leadership research and the leadership research in related industries. The review results in a broad understanding of existing knowledge regarding USFES leadership and highlights ways that this study contributes to the body of knowledge.

Related to this general purpose, there are four specific objectives for this chapter:

- offer historical background on all research in fire and emergency services,
- summarize the existing theoretical and empirical studies concerning USFES leadership,
- highlight the interest in leadership from fire service practitioners and educators who seek to both apply theory and call for further research of USFES leadership, and
- survey the theories and models that have been studied in related industries or professions, such as policing, but have not been studied in USFES.

Regarding the last objective, this chapter illuminates the theoretical frameworks and models that researchers and practitioners have considered relevant to the USFES. Inclusion and awareness of existing theoretical frameworks are important for an exploratory study, and none will be favorably presented over any others. Comparisons with the theoretical frameworks and models that have already been suggested or applied *a priori* for USFES Departments, and with theories studied in related industries will serve as a starting point for the analysis of self-referent data collected. This study is abductive, and as in grounded theory, my analysis remained open to all theoretical explanations for the data whether presented here in the literature review or not (Charmaz & Henwood, 2008).

My review of the literature demonstrated that research in USFES leadership has involved only one or a limited number of Departments within a single county or state – using variations of a case study approach. These studies considered a single aspect of USFES Departmental operations or did not specify the context or level of analysis for the study; many were limited to studying leadership at the scene of a fire incident – the *fire ground*. As explained in Chapter 1, there are areas of USFES Department operations that impact overall effectiveness that do not involve emergency responses, including governance, labor relations, training, and community risk reduction efforts. This review of the literature will demonstrate that there has not been research to examine *department-wide*, organizational leadership that is inclusive of all Department functions. This study contributes to the body of knowledge related to fire service leadership, specifically regarding department-wide, organizational leadership.

Fire Research since 1968 – Setting the Stage for USFES Leadership Research

While an organized fire service has existed in the United States for nearly 250 years, fire-service-related research has a shorter history. Interest in and support for research in USFES followed the 1968 Congressional Act for Fire Research and Safety and the creation of the National Commission on Fire Prevention and Control, charged with researching and proposing how the United States could provide more effective protection against the hazards of death, injury, and damage to property from fire. At the time, the United States led all industrialized nations in per capita deaths and property loss from fire (National Commission on Fire Prevention & Control, 1973). The Commission's report provided comprehensive data on the extent of the fire problem in the United States and its specific recommendations for how the federal government could play a role in reducing the fire losses through more effective support of local government efforts. The Commission recommended several federal initiatives to provide

guidance for the USFES goals and research related to fire safety and reducing the loss of lives and property to fire. Specific goals were to:

- reduce fires due to human behavior by 80%,
- increase in-hospital survival of burn patients by 50% through improved burn care,
- decrease firefighter injuries by 50% through better personal protective equipment (PPE),
- eliminate 90% of all large property losses through improvements in building design,
- reduce fire injuries and fatalities at the accident site by 80% through safe fabrics, design of products, detection devices, and on-the-scene care, and
- increase the fire control capability of fire Departments by 50% by faster response and more effective extinguishing methods (National Commission for Fire Prevention & Control, 1973, p. 136).

In 1974, Congress authorized the creation of the United States Fire Administration (USFA), the National Fire Academy (NFA), the National Fire Incident Reporting System (NFIRS), and National Bureau of Standards (renamed the National Institute for Standards and Technology [NIST] in 1988). These federal bodies were authorized to provide technical and educational assistance to local and state authorities, collect and analyze fire information, regulate material flammability, and provide financial assistance when necessary. They were also charged with improving communication across the disciplines involved in fire safety, and with research to provide more and better data that could help protect citizens, firefighters and property from the dangers of fire.

The National Commission on Fire Prevention and Control provided a vision of a broad coalition emerging to address the United States fire problem. National agencies would collect data for benchmarking and measuring, and develop training and education programs. Private

associations and enterprises such as the National Fire Protection Association (NFPA) and the Insurance Services Office (ISO) would analyze the data to direct and conduct research and development efforts and revise standards and protocols. Professional associations would disseminate information and create the relationships in other industries such as design and construction; and the burden of acting and implementing change to accomplish the goals would remain at the community level.

The coalition has accomplished much. Research both prior to 1973 and subsequent to the greater levels of funding for fire-related research since 1973 has filled gaps in the scientific knowledge of fire and firefighting. The industry has sought to keep pace with changing building materials, which has made the research overwhelmingly technically oriented and related to fire behavior and materials science, or tactical and operational related to incident response. Such research has contributed to progress in reaching the goals outlined by the 1973 National Commission on Fire Prevention and Control. Gaps between reality and the desired goals related to life safety remain, and there are statistical mysteries that warrant closer examination (Ahrens, 2013; Compton, 2005; Mueller, 2008).

Annual, unintentional, civilian fire deaths decreased from 6,503 in 1973 to 3,240 in 2013, a 50% decrease, but the civilian casualty rate has been at or near that level since 1999 (excluding deaths on September 11, 2001; NFPA, 2014). In the 2007-2009 statistics, the United States had moved closer the median rate of civilian casualties internationally, rather than among the worst industrialized nations, but fire and emergency service professionals continue to express frustration that there has not been greater progress. The number of fires that USFES Departments responded to during the same period, 2007 to 2009, decreased by 50%. Fire Departments have

increased the number of firefighters and invested in more equipment over the same period (Ahrens, 2013).

There are investigative journalists, economic researchers, and public policy analysts who have asked probing questions related to these statistics. Have the number of medical responses increased to an extent that justifies the additional equipment and personnel? Has fire prevention improved, through new fire retardant and fireproof materials, life safety education, building codes and enforcement, but not the quality of firefighting? Are the wrong kind of solutions (more equipment and firefighters) implemented without evaluation? Could USFES Department effectiveness improve if hiring practices ensured that Department demographics closely matched community demographics? Why should the public pay for costly, ineffective or inefficient service or accept that such questions go unanswered without critical examination? (Ash, 2014; Bowen, 2016; Gay, 2015; McChesney, 2002; Neyfakh, 2012; Tabarrok, 2013).

Close examination of fire fatality data and analysis by national organizations concerned with making improvements in the statistics points to a possible explanation for the inconsistent and sometimes puzzling statistics related to the American fire problem. USFES Departments have not continued to make progress toward the goals set in the 1970s in part because of incomplete implementation of recommended new codes and standards. The NFPA has a three-year cycle for codes and standards revisions related to firefighter safety, but many states do not follow the same cycle and skip multiple NFPA cycles (Brown, 2014). In between the release of planned revisions, the National Institute for Occupational Safety and Health's (NIOSH) Fire Fighter Fatality Investigation and Prevention Program and other reports make recommendations related to personal protective equipment, self-contained breathing apparatus, personal alert safety system devices, radio communications, incident command, and other topics. In spite of the

revision cycle and other recommendations related to fire fighter injury and fatality prevention, an evaluation of the Fire Fighter Fatality Investigation and Prevention Program showed that two-fifths of USFES Departments did not disseminate information from NIOSH to fire fighters at all (Peterson et al., 2008).

In some states, codes have been updated with the intention to reach the goals set by the National Commission on Fire Prevention and Control, without adequate monitoring and enforcement. Residential smoke detectors provide an example. An initial USFA survey of residential smoke detectors in 1977 found that only 22 % of US homes had one smoke detector; in 2010, a survey by the NFPA showed that 96 % of homes reported at least one smoke alarm, but many do not have enough smoke detector coverage for full effectiveness or current code compliance (Ahrens, 2013).

Another example is residential automatic sprinkler systems, except it is lobbyists *against* codes requiring them that has meant this highly effective technology to protect life and property goes largely un-implemented. The NFPA has recommended that states and local jurisdictions implement codes requiring residential automatic sprinkler systems since 2006. In 2014, 84%, of the 3,275 civilian fire deaths occurred in home fires. In 2008, it also became a recommendation of the International Code Council, yet in 2009, only 18.5% of homes built in the previous four years had sprinklers. In 2016, only two states, California and Maryland, have adopted the code recommendation. Homebuilders, housing industry trade groups and realtors have spent millions lobbying against the codes (Ahrens, 2016; Faturechi, 2016; Hall, 2013).

Of great concern in fire and emergency services are the statistics for firefighter line-of-duty deaths and firefighter injuries. Firefighter line-of-duty deaths and injuries have fluctuated and have had a downward trend as shown in Figure 2 and Figure 3.

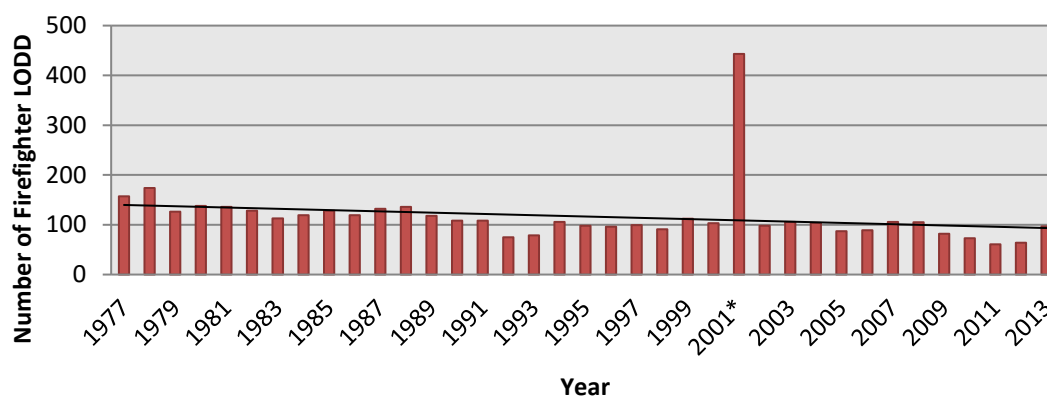


Figure 2. Annual firefighter line-of-duty deaths, including trend line. *2001 total includes 340 firefighters killed in terrorist attacks on September 11, 2001. Adapted from “The U.S. Fire Problem,” NFPA, 2014.

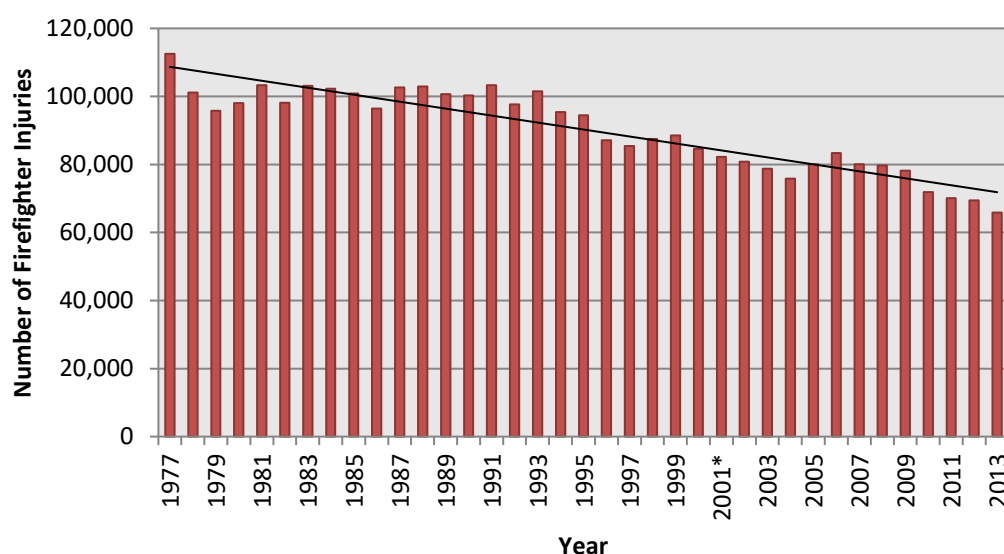


Figure 3. Annual firefighter injuries, including trend line. Adapted from “The U.S. Fire Problem,” NFPA, 2014.

Despite research and changes in PPE, understanding of fire behavior, and rigorous studies in materials science, the annual line-of-duty death rate for firefighters remains higher than the goal articulated, and the causes of death remain the same (Kline, 2011). In 1999, USFES professionals who participated in a workshop for the United States Fire Administration

recommended more and better research for developing and supporting policy positions that could impact firefighter safety, such as deployment staffing policies. The majority of fire-related research was through the Building and Fire Research Laboratory and the National Institute of Standards and Technology (NIST) and focused on building construction and materials science. There has been limited study from these entities on fire-loss reduction that relates to firefighter safety even though reducing the fire fighter fatalities and injuries has been a major goal within the industry (Moeller, 2007). When funding for fire service research was increased in 2003, it was directed toward several areas that impact fire fighter safety.

The Firefighting Research and Coordination Act of 2003 specifically mentioned more research on personal protection equipment, devices to locate victims, firefighters, and other rescue personnel, and evaluating the compatibility between new equipment and technology with existing technology. There is demonstrable need for this continuing technical research, in no small part because funding for fire service research has been historically small and disproportionate to the significant costs associated with providing fire and emergency services. USFA estimated the total funding available for USFES research at \$3.25 million in 2002; with the \$2.2 million added by the Firefighting Research and Coordination Act of 2003, the total research funding was only .02 per cent of the \$26 billion spent annually on career USFES Departments (Moeller, 2007). In contrast, the research agenda for a related industry, emergency medical services, called for funding equal to one per cent of the total spending in providing emergency medical services, or \$50 million in 2001, making apparent the trivial amount available for fire service research (Moeller, 2007).

Calling for USFES Leadership Research in Addition to Technical Research

All the technical research *has* contributed to improving the US fire problem, but the industry has not met the goals set in the 1970s despite the continued importance of them and attention to them. In the early 2000s, calls became louder for attention to areas besides fire behavior and materials or construction science and technology. Then Superintendent of the National Fire Academy, Dr. Denis Onieal, made the case for accelerating the “professionalization” of the industry, and results rolled out of committees for curriculum standardization and fire service educational expectations (Onieal, 2003a).

In 2004, the National Fallen Firefighters Foundation (NFFF) sponsored the Firefighter Life Safety Summit of fire service leaders to identify and then implement changes in the industry to reduce line-of-duty deaths that coincided with the attention to professionalization. The program, Everybody Goes Home®, has 16 Life Safety Initiatives, and builds on the technical research agenda to include calls for leadership research:

- Initiative 1, Cultural Change [to safety-driven organizations]
- Initiative 2, Accountability
- Initiative 3, Risk Management
- Initiative 4, Empowerment
- Initiative 5, Training & Certification
- Initiative 6, Medical & Physical Fitness
- Initiative 7, Research Agenda
- Initiative 8, Technology
- Initiative 9, Fatality, Near-Miss Investigations
- Initiative 10, Grant Support

- Initiative 11, Response Policies
- Initiative 12, Violent Incident Response
- Initiative 13, Psychological Support
- Initiative 14, Public Education
- Initiative 15, Code Enforcement & Sprinklers
- Initiative 16, Apparatus Design & Safety (NFFF, 2007)

In 2007, the NFFF first white paper of 16 (one for each initiative) discussed USFES safety culture and the importance of cultural change if the rate of firefighter deaths and injuries is to be substantially reduced. A specific recommendation in the paper called attention to the lack of research in the areas of organizational behavior, organizational culture, organizational change, and organizational leadership:

Recommendation # 7: Recognize and support the development of the following documents (through supplying articles and/or case studies):

- A primer for fire department leadership about organizational culture change, including what it is, its applicability to the fire service, its importance to fire department safety, its balance/coexistence with historical fire department traditions, and outlined strategies for implementation.
- An extensive, detailed academic study for fire department leadership that connects academic theory and applied science/industry case studies with fire department needs, including the integration of risk management/safety management/behavior-based systems within the fire department culture, strategic implementation considerations and its potential impact on the organization, fire service success stories with regard to culture change, a compendium of applicable literature on the subject, and so forth.

- Published leadership guides on how to achieve a holistic safety approach to each of the Everyone Goes Home® areas of LODD [line-of-duty death] focus (NFFF, 2007, p. 6).

There is progress toward achieving these recommendations. The website LifeSafetyInitiatives.com has been building an online article index for each of the 16 initiatives. The database includes articles from a variety of sources including trade publications, applied research from USFES practitioners, theses and dissertations, and peer-reviewed publications. It is updated frequently, and in the database for Initiative 1 there were over 300 articles tagged with the keywords *Initiative 1* and *leadership* at the end of 2015. While Initiative 1 specifically includes the call for academic study “that connects academic theory . . . with fire department needs,” in 2015, there were only 21 of more than 300 articles published in peer-reviewed, academic publications, revealing a significant gap in fully achieving the recommendation, and an opportunity for further research. Of the 21 peer-reviewed, academic studies tagged with *leadership*, four are concerned with other industries, and only 14 have been published since 2000, an average of less than one per year in the 21st Century. Greater than half (194) of the more than 300 articles listed are applied research papers from students in the National Fire Academy’s Executive Fire Officer Program (EFOP). The papers fulfill a program requirement to research and propose a solution to a challenge in their own Departments; they are secondary research, not research studies, since the students are not required to collect and analyze original data. To summarize, close analysis of the Lifesafetyinitiatives.com database indicates that it is not a source of academic research related to USFES leadership.

To begin filling the gap and encourage rigorous academic study of fire service leadership and management, Oklahoma State University’s Fire Protection Publications in the College of Engineering, Architecture, and Technology launched the *International Fire Service Journal of*

Leadership and Management (IFSJLM) in 2007. When it was launched, then National Fire Academy Superintendent Dr. Denis Onieal suggested specific research questions and studies for publication in the journal:

- Are successful service leaders and managers similar to or different from organizational leaders and managers in other public-sector professions?
- Can the use of psychological assessment instruments validate fire executive leadership effectiveness?
- Do successful fire service leadership skills vary among career, combination, and volunteer fire departments? If so, how and why?
- What leadership roles are required for the modern fire chief? What are the chief's responsibilities to the community, the department, and the profession? What are the expectations on the chief from the community, the department, and the profession?
- What impact does fire department culture have on leadership, managerial, and supervisory styles? Are there multiple types of fire cultures? If so, what are they?
- What is the impact of higher education on leadership in the fire service?
- Is there a correlation between fire executives who have advanced academic achievement and their perceived leadership effectiveness?
- Is there a correlation between fire executives who have advanced academic achievement and the fire losses (human and physical) experienced in their communities?
- How does the fire chief remain a student? Where/how/what are the most effective ways to continue the development of the chief fire executive?

- Are there empirical methods that can be used to examine the effectiveness of fire prevention and fire education activities? If so, can they be used in other fire and EMS areas?
- What is the influence of societal value shifts, generational shifts, and today's fire service work place for career and volunteer members?
- Is there a relationship between city and community demographics and fire loss? Can one use demographic data to predict and/or identify fire problems? If so, how and with what effectiveness?
- Are there communications techniques and/or processes used by other professions (e.g., commercial pilots/air traffic controllers, the military) that would have application in the fire service to improve communications, safety, or decision-making?
- What is the difference between the roles and functions served by leaders, managers, and supervisors in the modern fire service (Onieal, 2007b)?

In addition, he provided the suggested leadership research questions and problems from others:

- Define "success" in fire service leadership.
- With over 30,000 fire departments in the U.S., we need a taxonomy or taxonomies of "types" of departments. We need a way to talk "broadly" about various types of fire departments.
- Who will fund pure research for issues related to fire leadership and management?
- What methodologies can be used to examine fire leadership and management issues, problems, and concerns? How effective is the use of case studies, policy studies, systematic hypothesis testing, and meta-analyses as methodologies? Should studies be qualitative or quantitative, or both?

- What is the “ethics” of the fire service? Define it and analyze it.
- Fire leadership and management; is there an international skills set?
- Why hire as a fire chief a person who has risen up the fire leadership hierarchy instead of a private sector businessperson or a non-fire executive, such as a finance director, assistant city manager, or human resource director?
- What academic requirements are currently required for fire officers? What academic requirements should be required for fire officers and based on what criteria?
- Are the “reinventing government” and new public management movements impacting the modern fire service? If they are, how and with what level of impact?
- How is the issue of “diversity” affecting the modern fire service (Onieal, 2007b)?

It is an extensive list of research questions that would be useful and significant for USFES Departments. Doctoral and Masters students have studied a number of questions similar to those suggested related to higher education and officer development (Ditch, 2012; Jones, 2007, Standridge, 2012) and two have addressed issues related to fire service culture and leadership styles (Greco, 2014; Odom, 2011). While the suggested research questions listed above did not include fire ground leadership, there are recent studies in the USFES to study fire ground leadership (Dixon, 2014; Gasaway, 2008).

The IFSJLM has published both US and foreign research contributions and is the only journal solely focused on new leadership and management research specific to the fire service. It has aimed to raise the bar in the quality of research. One of the earliest articles assessed the quality of research used in policy development, concluding that the NFPA Standard 1710 for staffing and response times in career Departments was based on only four empirical studies, three that did not include original data collection and one that had methodological problems

(Moeller, 2007). This highlights an issue that the IFSJLM hopes to address through attention to the quality of the research: reduce the use of such problematic studies as evidence for policy formation and implementation. USFES Departments have been inconsistent in the implementation of new procedures and processes even when they *are* based in solid, undisputed, research-based, technical or tactical evidence that the change would contribute to more effective execution of the mission and attainment of goals (Buckman, 2014). The IFSJLM seeks to publish research of organizational leadership and management that can contribute in general to improved policy formation, revision, and implementation.

Since its inauguration, the IFSJLM has published dozens of studies on a number of specific fire-service issues, including line-of-duty deaths and injuries, education and training effectiveness, incident management, and general issues of concern in USFES Departments, including health and wellness and stress management, diversity, and dealing with fiscal stress. IFSJLM and trade publications have included overviews of leadership theories, models, or constructs and the possible implications for issues in the USFES. Since its inception, IFSJLM has not published research on USFES *department-wide* leadership. The next section reviews the studies that IFSJLM has published and additional theoretical and empirical studies in fire and emergency services of other countries.

A Beginning—Approaching Fire and Emergency Services Leadership with Academic Rigor

This section will provide detail regarding existing *leadership* research in fire and emergency services. This review of the scholarly work on leadership in USFES includes theoretical papers suggesting theories or models for application in the USFES through rigorous analysis of why or how the theory or model might be applied, but which have not yet been studied empirically. It will also include empirical studies in USFES leadership, additional

empirical leadership research in foreign fire and emergency services, and empirical research in non-departmental fire services (e.g., wild land firefighting). To organize the review of these theoretical and empirical studies and models, and to highlight the many different ways that researchers have investigated questions related to effective fire service leadership, this section follows the organization employed by David Day in *The Oxford Handbook of Leadership and Organizations* (2014), and separates the theories, approaches and issues based upon whether they are leader-centric, follower-centric, or dyad- and team-centric. Theoretical and empirical studies concerning fire and emergency services leadership span all of these categories. In addition to showing the variety of conceptual frameworks and models that have been applied to the USFES, the review of these studies will show that much of the research has focused on fire ground leadership and other types of incident command, or on leadership related to specific issues (e.g., fire fighter health initiatives), and that there have not been empirical studies of department-wide organizational leadership.

Leader-Centric Theories and USFES

Researchers have applied leader-centric theories in USFES leadership studies, including trait theories focused on leaders' cognitive resources, or an emphasis on leader behavior. Figuring importantly as a reason for research focused on leader traits and behaviors are the standards for fire service operations established by the National Fire Protection Association (NFPA), a not-for-profit founded in 1896. The NFPA Standard 1021 covers fire officer job performance requirements, and is the standard where there would likely be recommendations related to leadership, although the standard does not specifically mention leadership. It should be noted that the NFPA job performance requirements may be informed and influenced by research, through the knowledge of research from volunteers comprising its standards-development

committees, but Standard 1021 is *not* specifically research-oriented. Standard 1021 includes specifics important to the job performance requirements of fire officers such as intellectual abilities, experience, and expertise related to such things as human resource management and post-incident analysis (NFPA, 2014b). Standard 1021 is the basis for many training and education programs or succession planning in individual Departments or agencies, and Departments may identify additional specific emotional and social competencies for fire officers (Auernhammer, 1994; Onieal, 2003b).

A 1985 fire service leadership study for *Firehouse* magazine used the business framework presented in the national best-seller, *In Search of Excellence: Lessons from America's Best-Run Companies* (Peters & Waterman, 1982), which includes eight management principles that account for the success of the best-run companies:

1. Bias for action
2. Stay close to the customer
3. Autonomy and entrepreneurship
4. Productivity through people
5. Hands-on, value-driven
6. Stick to the knitting
7. Simple form, lean staff
8. Simultaneous loose/tight properties

The study developed a sample of 25 fire departments for the study, chosen by at least three of eight informed fire service insiders perceived as excellent. A group of 35 fire executive officers identified six measurable criteria to determine whether a fire department is excellent:

over a three to five-year period of time, an excellent fire department would show a trend in six areas:

- Decrease in fire loss
- Decrease in firefighter injury
- Financial responsibility
- Increase in performance standards
- Increase in amount and type of services offered
- Decrease in response time

For the departments among the original 25 perceived as excellent, 89% performed in three or more of the categories. The Chiefs of these departments were interviewed, and the interviews coded according to Peters and Waterman's eight management principles. The study provides evidence that five of the eight business management principles are also practiced in fire departments deemed excellent. One of the principles, stick to the knitting, was not present. Rather it was noted that leaders in excellent fire departments were diversifying to meet the communities' public safety needs. It is unknown if the remaining two principles are also practiced in the fire departments because they required on-site observation to verify (simple form, loose/tight management; Clark, 2015).

The leader behaviors identified in business as "five practices of exemplary leadership" (Kouzes & Posner, 2002) were the basis of a USFES study to determine if the quality of leadership practices improves organizational commitment levels (Lowe & Barnes, 2000). Kouzes and Posner's Leadership Practices Inventory (LPI) is an instrument for individuals to self-assess and/or others to rate the five practices: *challenging the process*, *inspiring a shared vision*, *enabling others to act*, *modeling the way*, and *encouraging the heart* (1988, 2010). In the USFES

study, participants in a single USFES Department completed the LPI about their supervisors' behaviors in each area, and participants self-assessed their personal levels of organizational commitment using the Organizational Commitment Questionnaire (Porter, Steers, Mowday, & Boulian, 1974). The study's authors did not specify the statistical methods employed and did not report significance levels, reporting only that the participants' perceptions of their supervisors' use of the five practices were all positively correlated with the self-reported organizational commitment levels; *enabling others to act* was most strongly correlated with self-reported organizational commitment and *inspiring a shared vision* was least strongly correlated with self-reported organizational commitment.

Leader behaviors were also the focus of a case study of a merger for a USFES Department and an emergency medical services organization (Greco, 2014). The study explored the impact of seven leader behaviors identified by Stephen Covey (1989) and areas of change, growth, and improvement for the merged organization. Covey's *seven habits of highly effective people* include:

1. Be proactive, both in roles and in relationships
2. Begin with the end in mind, have a vision
3. First things first, manage self and prioritize needs of others and the organization
4. Think win-win, to better achieve long-term sustainable results
5. Seek first to understand, then to be understood
6. Synergize, maximize teamwork
7. Sharpen the saw, by balancing and renewing energy, health, learning, and service

(Covey, 1989).

Members of the new, merged Department read *The Seven Habits of Highly Effective People* (Covey, 1989) and employed the principles during the organizational change. The data collected in interviews throughout the Department revealed four specific areas of growth and improvement for the merged department: the prevailing passive-aggressive, divisive spirit was replaced with a sense of department renewal and unification, improved communication, creating a new structure to provide better services, and solidifying and institutionalizing the new structure. The themes that emerged through the interviews were triangulated and validated with the data from the Insurance Standards Office (ISO) ratings improvement report and through the legal review and approval of updates to the Department's Standard Operating Guidelines.

Additional studies of single Departments have focused on leader-centric theoretical frameworks. One study involved a large metropolitan Department to determine if there were differences in preferred leadership style among generations (Odom, 2011). Participants made a choice from among three imaginary descriptions of a transformational leader, a transactional leader, and a *laissez faire* leader – the three styles of the full-range leadership model. All 330 firefighters participating in the study preferred the transformational leadership style, regardless of rank or generational cohort. The full-range leadership model emerged following James MacGregor Burns' seminal work on leadership published in 1978. He distinguished transformational and transactional leadership, and the full-range leadership model expands on that distinction with a continuum of leadership styles ranging from *laissez-faire* to transformational styles. At one end of the range is transformational leadership style, characterized by attention to subordinates to inspire them, consider them individually, and stimulate their intellectual needs. At the opposite end of the range is the *laissez-faire* style of leadership that avoids any involvement with subordinates. Between transformational and *laissez-*

faire is the transactional style, which involves specifying tasks and monitoring performance to achieve the tasks by providing a reward system (Northouse, 2013).

Transformational leadership theory is a leader-centric theoretical construct that has been studied in several USFES Departments. Transformational leadership theory contends that the leader's positive emotional, symbolic, and influence behaviors explain the motivation of others to exceed expectations (Bass & Riggio, 2010; Mhatre & Riggio, 2014). The specific behaviors of transformational leaders include idealized influence, inspirational motivation, intellectual stimulation, and individualized consideration. Three studies collected data from multiple Departments on the impact of transformational leadership behaviors for achieving specific operational goals such as reduced turnover or improved fitness for duty (Carter, 2005; Jahnke, Haddock, & Poston, 2014; Jahnke, Poston, Jitnarin, & Haddock, 2012; Kuehl et al., 2013). Only one employed the Multi-factor Leadership Questionnaire (MLQ) that quantitatively measures the transformational leadership behaviors (Carter, 2005). It was a mixed methods study that correlated the transformational leadership behaviors with attention to retention and reduced turnover based upon interview data in a New Jersey county's fire departments. Actual retention and turnover rates were not included in the study and there was no statistical analysis to determine what correlation there may be for the transformational leadership behaviors and actual retention and turnover rates.

Carter (2007) considered transformational leadership theory in his theoretical exploration of what theories of positive leadership are best matched with the leadership requirements in the fire service and concluded that a hybrid of many theories to create a model of ideal fire service leadership is the best option available because there is not a theory that is all-around well-suited for the fire service. In another study, the researcher identified central themes in a qualitative

study of fire ground leadership, identifying leading by example and leader accountability as central themes. The study involved only one Department, the participants were the leaders themselves, and there was not a specific conceptual framework (Dixon, 2014).

These studies of transformational leadership traits and behaviors in single Departments contradict an earlier, quantitative study of fire service officers showing that leader cognitive traits and behaviors do not correlate highly with organizational leadership performance (Fiedler, 1995). A broader study of multiple departments in the USFES could resolve the contradiction. Recent research in other types of organizations does correlate performance with the cognitive abilities, reflective judgment in particular, the highest level of cognitive capacity (Eigel & Kuhnert, 2005). Eigel and Kuhnert's 2005 study involved 21 executive officers of high-performing public companies from diverse industries, including manufacturing, technology, software, banking, distribution, textiles, insurance, and finance. Data on cognitive capacity were gathered in intensive interviews and the transcripts analyzed by cognition subject-matter-experts. These experts controlled for the possibility that references to position and authority would influence the analysis by removing references to titles, or positions, or type and number of direct-reports. Higher levels of cognitive capacity correlated positively with performance measures.

Two additional leader-centric USFES studies closely examined the specific cognitive ability of decision-making or judgment for fire ground incident commanders (Gasaway, 2008; Klein et al., 2010) and showed that recognition of situations, creating recognition-primed decision-making capacity, improved incident command when rapid decisions are necessary.

Follower-Centric Theories and USFES

Two studies have identified follower-centric principles important to fire officer development (Jones, 2007; Ditch 2012). These qualitative case studies identified self-identification as a learner and reflective attention to difficult experiential learning opportunities as key themes in the development of fire officers. Another study focused on the followers concerned additional, specific fire-service issues including fire fighter behavioral health and wellness, and recommended the application of researched-based protocols from other industries in response to potentially traumatic events (Gist et al., 2011). Woods and Branlat (2011) applied the follower-centric, adaptive-leadership conceptual framework in a theoretical study to explain failures in the USFES. In the adaptive conceptual framework, one possible cause of failure is decompensation – when the system exhausts its capacity to adapt as disturbances and challenges cascade. Another possible cause of failure within the adaptive conceptual framework is when the solutions proposed work at cross-purposes – when the solution is temporary or localized, but does not serve long-term success or a broader perspective. A third way that failure is a result of adaptive issues is when leaders and/or subordinates are stuck in outdated behaviors, often a result of people over-relying on past successes (Woods & Branlat, 2011).

Dyadic and Team-Centric Theories and USFES

Contingency theories address aspects related to the leader, the followers, and the situation (Hughes, Ginnett, & Curphy, 2010; Northouse, 2013). Leaders who effectively implement training in contingency theory accurately diagnose or assess information about the followers and the situation, and react accordingly to positively affect group or organizational outcomes. An empirical, quasi-experimental study of contingency theory using two USFES Departments provides evidence that contingency leadership theory training positively influences subordinates'

job satisfaction (Beaton et al., 2001). The study included repetition of the assessments for follower job satisfaction, including sources of occupational stress, and ratings of supervisory behavior to establish a pre-training baseline and measures at three and nine months following the leadership training. Subordinates of both the group completing the leadership training and the control group that did not receive the contingency leadership training completed both assessments all three times. On-the-job injury data was also included in the study. The training intervention for the experimental group of supervisors was an eight-hour workshop in a contingency-theory based model called *Leader Match*. There was improvement in the supervisor ratings of the experimental group for the three-month post-intervention that was statistically significant. At the nine-month post-intervention assessment, there were not statistically significant changes from the baseline measures. A limitation of this study is that it is not known if training in some other theoretical model might produce even greater improvements in job satisfaction ratings, or longer-lasting improvements. It also did not compare any differing lengths for the training intervention.

A study of homeland security Councils of Government that included USFES Departments highlighted the importance of relational leadership behaviors, a specific part of contingency leadership theory important when members of a team or group are highly capable (Hickman, 2010). The study concluded that influence relationships among collaborating colleagues were critical to Council success. The influence relationships developed among people who sought changes reflective of mutual interests, without any particular person responsible and without formal working groups (Duncan et al., 2012). The qualitative study focused on the experience of people in effective regional councils responsible for coordinating many federal, state, and local programs, with an all-hazards approach that involved various disciplines related

to public safety and security, including personnel from USFES Departments. Regional councils and fire departments have distinctions, including the scope and scale of the organizations, and similarities including the complexity of the organization and their operational environments.

The organizational learning theoretical framework is a dyadic/team conceptual framework applied to a mixed methods case study of the Charleston, South Carolina Fire Department following a multiple-fatality incident (Griffin, 2013). Organizational learning theory involves intentional collective analysis and learning processes. The organization implements individual and group-based learning experiences concerning the improvement of organizational performance goals, which result in changes to organizational processes, routines, policies, and structures (Schilling & Kluge, 2009). The Charleston case study used the Dimensions of Learning Organizations questionnaire and interviews. The intentional focus on lessons from the after-action review and subsequent training completed to address the deficiencies did produce improvements in the organizational learning measures.

Two other studies that considered dyadic/team theoretical frameworks examined how USFES leaders develop. One employed system-oriented, action research methodology for creating a prototype fire officer development program through exchanges with other Departments. The program was designed to increase experiential learning opportunities (Standridge, 2012). The other applied the experiential learning model to reveal the significance of experiential learning in the USFES (Jones, 2007). Kolb (1976) developed the experiential learning model to explain how individuals make sense of concrete experiences in different ways. The model includes four distinct learning styles for creating meaning from experiences; convergent, divergent, accommodating, and assimilating. Jones used the model as the theoretical framework to qualitatively examine the significance of experiential learning for 14 fire chiefs

who had been named Fire Chief of the Year, and to identify the types of experiences that produced the greatest learning that benefited their success as fire chiefs. Both of these studies acknowledge that there were limitations for their conclusions because of the influences of organizational dynamics and organizational politics that they did not control for or include in the studies. They suggested further research into USFES organizations to more fully explicate the influences on the development of individual leadership capacities.

To summarize the academic, empirical research in USFES, it is not extensive, and as a body of work, there have been leader-centric, follower-centric, and dyadic/team-centric theoretical frameworks applied in the industry. Empirical research has primarily applied a case study approach and focused on a particular sub-context, such as fire ground leadership, or a particular issue relevant to USFES, such as health and wellness or leader development.

USFES Leadership Publications from Practitioners and Educators

Training and textbooks for officers in the USFES address management practices in-depth and develop a bureaucratic ethos among them to make decisions based upon policies and procedures, numbers and costs (Compton, 2010; CPSE credentialing proficiencies, n.d.; IFSTA, 2011; NFPA Standard 1021, 2014; Russell, 2014; Smeby, 2006). Until recently, trade literature, web sites, and conferences for USFES officers focused on the technical execution of the incident command system, and on administrative functions. Scholarly leadership theories and models have entered the trade literature from fire service practitioner-authors who have learned about them. Theories and models have entered the training and educational programs and are presented at conferences. The trade literature, training and educational programs, and conference presentations are not backed by rigorous study to test the hypotheses that these theories and models correlate to performance and effectiveness in USFES. They illuminate that for industry

practitioners there are also many different theories and models considered relevant and applicable to the USFES.

Leader-Centric Theories in the Trade

The International Association of Fire Chiefs (IAFC) convenes a training event called the New Chief Leadership Symposium, with a curriculum that emphasizes developing critical thinking capacity. A model called SPECTRUM emphasizes the leader's role in organizational change (Wallace, 2014) and is a fire service application of a model developed in the 1980s by Neugarten for making changes in healthcare for seniors. In the SPECTRUM model, leaders consider Structure, Processes and/or products, the *Environment*, Culture, Technology, Resources, the *Ultimate* vision, and *Marketing*.

Authors in the trade literature focus extensively on leadership traits, charisma, and identity related to their own personal experiences (Brunacini, 2002; Coleman et al., 2010; Guzzi, 2011; Hunter et al., 2011; King, 2011; Maloney, 2014; Mumford, 2006; Rubin, 2012). This focus is reflective of charismatic and authentic leadership concepts that emphasize leaders' abilities to inspire, to motivate, and to have high expectations of performance. Traits and behaviors emphasized include being future-oriented, visionary communications, self-sacrifice, trustworthiness, and decisiveness (Northouse, 2013).

Follower-Centric Theories in the Trade

Servant leadership appears in the USFES trade publications in both a monograph and an international training symposium (Cochran, n.d.; Russell, 2014). Russell (2014) writes his personal testimony of becoming a servant leader because he lamented that too often, on-scene command and control carried over to the other contexts of the fire service and that he saw too much emphasis on processes and not enough attention to people. As Russell explains, he believes

that servant leadership is the appropriate theory to apply in USFES Departments because of the need for leaders to provide support for followers to function at their best, a summary of servant leadership theory supported in leadership literature (Hickman, 2010). The model of leadership through service originated with Greenleaf (1977), and includes a specific set of assumptions. Leaders must be attentive to the needs and concerns of their followers, empathize with them, and nurture them in order to bring out a similar focus on others and develop followers into servant leaders themselves. In addition, the servant leadership model emphasizes that servant leaders lead in ways that serve the greater good -- of the organization, the community, and society at large (Northouse, 2013).

Crew resource management (CRM) is a follower-centric theory receiving in-depth attention in the trade literature. CRM is a framework for organizational leadership originally designed for use in the aviation industry (then referred to as *cockpit resource management*) to optimize the “person-machine interface and the acquisition of timely, appropriate information, ...interpersonal activities including leadership, effective team formation and maintenance, problem-solving, decision-making, and maintaining situation awareness” (Wiener et al., 1993, p. 4). There are fire service leaders who advance it as transferable to the fire service (LeSage et al., 2010).

Adaptive leadership is a follower-centric theory advanced as applicable to USFES. Since 1985, the National Fire Academy’s Executive Fire Officer Program (EFOP) provides senior fire service leaders with a series of four courses in leadership and critical thinking, using the adaptive leadership theoretical framework. Leaders and educators in the National Fire Academy intentionally chose this theoretical framework with awareness of the pace of environmental change for fire and emergency services and the need for adaptive leadership that would focus on

learning and change throughout organizations. The adaptive leadership framework also focuses on maintaining values and positive organizational culture in the face of rapid change. The recently added National Fire Academy Managing Officer Program introduces the adaptive leadership theoretical framework. In this framework, leadership exists in the context of problems and challenges because that is when it is necessary for organizations to change their ways rather than continue to operate according to present practices if they are to sustain the organization (Heifetz, 2010). It requires the development of new approaches and it demands learning and changes in thinking and operating.

Team and Dyadic Theories in the Trade

Some of the opinions expressed in the industry reflect an understanding of complexity and systems theory. The oft-repeated phrase that the USFES has “a proud heritage of over 200 years, unimpeded by progress” reflects frustration with the inertia of systems (Thiel, 2013). More specifically, there are calls to address the systemic reasons for the failure to implement recent evidence-based recommendations, such as the NIST studies showing that external fire attack is an effective and safer strategy for fighting structure fires than internal attack (Buckman, 2014). There is a great deal of complexity to consider throughout the system in order to implement technical, operational change on the fire ground: establishing new staging procedures, acquiring new equipment and materials, pre-service education and training, and in-service education and training. There is legitimate concern that such systemic change could lead to deterioration of scene cohesion, the primary concern of first responders (Smeby, 2006).

While it is important that USFES professionals are intentionally analyzing and seeking to apply a variety of leadership theories and models, providing evidence of the broad interest in organizational leadership, there is an assumption of transferability that has not been examined

(Canyon & Morrison, 2010). Complex contexts, especially dangerous ones are “dimensional and findings from one typology cannot necessarily be generalized to another” (Hannah et al., 2009, p. 52). There is recent empirical research in non-fire-related, but similarly complex, intense, and dangerous public safety disciplines such as policing, emergency medicine, and the military that should be reviewed in order to begin examining the assumption of transferability.

Leadership Studies in Related Industries

Studies in organizations with missions that have similarities with USFES Departments, such as wildfire agencies, law enforcement agencies, and the military, have begun to acknowledge the complexity of these organizations that carry out dangerous and difficult missions. Recent studies moved beyond the mechanics of command and control and referred to the social construction of leadership processes when such mechanics do not exist and when a leader does not *become* a leader and followers do not *do* anything without the existence of relationships among them (Baran & Scott, 2010; Uhl-Bien & Ospina, 2012). They also acknowledged the importance of leadership in the supporting areas of the organizations and expanded the focus beyond incident command.

A study in wildfire management applied systems theory as the framework to examine mission alignment across federal and state agencies and local entities involved in wildfire management (Fleming, McCartha, & Steelman, 2015). Systems theory focuses on the interrelationships among parts of a greater whole and pays attention to the effects on the whole of changes in the interrelationships (Senge, 1994). One finding was that misalignment existed across the levels of government to a greater extent than within the same governmental level across different types of agencies. Related theoretical frameworks, referred to as integrative, networked, or collaborative leadership, have been used in studies of policing and in public

service agencies (Crosby & Bryson, 2014; Van Wart, 2013). Another framework applied in related, emergency management operations is diplomatic leadership. The diplomatic leadership model considers complexity arising from interactions of political, economic, environmental, and social issues that involve multiple and competing interests. Successful solutions for navigating the complexity and competing interests involve sophisticated multi-stakeholder initiatives and multi-sector governance (Mirvis, Hurley, & MacArthur, 2014).

Brunner (1997, as cited in Van Wart, 2003) called attention to one of the difficulties in leadership research in the public sector, and in extending or generalizing leadership research beyond small subsets, because of “contextual complexity” (Van Wart, 2003, p. 215). There are theorists researching the importance of context, extreme contexts, and shifting or multiple contexts for organizations (Antonakis, Avolio, & Sivasubramaniam, 2003; Comfort & Resodihardjo, 2013; Hannah & Parry, 2014; Hunter, Bedell-Avers, Mumford, 2009; Uhl-Bien & Marion, 2009; Waugh & Streib, 2006). These studies may provide useful comparisons to perspectives on USFES Department leadership because practitioners already acknowledge the different contexts and complexity (Russell, 2014, Compton, 2010). The Center for Public Safety Excellence, which provides an accreditation process involving benchmarking, acknowledged the multiple contexts important in USFES by establishing and examining several categories of operations in the accreditation process. The NFPA standards also make clear the multiplicity of contexts for USFES. These organizations include all of the following contexts in assessing USFES Departments or establishing standards for them:

- Trauma
- High Reliability
- Critical Action

- Training
- Administrative
- HR/union/associations
- Assessment and Planning
- Fiscal/Budgetary
- Governance, External Relationships
- Mergers e.g., FD/EMS (NFPA, CPSE, 2014)

Yet scholarly, theoretical works that “reflect sophisticated multifunctional, multilevel, and multi-situational models that were in evidence in the mainstream [leadership literature] by the 1990s are largely lacking in either monographs or the journal literature in the public sector” (Trottier et al., 2008; Van Wart, 2003, p. 223). In the public sector, including the USFES, there has been a debate about how much leadership mattered because of the bureaucratic structure and constraints on the influence of leadership (Moynihan, Pandey, & Wright, 2012). Hannah and Parry (2014) stated clearly that the “impact of organization context on leadership is an under-researched area” (p. 613). It is not possible to determine all the skills, knowledge, abilities and behaviors that members of an organization will need to meet the challenges they will face without an understanding of the context. Without an understanding of the context, preparation, learning, and development are ad hoc and unpredictable, reducing the likelihood of effectiveness and success at any endeavor (Day, Harrison, & Halpin, 2012). This is the situation that has existed in the public sector – little understanding of the context or of leadership within the contexts of the public sector. Examples of rigorous public-sector research demonstrate that leadership does matter, but the studies involve city or county government through the National

Administrative Studies Project and have not yet taken place across USFES Departments (Moynihan et al., 2012).

Public Administrative Leadership

Of all the leadership theories advanced as best by USFES practitioners, and among those studied in the USFES, notably absent are the theories of public administrative leadership. This study of perspectives on USFES department-wide leadership contributes to USFES industry knowledge, the field of public administration, and to organizational leadership research through an examination of whether or not the conceptual framework of public administrative leadership has relevance for USFES Departments.

Historical Background for Public Administrative Leadership Studies

Few studies of leadership in the public sector existed before the 21st century (Van Wart, 2003). This may be because of the perception that the public sector does not need leadership, but rather requires efficiently controlled management of processes and people for maximum productivity, underpinned by assumptions about management in the private sector that people are rational, competitive, and above all, self-interested, clear in the work of Frederick Winslow Taylor in the early 20th century (Terry, 1998). The bureaucratic nature of organizational structure and activities in the public sector created an assumption that leadership is unnecessary and therefore absent. Leadership impacts the public sector *from the outside*. For example, policy formulation and advocacy take place in the political realm, not *within* public agencies (Terry, 1998). While continuing to embrace managerial ideology and the assumption that human nature is selfish, in the 1980s public administration researchers and practitioners brought attention to the public sector with a new set of concepts. There were advocates for a systems-level quantitative and analytic approach for forecasting, aptly called *quantitative/analytic management*, and

advocates for political participation in the development of policy and strategy, referred to as *political management*, heralding a shift to conceptualizing public administration as having leadership roles (Moore, 1983).

In the 1990s, the concepts brought into public administration leadership theory continued to have the same managerial ideological foundation that people are self-interested and management is therefore important and good, but required freedom from excessive policies and regulations to be cost-effective and efficient, referred to as *liberation management*. Another model for managing public sector organizations is based upon the assumption that they should be run like businesses because the public sector is not so different from the private sector. Management of the public sector could be improved with the application of principles and practices of the private sector, particularly competition within and among agencies, to bring capitalist, market principles into the public sector to enhance efficiency – an approach called *market-driven management*.

Both of these 1990s frameworks for studying the public sector are in *the new public management* movement that called for dramatic reforms in government to enhance accountability, economy, and effectiveness (Terry, 1998). The new public management movement represented “a radical shift regarding how these important [managerial] values are achieved, demanding that managers abandon . . . the ‘engineering model’ of management” (p. 196). Instead of instruments of public policy and instruments of technique, “managers are (and should be) self-interested, opportunistic innovators and risk-takers who exploit information and situations to produce radical change” (p. 197). These new public management assumptions cultivated an entrepreneurial attitude for public administration leadership.

Since the early 2000s, the study of public administrative leadership has grown dramatically (Van Wart, 2013). The studies support public administrative leadership as a distinct subset of organizational leadership studies (Trottier et al., 2008; Van der Wal, de Graaf & Lasthuizen, 2008; Van Wart, 2013). Van der Wal, de Graaf, and Lasthuizen (2008) conducted a quantitative study that illuminated the differences in organizational values for the private and public sectors. The study showed that there are values that the private sector and the public share – accountability, expertise, reliability, effectiveness, and efficiency, for example – and also values that each is unique in giving primacy. For the private sector, profitability, innovativeness, and honesty were highly-placed values; and for the public sector, lawfulness, incorruptibility (to act without prejudice and bias toward private interests), and impartiality (to act without prejudice or bias toward specific group interests) were primary values. Scholars have been interested in both the similarities and the differences because of the importance of strategy in both sectors, and because of the increasing levels of inter-organizational coordination necessary. There are correspondingly rising levels of collaboration and shared responsibilities among governmental agencies and not-for-profit organizations involved in public issues (Crosby & Bryson, 2014; Joyce, 2012).

Research in Collaborative and Integrated Public Administrative Leadership

The most recent studies in public administrative leadership highlight the complexity and interconnectedness of the public sector and applied relational leadership theory and complexity leadership theoretical principles. When researchers applied these theories to the public sector, they gave attention to the systems within which an agency exists and to the processes that emerge in the systems as agents interact and adapt in complex, dynamic environments (Brooke & Grint; McGuire, Agranoff, & Silvia, 2011; Uhl-Bien & Marion, 2009). The complexities

studied in public administrative leadership models exist in multiple domains. There is complexity related to the level or jurisdiction -- federal, state, local, and inter- and intra-agency. There is complexity related to the necessity to execute multiple missions, such as during a natural disaster response that can involve the missions of evacuation, rescue, safety, and recovery. Another kind of complexity is that of dimensions, or realms of concern for a public administrative leader who must operate in the legal, political, fiscal, operational, and strategic realms. Complexity also arises in the problems and challenges that the public sector addresses because they are constantly evolving or changing. Change can occur as a problem grows or as the information available about the problem changes, which can vary in amount, source, and relevance. These kinds of complexity exist in the environments of USFES Departments.

The conceptual framework of *collaborative public management* specifically considers “the process of facilitating and operating in multi-organizational arrangements for solving problems that cannot be achieved, or achieved easily, by single organizations” (McGuire et al., 2011, p. 1). Leaders in these contexts develop networks, communicate effectively, build capacity, and bridge boundaries. It is also true of the similar conceptual framework of *public integrative leadership*, described as the “work of integrating people, resources, and organizations across various boundaries to tackle complex problems and achieve the common good” (Crosby & Bryson, 2014, p. 87). British researchers refer to *public leadership theory*, concerned with the sharing of learning across institutions, networking, and understanding, creating and demonstrating public value. Across each of these specialized conceptual frameworks there are six considerations, common to all of these types of public leadership:

- collectivity, horizontally across organizations or institutions and vertically within them
- multiple contexts – community, political, organizational, and individual

- uncertainty and complex, ‘wicked’ problems that require creative, adaptive capacities
- the advantages of collaboration over traditional leadership approaches
- the long-term and systems perspective
- the development of trust and legitimacy (Brookes & Grint, 2011).

Some public functions have a long history of multi-organizational arrangements and integrating people, resources, and organizations, such as safeguarding public health, education, land management, and emergency management, but USFES Departments have a shorter history of operating in such complexity. It is only in the second half of the 20th century that the missions of USFES Departments became more complex, adding emergency medical services, for example. The context and jurisdictions also became more complex after 1973 with the establishment of the United States Fire Administration and a new, federal level of regulation, training, and research. The earlier history of USFES Departments was characterized by a local, single organization and a decentralized approach to establishing operational and training standards and to executing the mission. Department leaders who include the principles of collaborative and integrative administrative leadership theory, adaptation to the complex systems and contexts may have greater Department effectiveness. Department performance may be enhanced from faster and more effective adoption of the latest safety and fire technology and operating standards, and access to federal and state funds for training and equipment, for example. Participation and performance in complex incidents requiring interoperability through the National Incident Management System may also be enhanced by attention to the processes emphasized in collaborative and integrative administrative leadership theory.

Characteristics of Public Administrative Leaders

There is still a strong tendency to attribute leadership to individuals in these conceptual frameworks of collaborative public management and integrative public leadership, even though the roots of these frameworks are in theoretical construct of complexity leadership theory, which focuses on leadership emerging collectively through interactions (Crosby & Bryson, 2014; Esteve, Boyne, Sierra, & Ysa, 2013; Hannah & Perry, 2014; Hazy & Uhl-Bien, 2014; McGuire & Silvia, 2010). Van Wart (2013) specifically called for the development of competency profiles for public sector leaders. Further, he said that such studies require grounded designs rather than using existing surveys and assessments, to avoid *a priori* assumptions of their applicability. This also assisted the development of administrative leadership theory as a subfield of leadership studies by identifying the differences and similarities between the public and private sectors, and within different parts of the very diverse public sector. A list of the key characteristics of public services leaders emphasized in recent studies and theoretical papers includes the following:

- Knowing the situation (including foresight)
- Knowing how to influence people
- Learning and personal development
- Challenging people to change
- Understanding and managing stakeholder expectations
- Building trust among stakeholders
- Maintaining and growing personal stamina, including resilience, self-confidence and self-efficacy
- Collaborating and building alliances and coalitions

- Assessing and evaluating (Esteve et al., 2012; Joyce, 2012; McGuire et al., 2011; Van Wart, 2004).

Modeling Public Administrative Leadership

In a detailed theoretical analysis, Van Wart (2004) proposed a *leadership action cycle* as a comprehensive model for the public sector. As shown in Figure 4, the leadership action cycle begins with an environmental assessment and self-assessment to set goals, then involves the application of traits, skills, and style in three areas of action related to tasks, people, or the organization, and concludes with an evaluation of personal and organizational effectiveness.

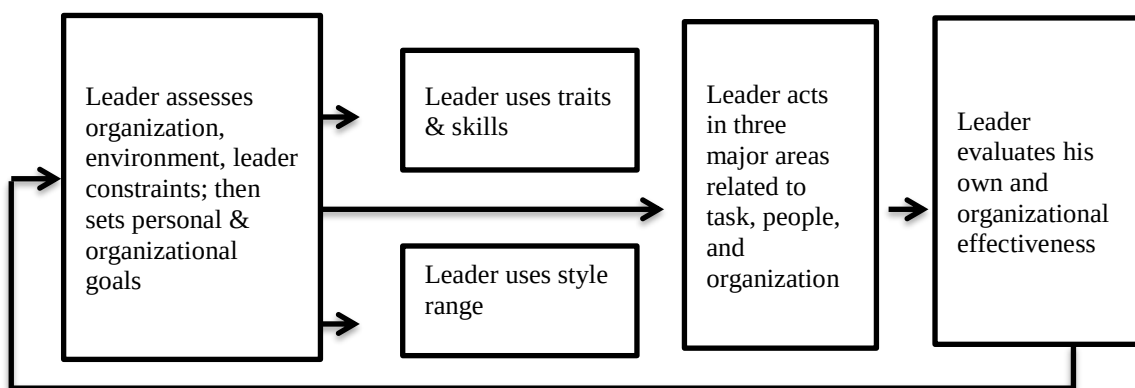


Figure 4. The public administrative leadership action cycle (Van Wart, 2004).

Within this model, there are five areas that are emphasized in public administrative leadership. First, leaders pay attention to the organization and coordination of work and frequently re-examine bureaucratic processes with vigilance to avoid over-formalization and excessive bureaucracy that can result in ineffectiveness. Second, leaders coordinate activities externally to ensure the needs and desires of stakeholders are met. Third, leaders focus on expanding or protecting resources and support and overcoming constraints peculiar to the public sector, such as legislative mandates or politically-motivated budget cuts. Fourth, leaders cultivate and value traits like fairness, honesty and integrity that build trust, and motivate followers

through service and seeking excellence. Finally, leaders are aware of the exercise of power – using mainly legitimate (position) power and expert power, but also referent power that stems from likeability and trust.

Conclusion

It is clear that USFES professionals have had interest in a variety of theories and approaches; there have been calls to apply theories or studies within every kind of positive leadership theory category: leader-centric, follower-centric, and dyadic/team-centric. Leader-centric theories imposed *a priori* in studies of single Departments include specific trait and behavior conceptual frameworks such as Covey's Seven Habits of Highly Successful People, cognitive resource theories like recognition-primed decision-making capacity, and Kolb's experiential learning model. Follower-centric theories imposed *a priori* on limited numbers of USFES officers from a single or small number of USFES Departments include identity theory to hypothesize the significance of self-identification as a learner, and the adaptive leadership conceptual framework. Practitioners have advocated for the relevance of two other follower-centric leadership theories: servant leadership theory, and the crew resource management model, but there has not been research applying them in the USFES. Dyadic- and team-centric theories used in several USFES case studies include contingency theories, the relational leadership model, and the organizational learning conceptual framework.

There are two theoretical frameworks that would seem to be relevant to the USFES that have not been applied to the USFES. Leadership researchers have recently acknowledged the importance of organizational context in understanding and studying leadership to develop theory, and there has been some research and theory developed specifically for unusual and extreme contexts such as those faced in related industries like the military and law enforcement. The

complex organizational environment of USFES Departments includes extreme contexts, and the latest research could be applicable in the USFES. Public administrative leadership theory is another that has been studied in other public-sector organizations, but not in USFES Departments. Leaders of public-sector organizations, including USFES Departments, must navigate some unique conditions and contexts such as the external political processes that can impact the organization in profound ways. Processes in public administrative leadership theory and the characteristics of effective public administrative leaders, as well as the priority areas of concern for them, may have relevance for the USFES.

Consensus regarding leadership in the USFES is absent among professionals, federal and state administrative organizations, associations, accreditors, and educators. This is in no small part because the studies applying so many different theories have been case studies of single departments or regions with significant limitations regarding generalizability. It is also possible that the theories imposed *a priori* are insufficient in some way, or that there may be a conceptual framework, such as public administrative leadership, that has not been applied in any empirical studies of USFES.

The researchers studying USFES leadership applied a variety of conceptual frameworks, including Kouzes and Posner's exemplary practices, servant leadership, transformational leadership, adaptive leadership, crew resource management, and authentic leadership, usually for a single context such as the scene of an incident, or for a single Department. The assumption of transferability has not been evaluated in these studies. Research based upon the theoretical constructs *a priori* minimize the opportunity for unexpected, new findings unique to a context different from origination contexts of those theoretical constructs (Rao & Woolcock, 2003; Van Wart, 2013). The instruments and assessments for these theoretical constructs were not

developed using insights from the environment of USFES, and they may not be meaningful, understood, or well-suited to studying leadership in the context of USFES Departments.

In the absence of generalizable studies and studies that corroborate a particular conceptual framework as contributing to USFES Department-wide leadership effectiveness, a study that does not assume a particular theory or framework in the industry could illuminate an applicable framework that would be useful to the industry and to further research. Improved understanding of effective USFES Department-wide leadership could contribute to more Department effectiveness within and across Departments, including reduced firefighter injuries and fatalities, civilian injuries and fatalities, and reducing property loss due to fires. An exploratory study of the perspectives on effective department-wide leadership from within USFES Departments could also provide important knowledge for succession planning, for assessment and development of department-wide leadership in educational programs such as the EFOP and programs of professional associations, and for a solid foundation to launch further studies. Chapter 3 describes the methodology and process for the exploratory study.

Chapter 3: Research Design and Method

This chapter describes in-depth the design and methodology for exploring the perceptions of USFES officers regarding Department-wide leadership. Q methodology, a self-referent approach that integrates qualitative and quantitative methods fully explored USFES Department officers' perceptions of what characterizes effective Department-wide leadership. This chapter explains the rationale for using Q methodology, describes the research design, and details the methodology, data collection and analysis procedures. Specifically, the study design sought to answer the following research questions:

1. What are the perspectives of USFES Department officers regarding effective *Department-wide* organizational leadership?
2. How do these perspectives differ and how are they similar?
3. How do these perspectives correspond to existing leadership constructs, concepts, theories and models?

The chapter concludes with a discussion of informed consent processes, potential ethical issues and how they were minimized.

Research Design

The study used an exploratory approach and social constructivist perspective using Q methodology. It focused on participants' personal, subjective interpretations and the meanings they have developed from their experiences with Department-wide leadership in USFES Departments. Q methodology identifies the varied or common perspectives of individuals within a group and is particularly suited to studying complex organizations, issues, and phenomena. The analysis of the individuals' points of view provided insight regarding the specific characteristics, knowledge, skills, abilities or competencies, behaviors, and processes that they perceive as

contributing to effective Department-wide leadership. The methodology can uncover shared understandings of elements of organizational leadership and social realities in the American fire service that are considered integral and exist in the environment that all fire service personnel encounter. This last point contains a constructionist logic, and indeed, Q methodology can be focused in either direction (Watts & Stenner, 2012).

Physicist and psychologist William Stephenson (1902–1989) originated Q methodology in 1935. He changed conventional factor analysis by shifting the focus of study from externally imposed metrics to *self-reference*; individuals perform the measuring by capturing their own perspective rather than being measured according to an instrument or assessment. In Q methodology, researchers explore how people give meaning to a set of stimuli presented on a topic or phenomenon. The participants' assessments of the stimuli are their personal measures of the phenomenon. Specifically, participants arrange the set of stimuli, and in so doing they express their own subjective viewpoints, allowing researchers to “discern the lay of the land in terms of the way stakeholders perceive the issue [they] are confronting” (Donner, 2001, p. 26). Q methodology provides the process for identifying detailed, subjective viewpoints on a matter and then uses statistical techniques to analyze these perceptions because, as Stephenson asserted, self-referent expressions of meaning – the inner experience of people – is as much a matter for study as is behavior (McKeown & Thomas, 2013).

Q methodology provides a means to both reveal and rigorously analyze multiple subjective viewpoints. The focus on self-referential meaning and interpretation means that Q methodology shares much with qualitative research. It is appropriate for the interpretive framework of social constructivism in that it is possible to learn and gain understanding of the predominant viewpoints relative to a particular social context (Watts & Stenner, 2012). It also

employs the type of statistical analyses typically used in quantitative studies and it is thus a mixed methodology approach (Brown, 2008, as cited in Ramlo & Newman, 2011; Cross, 2005). A key difference in the application of the statistical techniques is that they are not used to test any hypotheses based on *a priori* theories. As in other qualitative methodologies such as phenomenology, the researcher must intentionally set aside or suspend existing knowledge and experience of the topic of study to understand it more fully and deeply (Creswell, 2013).

Methodological Fit

The rationale for using Q methodology involves its unique, mixed methodological characteristics. The methodology allows for the exploration and discovery of shared, common perceptions from complex scenarios differently than quantitative confirmatory factor analysis does (Watts & Stenner, 2012). Q methodology can use centroid factor analysis that presents *all* possible correlative solutions for examination, rather than presenting only the best mathematical correlations. This is important when the subject matter is not heterogeneous – when it is possible that there are different social constructions and conflicting opinions or multiple perspectives on the subject.

Researchers have used Q methodology to study perceptions of complex political and environmental issues (Ellis, Barry, & Robinson, 2007; Focht, 2002). They have also employed the methodology to study leadership in several contexts by exploring perspectives on leadership in education (Brown, 2013; Militello & Janson, 2014; Taylor, 2012; Wagner, 2011; Warren, 2012), medicine (Beaver, 2001), business (Bunch, 2012; Phelan, 2012; Woods, 2011) and the military (Stewart, 2000). The review of literature on the topic of USFES leadership indicates that varied perspectives exist, possibly resulting from varied social constructions and a complex context, and the absence of any education and training industry standards related to

organizational leadership. As presented in Chapter 2, consensus or a single, normative definition of what constitutes effective leadership in USFES Departments does not seem to exist, nor has a grounded study been conducted that could determine what characteristics, behaviors, competencies, and processes characterize effective Department-wide leadership in USFES Departments. Q methodology is ideally suited for the purpose of such naturalistic inquiry for research questions where there is the possibility of multiple perspectives, and for exploring and describing the different perspectives that may be confounding the development of consensus. Q studies reveal key viewpoints and provide details about the viewpoints to shed light on the reasons that conflicting understandings of the phenomenon exist (Cross, 2005; Watts & Stenner, 2012).

Establishing the existence of specific viewpoints and a more detailed understanding of them through a Q study can be significant. Conventional measures and assessments in quantitative studies impose constructs on participants or contexts *a priori*. This is problematic when researchers use a measuring instrument on a sample from a population where the pre-defined construct or context is not part of the population's experiences, especially if there is no sound basis for testing and measuring something that may not exist (Watts & Stenner, 2012). Empirical research designed to explore, discover, and more completely understand complex phenomena must be the focus first, and Q methodology is well suited to this purpose.

Regarding administrative leadership, the conceptual framework in this study, Van Wart (2013) specifically called for grounded research that does not impose biases based upon any interest in a particular theory. He recommended research on the people at any and all levels and the processes and networks that manage, guide, and lead government and non-profit agencies, with a focus on civil service and appointed leaders rather than political, elected officials, and on

the development and implementation of policy rather than policy advocacy. A study grounded in USFES Departments will contribute to the body of knowledge regarding this subset of organizations and leaders that Van Wart recommended for further grounded study.

The need for studies of administrative leadership relate to a number of issues that are different for this type of organization than for others, including measuring performance, motivation and job satisfaction of employees. The nature of accountability is also different for government and non-profit organizations since public agencies are accountable to elected officials, taxpayers, and other citizens for decision outcomes related to the mission and objectives. Different missions and contexts exist across the broad organizational category of government agencies, presenting opportunities for discovering distinctions and similarities about leadership across multiple contexts, rather than assuming a single theory applies across these contexts (Crosby & Bryson, 2010; Van Wart, 2013). Van Wart (2013) issued an explicit call to study the different sectors of government and non-profit agencies, particularly regarding competencies and profiles of leadership, without using leadership surveys and questionnaires. Q methodology is an alternative to such instruments.

In USFES Departments, the self-referent viewpoints of the officers immersed in the context could clarify whether conventional measures, tests, and instruments are appropriate. The self-referent viewpoints can also undermine or reinforce established preconceptions or approaches, suggest a new approach or theory, or question professional practices. Any of these possibilities would be important to USFES as well as add to the body of knowledge related to leadership in complex organizations and administrative leadership (Crosby & Bryson, 2010; Van Wart, 2013; Watts & Stenner, 2012).

Bracketing

An important initial step to ensure that I completely present the self-referent viewpoints regarding effective Department-wide leadership in USFES municipal Departments is a reflective process, called bracketing in qualitative methodologies. The bracketing process involves setting aside personal beliefs, feelings, and perceptions. Bracketing facilitates openness to the lived experiences that individuals express during data collection. It is important throughout a qualitative study for a full exploration of a topic or phenomenon, and deeper understanding of it (Creswell, 2013; Smith, Flowers, & Larkin, 2009). In order to bracket, I first acknowledged my personal experiences and knowledge of the topic and made myself the focus of reflection to identify and isolate my own beliefs and feelings. I also completed the Q sorting process as part of the reflective process. Bracketing allowed me to subsequently engage with each participant openly for the interviews, and review the Q-sorts and interview transcripts without judgment.

I acknowledged experiences working with USFES professionals in educational settings. Affiliation with the International Public Safety Leadership & Ethics Institute instilled in me a sense of pride and deep appreciation for the professionals who put their lives at risk for public safety. It has also provided opportunities to learn about the organizational complexity in USFES Departments and to raise my awareness that the organizations may fulfill their difficult, dangerous mission mired in internal or external controversies and conflicts, or with well-coordinated relationships throughout the communities they serve. Awareness of the organizational complexity has brought to light specific, practical problems in the industry, described in chapter one, that have surprised, frustrated, concerned, or occasionally alarmed me. Qualitative researchers must set aside personal feelings such as the pride, appreciation, surprise, frustration, concern, and alarm I have felt, to listen to the individuals and read opinions from

individuals who have lived experiences with USFES Department leadership, giving all of their statements equal value (Creswell, 2013).

Overview of the Q Methodology Process

Q methodology involves five phases. The first, second, and third phases are the necessary preparatory phases prior to the data collection and analysis in the fourth and fifth phases of the study. First, Q researcher develop a collection of statements about the topic, called the *concourse*, as completely as possible to represent all the possible perspective components considered relevant to the subject. Q researchers use purposive, snowball sampling to interview individuals with knowledge and experience about leadership in USFES departments. In this study, I also searched trade literature for statements about the phenomenon of USFES department-wide leadership. Interviews and searches of trade literature continued until the concourse is referred to as *saturated*. Second, Q researchers sample the concourse to obtain a comprehensive collection of statements that fully captures the meanings inherent in the concourse, called the *Q-sample*. Third is the selection of participants for whom there is evidence through the initial concourse development that they collectively hold all of the varying perspectives on leadership, called the *P-sample*.

Data collection takes place in the fourth phase. Each participant sorts cards with the Q-sample statements printed on them according to a condition of instruction related to the personal saliency they perceive for each statement, such as “most like my point of view” and “most unlike my point of view” with respect to their view of organizational leadership in USFES departments. The final, sorted arrangement of the statements is called the Q-sort. Each Q-sort is recorded as the empirical representation of each participant’s perspective on the phenomenon. As part of the fourth phase, researchers conduct post-sort interviews to learn more about the reasons for the

placements in the Q-sort and the self-referent meanings of the statements the participants considered most salient.

In the fifth and final phase, researchers conduct factor analysis of the sorts across respondents, including both mathematical and manual rotation, to identify shared perspectives on leadership. Researchers interpret and validate the interpretations of these factors with those sorters whose sorts loaded most highly and purely on the factors. The objective of the interpretation and validation is to label and describe the perspectives on Department-wide leadership, and reveal the similarities and differences among them. Judgmental rotation allows researchers to abductively explore the nature of the differences among perspectives. For example, rotation can reveal the nature of conflict and consensus among perspectives held by various sorters. Figure 5 displays a summary of the sequence and flow of all phases in Q methodology research.

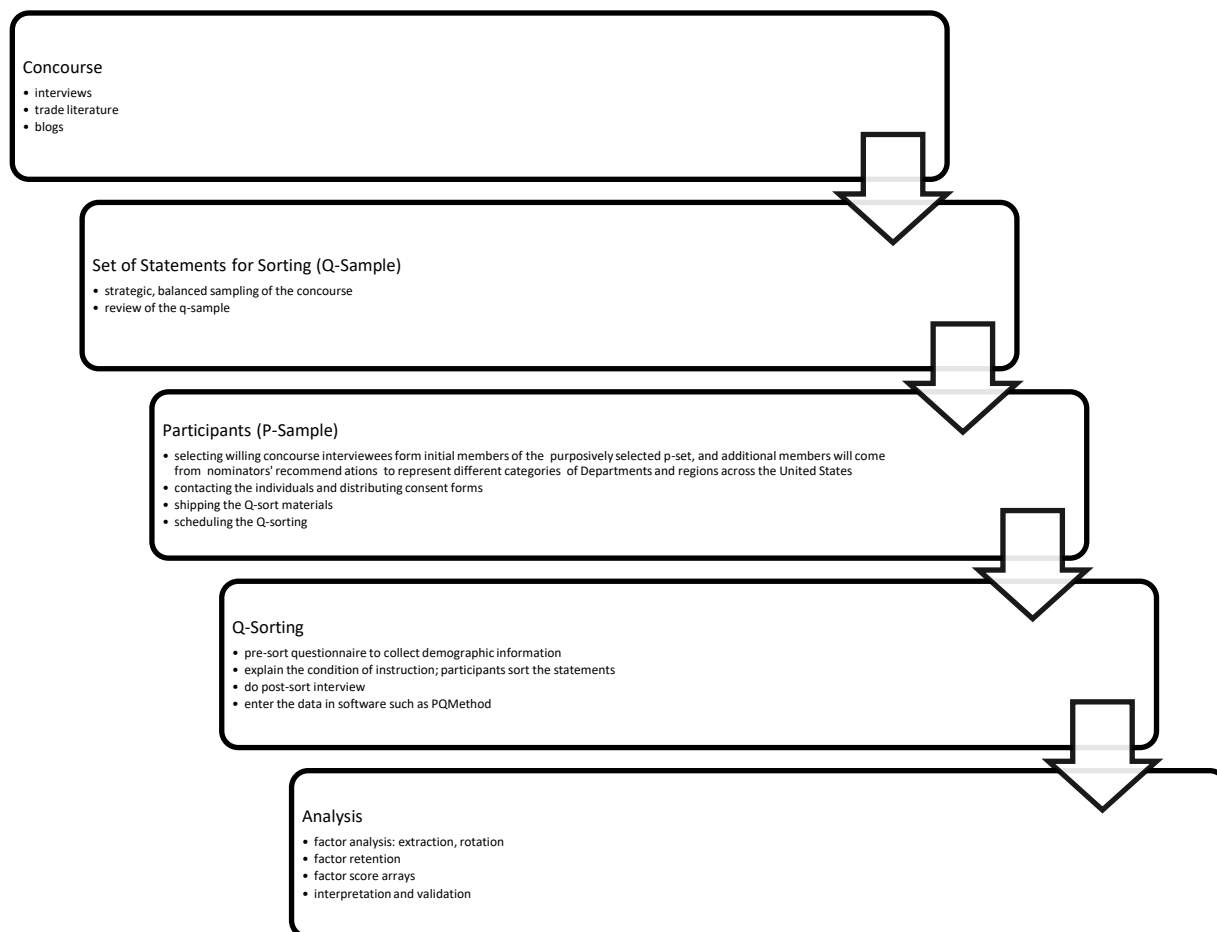


Figure 5. Sequential flow of Q methodology research.

The following sections on each of the phases of Q methodology explain the methodology in-depth, beginning with the development of the concourse, continuing with the Q-sample selection, identifying the P-sample, how the participants complete and record their Q-sorts, and the specific steps of data analysis that I completed in this study.

Q phase 1 in Detail – the Concourse

In Q methodology, the population and unit of analysis is different than in studies using other quantitative or qualitative methods. The sample is not of a population of people with individuals or groups as the units of analysis, but of a population of items, typically statements (or other objects such as pictures), related to a particular topic or phenomenon. The collection of statements is called a *concourse*. Participants complete a sorting process of a sample of

statements from the concourse. The arrangement of the statements that each participant completes makes operant the subjectivity of the sorter. It is important that the concourse is *saturated*, meaning that it is thorough and includes a complete collection of statements related to the topic of the study. Saturation is important to the credibility and validity of Q studies (McKeown & Thomas, 2013; Watts & Stenner, 2012). To ensure saturation, I included statements about USFES Department-wide leadership from written sources, such as trade publications and blogs. Statements from academic studies of leadership in USFES or related disciplines and industries (law enforcement, military, public administration) were not sources for the development of the concourse because it is not possible to know if the authors and participants of such studies are or have been USFES personnel, and it is important for this study that the statements were self-referent within the USFES (Focht, 2002; McKeown & Thomas, 2013; Shinebourne, 2009; Van Exel & de Graaf, 2005; Webler, Danielson, & Tuler, 2009). I conducted interviews from current or former USFES Department officers about effective Department-wide leadership to completely saturate the concourse. The interview protocol is detailed below.

Purposive and snowball sampling techniques were used to identify individuals who are highly capable of contributing to the concourse regarding leadership in USFES Departments because of current or former positions they have held (Creswell, 2013; Shinebourne, 2009; Watts & Stenner, 2012). The initial, purposively-selected participants for the concourse development are members of the United States Fire Administration National Fire Academy Board of Visitors (BOV) members and members of the International Public Safety Leadership and Ethics Institute (IPSLEI) Board of Directors and Curriculum Committee. They represent a diverse, perspectival cross-section of fire and emergency services interests. They are also a convenience sample in

that their contact information was available to me. Moreover, in their roles as advocates for the USFES, they expect to receive questions and requests for information, they have interest in the advancement of the professions in fire and emergency services, and they are likely to agree to participate. Authority for the BOV comes from Public Law 93-498, the Federal Fire Prevention and Control Act of 1974, specifying that the Secretary of the Interior shall appoint the Board of Visitors. The role of the BOV is to review NFA programs to ensure they fulfill the basic mission of the United States Fire Administration, to ensure that NFA facilities are adequate for their assigned purposes, to ensure appropriate funding levels for NFA programs, and to provide advice and recommendations to the USFA (USFA, 2015). IPSLEI is a not-for-profit educational organization with a mission to develop excellent leadership in service to the public safety (IPSLEI.org, n.d.).

The initial interviewees included National Fire Academy BOV members and IPSLEI Directors or Curriculum Committee members who are or were officers and managers in a position to form a perception of overall Department leadership (generally a lieutenant, captain, battalion chief, assistant chief, deputy chief, or administrative leader or manager, or chief). The initial interviewees were Mr. Robert Cumberland, member of the BOV (now retired) and the Director for the state of Maryland Cumberland Valley Volunteer Firemen's Association, Mr. Kevin Brame, IPSLEI Executive Director and former Assistant Chief for the North Las Vegas Fire Department, Mr. Randal Novak, Vice Chair of the BOV, Ms. Laurie Reinhardt, with the IPSLEI curriculum team and the former Human Resources administrator for the Orange County, California Fire Department, and Mr. Joseph Schum, also with the IPSLEI curriculum team and a Captain in the Reno, Nevada Fire Department. The Acting Superintendent of the National Fire Academy, Dr. Kirby Kiefer, provided permission to contact the members of the Board of

Visitors to participate in interviews. I am a member of the IPSLEI Board of Directors and had the contact information for those IPSLEI Board of Directors members who meet the criteria outlined. Appendix B contains the letter to Dr. Kiefer and his response. Appendix C contains the letters for the interviewees informing them about the study and the interviews for data collection. Participation was voluntary, and responses recorded, with permission, for transcription.

For the development of the concourse of statements related to effective Department-wide leadership in USFES municipal Departments, I conducted semi-structured interviews. I included all statements from the interviewees related to the phenomenon of department-wide leadership in the concourse. The initial interviews concluded with a question to expand the participation in the concourse development with a snowballing technique. This technique helped ensure that additional interviewees provided representation of the full variety of views on USFES department-wide leadership for inclusion in the concourse development process. The initial, purposive, convenience sample did not include all possible perspectives, and it was important that the concourse be saturated, or complete, with statements that come from all perspectives on USFES department-wide leadership. For this reason, the interviews concluded with a question, “Can you suggest others for an interview in this study who hold different views on effective leadership in USFES Departments?” If the interviewee was able to share contact information for additional people, I sent the e-mail participation letter (Appendix C) to the person or people suggested.

Forty to sixty sources for concourse development are common, although this is a suggestion only. Saturation may be attained with a smaller number of sources (Shinebourne, 2009). The study included 17 interviews providing statements for the concourse, and the additional sources for the concourse development were trade literature and blogs on the subject

of Department leadership from current or former fire officers. As explained above, only trade literature and blogs were used because the concourse should be self-referent from people within the industry.

When the interviews and review of the trade literature and blogs result in recurring, repeated statements, it is an indication that the concourse is saturated and further interviews or reviews of trade literature would not yield additional information about USFES Department leadership for inclusion in the concourse (Creswell, 2013; Watts & Stenner, 2012).

Interview questions to collect statements for the concourse were open-ended to facilitate the generation of self-referent, contextual thoughts on what is important for effective Department-wide leadership. The interviews began with a very brief preface to ensure that interviewees are considering *Department-wide* leadership:

Fire and emergency services Departments are complex organizations with many contexts that impact the Department's effectiveness. Examples of the different contexts include on-the-scene of an incident, training, stations, administrative offices, labor/management negotiations, government relations, community relations, prevention and education, and so on. Please think about leadership for the entire Department and all of these contexts, and then tell me your thoughts on what is important to Department-wide leadership in a fire and emergency services Department.

Following the preface, I asked interviewees the following questions:

1. What would you notice in a Department as visible evidence of effective USFES Department-wide leadership?
2. Is there an experience or experiences you can share that have shaped your view of effective Department-wide leadership?

- i. What do you consider important characteristics for a USFES Department leader to exercise effective, Department-wide leadership?
 - ii. Please provide an explanation about why you believe these characteristics are important.
3. What do you consider important knowledge for a USFES Department leader to exercise effective, Department-wide leadership?
 - i. Please provide an explanation about why you believe this knowledge is important.
4. What do you consider important actions for a USFES Department leader to exercise effective, Department-wide leadership? Actions include the skills a Department leader exercises, abilities he/she employs, behaviors he/she demonstrates, and processes he/she executes.
 - i. Please provide an explanation about why you believe these actions are important.
5. In your professional networking opportunities, what other views have you heard about important aspects of effective Department-wide leadership?
 - i. Why do you think others have or do not have different views?
6. In your professional networking opportunities, what differences have you discovered, if any, in Department-wide leadership across the categories of USFES Departments?
Categories of USFES Departments include distinctions of size and type of community – urban, suburban, rural – and also career, volunteer, and combination Departments?
7. Who would you suggest contacting for an interview in this study who thinks differently than you about what is important to effective Department-wide leadership in USFES Departments?

This protocol permitted the participants in the concourse-development interviews to explain fully their points of view. Participation in the interviews was voluntary and I obtained permission to record the interviews and transcribe them, as well as ensure confidentiality for those interviewed (Concourse interviewee's Informed Consent form is included in Appendix C). No identifying information about the person interviewed was stored with each recording or transcript because the statements become the "population" of the study, not the respondents, and personal identifying information is not necessary. Recordings were sent to and transcriptions received from a commercial transcription service, Rev, with a number and date identification (e.g., I1.020116), and securely stored in a password-protected computer file without personal identifying information of the interview participants. The transcriber completed a Confidentiality Agreement (Appendix H). I also took notes during the interview to record anything undetectable by audio recording. From the interview transcripts, I extracted concise statements important to USFES Department-wide leadership to include in the concourse.

Previous studies of leadership in American fire departments imposed a leadership construct on participants with a particular instrument, such as the Leadership Practices Inventory (Lowes & Barnes, 2000), a specific theory such as transformational leadership theory (Pillai & Williams, 2004), studied only one context within the department such as incident command (Klein et al., 2010) or focused on a single issue such as promoting firefighter health (Jahnke, et al., 2014). This study sought to describe perspectives on USFES Department-wide, *organizational* leadership, and required a grounded approach rather than imposing a particular construct. A semi-structured interview methodology provided participants with the opportunity to present their viewpoints without the constraints of questionnaire or survey based on a single leadership model or theory, consider organizational complexity that includes multiple contexts,

and take into account the many issues that fire department leaders handle. Eligible participants came from many departments across America, of varying sizes and community type.

Data from the semi-structured interviews provided a detailed and rich collection of self-referent statements about USFES Department-wide leadership. The data includes 17 interviews with USFES officers or supporting-area leaders from 10 states and the District of Columbia. Five interviewees were female. All of the major geographic regions of the United States are represented: East, South, Midwest, Northwest, and Southwest. The three types of American fire departments were also represented: 12 interviewees are from career fire departments, two are from volunteer fire departments, and one is from a combination department.

To ensure that the interviewees did not limit their responses to a particular context or a particular issue, all of the interviewees were provided with the topic and purpose of the study in the e-mail invitation and the Informed Consent document, and reminded to consider the entire organization in prelude to the interview questions. As an introduction for the interview and to begin thinking about leadership for the entire organization, interviewees were first asked to consider what they would notice in a department that provides evidence of effective department-wide leadership. Each of the interviewees mentioned that a positive morale about the mission and the work of the fire department would be evidence that leadership is effective, in one way or another. For example, participant I1 said does one see that “the attitudes and actions of people [are] congruent with the mission statement” and participant I3 said “spirits are good, people seem happy, they seem to like doing what they’re doing, and they like to go to work . . . I think the environment, what you see, atmosphere, I think all play a role in effective leadership.” Participant I14 described morale in terms of friendliness and openness for the mid-level managers in particular since they navigate in the middle of the organization. If they have positive

and motivated attitudes it is reflective of the leadership above them in the organizational hierarchy and can impact those below them in the organizational hierarchy. Other participants used words and phrases such as “engagement,” “energy,” “professionalism,” “esprit de corps” and “up-beat” to describe morale that is evidence of effective department-wide leadership. After the interviewees described what they might notice as evidence of effective department-wide leadership, the questions were specifically about the department leaders and their exercise of leadership that produces such an environment.

Q phase 2 in Detail – Concourse Analysis and Development of the Q-Sample

The interviews and the review of the trade literature provided a lengthy concourse of nearly 400 statements. From this concourse, the Q-sample of statements were selected. Q-sample selection involves paring down the concourse to a representative sampling of 40 – 60 statements. The Q-sample was not selected randomly, but by a rigorous, structured design process. The Q-sample must have adequate coverage of the possibilities in relation to the research question. In other words, the Q-sample must represent the complete concourse, or opinion domain. It cannot appear to be value-laden or biased toward a particular perspective or viewpoint. For such a balanced or structured Q-sample, the concourse is divided into themes or issues, and each theme will have approximately the same number of statements in the selected Q-sample (Focht, 2002; McKeown & Thomas, 2013, Watts & Stenner, 2012). The interview protocol followed a line of questioning to elicit different dimensions of leadership, specifically characteristics, knowledge, and skills, abilities, or behaviors, and the final Q-sample is also balanced according to the dimensions of leadership represented so that it is not biased toward personal traits, knowledge, or actions. To further categorize the dimensions of leadership, political and relational behaviors were categorized separately.

As a pre-requisite, preparation phase in a Q study, the actual concourse development and selection of the Q-sample is presented in the methodology chapter here. Interviewees and industry publications provided an expansive concourse of statements and several themes emerged. Broad Themes identified within the concourse were: Administration, Change, Command, Monitoring, Service, and Teambuilding. Appendix I contains a listing of 388 concourse statements categorized by these themes. The statements were further coded with narrower themes related to leadership, a step important for the Q-sample sampling process to ensure balance and that no single aspect of leadership was over-represented in the Q-sample. The sub-themes coded for the statements included:

- Accommodation
- Accountability
- Adaptation
- Building trust
- Creativity
- Decisiveness
- Diplomacy
- Discernment/selectivity
- Empathy
- Facilitation
- Fairness/justice
- Focus on the greater good
- High expectations
- Humility

- Inclusivity
- Macro viewpoint
- Networking
- Nurture
- Openness
- Reflection
- Spanning and crossing boundaries
- Strength
- Support
- Systems/complexity
- Transparency
- Vision

Because the Q-sample statements were self-referent and naturalistic, it was common for statements to integrate more than one of these sub-themes, which further demonstrates that they contained excess meaning. Tables 41, 42, 43 in Appendix I mark and total the sub-themes for each statement, and the leadership dimension each represents to illustrate this excess meaning and further show the balanced structure of the Q-sample. The complete discourse in Appendix I also noted those statements that every interviewee and every industry publication or blogger seemed to agree are salient to the topic so that they could be eliminated first, because one of the criteria for Q-sample statements is that they should be salient *and* discriminating or controversial; and should not include statements with broad consensus.

Table 1 is a listing of the balanced selection of statements for the Q-sample, showing one way that the sample is balanced by organizing the list of statements according to the broad theme

they represent. When applicable, a secondary theme is noted, and is an indicator of excess meaning in the statement. The statements included in the Q-sample adhere to all the following criteria:

1. Statements should be relatively short so that sorters can easily read and interpret each as a stand-alone statement. Statements should not have complex grammatical construction or difficult vocabulary. These are similar to requirements for survey statements or questions, but unlike ideal survey questions, Q statements should have excess meaning.
2. Statements should include only one element or concept so that sorters are not motivated to both agree and disagree with elements of the same statement.
3. Statements that have opposing meanings should not be included; this would offer no additional information in the sort.
4. Negative statements should not be included to help in avoiding confusion about sort placement.
5. Statements should be salient and discriminating or controversial; and should not include consensus statements.
6. The final Q-sample must have balanced representation of the themes and dimensions of leadership identified in the concourse; no themes or dimensions should be over- or under-represented (Focht, 2002; McKeown & Thomas, 2013).

Table 1

Q-Sample Statements Listed by Themes

Statement	Theme	Secondary Theme (if applicable)
effective fire department leaders should actively explore ways to partner with other public entities	Administration	
effective fire department leaders should treat human resource management is critical for the organization	Administration	
effective fire department leaders need to be smart about how to influence different individuals	Administration	Service
effective fire department leaders should value objectivity	Administration	Monitoring
effective fire department leaders' most important work is risk reduction	Administration	Service
effective fire department leaders should facilitate career development for all members	Administration	Service
effective fire department leaders should understand the whole system and consider the ripple effects of every action	Administration	Monitoring
effective fire department leaders should seek to be politically astute	Administration	
effective fire department leaders should create freedom for free thinkers	Change	Service
effective fire department leaders should embrace opportunities to do more functions that contribute to the public safety	Change	Service
effective fire department leaders should pursue grants to stay current and progressive	Change	Administration
effective fire department leaders should be future-oriented	Change	
effective fire department leaders should never be satisfied with the status quo	Change	
effective fire department leaders should guard against the need to always agree	Change	
effective fire department leaders need to create a safe zone for members to question processes and decisions	Change	Monitoring
effective fire department leaders need to break barriers to change	Change	
effective fire department leaders need to be decisive, sometimes without complete information	Command	
effective fire department leaders should act like corporate CEOs	Command	Monitoring
effective fire department leaders should hold their cards close until the support is there	Command	
effective fire department leaders should do the tough stuff first	Command	Administration
effective fire department leaders have to keep their personalities out of their leadership	Command	
effective fire department leaders need to tow the clear, hard line	Command	Monitoring
effective fire department leaders should maintain a stalwart demeanor	Command	
(continued)		

Statement	Theme	Secondary Theme (if applicable)
effective fire department leaders should operate at the 30,000' level	Command	
effective fire department leaders need to be vigilant stewards	Monitoring	Administration
effective fire department leaders should not accept excuses	Monitoring	
effective fire department leaders own the department's problems	Monitoring	Command
effective fire department leaders need to develop metrics that are meaningful for different stakeholders	Monitoring	Service
effective fire department leaders have to actively and regularly critique all practices and processes	Monitoring	Administration
effective fire department leaders should be wary of too much "buddy/buddy" behavior	Monitoring	
effective fire department leaders need to be strong disciplinarians	Monitoring	
effective fire department leaders should know that good SOPs run the department	Monitoring	Administration
effective fire department leaders should have empathy	Service	
effective fire department leaders give extra attention to community involvement	Service	
effective fire department leaders should let go of ego	Service	
effective fire department leaders should give away power	Service	
effective fire department leaders should share lots of information for the greater good	Service	
effective fire department leaders should believe everyone brings something to the table	Service	Change
effective fire department leaders should measure success through the betterment of department members	Service	Change
effective fire department leaders value equity	Service	Administration
effective fire department leaders form cross-division teams to address issues and challenges	Teambuilding	Change
effective fire department leaders seek informal opportunities to interact	Teambuilding	Service
effective fire department leaders need to facilitate organizational dialogue	Teambuilding	Change
effective fire department leaders should get things done collaboratively	Teambuilding	
effective fire department leaders should decentralize authority	Teambuilding	Service
effective fire department leaders need to pick the spots where to bend for the sake of morale	Teambuilding	
effective fire department leaders should give members time to correct issues on their own	Teambuilding	Change
effective fire department leaders should be democratic	Teambuilding	

Note: secondary themes are one indicator of excess meaning in the statements.

Staff members of the National Fire Academy reviewed the Q-sample and provided feedback for the sample of statements to ensure that they are readable and understandable, and judged to be open to individual interpretation. Kevin Brame, Curriculum Specialist in the Executive Fire Officer Program of the National Fire Academy was my contact at the National Fire Academy. Mr. Brame contacted five additional faculty members of the Executive Fire Officer Program to review the statements. The e-mail requesting the review of the Q-sample statements, Mr. Brame's acceptance message, and the message relating the NFA faculty assessment of the statements are located in Appendix D.

Q phase 3 in Detail – the P-Sample

Participants who complete the sorting process in a Q methodological study make up the *P-sample*. Individuals in the P-sample may be the same people who participated in Phase One to develop the concourse. In Q methodology, there are two key principles for consideration in selecting the P-sample. Participants must be likely to have well-formed views on the topic of the study, and they should also add to the breadth of views on the topic (Webler et al., 2009). In other words, P-sample members are selected purposively because they likely hold a view that is important to include in the study.

Q methodology experts do not agree on the ideal number of participants who complete the Q-sorting. Recommended ranges include 8 to 30 (Webler et al., 2009), and 30 to 50 (McKeown & Thomas, 2013). The number should be large enough to collect as many different views as may exist, but it should be smaller than the number of Q-statements (Webler et al., 2009). Q studies use fewer participants than survey research because the methodology is intended to discover patterning among perspectives, rather than across individuals, to identify factors. Q methodology discovers and provides detail on *points of view*, and on any given topic

there are generally three to five independent points of view (Brown, 1993). For this study, 25 sorters are in the P-sample.

Purposive and snowball sampling will involve identifying officers in USFES municipal Departments who would be knowledgeable and likely to be opinionated about what characterizes effective leadership in USFES Departments (McKeown & Thomas, 2013; Webler et al., 2009). The concourse interview participants represent different perspectives on department-wide leadership, so they will be the initial members of the P-sample. Within the industry, professionals refer to organizational differences, so it is important to ask participants if they are aware of different perspectives and use the snowball technique to ask sorters to suggest additional sorters who represent a different perspective than their own to ensure that P-sample includes the full range of possible perspectives.

Using the basic contact information provided (e-mail address or telephone number), I contacted each potential sorter to ask if they agree to receive detailed information about the study to consider participation. Upon confirming the e-mail address that should be used, I e-mailed the invitation to participate that disclosed the nature of the study, the Q-sort activity and post-sort interview, which I referred to as a specialized survey and interview about the survey since the invited participants were not likely be familiar with the terminology *Q-sort*. The invitation and informed consent form were clear that participation is voluntary and will be confidential, that the total time involved will not exceed an hour. It also described the minimal risk associated with participation (Appendix E). The participants completed a questionnaire on demographic information, including their age, gender, career/volunteer designation, Department type (>85% volunteer, >85% career, combination; urban, suburban, rural), rank, level of education and professional development opportunities they have completed. These demographic data were

important for the later factor analysis and interpretation related to the perspectives on Department-wide leadership because they are criteria that could influence the participants' perspectives (Watts & Stenner, 2012).

Q phase 4 in Detail – Q-Sorts and Post-Sort Interviews

Q sorting is central to Q methodology. During Q sorting, participants arrange a collection of items about a phenomenon or topic according to a subjective dimension such as *most agreement/most disagreement*, or *most like me/most unlike me*. When a participant has completed the Q-sort, the resulting arrangement of the items is self-referent and is the participant's operant viewpoint of the phenomenon of the study. The Q-sort captures a subjective perspective *in action* because the participant does not have to introspect, but produces and communicates a subjective point of view spontaneously in response to the stimuli of the Q-sample statements.

The Q-sort involves placing each item into a space in a quasi-normal, forced-choice distribution template according to a clearly-stated condition of instruction. The condition of instruction will include the same description of the phenomenon as presented to the concourse interviewees:

American fire and emergency services Departments (USFES) are complex organizations with many contexts that impact the Department's effectiveness. Examples of the different contexts include on-the-scene of an incident, training, stations, administrative offices, labor/management negotiations, government relations, community relations, prevention and education, and so on. Please think about leadership for the entire Department in all of these contexts as you complete this sorting exercise. Place each statement before you into 5 piles that correspond to agree, somewhat agree, disagree, somewhat disagree, or neutral/ambivalent/unsure. Then place the statements on the formboard in the spaces

Brown (1980, as cited in Watts & Stenner, 2012) provided guidelines for determining the width of the distribution (number of columns) and the narrowness of the distribution (steepness, determined by the number of rows) that will be followed to determine the configuration of the distribution template. For Q Samples of 40 statements or less, he suggested a nine-point distribution across the first row; for Q Samples of 40–60 statements, he suggested an 11-point distribution across the first row; and for Q Samples of greater than 60 items, he suggested a 13-point distribution across the first row. Distributions can be taller and narrower (leptokurtic) if there are more places in the columns at the middle of the distribution. Narrow distributions are recommended if participants are not likely to be very familiar with the subject matter, or if the subject matter is very complicated, because it allows participants to place more items in the middle part of the distribution indicating that the participant does not have strong opinions about more items or is indifferent about more items. The final distribution configuration for this study is shallower or flatter (platykurtic, as illustrated above in Figure 6), because the participants selected for the study will be knowledgeable about the subject matter and because it is possible that there are more than a few perspectives because of the perspectival differences and the number of categories of USFES departments that exist. The guidelines summarized above are not rules. Since the statements will reflect the complexity of the phenomenon and possible perspectives, researchers determine the final distribution configuration of the formboard in an iterative process that involves consideration of the concourse development and Q-sample selection.

Once the configuration of the formboard is set, the Q sorting phase can begin. If the participant thought that a statement is very similar to his view, he placed it to the extreme right on the distribution template; if he was ambivalent about a statement or is not sure what it means,

he placed the card somewhere in the middle of the distribution template; and if he thought a statement was very unlike his view, he placed the card to the extreme left of the distribution template. Participants received instructions and helpful hints for completing the sorting (Appendix G). For example, a tip to begin the sorting process suggested that participants read through all the statements completely, and sort them first into five groups, or piles, before placing them on the formboard. One group represented those statements that the participant had an immediate “take” on as similar to his or her view. It should be placed to the right to be near the “Most Like My View” side of the formboard. The second group is for statements that the participant had an immediate “take” on as different from his or her view. It should be placed to the left, to be near the “Most Unlike My View” side of the formboard. The tip recommended making three additional piles for statements that are “somewhat consistent” with the participant’s view, that are placed next to the far-right pile and closer to the middle of the form board; “somewhat inconsistent,” that are placed next to the far-left pile and closer to the middle of the form board; and a pile of statements that the participant considers neutral or lacking salience, ambivalent or ambiguous placed in the middle. Sorting instructions did not place any limits on the number of statements for each pile. Once the participant created the five piles recommended in the sorting tip, the next step involves placing the statement cards on the formboard.

Each participant placed statements on the form board one pile at a time, beginning with the highly salient “most like my view” pile and the “somewhat consistent with my view” pile. The participant spread out the statements in those two piles and reviewed them again to determine the two statements to place in the column labeled “most like my view” and continued placing statements on the form board working from right to left. The instructions made clear that vertical placement within each column was irrelevant. Each statement had a small number or

Condition of Instruction: What is your view of effective Department-wide leadership in USFES Departments?										
Most unlike my view			Indifference, Ambivalence, or Ambiguity					Most like my view		
-5	-4	-3	-2	-1	0	1	2	3	4	5
47	29	36	38	12	30	10	40	39	01	17
23	33	35	43	18	32	46	02	19	04	28
	24	27	11	34	44	03	26	25	08	
		20	22	09	05	31	41	45		
			15	06	37	16	42			
				21	07	14				
					13					
					48					

Figure 8. Illustration of a completed Q-sort recording sheet, with bold lines indicating the borders for the positive categories of statements, negative categories of statements, and indifferent/ambivalent/ambiguous statements.

It is important for individuals completing the Q sorts to engage physically with the statements of the Q-sample by actively manipulating, handling, and moving cards because the statements are the stimuli that will provoke the operant, behavioral expression of the sorter's viewpoint. During the process of actively sorting the statement cards, the sorter interprets each and imposes personal meaning on the statements (Watts & Stenner, 2009). Participants may also "think out loud" during the sorting process to explore the meaning of the statements. Sorters were in locations across the United States, so upon agreeing to participate, I requested the address to send the materials consisting of a set of statement cards, a formboard, a blank recording sheet to record the statement card placements, and instructions. I provided an internet video conference link to each participant and made myself available before and during the

sorting to answer questions or concerns, and conduct the post-sort interview. I asked participants to begin the sorting process only after starting the video conference so that I could observe the sorting process, listen to the participant during the sorting process, answer questions, and conduct the post-sort interview. The following section details the post-sort interview questions.

When the Q-sort was complete, the post-sort interview was an opportunity for me to ask questions about the sort. Questions were about the placements and the meanings that the participant gave to the highly salient statements of the Q-sample (“most like my view” and “most unlike my view”), and provided important, supporting qualitative data for the factor interpretations (Watts & Stenner, 2012). The post-sort interviews were audio recorded with permission from the participants, and later transcribed. As with the concourse interviews, participation was voluntary, and responses recorded, with permission, for transcription. Recordings were sent with a number and date identification (e.g., I1.020116) for transcriptions by a commercial transcription service, Rev, and securely stored in a password-protected Dropbox.com account without personal identifying information of the interview participants. The transcriber completed a Confidentiality Agreement (Appendix H). Post-sort interview questions included:

1. In your own words, how would you describe your point of view on department-wide leadership?
2. For the statements you indicated are most like your view on effective Department-wide leadership, please describe what these statements mean to you.
3. For the statements you indicated are most like your view on effective Department-wide leadership, please explain how these things affect Department effectiveness.

4. Please describe an experience that contributed to your choices about what is effective Department-wide leadership.
5. For the statements that you indicated are not like your view, can you explain why you think so?
6. Are there any characteristics that you consider important for a Department leader to have to exercise effective Department-wide leadership that were not included among the statements?
7. Is there any knowledge that you consider important for a Department leader to possess in order to exercise effective Department-wide leadership that was not included among the statements?
8. Are the any actions that you consider important for a Department leader to do in order to exercise effective Department-wide leadership that were not included among the statements? Actions include the skills a Department leader exercises, abilities he/she employs, behaviors he/she demonstrates, and processes he/she executes.

In order to effectively respond to the last research question, about how the perspectives revealed in the Q-sorts correspond to existing leadership constructs, concepts, theories and models, I included theoretical Q-sorts. A theoretical Q-sort is an estimation of how a person who thinks purely and completely in terms of a particular construct, concept, theory, or model might sort the Q-sample statements. I completed a total of seven theoretical Q-sorts by first preparing and developing the mindset of the particular construct, concept, theory, or model by reviewing a leadership theory textbook (Northouse, 2012) and two academic studies based upon it. Five of the theoretical Q-sorts were theories or models that have been advanced as relevant or studied *a priori* in USFES: adaptive leadership theory, crew resource management, servant leadership

theory, and two types of leadership from the full-range leadership model, transformational and transactional leadership. Two additional theoretical Q-sorts for public administrative leadership theory and complexity leadership theory were also included these have been studied in related industries. To contribute to USFES industry knowledge, the field of public administration, and to organizational leadership research through an examination of whether or not the conceptual framework of public administrative leadership has relevance for USFES Departments it was important to include a theoretical Q-sort based upon public administrative leadership. Recent scholarship indicates that complexity leadership theory has relevance in organizations that have complex and dynamic relationships among contexts, including emergency and critical incidents, administrative, and community and political contexts (Hannah & Parry, 2014; Osborn, Uhl-Bien, & Milosevic, 2014; Uhl-Bien, Marion, & McKelvey, 2007) and across levels of concern for leadership, from the personal to the group to the cultural and organizational (DeJoy, Smith, & Dyal, 2017; Griffin & Talati, 2014).

Q Phase 5 in Detail – Data Analysis

The final phase of the Q study consists of the statistical analysis of the Q-sorts. Each Q-sort was entered into a dedicated computer software package, PQMethod (Schmolck, 2014), that facilitated analysis. There are three sections for the statistical analysis of the Q-sorts involving methodological transitions. The first statistical analysis section involved transitioning from Q-sorts to Q-factors by sort intercorrelation and subjecting the Q-sorts to a by-person (by-Q-sort) factor analysis. In this first statistical step, the software analyzed the level of variation across the Q-sorts.

Q-sorts that are highly correlated have portions that are arranged similarly indicating an area of shared meaning. The first step of the factor analysis identified these “families” or clusters

of highly-correlated sorts using principal components analysis (PCA) and showed how many different families (factors) there are among the sorts and how closely each was correlated to each other. The factor analysis also indicated the loading each sort has on each factor, indicating the extent that each individual sort was associated with each factor. Orthogonal factors extracted were first rotated using Varimax rotation to achieve simple structure. Traditionally, Q methodologists initially extract seven factors, although the number of factors retained may be different as explained below. Q researchers use manual (judgmental) rotation of the factors to explore the perspectives. Judgmental rotation allows researchers to abductively discover the nature of the differences among perspectives. For example, rotation can reveal and contribute to understanding the nature of conflict and consensus among perspectives held by various sorters (Focht, 2002). Judgmental rotation in this study also contributed to the examination of how closely any of the perspectives matched existing theories or models.

The second transition moved from factors to factor score arrays via weighted averaging of the significantly-loading Q sorts to identify the degree of pattern multiplication among the sorts indicating shared perspectives (Watts & Stenner, 2012). In this second section of statistical analysis, researchers determine the number of factors to retain for interpretation, guided by a set of criteria including:

1. The overall factor solution must be stable
2. There must be at least two Q sorts loading on each factor that is retained
3. The eigenvalue for each factor retained is greater than unity
4. The factors retained minimize the number of non-significant Q sorts
5. Factor correlations are minimized
6. Factors that produce only bipolar splits of others are not retained

7. Each factor retained captures a perspective on the phenomenon of study (Focht, 2002)

For the Q sorts that loaded significantly on a factor, the participants have sorted the statements similarly and the software created a common factor score array that represents an average sort for the factor, or common sort. Some Q-sorts had statistically significant loading on more than one factor, and are called *confounded*. These were also included in the common factor score array. Significant loadings were defined as a load exceeding the two-tailed z-score associated with a preferred significance level divided by the square root of the number of statements. For example, for $\alpha = 0.01$ and 49 statements in the Q sample, the minimum significant loading is $3.09/\sqrt{49} = 0.441$.

The third section of the analysis transitioned from factor score arrays to factor interpretation and validation. Interpretation of each factor (common perspective) began by examining the common factor score array for the perspective, focusing on those statements of highest salience located at the extremes of the arrays. The statements in the middle of the factor score array also contributed to clarifying the potential meaning of the perspective associated with a factor.

Next the factor score arrays were compared to each other to identify the statements that are arranged similarly (indicating consensus statements) and differently (indicating discriminating statements). The software identifies the discriminating features of the perspectives based upon the orthogonal factors. These unique features of a perspective are emphasized in a descriptive paragraph that is the preliminary interpretation of the perspective, and the meaning of the perspective is captured with a shorter label for the perspective. The factor interpretations were triangulated with the qualitative interview data from those Q-sorters whose sorts loaded most highly and purely on the factor (their sorts most closely resembled the common sort and did

not also closely resemble any other factors), refining the interpretation and establishing credibility. In the final validation step, the significant, pure loading Q sorters reviewed the factor labels and descriptive paragraphs (Creswell, 2013; Focht, 2002).

Ethical Assurances

This study followed the requirements of the National Institutes of Health, the American Psychological Association, and The Chicago School of Professional Psychology regarding human subjects. There are three areas of ethical concern: research cannot be conducted without the legitimate consent of the participants; the research design must not involve unreasonable risk of harm to participants; and researchers must protect the confidentiality of the participants (Creswell, 2013). The next sections provide details followed in the study.

Informed Consent

I contacted potential participants via e-mail with information about the study and their participation in it in both an e-mail letter inviting their participation and in the attached informed consent document developed with the template provided by The Chicago School of Professional Psychology (Appendix F). I made them aware of the purpose of the study, their role in the data collection, the method of data collection, the amount of time that they could expect their participation to involve, and the confidentiality of their responses. The informed consent document also disclosed the minimal risk of harm. It was clear that participation would be voluntary and that participants may leave the study at any time. Participants provided consent and provided permission to record their interviews.

Minimizing Risk of Harm

The primary risks of harm to participants included the possibilities that participating USFES officers could be affected negatively professionally, economically, psychologically,

and/or socially by their participation. Professionally and economically, the opinions expressed through participation in the study could affect a promotional process or credentialing, if known. Psychologically, a participating USFES officer could experience concern or questions related to their opinions about Department-wide leadership expressed in the study that persist beyond the period of the study. Socially, a participating USFES officer whose view expressed in the study was shared or became public could experience stigmatization or marginalization if they have a perspective that differs from others' perspectives. To avoid and minimize these risks, the only information available to the public from this dissertation research assures confidentiality and anonymity for participants and Departments. I also confirmed with each participant that he or she was in a safe environment for completing the sort and the interview and assured them that they could choose to stop participating at any time.

Protecting Confidentiality

There were measures to ensure the participants that their participation and the information provided during the study will remain confidential. Data collected in the study was coded to exclude names and identifying descriptions to protect the participants' privacy. Interview and Q-sort data is stored in a password-protected electronic file. My research supervisor, Dr. Kathleen Kleinfeldt, and Q methodology expert, Dr. Will Focht, and I were the only people who could access or review the data. I stored back-up files of the data on an external drive that is also password-protected and kept in a locked safe box. I deleted e-mails and other written correspondence at the conclusion of the study. Interview recordings include only the codes and no names or identifying descriptions, and the audio files were deleted after transcription. Q-sort data and transcribed interviews will be secured for a period of five years following the completion of the proposed study and then deleted.

Assumptions, Delimitations, and Limitations

The study has assumptions, delimitations, and limitations. Both an assumption and a delimitation, only internal perspectives from USFES officers were included. This perspective is assumed to be relevant to effective organizational leadership in the USFES. The study did not include the perspectives of other stakeholders such as those from other public safety entities, medical institutions, community organizations, or city government. It was concerned with a single organizational type in USFES, municipal Departments, and not with other types of organizational and delivery structures that exist in fire and emergency services, such as wild land protection agencies, industry and airport fire protection services, and military fire brigades.

The choice of Q methodology presents limitations. The Q-sample, because it is pre-determined for the participants, may limit the expression of opinion available to them (Militello & Benham, 2010). This limitation has its roots in the concourse development and could result if the concourse is not fully saturated and thus does not include representations that resonate with all participants, perhaps because there are variations of municipal departments that are unrepresented in the concourse, or if the researcher responsible for developing the concourse and Q-sample allows personal biases to influence the process. Due to the small, non-random participant sample, a Q methodological study does not indicate the proportions of a wider population that may hold the perspectives elucidated in the Q study. Generalization of other kinds may not be precluded, however, depending upon the results and interpretations possible from the results. The proposed study may or may not have wider implications for USFES of a conceptual nature by bringing to light a theoretical proposition for further study (Boon-itt & Pongpanarat, 2011; Watts & Stenner, 2012).

A potential limitation related to Q methodology can result when the researcher conducting the analysis and interpretation of the Q-sorts names the factors after reviewing the data supplied through the statistical program. It is possible that the factor names will not represent the participants' viewpoints effectively or not resonate conceptually for the audiences reviewing the study. I am not a fire service professional, both a potential strength and limitation because a person who is not a USFES professional will be more likely to have an objective stance related to the industry and can objectively interpret the data as it is presented, but such a person may not have the background, training, and experience to effectively name and describe the factors so that they are useful in the industry. I minimized this limitation by validating the factor interpretations through follow-up discussions with participants whose Q-sorts significantly and purely loaded on each factor to verify interpretation and factor labels, and revise them if necessary.

Another potential limitation of this study is related to my lack of extensive experience with statistical analyses in general and Q methodology statistical analyses in particular. I have completed one graduate-level statistical methods course, in 2012, and have never before completed a Q methodology study. A statistical software package such as PQMethod will provide the mathematical and statistical support, but it does not address inexperience with the analysis and interpretation of the results. I minimized this limitation through extensive consultation with Dr. Will Focht, a Q methodology expert.

Chapter Summary

This exploratory study employed Q methodology, a mixed method for collecting and analyzing data that can begin to address the gaps in understanding of USFES Department-wide leadership. Q methodology is ideally suited for the purposes of inquiry and research into

questions where there is the possibility of multiple perspectives, and can be effective to discover and more fully understand the perspectives on effective USFES organizational leadership, which are not presently clear. The specific research questions investigated with Q methodology were:

1. What are the perspectives of officers in the USFES Departments regarding effective *Department-wide* leadership?
2. How do these perspectives differ and how are they similar?
3. How do these perspectives correspond to existing leadership constructs, concepts, theories, and models?

Q methodology involves five phases that the chapter described in-depth to show how each phase proceeds. The first phase, to develop a saturated concourse related to the phenomenon of the study (effective Department-wide leadership in USFES departments), involved a combination of purposive and snowball sampling beginning with the members of the United States Fire Administration's National Fire Academy Board of Visitors and leaders with the International Public Safety Leadership and Ethics Institute, and resulted in a concourse of nearly 400 self-referent statements about organizational leadership in USFES departments. In the second phase, thematic coding and analysis of the statements allowed for a balanced selection from among the concourse statements to develop the Q-sample, a series of statements that are representative of the entire concourse. The Q-sample selected for this study consisted of 48 statements representing all of the themes in the concourse equally. The third phase involved the selection of the P-sample, the participants who complete the Q-sorting of the Q-sample statements to express their perspective on USFES organizational leadership, which constitutes the fourth phase. The fifth phase was the statistical analyses to determine the factors characterizing the various perspectives that exist. The fifth phase involved detailed interpretation

and description of each grouping of perspectives. The study concerned only USFES municipal departments because the participants involved was limited to those types of departments. The next chapter presents the results of the fourth and fifth phases of the methodology.

Chapter 4: Presentation of Results

Introduction

This chapter presents the analyses of the data collected as Q-sorts. The Q-sort data collected are the raw information for the study's overarching research goals to describe the perspectives within the American fire service on what characterizes effective USFES Department-wide, organizational leadership, and discover what elements, if any, are shared. Each Q-sort is an individual participant's arrangement of the Q-sample statement cards according to his or her subjective perspective and thus captures their perception of the topic with a consistent, operant process that allows for a statistical analysis of the differences and similarities among the perspectives. Q-sort data and analysis from the participants (N = 25) and post-sort interviews provided details to discuss in response to the research questions:

1. What are the perspectives of officers in the USFES Departments regarding effective *Department-wide* leadership?
2. How do these perspectives differ and how are they similar?

The inclusion of seven theoretical Q-sorts, or sorts of the Q-sample according to the principle tenets of theories that have been promoted in the USFES, previously studied in the USFES, or studied in related industries, provided data to answer the third research question:

3. How do these perspectives correspond to existing leadership constructs, concepts, theories, and models?

This chapter first describes the demographics of the participants completing the Q-sorts, and provides the listing of the concourse statements that all concourse development sources agreed were important to effective organizational leadership. It then describes the statistical processes completed on the Q-sort data, and presents the analysis of the Q-sorts. Following the

presentation of the results for all the Q-sorts, the analysis concludes an interpretation of each of the groups of perspectives emerging from the statistical analysis.

Data Collection

Participants from the concourse development interviews were invited to continue participation by completing a Q-sort. Seven agreed to complete Q-sorts, and the remaining participants were identified through snowballing from these seven by asking them to suggest a participant who has a different perspective on organizational leadership than their own.

The mean age of the participants was 45.4 years, with a standard deviation of 9.2. Four of the participants were female, or 16%, and 21, or 84% were male. Average years of service in a fire department was 21.8 years with a standard deviation of 9.3. Participants came from four regions of the United States identified as East, South, Midwest, West as shown in Table 2. All three major types of municipal fire departments were represented, career (13), volunteer (4), and combination (8), with the range in department size from 20 to 4,900 and a mean size of 430 (standard deviation of 970), as shown in Table 3. Additional demographic information collected included rank and level of education. Four levels of rank were represented including Chief (Level 1, N=3)), Deputy or Assistant Chief (Level 2, N=8), Battalion Chief (Level 3, N=5), and Captain (Level 4, N=9). Four levels of education were also represented, including Associates (Level 1, N=1), Bachelors (Level 2, N=12), Masters (Level 3, N=11), and Doctoral (Level 4, N=1). Table 4 displays the number of participants for each rank and educational level.

Table 2

Geographic Regions Represented by Participants

Region	N	Percentage of Participants
East	7	28%
South	6	24%
Midwest	4	16%
West	8	32%

Table 3

Fire Department Type, Range of Department Size, Represented by Participants

Department Type	Size Range	N
Career	20 – 4,900	13
Volunteer	25 – 70	4
Combination	32 - 454	8

Table 4

Number of Participants by Rank and Education Level

Rank	N	Level of Education	N
Level 1 (Chief)	3	Level 3 (Masters)	3
Level 2 (Deputy, Assistant, Division Chief)	8	Level 2 (Bachelors)	5
		Level 3 (Masters)	2
		Level 4 (Doctoral)	1
Level 3 (Battalion Chief)	5	Level 2 (Bachelors)	2
		Level 3 (Masters)	3
Level 4 (Captain, Lieutenant)	9	Level 1 (Associates)	1
		Level 2 (Bachelors)	3
		Level 3 (Masters)	5

While the sampling technique was purposive to represent as many different perspectives as may exist in the USFES rather than to represent different ages, years of service, rank, department type, department size, and regions, the P-Sample does show representation of all the categories within these variables. This allows for an examination of the Q-sort perspectives in light of these variables and an assessment whether these variables seem to influence any of the perspectives represented in the Q-sort data.

Elements that All Sources Agree Contribute to Effective Organizational Leadership

A partial response to the research questions on how the perspectives of officers in the USFES Departments regarding effective organizational leadership are similar is provided by those statements collected in the concourse that all sources agree contribute to effective organizational leadership. Of the nearly 400 statements in the concourse, 54 are statements that recurred and are shared across perspectives on effective organizational leadership:

1. Effective fire department leaders should communicate, communicate, communicate
2. Effective fire department leaders should vary their types of communications
3. Effective fire department leaders expect rigorous, tough, and regular training
4. Effective fire department leaders minimize interference of their communications
5. Effective fire department leaders “have two buckets -- one with water and one with gas; which one do they prefer to use?”
6. Effective fire department leaders should “lead by walking around”
7. Effective fire department leaders need to be efficient and do not “have meetings about meetings”
8. Effective fire department leaders are honest about what they cannot do well and what they need from others
9. Effective fire department leaders should not be too busy keeping their job rather than busy doing their job
10. Effective fire department leaders should consider strong, positive labor/management relations a top priority
11. Effective fire department leaders should convey that it is an honor and a privilege to serve the department

12. Effective fire department leaders should leaders admit mistakes and address them
13. Effective fire department leaders are “on” all the time, whether on the job or off
14. Effective fire department leaders should be calm under pressure
15. Effective fire department leaders should be consistently fair
16. Effective fire department leaders should be good at personal and professional writing as well as technical writing
17. Effective fire department leaders should be good public speakers
18. Effective fire department leaders should be positive role models
19. Effective fire department leaders should be self-aware
20. Effective fire department leaders should communicate effectively
21. Effective fire department leaders should not get too emotional
22. Effective fire department leaders should not whine
23. Effective fire department leaders shouldn’t just tell, they should show
24. Effective fire department leaders should be encouraging
25. Effective fire department leaders should get out from behind their desks
26. Effective fire department leaders should be consistent
27. Effective fire department leaders should have exceptional organizational skills
28. Effective fire department leaders should know more than just firefighting
29. Effective fire department leaders should “walk the talk”
30. Effective fire department leaders should maintain the bridge between the leadership and members
31. Effective fire department leaders should maintain unity in the organization

32. Effective fire department leaders should separate what is right for the department from what is right for them as individuals
33. Effective fire department leaders should study about leadership
34. Effective fire department leaders should support a respectful environment
35. Effective fire department leaders should value professionalism
36. Effective fire department leaders should not “fake it till they make it” (not self-serving)
37. Effective fire department leaders should not hesitate to un-do their own bad decisions and mistakes
38. Effective fire department leaders should expect and return loyalty
39. Effective fire department leaders should emulate and embody the core values of the department
40. Effective fire department leaders should understand crisis management
41. Effective fire department leaders should understand and focus on public relations
42. Effective fire department leaders should serve with integrity
43. Effective fire department leaders should have knowledge of the civil service and law
44. Effective fire department leaders should have presence that is apparent throughout the organization so leadership is not an abstraction
45. Effective fire department leaders should have tenacity for the long haul
46. Effective fire department leaders should be able to operate in all the realms that impact departments - legal, fiscal, political, strategic, technical
47. Effective fire department leaders should interface well with other agencies and state-level fire officials

48. Effective fire department leaders should be authentic and never just “clone” another leader
49. Effective fire department leaders should seek to comply and contribute to the 16 National Fire Service Life Safety Initiatives
50. Effective fire department leaders should set priorities according to the public interest
51. Effective fire department leaders should stay consistent on the priorities and issues until resolved
52. Effective fire department leaders should stay on point, are not easily sidetracked
53. Effective fire department leaders should keep clear lines of authority and uphold the chain of command
54. Effective fire department leaders should keep people focused on the public mission

The in-depth factor analysis of the Q-sorts used a sample of statements that did not include the concourse statements above that all would agree upon and are not salient, discerning or controversial. This allows for the discovery of factors that distinguish different perspectives that exist related to organizational leadership in the USFES and how the perspectives are similar and different related to the salient concourse statements.

Factor Extraction from Q-Sort Data

Statistical Processes

Each Q-sort was entered into the Q-methodology-dedicated computer software package, PQMethod (Schmolck, 2014), that facilitates analysis. As explained in Chapter 3, there are three sections for the statistical analysis of the Q-sorts. The first statistical analysis section involves transitioning from Q-sorts to Q-factors by sort intercorrelation and subjecting the Q-sorts to a by-person (by-Q-sort) factor analysis. Principal component analysis (PCA) provides a correlation

matrix for the sorts indicating the degree of similarity of each sort to every other sort. PCA also completes the statistical processes to cluster the Q-sorts into eight, unrotated factors and provides the analysis of variance. As Brown (1980) explained, “One of the positive claims of the principal components method is that the factors extracted account for the maximum amount of variance” (p. 222). Based upon the mathematical factor extraction with PCA and additional, mathematical Varimax rotations, and consideration of the Q-sort loadings on the factors, I next decided the number of factors to retain for in-depth analysis. Six factors were retained for description and interpretation.

Varimax rotation provided details about how each Q-sort loads on the factors with a correlation score. As Brown explained, “the rotation does really not affect the relationship among the facts – i.e., the data points are not moved around – only the vantage point from which the relationships are observed” (Brown, 1980, p. 230). To add new vantage points, PQMethod allows for manual rotation to abductively explore the relationships among the Q-sorts. In this study, several manual rotations were executed to see the relationships among the Q-sorts from new vantage points, but only one of these manual rotations uncovered new relationships among the Q-sorts, resulting in two additional factors for analysis.

In the second section of statistical analysis, common factor score arrays were produced based upon the Q-sorts that load significantly on each factor, including the manually rotated factors as additional, different factors from the six retained in the Varimax rotation. Common factor score arrays represent an average Q-sort of the statements for each factor. Each model array is presented as a list of the statement rankings for that array in descending order by the statements’ Q-scores expressed as Z-scores (normalized). Output from the program also includes

lists of the differences in statement rankings between factors, consensus statements across the factors, and distinguishing statements for each factor.

The third phase of the analysis transitions from factor score arrays to factor interpretation and validation. The interpretation of each factor, representing a group of Q-sorts or a single Q-sort that does not have similarities with any others, results in a label and descriptive paragraph that captures the distinct elements of each perspective. Participants whose Q-sorts loaded most significantly and purely on each factor review and validate the labels and descriptions.

The results of these processes are presented in order in the remainder of the chapter: mathematical factor extraction and number of factors retained; factor correlation matrix and the manual rotation details; model factor score arrays; and their interpretation and validation.

Mathematical Factor Extraction, and Factor Retention Decision

A six-factor solution emerged from the Q-sort data from an N of 25 Q-sorts and N of 7 theoretical Q-sorts. Five considerations contributed to the decision to retain six factors: 1) the eigenvalues for the eight-factors of the un-rotated PCA, 2) the Q-sort factor loadings at the $p < 0.01$ level in the six-factor Varimax-rotated solution, 3) the lowest number of non-significantly loading Q-sorts and confounded loading Q-sorts in the six-factor Varimax-rotated solution, 4) the percentage of variance explained by the retained factors, and 5) the differences across the distinguishing statements for the factors and low number of consensus statements.

Eigenvalue analysis alone was insufficient to determine the number of factors to retain because it indicated the possibility of retaining as many as eight factors and as few as three. As displayed in Table 5, all eight factors in the un-rotated, PCA of the Q-sort data were greater than 1, indicating that all of them explain a significant amount of variance, but factors G and H are minimally above 1. Greater than 50%, or most of the variance is explained with just Factors A,

B, and C. There is flattening of the Eigenvalue scree plot (Figure 9) after Factor D and after Factor G, also shown by the smaller difference between Eigenvalues from Factor D to E and Factor G to H in Table 6. This could provide the rationale for retaining four or seven factors. Factor retention based upon analysis of the eigenvalues is common, but also arbitrary, and in this case, does not provide a clear indication of the number of factors to retain.

Table 5

Un-rotated Factor Eigenvalues and Explained Variance

Factor	A	B	C	D	E	F	G	H
Eigenvalue	11.691	2.438	2.137	1.762	1.657	1.433	1.204	1.146
Explained Variance	37%	8%	7%	6%	5%	4%	4%	4%
Cumulative Explained Variance	37%	45%	52%	58%	63%	67%	71%	75%

Note. PQMethod 2.35

Table 6

Unrotated PCA Factor Eigenvalues, and Difference in Eigenvalues between Factors

Factor	Eigenvalue	Difference
A	11.691	
B	2.438	9.252
C	2.137	0.302
D	1.762	0.375
E	1.657	0.105
F	1.433	0.224
G	1.204	0.229
H	1.146	0.058

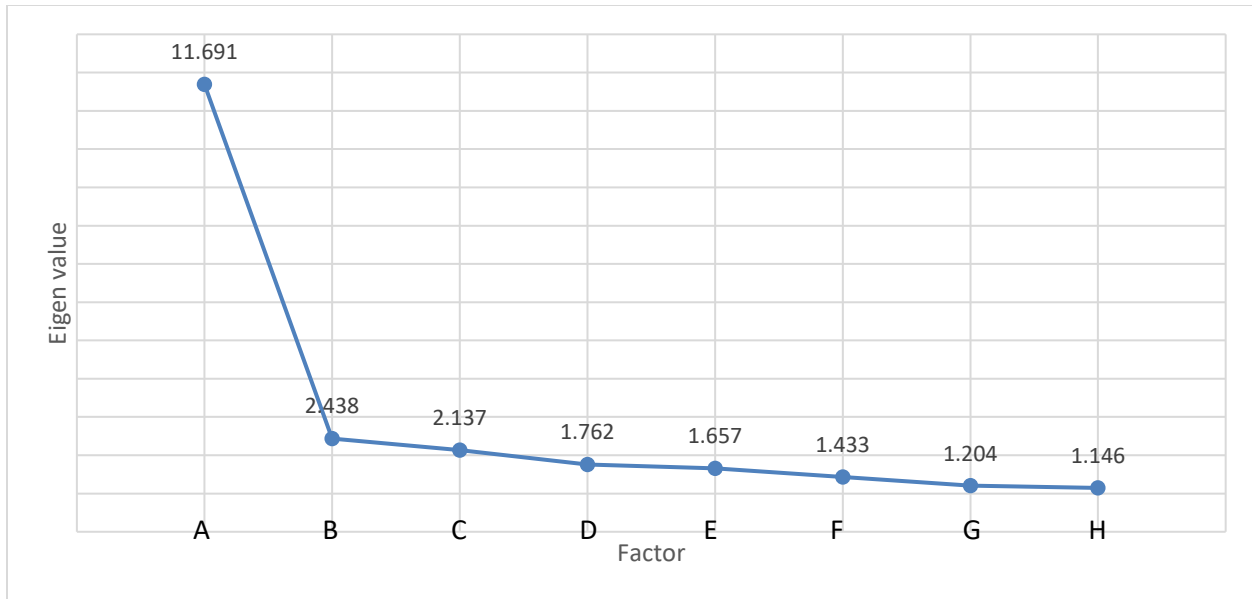


Figure 9. Scree plot of un-rotated, PCA factor Eigenvalues.

Comparing Varimax rotation output for three, four, five, and six factors showed that retaining six factors resulted in the highest number of Q-sorts that significantly load on a factor at the $p < .01$ level and the lowest number of sorts that are confounded or do not significantly load on any factor. The retention of six factors results in 22 Sorts significantly loading at the $p < .01$ level, two confounded-loading Sorts, and one non-significantly loading Sort. Table 7 displays this information. This supports the decision to retain the mathematical factor extraction of six factors.

Table 7

Summary of Changes for the 25 Participant Sorts over Four Varimax Rotations Retaining Three, Four, Five, and Six Factors

Sort	Factor Loadings, retaining 3 Factors			note	note on change for Rotation retaining 4 Factors	note on change for Rotation retaining 5 Factors	note on change for Rotation retaining 6 Factors
	A	B	C				
16	0.855	-0.169	-0.097				
2	0.745	0.143	0.050		moves to D	moves back to A	stays on A
14	0.727	0.136	0.425		confounded on A	confounded on A	confounded on A
12	0.715	-0.191	0.254				
8	0.688	-0.398	0.101		confounded on A	confounded on A	confounded on A
1	0.682	0.060	0.101		moves to D	stays on D	stays on D
24	0.670	-0.143	0.222			confounded on A	moves back to A
7	0.656	0.189	0.238				
22	0.622	0.047	0.407			moves to E	stays on E
18	0.603	0.084	0.282		moves to D	stays on D	stays on D
25	0.580	0.097	0.379		unloaded	Loads on E	stays on E
4	0.567	0.124	0.305				
23	0.564	-0.364	0.321		moves to B	stays on B	stays on B
20	0.536	-0.170	0.422		unloaded	loads to E	stays on E
3	0.465	0.127	0.220				moves from A to F
9	0.121	0.796	0.155		changes to bi-polar	stays bi-polar	stays bi-polar
17	0.318	0.525	-0.038		moves to D	stays on D	stays on D
15	0.075	0.150	0.748			confounded on C	loads on C
10	-0.025	-0.232	0.654				
19	0.415	-0.156	0.591			moves to E	stays on E
11	0.288	0.037	0.553			moves to E	stays on E
21	0.375	-0.450	0.252	bi-polar	loads on B	stays on B	stays on B
13	0.616	-0.049	0.453	confounded	loads on A	moves to E	stays on E
5	0.511	0.028	0.528	confounded	loads on C	unloaded	now loaded on F
6	0.321	-0.425	0.435	unloaded	Loads on B	Stays on B	Stays on B
# of Sorts that are NOT significantly, positively loading				4	5	6	3

Note: Sorts are re-ordered according to significant loading in the 3-factor rotation. Significant loading is indicated in bold font.

Table 8 displays the factor loadings for all the Q-sorts (N = 25) and the theoretical Q-sorts (N = 7) in the six-factor Varimax rotation. Only Q-sort 3 does not load significantly on any factor. Only two sorts are confounded: Sort 2 on loads significantly on both Factor A and Factor

B; Sort 9 is negatively and significantly loaded on Factor C and also significantly loaded on Factor D.

Table 8

Rotated Factor Loadings Matrix, Six-Factor Solution, Reordered by Factor from Higher to Lower

Sort	Theoretical Sort	A	B	C	D	E	F
11*		0.774*	0.072	0.074	-0.203	-0.034	0.027
22		0.670	0.200	0.176	0.007	0.166	0.280
20		0.669	0.301	0.210	0.042	0.071	-0.050
5		0.650	0.184	0.221	0.158	0.109	0.094
14		0.641	0.179	0.252	0.297	0.114	0.363
13		0.636	0.174	0.349	0.176	-0.012	0.239
25		0.611	0.190	0.151	0.209	0.054	0.204
27 ^a	Adaptive	0.586^a	0.353	0.095	0.497^a	-0.256	0.117
19		0.540	0.283	0.359	-0.166	0.381	0.086
2 ^a		0.447^a	0.486^a	-0.092	0.387	0.156	0.101
31 ^b	Transact	-0.120	-0.882^b	0.076	0.160	0.072	0.148
16*		0.277	0.783*	0.146	0.287	-0.048	0.159
26	Servant	0.012	0.690	0.259	0.265	0.333	0.308
29	Transfor	0.314	0.647	0.155	-0.020	-0.064	0.288
8		0.318	0.595	0.402	0.166	-0.116	0.034
32	Complex	0.361	0.590	0.252	0.096	-0.435	0.312
12		0.350	0.547	0.352	0.257	0.233	0.109
24		0.432	0.457	0.250	0.284	0.115	0.006
23		0.330	0.156	0.663	0.224	-0.080	0.194
6*		0.199	0.202	0.635*	-0.015	0.147	0.093
21		0.199	0.141	0.611	-0.004	-0.020	0.107
10		0.178	-0.301	0.601	0.299	0.297	-0.216
9 ^{ab}		0.292	-0.250	-0.506^{ab}	0.496^a	0.143	0.292
28 ^{ab}	Pubadmin	0.335	0.067	0.447^a	0.158	-0.555^{ab}	-0.094
30	CRM	-0.187	0.057	0.102	0.758	-0.140	-0.022
4*		0.405	0.270	0.110	0.572*	0.291	-0.043
7		0.220	0.369	0.178	0.458	0.442	0.321
15*		0.359	-0.057	0.191	0.008	0.701*	0.062
17*		0.053	-0.001	-0.128	-0.039	0.083	0.849*
1		0.330	0.253	0.293	0.105	-0.132	0.637
18		0.314	0.196	0.299	0.159	0.157	0.559
3 ^c		0.397	0.128	0.069	0.394	-0.069	0.163
% Explained Variance		18	15	10	9	6	8
Cumulative Explained Variance		18	33	43	52	58	66

Note. Statistically significant sort loadings > .446 shown in bold. Sorts by participants shown in black; theoretical sorts shown in red. $p < 0.01$, Critical Value = 0.446; Critical Value = $\frac{1}{\sqrt{N}} \times Z\text{Score Value for } p < 0.01 =$

$$\frac{1}{\sqrt{48}} \times 3.09 = .446$$

* = significantly loading sorts for each factor that are also non-significantly loading on other factors; sorters who validated factor descriptions and labels

^a = confounded loadings on two factors

^b = bi-polar loading

^c = non-significantly loading sort

The six factors together explain 66% of the variance among the Q-sorts, which is a greater level of variance explained than with fewer factors retained. In addition, the explained variance for any one factor did not drop significantly as a greater number of factors were retained for a mathematical rotation, indicating that added factors were in fact distinct and not contributing to explained variance in the factor solution with fewer factors retained. In this study, the explained variance for Factors A, B, and C fluctuated by less than two percentage points as the number of factors retained increased.

The differences in Q score values for the same statements across Q-sorts were also examined. A high frequency of statements with similar Q scores indicates that two or more factors are similar. Table 9 displays the Q-sort ranking and Z-Score for each statement in each factor. The Z-Scores are the normalized factor scores for each statement that allow cross-factor comparison of the statement rankings.

Table 9

Q-Set (Statements) Factor Z-Scores with Corresponding Statement Ranks within the Factor

	Statement (truncated) Effective fire department leaders should...	Factor A		Factor B		Factor C		Factor D		Factor E		Factor F	
		Z-Score	Rank	Z-Score	Rank	Z-Score	Rank	Z-Score	Rank	Z-Score	Rank	Z-Score	Rank
1	actively explore ways to partner with other public entities	0.08	25	0.07	27	1.13	7	-0.3	32	-1.23	41	0.19	21
2	create freedom for free thinkers	0.68	11	1.31	4	0.06	24	0.01	26	-0.45	33	-0.47	34
3	embrace opportunities to do more functions that contribute to the public safety	0.44	17	-0.4	33	1.89	1	-0.12	28	-0.29	31	0.65	13
4	decisive, with or without complete information	-1.14	41	-1.09	38	-1.21	44	-0.44	33	0.55	16	0.37	18
5	pursue grants to stay current and progressive	-0.24	30	0.09	25	-0.19	29	0.27	23	0.31	19	0.08	26
6	know that human resource management is critical for the organization	1.09	7	0.02	28	0.24	21	0.47	15	1.75	3	1.25	6
7	vigilant steward	0.56	16	-1.1	39	0.8	9	0.99	6	-0.1	26	-1.51	43
8	Empathetic	0.13	24	0.65	15	0.18	22	-0.46	34	0.18	20	0.98	8
9	do not accept excuses	-1.79	46	-1.51	45	-2.36	48	0.27	22	-0.63	38	1.13	7
10	form cross-division teams to address issues and challenges	-0.01	27	0.38	20	0.97	8	0.84	9	-1.99	47	0.41	17

(continued)

	Statement (truncated) Effective fire department leaders should...	Factor A		Factor B		Factor C		Factor D		Factor E		Factor F	
		Z-Score	Rank	Z-Score	Rank	Z-Score	Rank	Z-Score	Rank	Z-Score	Rank	Z-Score	Rank
11	seek informal opportunities to interact	0.43	18	0.96	10	-0.54	32	-0.74	40	1.18	8	1.37	5
12	own the department's problems	1.6	3	-0.92	35	0.78	11	1.95	2	1.31	5	2.13	1
13	give extra attention to community involvement	0.23	22	-0.21	31	0.38	18	-0.77	41	0.73	12	0.08	26
14	develop metrics that are meaningful for different stakeholders	0.15	23	0.54	16	0.58	13	-0.55	36	-0.5	36	0.71	12
15	future-oriented	1.64	2	0.3	22	-0.59	34	1.22	4	0.68	14	0.58	16
16	act like a corporate CEO	-1.25	42	-1.41	44	-1.14	43	-2.04	46	-1.39	45	0.17	22
17	hold their cards close until the support is there	-1.65	45	-1.32	43	-0.73	39	-2.31	48	-1.62	46	-1.01	38
18	Be smart about how to influence different individuals	1.08	8	0.28	23	0.55	14	-1.41	43	0.13	21	1.71	2
19	let go of ego	1.05	9	0.91	11	1.77	2	-0.2	30	1.12	9	0.26	19
20	never satisfied with the status quo	0.72	10	0.53	17	-0.61	35	0.47	16	0	24	1.49	4
21	do the tough stuff first	-0.65	35	-1.14	42	-0.69	38	-0.55	37	0.73	12	-0.46	33
22	keep their personalities out of their leadership	-0.78	37	-0.65	34	0.33	19	-0.71	39	-1.31	44	-1.6	47
23	value objectivity	0.3	19	0.49	18	0.17	23	0.8	10	0.31	19	0.72	11
24	give away power	-0.63	34	1.5	2	-0.25	30	-0.28	31	-0.26	30	-0.11	29
25	share lots of information for the greater good	0.25	21	1.34	3	-0.08	27	1.21	5	-2.12	48	0.83	9
26	facilitate organizational dialogue	1.52	4	1.19	6	0.43	17	0.43	17	-1.31	44	0.59	15
27	get things done collaboratively	0.66	12	0.77	12	1.73	4	0.76	11	-0.23	29	0.8	10
28	actively and regularly critique all practices and processes	-0.19	29	-1.05	37	1.17	6	0.19	24	0.63	15	-1.18	40
29	Be wary of too much "buddy/buddy" behavior	-0.93	39	-1.14	41	0.79	10	-1.84	45	0.05	22	-1.9	48
30	guard against the need to always agree	0.62	14	0.34	21	-0.93	42	-1.51	44	0	24	-1.58	45
31	tow the clear, hard line	-1.89	47	-1.95	48	-1.66	45	-0.99	42	1.85	1	-0.17	30
32	stalwart demeanor	-0.95	40	-1.13	40	-2.24	47	-0.01	27	-1.26	42	-1.43	42
33	the most important work is risk reduction	-0.36	32	-0.2	29	1.77	3	0.87	8	-0.99	39	0.2	20
34	facilitate career development for all members	0.58	15	0.76	13	-0.57	33	0.39	20	1.8	2	1.59	3
35	believe everyone brings something to the table	1.4	5	1.81	1	0.26	20	2.12	1	1.2	7	-1.52	44
36	decentralize authority	-0.65	36	1.17	7	0.02	25	0.91	7	-0.58	37	-0.03	28
37	pick the spots where to bend for the sake of morale	-0.79	38	0.38	19	-0.68	37	-0.7	38	0.73	12	-1.6	47
38	strong disciplinarian	-2.01	48	-1.82	47	-1.72	46	-0.51	35	-0.03	25	-0.41	32
39	believe that good SOPs run the department	-1.36	44	-1.74	46	-0.81	41	0.47	14	1.49	4	-1.27	41
40	create a safe zone for members to question processes and decisions	-0.16	28	1.11	8	-0.06	26	0.28	21	1.26	6	0.11	24
41	operate at the 30,000' level	-0.39	33	-0.94	36	-0.19	28	0.4	19	-1.18	40	-0.5	35
42	understand the whole system and consider the ripple effects of every action	2.04	1	0.28	24	0.63	12	0.42	18	-0.42	32	0.16	23
43	seek to be politically astute	0.27	20	-0.32	32	0.46	16	-2.09	47	0.71	13	0.07	27

(continued)

	Statement (truncated) Effective fire department leaders should...	Factor A		Factor B		Factor C		Factor D		Factor E		Factor F	
		Z-Score	Rank	Z-Score	Rank	Z-Score	Rank	Z-Score	Rank	Z-Score	Rank	Z-Score	Rank
44	measure success through the betterment of department members	0.65	13	-0.2	30	-0.63	36	0.52	13	0.45	17	-0.65	37
45	value equity give members time to correct issues on their own	-0.26	31	0.74	14	1.35	5	0.12	25	-0.5	36	-1.04	39
46	democratic	0.01	26	1.04	9	-0.81	40	1.7	3	-0.18	28	-0.2	31
47	break barriers to change	-1.31	43	0.09	26	-0.29	31	-0.19	29	-0.13	27	-0.6	36
48		1.25	6	1.19	5	0.53	15	0.63	12	-0.5	36	0.61	14

Table 10 shows the Q-sort values for each statement, corresponding to the column of the formboard where it would be placed, in order from consensus to distinguishing statements as defined by the range and difference of the Q-sort values across the factors. Only two statements are consensus statements, 5 and 23, with a small range from the low to high Z-score of -1 to 1 and 0 to 2 respectively, and a low spread of only 2 for both. For all the other statements, the perspectives represented by the factors have a greater degree of variation in the Q-sort values as indicated by a larger range and greater spread.

Table 10

Factor Q-Sort Values for Statements, Sorted from Consensus to Distinguishing

No.	Statement (truncated) Effective fire department leaders should...	Statement Q-Score for Each Factor						Range	Spread
		A	B	C	D	E	F		
5	pursue grants to stay current and progressive	-1	0	-1	0	1	0	-1 to 1	2
23	value objectivity	1	1	0	2	1	2	0 to 2	2
17	hold their cards close until the support is there	-4	-3	-2	-5	-4	-2	-5 to -2	3
40	create safe zone for members to question processes, decisions	0	3	0	0	3	0	0 to 3	3
47	are democratic	-3	0	-1	-1	0	-2	-3 to 0	3
1	actively explore partnering with other public entities	0	0	3	-1	-3	0	-1 to 3	4
6	know human resource management is critical	3	0	0	1	4	3	0 to 4	4
8	have empathy	0	1	0	-1	1	3	-1 to 3	4
14	develop meaningful metrics for different stakeholders	0	1	2	-2	-2	2	-2 to 2	4
16	act like a corporate CEO	-3	-4	-3	-4	-4	0	-4 to 0	4
41	operate at the 30,000' level	-1	-2	0	1	-3	-2	-3 to 1	4
44	measure success through the betterment of department members	2	-1	-2	2	1	-2	-2 to 2	4
2	create freedom for free thinkers	2	4	0	0	-1	-1	-1 to 4	5
4	are decisive, with or without complete information	-3	-2	-4	-1	1	1	-4 to 1	5
13	give extra attention to community involvement	0	-1	1	-3	2	0	-3 to 2	5
21	do the tough stuff first	-2	-3	-2	-2	2	-1	-3 to 2	5
27	get things done collaboratively	2	2	4	2	-1	2	-1 to 4	5
32	have stalwart demeanor	-3	-3	-5	0	-3	-3	-5 to 0	5
36	decentralize authority	-2	3	0	3	-2	0	-2 to 3	5

(continued)

No.	Statement (truncated) Effective fire department leaders should...	Statement Q-Score for Each Factor						Range	Spread
		A	B	C	D	E	F		
38	are strong disciplinarians	-5	-5	-4	-2	0	-1	-5 to 0	5
46	give members time to correct issues on their own	0	3	-3	4	0	-1	-1 to 4	5
3	embrace opportunities to do more for the public safety	1	-1	5	0	-1	2	-1 to 5	6
7	are vigilant stewards	1	-2	3	3	0	-3	-3 to 3	6
15	are future-oriented	5	0	-1	4	2	1	-1 to 5	6
19	let go of ego	3	2	5	-1	3	1	-1 to 5	6
20	are never satisfied with the status quo	2	1	-2	1	0	4	-2 to 4	6
22	keep their personalities out of their leadership	-2	-1	1	-2	-4	-5	-5 to 1	6
24	give away power	-1	5	-1	-1	-1	-1	-1 to 5	6
28	actively, regularly critique all practices, processes	-1	-2	3	0	1	-3	-3 to 3	6
30	guard against the need to always agree	2	0	-3	-4	0	-4	-4 to 2	6
33	most important work is risk reduction	-1	-1	4	3	-2	1	-2 to 4	6
34	facilitate career development for all members	1	2	-1	1	5	4	-1 to 5	6
42	understand whole system, consider ripple effects of actions	5	0	2	1	-1	0	-1 to 5	6
45	value equity	-1	2	4	0	-2	-2	-2 to 4	6
48	break barriers to change	3	4	1	2	-2	2	-2 to 4	6
11	seek informal opportunities to interact	1	2	-1	-3	3	4	-3 to 4	7
12	own the department's problems	4	-2	2	5	4	5	-2 to 5	7
29	are wary of too much "buddy/buddy" behavior	-2	-3	2	-4	0	-5	-5 to 2	7
37	pick the spots where to bend for the sake of morale	-2	1	-2	-2	2	-5	-5 to 2	7
43	seek to be politically astute	1	-1	1	-5	2	0	-5 to 2	7
9	do not accept excuses	-4	-4	-5	0	-2	3	-5 to 3	8
10	form cross-division teams to address issues, challenges	0	1	3	3	-5	1	-5 to 3	8
18	be smart about how to influence individuals	3	0	2	-3	0	5	-3 to 5	8
26	facilitate organizational dialogue	4	3	1	1	-4	1	-4 to 4	8
39	believe that good SOPs run the department	-4	-4	-3	2	4	-3	-4 to 4	8
25	share lots of information for the greater good	0	4	0	4	-5	3	-5 to 4	9
35	believe everyone brings something to the table	4	5	1	5	3	-4	-4 to 5	9
31	tow the clear, hard line	-5	-5	-4	-3	5	-1	-5 to 5	10

Factor Correlations

In the six-factor solution, Factor A is strongly correlated with Factor B (0.602) and Factor C (0.519) indicating that there are similarities between Factor A and Factor B, and similarities between Factor A and Factor C. There is not as strong a correlation between Factors B and C (.0364), indicating that they are distinct from each other and the ways that each is similar to Factor A are distinct. The remaining correlations between factor scores range from 0.086 to 0.436, indicating that they are not as similar. All the factor correlations are displayed in Table 11.

Table 11

Correlations Between Factor Scores

Factor	A	B	C	D	E	F
A	1.000	0.602	0.519	0.436	0.159	0.394
B		1.000	0.364	0.374	-.038	0.259
C			1.000	0.183	-.086	0.114
D				1.000	0.064	0.214
E					1.000	0.126
F						1.000
Explained Variance	18%	15%	10%	9%	6%	8%
Cumulative Explained Variance	18%	33%	43%	52%	58%	66%

Manual Rotation

Watts and Stenner (2012) explained that “access” to the subject matter of the Q-sorts is limited by the “number of viewpoints made available to us by the extracted factors,” and there is no special privilege for mathematically extracted factors in Q methodology (Watts & Stenner, 2012, p. 117). The manual rotation option allows researchers to explore additional vantage points for observing the Q-sort data and ensuring that the factors reported offer the most informative explanations of the viewpoints on the topic. Only one manual rotation that I performed showed significant change in the relationships among the sorts from the new vantage point. The adaptive leadership theoretical sort (Sort 27) was confounded on Factors A and D, as shown graphically in Figure 10. The problem represented in the diagram and by the significant loading on both Factors A and D is that neither of these factors fully represents this theoretical Q-sort. Watts and Stenner describe this situation as a *blurred image* for a Sort or group of Sorts that may be significantly loading on two factors. Manually rotating the factor axes brings the image of this theoretical sort perspective into focus, as shown in Figure 11 with Sort 27 on the positive end of the Factor A¹. From the vantage point of Sort 27, Sorts 2, 3, 4, 7, and 9 are also more significantly loaded on Factor A¹ than with the unrotated Factors A or D. Factor D¹ now has Sorts on both the positive

and the negative ends of it and is thus a bi-polar factor, meaning that it actually represents two opposing viewpoints. Both of these rotated Factors, A^1 and D^1 , will be described in detail along with Factors A, B, C, D, E, and F in the following section.

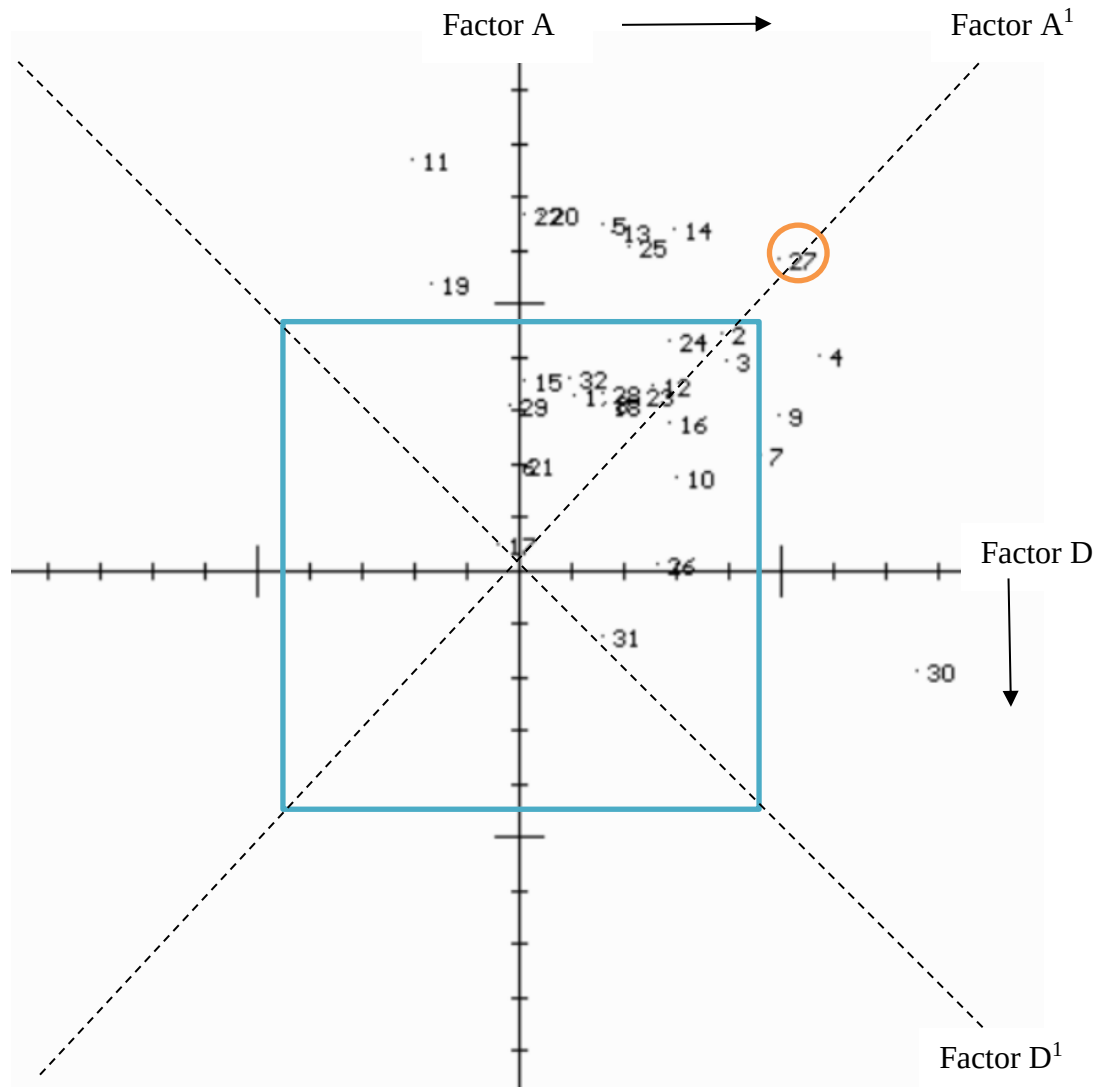


Figure 10. The relative positions of the Q-sorts in relation to Factor A and Factor D. The orange circle indicates the position of Q-sort 27. The blue square indicates the approximate boundary between insignificantly loading Q-sorts (inside the square, < 0.446 on the factor axes) and significantly loading Q-sorts (outside the square, > 0.446 on the factor axes). The dotted lines represent the factor axes rotated orthogonally and clockwise by 40° to align Factor A^1 with Q-sort 27.

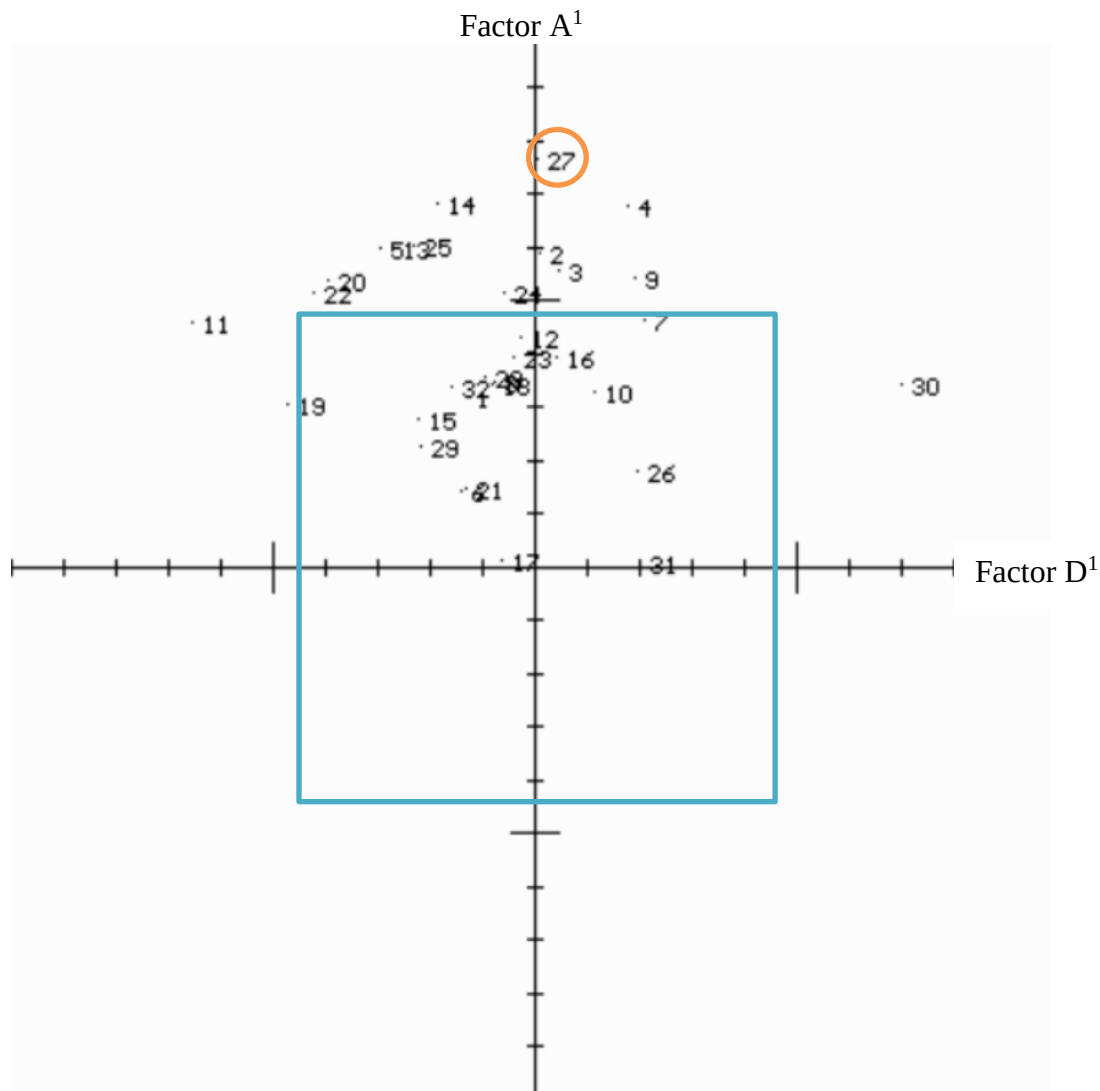


Figure 11. The relative positions of the Q-sorts in relation to Factor A¹ and Factor D¹ following the 40° orthogonal rotation of the Factors to Q-sort 27, indicated by the orange circle. The blue square indicates the approximate boundary between insignificantly loading Q-sorts (inside the square, < 0.446 on the factor axes) and significantly loading Q-sorts (outside the square, > 0.446 on the factor axes).

Model Factors: Shared Perspectives on USFES Organizational Leadership

As described above, it was determined that six factors retained from the PCA and Varimax rotation of the data provide detail on six perspectives on USFES organizational leadership among the 25 participants completing the Q-sorts. A manual rotation of two factors

provided data to analyze for two additional perspectives. There are two Q-sorts that bear analysis independently because they are idiosyncratic and represent separate, distinct perspectives. This section of the chapter will describe these different perspectives on organizational leadership in USFES departments.

Each perspective has a short label and description, appearing below as headings for each perspective and the first paragraph following each perspective's heading. The label and description were developed from analysis of the model factor array, which can be thought of as an average Q-sort arrangement of the statements based upon those participant Q-sorts that loaded significantly on the factor. The software displayed the model factor array as a listing of the statements in descending order by Z-scores (normalized Q-scores), and it is important to note those statements that are highly salient on both ends of the list – salient in the positive “most like my view” end of the model array, and salient in the negative “most unlike my view” end of the model array. It can also shed light on the unique nature of a perspective to note the statements in between that have the lowest Z-scores indicating that they are statements for the model factor array that are not salient.

The analysis of each model factor array also included attention to the factor's distinguishing statements, those statements that have statistically significant different Z-scores from the Z-scores for the same statements in the other model factor arrays, and the listings of the differences between each pair of factors. Reference to the participant interviews and post-sort interviews for those participants' Q-sorts that load significantly on each factor also informed the development of the label and description for each perspective. As a final step in the development of the labels and descriptions, a participant whose Q-sort was one of the highest significantly loading sorts for the factor and that had the lowest loadings on all the other factors reviewed and

validated the label and description. Any feedback contributed to the final versions of labels and descriptions.

Perspective A – Leading as Facilitation for Increasing Individual and Organizational Capacities to Meet Dynamic Challenges

“Leaders improve the department one person at a time” might be the motto for those who share this perspective on USFES organizational leadership. In this perspective, leaders are acutely attuned to the individuals of the department, their interactions across the department, as well as dynamic influences on the department from the community and beyond. Attention to the whole system is important to identifying problems and challenges to overcome and opportunities for learning, growth, and improvement. Leaders express care and accountability for the people who must do the work. Leaders must take charge of important decisions with a long view toward the future implications for the members of the department and the community, and communicate effectively about the big picture to provide problem-solving guidance.”

When explaining the placement of statements in the formboard during post-sort interviews, participants whose sorts loaded significantly on Factor A said a number of things that support this label and description:

Sorter 11 – “Ineffective leaders avoid conflicts or arguments or don’t want to explore other perspectives; effective leaders consider others’ perspectives and priorities and make sure there is a good process for decisions.”

Sorter 5 – “Your people are the ones who get the work done so you have to take care of them in order to complete our mission.”

Sorter 20 – “Leadership has to facilitate connections and dialogue across parts of the organization to guide toward improved performance, learning and growth as an

organization.” “Leaders are expected at all levels to be vigilant stewards. It means ‘to care for.’ Stewardship is at the heart of the mission of public safety – caring for the people in the organization and caring for the people in the community.”

Sorter 22 – “Everyone in the department holds a key to service delivery. Leaders need the people skills to motivate, develop, understand, capitalize on strengths, help people grow beyond limitations, and communicate with all of them. You have to really care; people need to know that you, and they, are more than a title or rank. A Chief has to be real and realize that it is one role on the team, one piece of the puzzle, identifying the big issues facing us and lean into them, and realize that the people will work together to address challenges and get things done.”

The model factor score array also explains elements of the Perspective A description. Bold font in Table 12 indicates those statements of high salience in the Factor A model array in both the positive, “most like my view” and the negative, “most unlike my view” sense. The positive Z-scores for statements 42, 15, and 48 support the elements in the description related to attention to the system within which the Department exists, and related to improving for the future. Statement 42 is a distinguishing statement for Factor A, as shown in Table 13, further supporting the description’s emphasis on attention to the system. The positive Z-Scores for statements 12 and 26 relate to problem-solving. Statements 35, 6, and 18 support the emphasis in the description on attention to and engaging individuals within the department because of their positive Z-scores, and the negative Z-scores for statements 29, 16, and 17 also support the emphasis on engaging with individuals. Statement 18 is another distinguishing statement as shown in Table 13, providing additional support for the emphasis on engaging individuals. In this perspective, department leaders are in charge and set the priorities without being

authoritarian, as evidenced by the negative Z-scores for statements 47, 31, and 38, as well as the neutral Z-score for statements 46 and 10.

Table 12

Descending Factor Score Array for Factor A

No.	Statement (truncated)	Z-Score
42	understand whole system, consider ripple effects of actions	2.038
15	are future-oriented	1.639
12	own the department's problems	1.601
26	facilitate organizational dialogue	1.521
35	believe everyone brings something to the table	1.404
48	break barriers to change	1.246
6	know human resource management is critical	1.093
18	be smart about how to influence individuals	1.077
19	let go of ego	1.050
20	are never satisfied with the status quo	0.716
2	create freedom for free thinkers	0.685
27	get things done collaboratively	0.655
44	measure success through the betterment of department members	0.651
30	guard against the need to always agree	0.617
34	facilitate career development for all members	0.584
7	are vigilant stewards	0.563
3	embrace opportunities to do more for the public safety	0.437
11	seek informal opportunities to interact	0.434
23	value objectivity	0.297
43	seek to be politically astute	0.265
25	share lots of information for the greater good	0.253
13	give extra attention to community involvement	0.227
14	develop meaningful metrics for different stakeholders	0.154
8	have empathy	0.125
1	actively explore partnering with other public entities	0.079
46	give members time to correct issues on their own	0.013
10	form cross-division teams to address issues, challenges	-0.009
40	create safe zone for members to question processes, decisions	-0.165
28	actively, regularly critique all practices, processes	-0.192
5	pursue grants to stay current and progressive	-0.245
45	value equity	-0.258
33	most important work is risk reduction	-0.358
41	operate at the 30,000' level	-0.393
24	give away power	-0.635
21	do the tough stuff first	-0.651
36	decentralize authority	-0.654
22	keep their personalities out of their leadership	-0.782
37	pick the spots where to bend for the sake of morale	-0.794
29	are wary of too much "buddy/buddy" behavior	-0.925

(continued)

No.	Statement (truncated)	Z-Score
32	have stalwart demeanor	-0.950
4	are decisive, with or without complete information	-1.141
16	act like a corporate CEO	-1.250
47	are democratic	-1.311
39	believe that good SOPs run the department	-1.355
17	hold their cards close until the support is there	-1.654
9	do not accept excuses	-1.794
31	tow the clear, hard line	-1.893
38	are strong disciplinarians	-2.013

Note. Bold font indicates statements of high salience for the factor array

Table 13

Distinguishing Statements for Factor A

No.	Statement	Factor A		Factor B		Factor C		Factor D		Factor E		Factor F	
		Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score
42	understand whole system, consider ripple effects of actions	5	2.04*	0	0.28	2	0.63	1	0.42	-1	-0.42	0	0.16
18	be smart about how to influence individuals	3	1.08	0	0.28	2	0.55	-3	-1.41	0	0.13	5	1.71
2	create freedom for free thinkers	2	0.68	4	1.31	0	0.06	0	0.01	-1	-0.45	-1	-0.47
11	seek informal opportunities to interact	1	0.43	2	0.96	-1	-0.54	-3	-0.74	3	1.18	4	1.37
47	break barriers to change	-3	-1.31	0	0.09	-1	-0.29	-1	-0.19	0	-0.13	-2	-0.60

Note. $p < 0.05$; Asterisk (*) Indicates Significance at $p < 0.01$

The demographic information about those participants whose Q-sorts loaded significantly on Factor A is shown in Table 14. The nine participants come from all four regions of the U.S. None of the participants with Q-sorts that loaded significantly on Factor A come from rural USFES departments, but do come from all three types of departments, career, volunteer and combination. Department size ranges from 20 members to 458 members. Rank ranges from Captain to Chief. Years of service ranges from eight years to 36 years. The highest level of education of the participants includes Associate's, Bachelor's, and Master's degree levels, and three of the nine participants with Q-sorts loading significantly on this Factor completed the Executive Fire Officer Program at the National Fire Academy. All are males between the age of 24 and 59.

Table 14

Demographics for the Q-Sorters with Significant Loading on Factor A

	Factor	Region	Comm Type	Dept Type	Dept Size	Rank	Service	Education	EFO	Gender	Age
2	AB	East	rural	Volunteer	70	Lieut.	8	AA		M	25
5	A	East	urban	Career	20	Chief	16	BA, MPA	Y	M	40
11	A	Midwest	suburban	combination	32	Captain	17	BS		M	41
13	A	Midwest	suburban	combination	454	Captain	9	MPA		M	37
14	A	South	suburban	Career	82	Asst Chief	22	BS		M	38
19	A	West	urban	Career	458	Asst Chief	22	BS, 2 AAs		M	50
20	A	West	urban	Career	150	Dep. Chief	32	MA	Y	M	56
22	A	West	urban	Career	200	Chief	36	MS	Y	M	58
25	A	West	urban	combination	111	Captain	11	MS		M	35

Perspective B – Leadership Success Comes Through Empowering Others

This perspective on effective fire department organizational leadership considers it important for leaders to create a positive professional and social/emotional climate and involve members in initiatives that solve challenges and enable the department to improve performance. Leaders provide information, listen, and delegate and empower to engage members to think critically and dialogue about the challenges. In this perspective, maintaining the status quo in the department would be like falling behind; continuous improvement through empowerment of individual members is the objective.

Support for this description of the perspective comes from the interviews with those participants whose sorts load significantly on the factor:

Sorter 8: “Performance comes from bringing people together in the right environment.”

Sorter 12: “Leaders need to take the time to know people – their needs, skill sets – and be open to where people are at the time you are interacting.” “[Fire department leaders] should not act like corporate CEOs, because they are only externally-focused, on projections, customers, shareholders, for example, but in the fire service we have to have internal focus because WE are as important as the people we serve; CEOs are about money, not people.”

Sorter 16: “Sharing power, for decisions and working on solutions, creates engagement and cooperation. There shouldn’t be selfishness in the fire department because it prevents working together for the citizens.”

Sorter 2: “I hate it when people say ‘we’ve always done it that way.’ If leadership doesn’t think there are problems to fix and changes to make, they are ‘in a crystal palace’ thinking everything is great and are not doing their jobs. They aren’t involved and aware enough.”

The emphasis on a positive professional and social environment is also supported by the positive Z-scores in the model factor array for Factor B, shown in Table 15, for statements 35, 24, 25, 2, 26, 11, and the negative Z-scores for statements 9, 31, 38. The emphasis on involving members in solving challenges and improving performance is related to the positive Z-scores for statements 36, 48, 40, and the negative Z-scores for statements 17, 16, 39. Both of these things that are emphasized in this perspective are supported by the fact that statement 24, related to effective leaders giving away power, is a distinguishing statement at the $p < .01$ level as indicated in Table 16.

Table 15

Descending Factor Score Array for Factor B

No.	Statement (truncated)	Z-Score
35	believe everyone brings something to the table	1.813
24	give away power	1.497
25	share lots of information for the greater good	1.337
2	create freedom for free thinkers	1.310
48	break barriers to change	1.188
26	facilitate organizational dialogue	1.187
36	decentralize authority	1.166
40	create safe zone for members to question processes, decisions	1.110
46	give members time to correct issues on their own	1.041
11	seek informal opportunities to interact	0.956
19	let go of ego	0.906
27	get things done collaboratively	0.766
34	facilitate career development for all members	0.758
45	value equity	0.738
8	have empathy	0.654
14	develop meaningful metrics for different stakeholders	0.544
20	are never satisfied with the status quo	0.534
23	value objectivity	0.492
37	pick the spots where to bend for the sake of morale	0.385
10	form cross-division teams to address issues, challenges	0.384
30	guard against the need to always agree	0.344
15	are future-oriented	0.303
18	be smart about how to influence individuals	0.281
42	understand whole system, consider ripple effects of actions	0.281
5	pursue grants to stay current and progressive	0.093
47	are democratic	0.088
1	actively explore partnering with other public entities	0.074
6	know human resource management is critical	0.019
33	most important work is risk reduction	-0.200
44	measure success through the betterment of department members	-0.203
13	give extra attention to community involvement	-0.210
43	seek to be politically astute	-0.322
3	embrace opportunities to do more for the public safety	-0.397
22	keep their personalities out of their leadership	-0.646
12	own the department's problems	-0.917
41	operate at the 30,000' level	-0.944
28	actively, regularly critique all practices, processes	-1.051
4	are decisive, with or without complete information	-1.089
7	are vigilant stewards	-1.103
32	have stalwart demeanor	-1.132
29	are wary of too much "buddy/buddy" behavior	-1.137
21	do the tough stuff first	-1.140
17	hold their cards close until the support is there	-1.321
16	act like a corporate CEO	-1.410
9	do not accept excuses	-1.515
39	believe that good SOPs run the department	-1.744
38	are strong disciplinarians	-1.823
31	tow the clear, hard line	-1.946

Note. Bold font indicates statements of high salience for the factor array

Table 16

Distinguishing Statements for Factor B

No.	Statement	Factor A		Factor B		Factor C		Factor D		Factor E		Factor F	
		Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score
24	give away power	-1	-0.63	5	1.50*	-1	-0.25	-1	-0.28	-1	-0.26	-1	-0.11
2	create freedom for free thinkers	2	0.68	4	1.31*	0	0.06	0	0.01	-1	-0.45	-1	-0.47
46	give members time to correct issues on their own	0	0.01	3	1.04	-3	-0.81	4	1.70	0	-0.18	-1	-0.20
45	value equity	-1	-0.26	2	0.74	4	1.35	0	0.12	-2	-0.50	-2	-1.04
12	own the department's problems	4	1.60	-2	0.92*	2	0.78	5	1.95	4	1.31	5	2.13

Note. $p < .05$; Asterisk (*) Indicates Significance at $p < .01$

The demographic information about those participants whose Q-sorts loaded significantly on Factor B is shown in Table 17. The four participants come from two regions of the U.S, the Midwest and West. None of the participants with Q-sorts that loaded significantly on Factor B come from rural USFES departments, and none come from a combination type of department. Department size ranges from 35 members to 1200 members. Rank ranges from Lieutenant to Deputy Chief. Years of service ranges from five years to 36 years. The highest level of education of the participants includes Bachelor's, Master's degree, and one participant with a Ph.D. who has also completed the Executive Fire Officer Program. Two are females and two are males. The ages of the participants range from 42 to 59.

Table 17

Demographics for the Q-Sorters with Significant Loading on Factor B

Factor	Region	Comm Type	Dept Type	Dept Size	Rank	Service	Education	EFO	Gender	Age	
8	B	Midwest	suburban	career	135	Dep. Chief	24	MPA, Ph.D.	Y	M	47
12	B	Midwest	suburban	volunteer	35	Lieut.	5	MA, MS		F	47
16	B	South	urban	Career	650	Bat. Chief	20	BS		M	42
24	B	West	urban	Career	1200	Asst. Chief -	36	AS, BA		F	59

Perspective C – Leading for Mission-driven Consistency to Advance the Public Safety

Description: This perspective on effective USFES organizational leadership prioritizes the focus on the common and greater good in service to the community. Leaders support collaboration and teamwork within the organization, and develop external relationships across stakeholders to build support, address issues, and improve or expand service. The department represents as well as protects the community, and department leaders therefore value equity and consistency in the treatment of people both inside and outside of the organization.

When explaining the placement of statements in the formboard during post-sort interviews, participants whose sorts loaded significantly on Factor C said a number of things that support this label and description:

Sorter 6 – “Personality or background or whether a leader likes or dislikes someone should not exclude anyone or minimize their contributions.” “If we are really doing our job to serve the public, we should be looking for ways to keep the public as safe as possible very actively. We have to stay relevant.”

Sorter 10 – “Leadership has to cut down on the risks to personnel, and the public. If we teach the public to reduce their risks, we reduce our risks too.”

Sorter 21 – “Effective leaders set priorities according to the public interest.” “Camaraderie is NOT a sign of effective leadership; looking out for each other (buddy-buddy behavior) does not breed excellence; there has to be accountability to the public, not just to each other.”

Sorter 23 – “Ultimately, we are here to serve the community.”

Emphasis on community is supported by positive Z-scores for statements 3, 33, and 1, shown in Table 18. Statement 3 about embracing opportunities to do more for the public safety,

and statement 1 about active exploration of partnerships with other public entities are also distinguishing statements for this factor at the $p < .01$ level, displayed in Table 19. Emphasis on teamwork is supported by the high positive salience of statements 27 and 10. Emphasis on representation supported by high positive salience of statement 45, another distinguishing statement for Factor C, and statement 29.

Table 18

Descending Factor Score Array for Factor C

No.	Statement (truncated)	Z-Score
3	embrace opportunities to do more for the public safety	1.894
19	let go of ego	1.773
33	most important work is risk reduction	1.772
27	get things done collaboratively	1.735
45	value equity	1.348
28	actively, regularly critique all practices, processes	1.174
1	actively explore partnering with other public entities	1.132
10	form cross-division teams to address issues, challenges	0.973
7	are vigilant stewards	0.801
29	are wary of too much "buddy/buddy" behavior	0.794
12	own the department's problems	0.776
42	understand whole system, consider ripple effects of actions	0.626
14	develop meaningful metrics for different stakeholders	0.581
18	be smart about how to influence individuals	0.551
48	break barriers to change	0.527
43	seek to be politically astute	0.465
26	facilitate organizational dialogue	0.430
13	give extra attention to community involvement	0.381
22	keep their personalities out of their leadership	0.328
35	believe everyone brings something to the table	0.257
6	know human resource management is critical	0.242
8	have empathy	0.181
23	value objectivity	0.169
2	create freedom for free thinkers	0.059
36	decentralize authority	0.022
40	create safe zone for members to question processes, decisions	-0.060
25	share lots of information for the greater good	-0.075
41	operate at the 30,000' level	-0.190
5	pursue grants to stay current and progressive	-0.194
24	give away power	-0.252
47	are democratic	-0.291
11	seek informal opportunities to interact	-0.541
34	facilitate career development for all members	-0.575
15	are future-oriented	-0.588
20	are never satisfied with the status quo	-0.606
44	measure success through the betterment of department members	-0.632
37	pick the spots where to bend for the sake of morale	-0.681

(continued)

No.	Statement (truncated)	Z-Score
21	do the tough stuff first	-0.694
17	hold their cards close until the support is there	-0.731
46	give members time to correct issues on their own	-0.807
39	believe that good SOPs run the department	-0.812
30	guard against the need to always agree	-0.931
16	act like a corporate CEO	-1.143
4	are decisive, with or without complete information	-1.206
31	tow the clear, hard line	-1.663
38	are strong disciplinarians	-1.722
32	have stalwart demeanor	-2.236
9	do not accept excuses	-2.357

Note. Bold font indicates statements of high salience for the factor array

Table 19

Distinguishing Statements for Factor C

No.	Statement	Factor A		Factor B		Factor C		Factor D		Factor E		Factor F	
		Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score
3	embrace opportunities to do more for the public safety	1	0.44	-1	-0.40	5	1.89*	0	-0.12	-1	-0.29	2	0.65
33	most important work is risk reduction	-1	-0.36	-1	-0.20	4	1.77*	3	0.87	-2	-0.99	1	0.20
27	get things done collaboratively	2	0.66	2	0.77	4	1.73*	2	0.76	-1	-0.23	2	0.80
45	value equity	-1	-0.26	2	0.74	4	1.35	0	0.12	-2	-0.50	-2	-1.04
1	actively explore partnering with other public entities	0	0.08	0	0.07	3	1.13*	-1	-0.30	-3	-1.23	0	0.19
22	keep their personalities out of their leadership	-2	-0.78	-1	-0.65	1	0.33*	-2	-0.71	-4	-1.31	-5	-1.60
35	believe everyone brings something to the table	4	1.4	5	1.81	1	0.26	5	2.12	3	1.20	-4	-1.52
34	facilitate career development for all members	1	0.58	2	0.76	-1	0.57*	1	0.39	5	1.80	4	1.59
15	are future-oriented	5	1.64	0	0.30	-1	0.59*	4	1.22	2	0.68	1	0.58
32	have stalwart demeanor	-3	-0.95	-3	-1.13	-5	-2.24	0	-0.01	-3	-1.26	-3	-1.43
9	do not accept excuses	-4	-1.79	-4	-1.51	-5	-2.36	0	0.27	-2	-0.63	3	1.13

Note. $p < .05$; Asterisk (*) Indicates Significance at $p < .01$

The demographic information about those participants whose Q-sorts loaded significantly on Factor C is shown in Table 20. The five participants come from three regions of the U.S., all but the South. They represent all three of the community types, urban, suburban, and rural, and from all three department types, career, volunteer, and combination. Department size ranges from 25 members to 4900 members. Rank ranges from Captain to Assistant Chief. Years of service ranges from 17 years to 37 years. The highest level of education of the participants

includes Bachelor's, and Master's degree levels, and three of the five participants with Q-sorts loading significantly on this Factor completed the Executive Fire Officer Program at the National Fire Academy. One is a female, age 59, and the other four are males between the age of 34 and 59.

Table 20

Demographics for the Q-Sorters with Significant Loading on Factor C

	Factor	Region	Comm Type	Dept Type	Dept Size	Rank	Service	Education	EFO	Gender	Age
6	C	East	Urban	career	300	Bat. Chief	39	BS		M	58
9	CD	Midwest	Rural	volunteer	25	Admin Chief	17	2 BS		M	34
10	C	Midwest	suburban	combination	50	Bat. Chief	31	AA, BS	Y	M	59
21	C	West	Urban	career	4900	Asst Chief	37	MBA	Y	F	59
23	C	West	Urban	career	200	Captain	19	MPA	Y	M	43

Perspective D – Leaders Optimize Contributions from All Members

Description: In this perspective on organizational leadership, leaders make it clear that every person is valued and is expected to pay attention to the department's challenges and contribute to risk-reduction from their vantage point. Leaders foster open and effective communication among teams and across levels so that information is shared in a timely way because it is essential for consistently high-performance, problem-solving, and decision-making. Information sharing contributes to collaboration and learning for individuals and teams, which is important to the organization's continued success into the future.

When explaining the placement of statements in the formboard during post-sort interviews, participants whose sorts loaded significantly on Factor D said a number of things that support this label and description:

Sorter 4 – “Ideas can come from anywhere.”

“To do better as an organization, individuals have to get better; when you see individuals getting better, you have a sense for how well the organization will move forward.”

Sorter 7 – “To solve problems, you have to get different perspectives and make it a team effort. Value what everyone sees.”

A focus on open communication throughout the organization is indicated by the inclusion of five among those highly, positively salient in the model factor: 35, 25, 10, 36 and 27, and the inclusion of statements 30, 16, and 17 among those highly negatively salient in the model factor, highlighted in Table 21. Highly-positively salient statements that point to the focus on high-reliability and performance are 7 and 33. Statement 46 is a distinguishing statement for Factor D, listed in Table 22, and supports the description’s emphasis on the expectation that everyone should be paying attention and contributing.

Table 21

Descending Factor Score Array for Factor D

No.	Statement (truncated)	Z-Score
35	believe everyone brings something to the table	2.120
12	own the department’s problems	1.947
46	give members time to correct issues on their own	1.697
15	are future-oriented	1.217
25	share lots of information for the greater good	1.212
7	are vigilant stewards	0.985
36	decentralize authority	0.908
33	most important work is risk reduction	0.870
10	form cross-division teams to address issues, challenges	0.842
23	value objectivity	0.803
27	get things done collaboratively	0.764

(continued)

No.	Statement (truncated)	Z-Score
48	break barriers to change	0.627
44	measure success through the betterment of department members	0.519
39	believe that good SOPs run the department	0.471
6	know human resource management is critical	0.470
20	are never satisfied with the status quo	0.466
26	facilitate organizational dialogue	0.434
42	understand whole system, consider ripple effects of actions	0.420
41	operate at the 30,000' level	0.399
34	facilitate career development for all members	0.393
40	create safe zone for members to question processes, decisions	0.284
9	do not accept excuses	0.274
5	pursue grants to stay current and progressive	0.269
28	actively, regularly critique all practices, processes	0.188
45	value equity	0.120
2	create freedom for free thinkers	0.006
32	have stalwart demeanor	-0.005
3	embrace opportunities to do more for the public safety	-0.125
47	are democratic	-0.193
19	let go of ego	-0.203
24	give away power	-0.283
1	actively explore partnering with other public entities	-0.302
4	are decisive, with or without complete information	-0.438
8	have empathy	-0.458
38	are strong disciplinarians	-0.505
14	develop meaningful metrics for different stakeholders	-0.548
21	do the tough stuff first	-0.553
37	pick the spots where to bend for the sake of morale	-0.698
22	keep their personalities out of their leadership	-0.707
11	seek informal opportunities to interact	-0.735
13	give extra attention to community involvement	-0.770
31	tow the clear, hard line	-0.991
18	be smart about how to influence individuals	-1.409
30	guard against the need to always agree	-1.514
29	are wary of too much "buddy/buddy" behavior	-1.836
16	act like a corporate CEO	-2.039
43	seek to be politically astute	-2.086
17	hold their cards close until the support is there	-2.308

Note. Bold font indicates statements of high salience for the factor array

Table 22

Distinguishing Statements for Factor D

No.	Statement	Factor A		Factor B		Factor C		Factor D		Factor E		Factor F	
		Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score
46	give members time to correct issues on their own	0	0.01	3	1.04	-3	-0.81	4	1.7	0	-0.18	-1	-0.2
39	believe that good SOPs run the department	-4	-1.36	-4	-1.74	-3	-0.81	2	0.47	4	1.49	-3	-1.27
41	operate at the 30,000' level	-1	-0.39	-2	-0.94	0	-0.19	1	0.4	-3	-1.18	-2	-0.5

(continued)

No.	Statement	Factor A		Factor B		Factor C		Factor D		Factor E		Factor F	
		Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score
9	do not accept excuses	-4	-1.79	-4	-1.51	-5	-2.36	0	0.27	-2	-0.63	3	1.13
32	have stalwart demeanor	-3	-0.95	-3	-1.13	-5	-2.24	0	-0.01*	-3	-1.26	-3	-1.43
4	are decisive, with or without complete information	-3	-1.14	-2	-1.09	-4	-1.21	-1	-0.44	1	0.55	1	0.37
13	give extra attention to community involvement	0	0.23	-1	-0.21	1	0.38	-3	-0.77	2	0.73	0	0.08
31	tow the clear, hard line	-5	-1.89	-5	-1.95	-4	-1.66	-3	-0.99	5	1.85	-1	-0.17
18	be smart about how to influence individuals	3	1.08	0	0.28	2	0.55	-3	-1.41*	0	0.13	5	1.71
43	seek to be politically astute	1	0.27	-1	-0.32	1	0.46	-5	-2.09*	2	0.71	0	0.07

Note. $p < .05$; Asterisk (*) Indicates Significance at $p < .01$

The demographic information about those participants whose Q-sorts loaded significantly on Factor D is shown in Table 23. The two participants both come from the East region of the U.S. and represent a suburban combination department and an urban career department. Department sizes are 38 and 150 members. Both participants are Captains, and both have Bachelor's degrees. One is male and 45 years old, and one is Female who is 49 years old. Years of service ranges from 22 to 23 years.

Table 23

Demographics for the Q-Sorters with Significant Loading on Factor D

	Factor	Region	Comm Type	Dept Type	Dept Size	Rank	Service	Education	EFO	Gender	Age
4	D	East	suburban	combination	38	Captain	22	AS, BS	M	45	4
7	D	East	Urban	career	150	Captain	23	BFA		F	7

Perspective E – Leading for Consistent Structure to Foster Performance

This perspective regards organizational leadership in fire departments as essential for establishing a climate of mutual support to fulfill the mission, and conveying conviction and the

importance of vigilance regarding the mission. Leaders provide clear structure, establish expectations, and organize tasks and people while remaining approachable and strongly connected to people, coaching to achieve high standards of performance.

Sort 15 is the only significantly-loading sort on Factor E, with a correlation to it of 0.701. One other sort is close to the significant-loading threshold – Sort 7 has a correlation of 0.442 (the significant-loading threshold is 0.446).

Sorter 15 – “It is important to set standards or nobody will work toward a high standard. We need good SOPs and to educate people and establish how things should happen or people will freelance and there is no accountability. There will still be issues and questions, so be approachable and work on the issues. The leader has to be “real” and authentic in order to be approachable, and serious about the mission he/she has been entrusted with.”

The following quote from Sorter 7 demonstrates the similarity with this perspective – “In a diverse department, leaders need to remove their own feelings from situations and empathize so people feel supported, and the support will be mutual.”

The importance of structure is supported by the high, positive salience for statements 31, 39, and 40, shown in Table 24. Statements 31 and 39 are distinguishing statements for Factor E as shown in Table 25 (Statement 31 is statistically significant as a distinguishing statement at the $p > .01$ level). The importance of approachability and connecting with individuals is supported also by high, positive salience for statement 11 and high, negative salience for statement 41.

Table 24

Descending Factor Score Array for Factor E

No.	Statement (truncated)	Z-Score
31	tow the clear, hard line	1.854
34	facilitate career development for all members	1.803
6	know human resource management is critical	1.752
39	believe that good SOPs run the department	1.489
12	own the department's problems	1.307
40	create safe zone for members to question processes, decisions	1.255
35	believe everyone brings something to the table	1.204
11	seek informal opportunities to interact	1.175
19	let go of ego	1.124
13	give extra attention to community involvement	0.730
21	do the tough stuff first	0.730
37	pick the spots where to bend for the sake of morale	0.730
43	seek to be politically astute	0.708
15	are future-oriented	0.679
28	actively, regularly critique all practices, processes	0.628
4	are decisive, with or without complete information	0.548
44	measure success through the betterment of department members	0.445
23	value objectivity	0.314
5	pursue grants to stay current and progressive	0.314
8	have empathy	0.183
18	be smart about how to influence individuals	0.131
29	are wary of too much "buddy/buddy" behavior	0.051
20	are never satisfied with the status quo	0.000
30	guard against the need to always agree	0.000
38	are strong disciplinarians	-0.029
7	are vigilant stewards	-0.103
47	are democratic	-0.131
46	give members time to correct issues on their own	-0.183
27	get things done collaboratively	-0.234
24	give away power	-0.263
3	embrace opportunities to do more for the public safety	-0.285
42	understand whole system, consider ripple effects of actions	-0.416
2	create freedom for free thinkers	-0.445
45	value equity	-0.496
14	develop meaningful metrics for different stakeholders	-0.496
48	break barriers to change	-0.496
36	decentralize authority	-0.576
9	do not accept excuses	-0.628
33	most important work is risk reduction	-0.993
41	operate at the 30,000' level	-1.175
1	actively explore partnering with other public entities	-1.227
32	have stalwart demeanor	-1.255
26	facilitate organizational dialogue	-1.307
22	keep their personalities out of their leadership	-1.307
16	act like a corporate CEO	-1.387
17	hold their cards close until the support is there	-1.621
10	form cross-division teams to address issues, challenges	-1.986
25	share lots of information for the greater good	-2.117

Note. Bold font indicates statements of high salience for the factor array

Table 25

Distinguishing Statements for Factor E

No.	Statement	Factor A		Factor B		Factor C		Factor D		Factor E		Factor F	
		Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score
31	tow the clear, hard line	-5	-1.89	-5	-1.95	-4	-1.66	-3	-0.99	5	1.85*	-1	-0.17
39	believe that good SOPs run the department	-4	-1.36	-4	-1.74	-3	-0.81	2	0.47	4	1.49	-3	-1.27
21	do the tough stuff first	-2	-0.65	-3	-1.14	-2	-0.69	-2	-0.55	2	0.73*	-1	-0.46
27	get things done collaboratively	2	0.66	2	0.77	4	1.73	2	0.76	-1	-0.23	2	0.80
48	break barriers to change	3	1.25	4	1.19	1	0.53	2	0.63	-2	-0.59	2	0.61
9	do not accept excuses	-4	-1.79	-4	-1.51	-5	-2.36	0	0.27	-2	-0.63	3	1.13
1	actively explore partnering with other public entities	0	0.08	0	0.07	3	1.13	-1	-0.30	-3	-1.23	0	0.19
26	facilitate organizational dialogue	4	1.52	3	1.19	1	0.43	1	0.43	-4	-1.31*	1	0.59
10	form cross-division teams to address issues, challenges	0	-0.01	1	0.38	3	0.97	3	0.84	-5	-1.99*	1	0.41
25	share lots of information for the greater good	0	0.25	4	1.34	0	-0.08	4	1.21	-5	-2.12*	3	0.83

Note. $p < .05$; Asterisk (*) Indicates Significance at $p < .01$

The demographic information about the participant whose Q-sort loaded significantly on Factor E is shown in Table 26. The participant is from the South region of the U.S. and represents a suburban volunteer department of 42 members. He is the Chief with 18 years of service and a Master's degree.

Table 26

Demographics for the Q-Sorter with Significant Loading on Factor E

	Factor	Region	Comm Type	Dept Type	Dept Size	Rank	Service	Education	EFO	Gender	Age
15	E	South	suburban	volunteer	42	Chief	18	MS		M	42

Perspective F – Leaders Foster Engagement to Maintain Performance

This perspective of organizational leadership in fire departments emphasizes accountability and engagement – engagement of leadership with department members and among department members, and accountability for the mission. Leaders are discerning about individuals’ strengths, listen with empathy to coach, advise, and assist department members to reach their individual potential. Leaders cultivate relationships and open communication, and consider this role essential for the entire department to confront challenges and issues effectively and for top performance.

When explaining the placement of statements in the formboard during post-sort interviews, participants whose sorts loaded significantly on Factor F said a number of things that support this label and description:

Sorter 17: “Help people learn, nobody should feel like they have to “fake it until they make it” and you have to help people develop courage to think and make a call, avoid “the dead fish syndrome” of going whichever way the current is flowing.”

Sorter 18: “Good human resources practices are critical so that people engage with the mission, not BS.”

Sorter 1: “Involve people in decision making so they learn, take ownership, and are empowered. Leadership should “trust but verify” and you can’t do much of that if you do not know your peoples.”

The emphasis on individualized attention in this perspective is also supported by the high, positive salience of statements 18, 34, 11, 6, and 8, listed in bold in Table 27. The emphasis on accountability is supported by the high, positive salience for statements 12 and 9. The high, negative salience for statement 35 supports the inclusion of leaders’ discernment about

individuals' capabilities. Three of these highly-salient statements, 18, 9, and 35, are all distinguishing statements for this perspective, as shown in Table 28. Engagement of leaders to provide the individualized attention is supported by the neutral placement of statements 36, 24, and 41.

Table 27

Descending Factor Score Array for Factor F

No.	Statement (truncated)	Z-Score
12	own the department's problems	2.132
18	be smart about how to influence individuals	1.714
34	facilitate career development for all members	1.588
20	are never satisfied with the status quo	1.486
11	seek informal opportunities to interact	1.373
6	know human resource management is critical	1.251
9	do not accept excuses	1.133
8	have empathy	0.983
25	share lots of information for the greater good	0.832
27	get things done collaboratively	0.800
23	value objectivity	0.715
14	develop meaningful metrics for different stakeholders	0.711
3	embrace opportunities to do more for the public safety	0.654
48	break barriers to change	0.613
26	facilitate organizational dialogue	0.593
15	are future-oriented	0.577
10	form cross-division teams to address issues, challenges	0.414
4	are decisive, with or without complete information	0.370
19	let go of ego	0.256
33	most important work is risk reduction	0.203
1	actively explore partnering with other public entities	0.187
16	act like a corporate CEO	0.174
42	understand whole system, consider ripple effects of actions	0.163
40	create safe zone for members to question processes, decisions	0.110
13	give extra attention to community involvement	0.077
5	pursue grants to stay current and progressive	0.077
43	seek to be politically astute	0.069
36	decentralize authority	-0.032
24	give away power	-0.110
31	tow the clear, hard line	-0.171
46	give members time to correct issues on their own	-0.203
38	are strong disciplinarians	-0.407
21	do the tough stuff first	-0.463
2	create freedom for free thinkers	-0.467
41	operate at the 30,000' level	-0.503
47	are democratic	-0.601
44	measure success through the betterment of department members	-0.646
17	hold their cards close until the support is there	-1.011
45	value equity	-1.044

(continued)

No.	Statement (truncated)	Z-Score
28	actively, regularly critique all practices, processes	-1.178
39	believe that good SOPs run the department	-1.267
32	have stalwart demeanor	-1.433
7	are vigilant stewards	-1.511
35	believe everyone brings something to the table	-1.519
30	guard against the need to always agree	-1.580
37	pick the spots where to bend for the sake of morale	-1.604
22	keep their personalities out of their leadership	-1.604
29	are wary of too much "buddy/buddy" behavior	-1.901

Note. Bold font indicates statements of high salience for the factor array

Table 28

Distinguishing Statements for Factor F

No.	Statement	Factor A		Factor B		Factor C		Factor D		Factor E		Factor F	
		Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score	Value	Z-Score
18	be smart about how to influence individuals	3	1.08	0	0.28	2	0.55	-3	-1.41	0	0.13	5	1.71
20	are never satisfied with the status quo	2	0.72	1	0.53	-2	-0.61	1	0.47	0	0.00	4	1.49
9	do not accept excuses	-4	-1.79	-4	-1.51	-5	-2.36	0	0.27	-2	-0.63	3	1.13
16	act like a corporate CEO	-3	-1.25	-4	-1.41	-3	-1.14	-4	-2.04	-4	-1.39	0	0.17*
31	tow the clear, hard line	-5	-1.89	-5	-1.95	-4	-1.66	-3	-0.99	5	1.85	-1	-0.17
35	believe everyone brings something to the table	4	1.40	5	1.81	1	0.26	5	2.12	3	1.20	-4	-1.52*
37	pick the spots where to bend for the sake of morale	-2	-0.79	1	0.38	-2	-0.68	-2	-0.70	2	0.73	-5	-1.60

Note. $p < .05$; Asterisk (*) Indicates Significance at $p < .01$

The demographic information about three participants whose Q-sorts loaded significantly on Factor F is shown in Table 29. The three participants come from the East, South, and West regions of the U.S. and represent all three types of communities, and two department types—one career department and two combination departments. Department size ranges from 70 to 1,250 members. All three are males; two are 40, and one is 59.

Table 29

Demographics for the Q-Sorters with Significant Loading on Factor F

	Region	Comm Type	Dept Type	Dept Size	Rank	Service	Education	EFO	Gender	Age
1	East	all	Combination	90	Deputy Chief	15	BS	Y	M	40
17	South	urban	Career	1,250	District Chief	32	MS	Y	M	55
18	West	Rural	Combination	70	Captain	18	BA, BS	Y	M	40

Perspective A¹ – Leaders Drive Adaptation and Change

This perspective on organizational leadership in the USFES closely mirrors adaptive leadership since the A¹ Factor was intentional aligned with the adaptive leadership theoretical sort. Adaptive leaders understand the whole system “from the balcony view” and identify the challenge(s) facing the organization and then remove barriers to change and give the challenges to members of the organization to develop the response, since there is no prior experience with adaptive challenges. Leaders regulate the distress that working on adaptive challenges can produce and make sure that everyone is engaged in the development of the response.

The factor array in Table 30 supports these aspects of adaptive leadership with the high, positive salience of statements 42, 35, and 26:

Table 30

Descending Factor Score Array for Factor A¹

No.	Statement (truncated)	Z-Score
15	are future-oriented	1.881
12	own the department's problems	1.858
35	believe everyone brings something to the table	1.735
42	understand whole system, consider ripple effects of actions	1.565
48	break barriers to change	1.430
26	facilitate organizational dialogue	1.358
6	know human resource management is critical	1.102
20	are never satisfied with the status quo	0.885
44	measure success through the betterment of department members	0.761
		(continued)

No.	Statement (truncated)	Z-Score
19	let go of ego	0.594
34	facilitate career development for all members	0.593
23	value objectivity	0.592
2	create freedom for free thinkers	0.569
46	give members time to correct issues on their own	0.560
7	are vigilant stewards	0.526
27	get things done collaboratively	0.508
25	share lots of information for the greater good	0.507
18	be smart about how to influence individuals	0.463
41	operate at the 30,000' level	0.432
3	embrace opportunities to do more for the public safety	0.384
8	have empathy	0.344
11	seek informal opportunities to interact	0.246
45	value equity	0.097
14	develop meaningful metrics for different stakeholders	0.077
10	form cross-division teams to address issues, challenges	0.058
1	actively explore partnering with other public entities	0.024
40	create safe zone for members to question processes, decisions	-0.042
5	pursue grants to stay current and progressive	-0.081
13	give extra attention to community involvement	-0.245
36	decentralize authority	-0.287
33	most important work is risk reduction	-0.335
28	actively, regularly critique all practices, processes	-0.351
37	pick the spots where to bend for the sake of morale	-0.399
30	guard against the need to always agree	-0.405
21	do the tough stuff first	-0.504
24	give away power	-0.759
43	seek to be politically astute	-0.807
22	keep their personalities out of their leadership	-0.816
32	have stalwart demeanor	-0.832
47	are democratic	-0.833
4	are decisive, with or without complete information	-1.092
9	do not accept excuses	-1.160
39	believe that good SOPs run the department	-1.219
29	are wary of too much "buddy/buddy" behavior	-1.292
16	act like a corporate CEO	-1.668
17	hold their cards close until the support is there	-1.852
38	are strong disciplinarians	-2.024
31	tow the clear, hard line	-2.146

Note. Bold font indicates statements of high salience for the factor array

The demographic information about those participants whose Q-sorts loaded significantly on Factor A¹ is shown in Table 31. The 13 participants come from all four regions of the U.S. and represent all three community types, urban, suburban, and rural, and all three department types, career, volunteer, and combination departments. Department sizes range from 20 to 1200 members. Ranks range from Captains to Chiefs, with educational level ranging from Associate's

degrees to Master's degrees. There are two females and 11 males represented, ranging in age from 25 to 59. Years of service ranges from eight to 36 years.

Table 31

Demographics for the Q-Sorters with Significant Loading on Factor A¹

	Region	Comm Type	Dept Type	Dept Size	Rank	Service	Education	EFO	Gender	Age
2	East	rural	volunteer	70	Lieutenant	8	AA		M	25
3	East	suburban	combination	35	Captain	16	MA, BS		M	36
4	East	suburban	combination	38	Captain	22	AS, BS		M	45
5	East	urban	Career	20	Chief	16	BA, MPA	Y	M	40
7	East	urban	Career	150	Captain	23	BFA		F	49
9	Midwest	rural	volunteer	25	Admin Chief	17	2 BS		M	34
11	Midwest	suburban	combination	32	Captain	17	BS		M	41
13	Midwest	suburban	combination	454	Captain	9	MPA		M	37
14	South	suburban	Career	82	Asst Chief	22	BS		M	38
20	West	urban	Career	150	Dep. Chief	32	MA	Y	M	56
22	West	urban	Career	200	Chief	36	BS, MS	Y	M	58
24	West	urban	Career	1200	Asst. Chief	36	AS, BA		F	59
25	West	urban	combination	111	Captain	10	MS		M	35

Perspective D¹ – Everyone is Responsible

This perspective on leadership believes the cliché that a group is only as strong as its weakest link. All members of the department must know and adhere to the policies and procedures and hold each other accountable for high performance in service to the mission. There is little flexibility regarding discipline or performance evaluations.

The factor array in Table 32 supports these tenets for Perspective D¹ with the high, positive salience of statements 9, 31, 38, and 39:

Table 32

Descending Factor Score Array for Factor D¹

No.	Statement (truncated)	Z-Score
36	decentralize authority	1.844
39	believe that good SOPs run the department	1.434
9	do not accept excuses	1.417
32	have stalwart demeanor	1.417
33	most important work is risk reduction	1.266
38	are strong disciplinarians	1.161
28	actively, regularly critique all practices, processes	1.097
24	give away power	1.058
25	share lots of information for the greater good	1.034
31	tow the clear, hard line	1.032
10	form cross-division teams to address issues, challenges	0.890
4	are decisive, with or without complete information	0.785
22	keep their personalities out of their leadership	0.751
40	create safe zone for members to question processes, decisions	0.747
47	are democratic	0.730
41	operate at the 30,000' level	0.711
45	value equity	0.667
46	give members time to correct issues on their own	0.594
7	are vigilant stewards	0.539
6	know human resource management is critical	0.440
21	do the tough stuff first	0.216
27	get things done collaboratively	0.169
35	believe everyone brings something to the table	0.160
48	break barriers to change	0.144
12	own the department's problems	-0.013
37	pick the spots where to bend for the sake of morale	-0.046
2	create freedom for free thinkers	-0.193
34	facilitate career development for all members	-0.207
8	have empathy	-0.224
5	pursue grants to stay current and progressive	-0.425
16	act like a corporate CEO	-0.427
23	value objectivity	-0.489
29	are wary of too much "buddy/buddy" behavior	-0.530
26	facilitate organizational dialogue	-0.592
1	actively explore partnering with other public entities	-0.618
17	hold their cards close until the support is there	-0.852
14	develop meaningful metrics for different stakeholders	-0.865
3	embrace opportunities to do more for the public safety	-0.880
20	are never satisfied with the status quo	-0.939
11	seek informal opportunities to interact	-1.083
44	measure success through the betterment of department members	-1.121
19	let go of ego	-1.136
42	understand whole system, consider ripple effects of actions	-1.185
15	are future-oriented	-1.195
13	give extra attention to community involvement	-1.250
43	seek to be politically astute	-1.908
18	be smart about how to influence individuals	-2.051
30	guard against the need to always agree	-2.075

Note. Bold font indicates statements of high salience for the factor array

The two participants whose sorts load significantly on Factor D¹ have one demographic characteristic in common as shown in Table 33. The participants both have medical education.

Table 33

Demographics for the Q-Sorters with Significant Loading on Factor D¹

	Region	Comm Type	Dept Type	Dept Size	Rank	Service	Education	EFO	Gender	Age
11	Midwest	suburban	combination	32	Captain	17	BS		M	41
19	West	Urban	career	458	Asst Chief	22	BS,2 AAs		M	50

Idiosyncratic Q-Sorts

The study includes two Q-sorts as distinct perspectives, Q-sort 8 and Q-sort 9. Q-sort 8 is distinct because while it significantly loads on only one factor, B, it is unlike any other sorts because it has higher loadings on two additional factors: 0.318 on Factor A and 0.402 on Factor C. Q-sort 9 is distinct because it is the only sort with a significant, negative loading on Factor C (-0.506), while also significantly loading on Factor D (0.495).

Perspective Q-Sort 8 – Bring the Right People to the Table

In this perspective on organizational leadership, dialogue with freedom to question, think critically, and propose creative ideas is important to address challenges and improve performance. Leaders unite and support teamwork at the same time as they seek such divergent thought. Leaders also need to be active in three spheres of the organization to be able to bring the right people to the table – the central sphere of the people who do the work, another sphere of the people who manage and administrate the organization, and the third sphere of the governing authority and community the organization serves.

In the sort, the statements placed with high, positive salience include 2, 8, 26, and 40 that support the emphasis in the description on dialogue with the freedom to question. Statements 10 and 35 also have high, positive salience relating to teams.

Perspective Q-Sort 9 – Taking Charge

Leaders are the firm guiding hand for the department members, expressing care for them, while expecting the job to get done. Leaders build department members up through support so that there is success. Failures are on the leaders' shoulders.

In the sort, this description is supported by the high, positive salience for statements 9 and 12, and high, negative salience for statement 24 about giving away power.

Chapter Summary

The chapter presented the statements that all perspectives share, and statistical analysis of the Q-sorts regarding perspectives on organizational leadership in USFES departments. Principal component analysis and Varimax mathematical rotation extracted factors to explain the variance among the Q-sorts, identifying six perspectives on organizational leadership in USFES departments. Manual rotation of two factors to look at the relationships among the sorts from a new vantage point elicited two additional perspectives. Two idiosyncratic Q-sorts represent additional distinct perspectives. Included in the chapter are labels and descriptions of each of the 10 perspectives:

1. Perspective A - Leading as Facilitation for Increasing Individual and Organizational Capacity to Meet Dynamic Challenges
2. Perspective B - Leadership Success Comes through Empowering Others
3. Perspective C - Leading for Mission-Driven Consistency to Advance Public Safety
4. Perspective D - Leaders Optimize Contributions from All Members

5. Perspective E - Leading for Consistent Structure to Foster Performance
6. Perspective F - Leaders Foster Engagement to Maintain Performance
7. Perspective A¹ - Leaders Drive Adaptation and Change
8. Perspective D¹ - Everyone is Responsible
9. Perspective 8 - Bring the Right People to the Table
10. Perspective 9 - Taking Charge

Demographic characteristics of the participants with significantly loading Q-sorts for each factor are also included in the chapter.

Chapter 5 will discuss the findings and conclusions related to the research questions, concerning the perspectives on organizational leadership in the USFES, how they are similar and how they are different, and how they correspond to existing theories and constructs. Chapter 5 also includes delimitations and limitations of the study, and the recommendations for future study and implications of this study.

Chapter 5: Discussion of Findings

Introduction

Chapter 4 presented in detail the Q-sort data and the analysis of it describing 10 perspectives on organizational leadership in USFES Departments. It was not unexpected that there are multiple perspectives, based upon the Chapter 1 background for the study and the literature review of Chapter 2. There are limited studies previously completed using various theoretical frameworks and leaders writing in trade publications espousing a variety of theoretical concepts, leadership traits and behaviors as key to effective fire service organizational leadership. This chapter will discuss how the perspectives differ and how they are similar, and how the perspectives correspond to existing leadership constructs, concepts, theories, and models. Implications of the discussion and recommendations are included. The chapter also notes the delimitations and limitations for the study and suggests opportunities for future research.

Summary of Findings

There are characteristics, knowledge, skills, abilities, and behaviors that all the perspectives on organizational leadership in USFES departments agree upon described in detail below. All the theoretical Q-sorts load significantly on a perspective, although not always with a high level of loading, indicating that theories do not map perfectly to any of the perspectives. Each perspective does share some commonality with existing leadership theories, either one of the theories represented in the data with a theoretical Q-sort or another theory or model, but there are not exact matches. The theoretical Q-sorts included are for those theories and models described in Chapter 2, consisting of theories and models that have been studied in USFES

organizations or closely-related industries: adaptive leadership theory, complexity leadership theory, crew resource management, public administrative leadership theory, servant leadership, transactional leadership, and transformational leadership. Because there are not exact matches with the theories included as theoretical Q-sorts, I considered additional theories or models that may be more closely represented by the perspectives; these include the learning organization model, authentic leadership theory, leader-member exchange theory, boundary-spanning leadership, regulatory focus theory, the managerial grid model, and Mintzberg's model of management.

Among the perspectives described in this study, there is overlap in terms of the underlying leadership constructs they express. There is only one pair of perspectives where there is a true disagreement revealed. The manual rotation of two factors to look at the shared perspectives from the vantage point of the adaptive theoretical sort exposed a bi-polar factor with Q-sorts significantly loading on both the positive and negative poles of the factor. The nature of the conflict is related to how leader-centric the organization should be, to the flexibility within the organization regarding the interactions among individuals, and the regulatory focus of leadership.

Examination of the demographics of the sorters loading significantly on the different factors does not indicate that the demographic characteristics collected for the participants are predictors for a participant to hold a particular perspective. The perspectives are distributed across geographic regions and across types of departments and types of community. There is one possible exception, the level of education.

Shared Characteristics, Knowledge, Skills, Abilities, and Behaviors Revealed in the Concourse Development

There were statements in the complete concourse on organizational leadership in the USFES departments that were repeated often and can be assumed to be shared across perspectives, listed in the previous chapter (pp. 112-115). Several additional statements were often repeated in the Q-sort post-interviews and are included here too. These statements included traits and characteristics, knowledge, skills and abilities, and behaviors deemed significant across all the perspectives.

- Characteristics that are agreed as important: The participants and trade literature often mentioned the following characteristics as important to effective department-wide leadership: integrity, honesty, confidence, decisiveness, fairness, and professionalism. Expressions reflecting these traits included “walk the talk” and “grace under pressure” with “emotions under control” and that leaders must “set aside favoritism.” There was also agreement that effective department-wide leaders are self-aware, particularly about their own response patterns to challenging people or situations. Participant 17 put it in a unique way, explaining that every leader “carries two buckets, one with water and one with gas, and the good ones know which one to use.”
- Knowledge that is agreed as important: Interviewees and trade literature agreed on several areas of knowledge that are important for effective organizational leadership. All department leaders must be well-versed in the National Incident Management System, regardless of department size or type, or community size or type. Department leaders should know the elements of effective public relations and *communications*

theory because of the universality of the necessity for effective communications throughout the contexts of the fire service, labor/management relations, the roles of the divisions within the department, the community governance structure and people, and the 16 National Fire Service Life Safety Initiatives They should also know some psychology and make an effort to gain knowledge of leadership, although the participants did not necessarily agree on when or how department leaders should study leadership – through experience and mentorship, formal education, or self-study.

- Skills, abilities, and behaviors that are agreed to be important: Skills and abilities that were consistently mentioned or written in the industry as important to leadership in the fire service include communications skills, organizational skills, the ability to run effective meetings, and public speaking. Behaviors consistently mentioned include supporting rigorous, regular training for Operations, providing the example and role-modeling the core values and mission of the fire service, and professionalism.

Because these characteristics, knowledge, skills and abilities recurred in the discourse and trade literature and because they are not distinct to any perspective, they were not included in the Q-sample of statements and are assumed to be integrated within all 10 of the perspectives.

Similarities Revealed in the Q-Sorts

There are additional ways several perspectives on organizational leadership are similar, that the Q-sorts revealed. Two statements are not salient in any of the perspectives. As shown in Table 10 (p.124), the statement concerning pursuit of grants and the statement concerning valuing objectivity have very close Q-values across all the model factor arrays. The statement about pursuing grants had Q-values in the model factor arrays ranging from -1 to 1. The low

absolute value of the Q-values and small spread indicates that all the perspectives agree that this is not a strongly salient part of the model views on organizational leadership in either the negative sense (most unlike the view on organizational leadership) or the positive sense (most like the view on organizational leadership). The model factor arrays also have similar Q-values for the statement that effective fire department leaders value objectivity, in the slightly positively salient sense with Q-values of 1 and 2. The PQMethod report listed these two statements as consensus statements across all factors in the six-factor solution, with the statement concerning grants significant at the $p < .05$ level, and the statement concerning objectivity significant at the $p < .01$ level.

Four perspectives agreed that organizational leadership should focus on change: Perspective A: Leading as Facilitation for Increasing Individual and Organizational Capacity to meet Dynamic Challenges, Perspective B: Leadership Success comes through Empowering Others, Perspective D: Leaders Optimize Contributions from All Members, and Perspective A¹, Leaders Drive Adaptation and Change. These perspectives also agreed that an authoritarian, task-focused style is not effective in organizational leadership. Except for Perspective B, these perspectives also agreed that organizational leadership should be future-oriented. Note that these elements of agreement in this group of perspectives does not mean that the perspectives are similar to each other overall: in Perspective A, effective organizational leadership is concerned with both individuals in the department and the broader system, whereas in Perspective B, attention is greatest to individual members of the department, and in Perspective A¹, attention is greatest to the broader system.

Three other perspectives agreed that asserting structure and upholding effective standard operating procedures (SOPs) need to be strong components of effective organizational

leadership: Perspective E: Leading for Consistent Structure to Foster Performance, Perspective D¹: Everyone is Responsible, and Perspective 9: Taking Charge. These three perspectives share a focus on individuals, but with differences in the reason or purpose for the focus on individuals; for task-focused consistency and performance in Perspective E, for accountability in Perspective D¹, and for supporting individual effort in Perspective 9. Perspective F also focuses on individuals, but with another distinction in the purpose for the focus, to foster engagement and performance.

10 Perspectives on USFES Organizational Leadership

Despite all of the characteristics, the knowledge, and the skills, abilities and behaviors that are agreed to be important to organizational leadership in USFES departments, and the commonality that groups of perspectives have with each other, this study shows that there are 10 separate perspectives on what constitutes effective organizational leadership. Chapter 4 detailed the analysis of the Q-sort data that resulted in labels and descriptions for the perspectives that show the distinctions. The labels and descriptions are reprinted here for ease of reference:

- Perspective A: Leading as Facilitation for Increasing Individual and Organizational Capacity to Meet Dynamic Challenges
 - Description: “Leaders improve the department one person at a time” might be the motto for those who share this perspective on USFES organizational leadership. In this perspective, leaders are acutely attuned to the individuals of the department, their interactions across the department, as well as dynamic influences the department from the community and beyond. Attention to the whole system is important to identifying problems and challenges to overcome and opportunities for learning, growth, and improvement. Leaders

express care and accountability for the people who must do the work. Leaders must take charge of important decisions with a long view toward the future implications for the members of the department and the community, and communicate effectively about the big picture to provide problem-solving guidance.”

- Perspective B: Leadership Success Comes through Empowering Others
 - Description: This perspective on effective fire department organizational leadership considers it important for leaders to create a positive professional and social/emotional climate and involve members in initiatives that solve challenges and enable the department to improve performance. Leaders provide information, listen, and delegate and empower to engage members to think critically and dialogue about the challenges. In this perspective, maintaining the status quo in the department would be like falling behind; continuous improvement through empowerment of individual members is the objective.
- Perspective C: Leading for Mission-driven Consistency to Advance Public Safety
 - Description: This perspective on effective USFES organizational leadership prioritizes the focus on the common and greater good in service to the community. Leaders support collaboration and teamwork within the organization, and develop external relationships across stakeholders to build support, address issues, and improve or expand service. The department represents as well as protects the community, and department leaders

therefore value equity and consistency in the treatment of people both inside and outside of the organization.

- Perspective D: Leaders Optimize Contributions from All Members
 - Description: In this perspective on organizational leadership, leaders make it clear that every person is valued and is expected to pay attention to the department's challenges and contribute to risk-reduction from their vantage point. Leaders foster open and effective communication among teams and across levels so that information is shared in a timely way because it is essential for consistently high-performance, problem-solving, and decision-making. Information sharing contributes to collaboration and learning for individuals and teams, which is important to the organization's continued success into the future.
- Perspective E: Leading for Consistent Structure to Foster Performance
 - Description: This perspective regards organizational leadership in fire departments as essential for establishing a climate of mutual support to fulfill the mission, and conveying conviction and the importance of vigilance regarding the mission. Leaders provide clear structure, establish expectations, and organize tasks and people while remaining approachable and strongly connected to people, coaching to achieve high standards of performance.
- Perspective F: Leaders Foster Engagement to Maintain Performance
 - Description: This perspective of organizational leadership in fire departments emphasizes accountability and engagement – engagement of leadership with department members and among department members, and accountability for

the mission. Leaders are discerning about individuals' strengths, listen with empathy to coach, advise, and assist department members to reach their individual potential. Leaders cultivate relationships and open communication, and consider this role essential for the entire department to confront challenges and issues effectively and for top performance.

- Perspective A¹: Leaders Drive Adaptation and Change
 - Description: This perspective on organizational leadership in the USFES closely mirrors adaptive leadership since the A¹ Factor was intentional aligned with the adaptive leadership theoretical sort. Adaptive leaders understand the whole system “from the balcony view” and identify the challenge(s) facing the organization and then remove barriers to change and give the challenges to members of the organization to develop the response, since there is no prior experience with adaptive challenges. Leaders regulate the distress that working on adaptive challenges can produce and make sure that everyone is engaged in the development of the response.
- Perspective D¹: Everyone Is Responsible
 - Description: This perspective on leadership believes the cliché that a group is only as strong as its weakest link. All members of the department must know and adhere to the policies and procedures and hold each other accountable for high performance in service to the mission. There is little flexibility regarding discipline or performance evaluations.

- Perspective 8 – Bring the Right People to the Table
 - Description: In this perspective on organizational leadership, dialogue with freedom to question, think critically, and propose creative ideas is important to address challenges and improve performance. Leaders unite and support teamwork at the same time as they seek such divergent thought. Leaders also need to be active in three spheres of the organization to be able to bring the right people to the table – the central sphere of the people who do the work, another sphere of the people who manage and administrate the organization, and the third sphere of the governing authority and community the organization serves.
- Perspective 9 – Taking Charge
 - Description: Leaders are the firm guiding hand for the department members, expressing care for them, while expecting the job to get done. Leaders build department members up through support so that there is success. Failures are on the leaders' shoulders.

USFES Perspectives and Leadership Constructs, Theories, and Models

Seven theories were included in the data as theoretical Q-sorts, so it is possible to assess how the USFES perspectives correlate with these theories: adaptive leadership theory, complexity leadership theory, crew resource management framework, public administrative leadership theory, servant leadership theory, transformational leadership theory, and transactional leadership theory.

Four of the theoretical Q-sorts significantly load on a single perspective, B, Leadership Success comes through Empowering Others: transformational (0.647), transactional (negative

significant loading, -0.882), servant (0.690), and complexity (0.590) leadership theories. There are two possible explanations. It may be that Perspective B incorporates distinct aspects from all of the four theories, or it may be that the four theories have much in common and overlap. This second possibility is supported by a multitrait-multimethod matrix (MTMM) meta-analysis of empirical studies involving transformational, transactional, consideration, initiating structure, and leader-member exchange theories and constructs that found statistically significant correlation and convergence across them at the $p < .01$ level (Rowold & Borgmann, 2013).

While it was not among the research objectives of this study to show convergence of leadership theories or constructs, the significant loading of four theoretical Q-sorts on a single factor does provide support for the possibility that the four theories are convergent. (It also brings to light the possibility of Q-methodology as a research approach to validate results of studies like Rowold and Borgmann's.). The servant leadership theoretical sort has the highest significant loading on Perspective B at 0.690, indicating that it is the most closely aligned with Perspective B. Both Perspective B and servant leadership theory are focused on growth and empowerment for followers.

Researchers of complexity leadership theory, one of the theoretical Q-sorts significantly loading on Perspective B, have also concluded that there are three leadership constructs “entangled” within it – adaptive leadership, administrative leadership, and enabling leadership (Uhl-Bien et al., 2007) – further convoluting the picture regarding the correlations with theory for Perspective B, because two of these “entangled” leadership constructs were included in the study as theoretical Q-sorts, but adaptive and administrative theoretical sorts did NOT load significantly on Perspective B. Avolio and Gardner (2005) have proposed that there is convergence across servant, charismatic, transformational, and spiritual leadership theories that

exists because they share leadership constructs with authentic leadership theory (ALT), each adding distinct elements to the foundational ALT constructs they hold in common. Two of these included in Avolio and Gardner's study, servant and transformational leadership theories, were included in the Q-data as theoretical sorts that loaded significantly on Perspective B, indicating that Perspective B has commonality with ALT in addition to servant and transformational leadership theories. ALT is defined as "a process that draws from both positive psychological capacities and a highly developed organizational context, which results in both greater self-awareness and self-regulated positive behaviors on the part of leaders and associates, fostering positive self-development" (Avolio, Walumbwa, & Weber, 2009, p. 424). This definition of ALT is very leader-centric, however, whereas the perspective represented by Perspective B is follower-centric.

The theoretical sort for adaptive leadership theory was confounded, loading significantly on both Perspective A and Perspective D, at levels 0.586 and 0.497 respectively. Both perspectives share things in common with each other and with adaptive leadership theory, described in Chapter 2, such as attention to collaborative work to ensure success into the future. Each is slightly different from adaptive leadership theory in separate ways. Perspective A does not have a strong emphasis on inclusivity, whereas adaptive leadership theory does, as demonstrated by the adaptive leadership theory inclusion of *protection for voices*. Perspective D shares the emphasis on inclusivity with adaptive leadership theory, but does not have a clear emphasis on scanning the environment and influences on the organization as adaptive leadership theory does, which Perspective A does share with adaptive leadership theory. Perspective A shares this focus on the system with an additional model that was not included as a theoretical Q-sort -- the learning organization model. The learning organizational model may be more

representative of Perspective A because it shares all the components essential to the learning organization model: building a shared vision, personal mastery for members, team learning, systems thinking, and awareness of mental models or assumptions (Senge, 1994).

Perspective A¹ is the adaptive leadership theoretical perspective because it was the result of manually rotating factors A and D to view the relationships among the Q-sorts and their loadings from the perspective of the adaptive leadership theoretical Q-sort. All of the Q-sorts that had significant loading on Perspective A except two also significantly loaded on Perspective A¹ (Q-sorts 11 and 19 significantly negatively loaded on D¹ as discussed below). This is further support in addition to the adaptive leadership theoretical sort significantly loading on Perspective A for the conclusion that Perspective A shares elements with adaptive leadership theory. Six additional Q-sorts besides Q-sort 27 beyond those that had been significantly loading on Perspective A also significantly loaded on Perspective A¹. Two of these, Q-sorts 2 and 9, had been confounded prior to the manual rotation, on Perspectives A and B for Q-sort 2, and Perspective C (bi-polar, negative significant loading) and Perspective D for Q-sort 9. Q-sort 3 that did not load significantly on any factors prior to the manual rotation also significantly loads on Perspective A¹. Finally, Q-sorts 4 and 7 that had significantly loaded on Perspective D, have significant loading on Perspective A¹, further supporting that Perspective D has commonality with adaptive leadership, as mentioned above. The manual rotation cleared up the view regarding both the adaptive leadership and CRM theoretical Q-sorts and provides clarity for the perspectives of other participants whose Q-sorts were confounded or not loading significantly prior to the manual rotation, adding their perspectives to those sharing commonality with adaptive leadership theory.

The theoretical sort for public administrative leadership theory (PAL) loaded significantly on Perspective C, but at 0.447, it is barely significant (> 0.446). PAL is described in Chapter 2 (pp. 58-65). Perspective C has a clear emphasis on improving and expanding the organization's service to the public good through attention to a broad definition of stakeholders, also an important part of PAL. Perspective C shares concern for building and maintaining trust for individuals both within and outside the organization with PAL. The reasons for the lower value of the correlation of PAL and Perspective C relate to the components of PAL that are not included in Perspective C. PAL includes an awareness of the system and influences on the organization from the system, and includes an emphasis on assessment and evaluate, but these are not emphasized in Perspective C. Also of note regarding the PAL theoretical sort is that it has a negative significant loading on Perspective E, meaning that there is an element or elements in Perspective E that are oppositional to PAL. Perspective E does not include focus beyond the organization, which is an important part of PAL. Another leadership model besides PAL has shared characteristics with Perspective C, boundary-spanning leadership. Boundary-spanning leadership emphasizes the capability to create direction, alignment, and commitment across boundaries in service to a vision or goal. This capacity can exist in and between individuals, groups and teams, and organizations (Ernst & Yip, 2009).

The theoretical sort for crew resource management (CRM) loaded significantly on Perspective D (0.758) and on Perspective D¹ (0.700). Fire service trade publications promote CRM as transferable from the aviation industry, and the theoretical Q-sort loading highly significantly on both Perspectives D and the manually-rotated D¹ indicate that this is indeed one of the perspectives on organizational leadership in USFES departments, although there is not

empirical research related to CRM in USFES departments. CRM and Perspectives D and D¹ share the emphasis on team collaboration, communication, and risk reduction.

Perspective D¹ was bi-polar, with the CRM theoretical sort as the only Q-sort that had positive significant loading and with Q-sorts 11 and 19 that had negative significant loading. The views expressed in Q-sorts 11 and 19 are therefore in opposition to CRM. The nature of the disagreement or opposition within Perspective D¹ is that CRM is very team-centric and inflexible regarding a communications protocol, whereas Q-sorts 11 and 19 are leader-centric and reflect more flexibility because of an emphasis on adaptation. The nature of the disagreement can also be interpreted in light of regulatory focus theory, which posits that security-related leadership differs from nurturance-related leadership. Security-related leadership is characterized by a prevention focus involving safety, responsibilities, duties, and obligations. Nurturance-related leadership involves a focus on promotion of advancement, growth, and aspirations for individuals (Gorman et al., 2012). CRM has security as the regulatory focus, and adaptive leadership theory has a greater nurturance regulatory focus.

The negatively-loading PAL theoretical sort is the only theoretical sort significantly loading on Perspective E. In addition to the opposition to PAL that Perspective E has related to focus beyond the organization, it is possible that it correlates with another leadership theory or model that was not included as a theoretical sort, or that it does not correlate with existing theories and models. Perspective E is focused on the department members and includes both a task-orientation and a relationship-orientation. An additional model that this bears resemblance to is the situational leadership model (Blanchard, 1985). In this model, the application of each type of behavior in the right amounts on the part of the leader, based upon the developmental level of the follower determines the appropriate leadership style. Leadership effectiveness

depends upon accurate diagnosis of the follower's developmental level and the ability of the leader to implement each style with skill. Perspective E's identification of coaching as an important leadership capacity to achieve high performance could be seen as related to the situational leadership model.

Perspective F does not have a theoretical Q-sort that loads significantly on it. This may be because another theory or model other than those included as theoretical Q-sorts in this study may correlate with it. The theoretical Q-sorts included were only seven among those that have been applied in studies of leadership in American fire departments or advanced as applicable in trade literature. It could also be that Perspective F is a perspective that does not correlate with any existing leadership theories or models. Perspective F, labeled Leaders Foster Engagement to Maintain Performance does, however, have some similarities with existing theory. The emphasis on accountability and engagement of department members that the leader drives through discernment about individuals' strengths, and cultivating relationships to coach based upon strengths has some similarities with the leader-member exchange model of leadership (LMX). LMX emerged in the 1970s and concentrates on the quality of relationships between a leader and members of the organization. Leaders treat members differently based upon their relationships and affect the level of engagement and performance (Northouse, 2013). LMX has not been studied in American fire departments.

The idiosyncratic sorts, Q-sorts 8 and 9, represent perspectives with similarity to existing theoretical frameworks. Q-sort 8 and the complexity theoretical sort load on Perspective B at 0.595 and 0.590 respectively. The similar correlation with Perspective B for Q-sort 8 and the complexity theoretical sort can mean that the perspective represented by Q-sort 8 and the complexity leadership theory perspective are more similar than Q-sort 8 is similar to the other

two theoretical Q-sorts that significantly load on Perspective B, servant leadership and transformational leadership at 0.690 and 0.647 loadings respectively. The correlation matrix of the unrotated Q-sorts bears this out as well, indicating that Q-sort 8 is correlated with the complexity theoretical sort at 0.64 (see the correlation matrix in Appendix J).

A key observation that made Q-sort 8 unique is that it also loads on Perspective A at 0.318 and Perspective C at 0.402, and while not significant levels of loading, it is notable that it loads at levels approaching significant (0.446) on two additional factors besides significantly loading on Perspective B. As already discussed above, adaptive leadership theory has similarity with Perspective A, and public administrative leadership with Perspective C, and Q-sort 8 does share some commonality with these theories in addition to those theoretical sorts with positive significant loading on Perspective B, but none of these theories have exceptionally strong loadings on the perspectives, nor do they strongly correlate with Q-sort 8 (see the correlation matrix in Appendix J; Q-sort 8 correlates with adaptive leadership theoretical sort at 0.59, servant leadership theoretical sort at 0.52, transformational leadership theoretical sort at 0.44, and the public administrative leadership sort at 0.36). Q-sort 8 may correspond with another theory or model not included among the theoretical sorts in this study, or it may represent a unique perspective that does not fit any existing theories or models.

One model that was not included among the theoretical sorts in this study and that has not been studied in USFES organizations that has similarity to Q-sorter 8's perspective is Henry Mintzberg's (1998, 2009) model of managing. Q-sorter 8 explained a view on organizational leadership in terms of spheres of concern – the “front line” sphere, the middle-management sphere, and the external sphere – all involving communication and building collaboration. This is similar to the specifics included in Mintzberg's model he developed after observing hundreds of

organizational managers and leaders. Mintzberg's model includes three directions of leadership and management – into the unit, into the rest of the organization, and beyond the organization, and three planes of action within each of these directions – information, people, and action. However, both Mintzberg's model and the perspective of Q-Sort 8 could be said to represent the principles of interactive dynamics and emergence that are keys to complexity leadership theory, and this closer analysis of the perspective of Q-Sort 8 further bolsters the case for its alignment with complexity leadership theory (Osborn et al., 2014; Uhl-Bien & Marion, 2009; Uhl-Bien et al., 2007).

Q-sort 9 is also unique, with confounded significant loadings on Perspectives C (-0.506) and D (0.496). It is the only Q-sort with a negative significant level of loading on Perspective C, indicating that it is a perspective that disagrees with Perspective C, Leading for Mission-driven Consistency to Advance the Public Safety. The nature of the opposition to Perspective C is in where leaders focus their attention. In the view of Q-sort 9, effective organizational leaders should focus their attention on the members of the organization and not outside the organization. Q-sort 9's perspective shares the focus on members and supporting them to improve performance with Perspectives D and E, but Perspective D, Leaders Optimize Contributions from All Members, and Perspective E, Leading for Consistent Structure to Foster Performance, do not express the same level of opposition to giving attention to the community as Q-sort 9 does. Q-Sort 9's strong focus on individual consideration, along with structure/task and some future-orientation—that for Q-Sorter 9 is to focus on future results and performance—is similar to the quadrant of the Blake and Mouton managerial grid they call *team management* which has a high focus-on-people and high focus-on-results (Northouse, 2012).

Summary of Theoretical Similarities and Differences

To make these similarities and differences with leadership theories and models clearer, and because the leadership theories and models have some overlapping underlying constructs, Table 34 simplifies the information in Appendix I on the themes and sub-themes in the concourse statements, displaying the Q-sample statements, the initial broad-theme-coding of the statements in the column labeled Concourse Theme, and adds the theoretical leadership construct or constructs that the statements represent.

Table 34

Q-Sample Statements, Concourse Theme, Leadership Constructs

	Statement	Concourse Theme	Leadership Construct	Secondary Leadership Construct
1	effective fire department leaders should actively explore ways to partner with other public entities	Administration	Community	System
2	effective fire department leaders should create freedom for free thinkers	Change	Individualized consideration	Change
3	effective fire department leaders should embrace opportunities to do more functions that contribute to the public safety	Change	Community	System
4	effective fire department leaders need to be decisive, sometimes without complete information	Command	Structure	
5	effective fire department leaders should pursue grants to stay current and progressive	Change	Change	
6	effective fire department leaders should treat human resource management is critical for the organization	Administration	Individualized consideration	Relationships
7	effective fire department leaders need to be vigilant stewards	Monitoring	Community	System
8	effective fire department leaders should have empathy	Service	Individualized consideration	Diversity
9	effective fire department leaders should not accept excuses	Monitoring	Structure	
10	effective fire department leaders form cross-division teams to address issues and challenges	Teambuilding	Team	
11	effective fire department leaders seek informal opportunities to interact	Teambuilding	Team	individualized consideration
12	effective fire department leaders own the department's problems	Monitoring	Structure	Reflection
13	effective fire department leaders give extra attention to community involvement	Service	Community	System
14	effective fire department leaders need to develop metrics that are meaningful for different stakeholders	Monitoring	Community	System

(continued)

	Statement	Concourse Theme	Leadership Construct	Secondary Leadership Construct
15	effective fire department leaders should be future-oriented	Change	Future	
16	effective fire department leaders should act like corporate CEOs	Command	Structure	Future
17	effective fire department leaders should hold their cards close until the support is there	Command	Community	Relationships
18	effective fire department leaders need to be smart about how to influence different individuals	Administration	Individualized consideration	
19	effective fire department leaders should let go of ego	Service	Relationships	diversity
20	effective fire department leaders should never be satisfied with the status quo	Change	Change	reflection
21	effective fire department leaders should do the tough stuff first	Command	Structure	
22	effective fire department leaders have to keep their personalities out of their leadership	Command	Relationships	structure
23	effective fire department leaders should value objectivity	Administration	Structure	individualized consideration
24	effective fire department leaders should give away power	Service	Team	diversity
25	effective fire department leaders should share lots of information for the greater good	Service	System	community
26	effective fire department leaders need to facilitate organizational dialogue	Teambuilding	Team	relationships
27	effective fire department leaders should get things done collaboratively	Teambuilding	Team	relationships
28	effective fire department leaders have to actively and regularly critique all practices and processes	Monitoring	Reflection	change
29	effective fire department leaders should be wary of too much "buddy/buddy" behavior	Monitoring	relationships	
30	effective fire department leaders should guard against the need to always agree	Change	Diversity	
31	effective fire department leaders need to tow the clear, hard line	Command	Structure	
32	effective fire department leaders should maintain a stalwart demeanor	Command	Structure	- relationships
33	effective fire department leaders' most important work is risk reduction	Administration	System	community
34	effective fire department leaders should facilitate career development for all members	Administration	Individualized consideration	diversity
35	effective fire department leaders should believe everyone brings something to the table	Service	Diversity	team
36	effective fire department leaders should decentralize authority	Teambuilding	Team	
37	effective fire department leaders need to pick the spots where to bend for the sake of morale	Teambuilding	Team	individualized consideration
38	effective fire department leaders need to be strong disciplinarians	Monitoring	Structure	relationships
39	effective fire department leaders should know that good SOPs run the department	Monitoring	Structure	

(continued)

	Statement	Concourse Theme	Leadership Construct	Secondary Leadership Construct
40	effective fire department leaders need to create a safe zone for members to question processes and decisions	Change	Diversity	reflection
41	effective fire department leaders should operate at the 30,000' level	Command	System	
42	effective fire department leaders should understand the whole system and consider the ripple effects of every action	Administration	System	relationships
43	effective fire department leaders should seek to be politically astute	Administration	Community	system
44	effective fire department leaders should measure success through the betterment of department members	Service	Individualized consideration	diversity
45	effective fire department leaders value equity	Service	Individualized consideration	diversity
46	effective fire department leaders should give members time to correct issues on their own	Teambuilding	Team	reflection
47	effective fire department leaders should be democratic	Teambuilding	Community	diversity
48	effective fire department leaders need to break barriers to change	Change	Change	system

Table 35 further clarifies how the 10 USFES perspectives embody these leadership constructs based upon the salient statements for each perspective. In this way, Table 35 provides a visual way to see the perspectives vis a vis the leadership constructs, or general areas of attention and focus from leadership theories and models, and constructs the perspectives share and the constructs that differ for them.

Table 35

Leadership Constructs from Theories Corresponding to Perspectives on USFES Organizational Leadership

Constructs	USFES Perspective									
	A	B	C	D	E	F	A ¹	D ¹	Q-Sort 8	Q-Sort 9
Individual Consideration	X	X	X	X	X	X	X	X	X	X
System	X	X	X				X		X	
Community	X		X						X	
Structure, task orientation				X	X			X		X
Change	X	X	X	X			X		X	
Future (vision)	X			X			X		X	X
Teams/collaboration	X		X	X						
Relationships		X	X		X	X			X	
Reflection, assessment	X	X		X			X		X	
Diversity/Inclusivity		X	X	X			X			

Table 35 shows that all ten perspectives share the underlying leadership construct of individual consideration. It is the only underlying construct shared across every perspective.

Additional observations made clear in Table 35 include:

- five perspectives share the underlying construct of focus on the system, A, B, C, A¹, and Q-sort 8
- only three perspectives share the underlying construct of community, A, C, and Q-sort 8.
- four perspectives share the underlying construct of structure and task orientation, D, E, D¹, and Q-sort 9
- six perspectives share the underlying construct of change, A, B, C, D, A¹, and Q-sort 8.
- five perspectives – not the same five that share a focus on the system – share the underlying construct of being future-oriented, A, D, A¹, Q-sort 8, and Q-sort 9
- three perspectives – not the same three that shared a focus on community – share the underlying construct focused on teams and collaboration, A, C, and D
- five perspectives different than the five sharing a focus on the system and the five sharing a future-orientation share the underlying construct of relationships, B, C, E, F, and Q-sort 8
- a different combination of five perspectives share the underlying construct related to reflection and assessment, A, B, D, A¹, and Q-sort 8
- a different combination of four perspectives than the four that focus on structure and task orientation share the underlying construct focused on diversity and inclusivity
- Perspectives A, B, C, D, A¹, and Q-sort 8 are more complex perspectives including six or seven underlying constructs

- Perspectives E, F, D¹, and Q-sort 9 are less complex perspectives including two or three underlying constructs
- each perspective represents a unique combination of the underlying constructs.

Significance of Theoretical Findings

It is important to note that the statistical processes in Q methodology are not for the purposes of confirming a theory or any other positivist purpose. Nor did this study include theoretical sorts for such a purpose. The purpose in any positivist approach is to reduce and/or eliminate the qualitative and subjective. Instead, Q methodology maximizes these and makes subjectivity the main focus because the Q sort data represent expressions of qualitative intensity related to feeling or value (Stenner, 2011). Q methodologists execute the statistical processes for a deep understanding of subjectivity, not to eliminate it. The operationalized, subjective perspectives (the Q-sorts) of participants are the data to be analyzed, and the statistical processes allow researchers to see underneath the surface of their points of view and look at the complexities of personal perspectives in statistically-rigorous detail.

The detailed interpretation about the way a participant or group of participants completed the sorting activity is the goal, providing an explanation and deep understanding of each perspective. Brown (2008, as cited in Ramlo, 2016) asserted that Q is qualitative in both practice and theory, despite its use of factor analysis. In Q methodology, the factor extraction separates groups based upon *qualitative* differences in perspectives of the participants. Q methodology “uses subjective data to objectively group people, using factor analysis, with the explicit purpose to scientifically group people to best study their subjectivity about a specific topic” (Ramlo, 2016 p. 33). Q methodology “operates with an ontology in which the ultimate realities are neither objects nor subjects, but actual occasions of experience” (Stenner & Stainton-Rogers, 2004, p.

201, as cited in Ramlo, 2016). Meaning that the methodology as conceived by Stephenson is definitively *not* positivist, in which research involves the objective measurement of the natural world, because it is *subjectivity*, made operant and measurable through Q-sorts, that is measured and treated objectively with quantitative factor analysis.

The study included the seven theoretical sorts because of the assertions and recommendations of theories as effective in both academic and practitioners' circles, and my interest therefore in knowing if the theories are represented in the perspectives and to what extent. Q methodology is able to do this through making the participants' perspectives operant and because I made the theoretical perspectives operant using the same sample of statements regarding effective organizational leadership, and completing the factor analysis on both the participants' and the theoretical perspectives. In this study, the theories represent additional "participants" that have a view on effective organizational leadership. Q methodology provides a pragmatic way to include multiple theories and measure to what extent they correlate with existing perspectives among fire service professionals, and allowed for the abductive pursuit of new insights related to the alignment or lack thereof of theories with the real perspectives in the industry.

Demographic Characteristics and Perspectives on USFES Organizational Leadership

Examination of the demographics of the sorters included in each shared perspective does not indicate that any of the demographic characteristics are predictors for a particular perspective. For example, the four females' Q-sorts are included in three different perspectives: two significantly loaded on Perspective B, one on Perspective C, and one on Perspective D. Gender and the additional demographic characteristics by perspective are displayed in Table 36.

Table 36

Factors and Demographic Characteristics

Factor	Region	Comm Type	Dept Type	Dept Size	Rank	Yrs Service	Education	EFO	Gender	Age
A	East	urban	Career	20	1	16	2	Y	M	40
A	Midwest	suburban	combination	32	4	17	3		M	41
A	Midwest	suburban	combination	454	4	9	2		M	37
A	South	suburban	Career	82	2	22	3		M	38
A	West	urban	Career	458	2	22	3		M	50
A	West	urban	Career	150	2	32	2	Y	M	56
A	West	urban	Career	200	1	36	2	Y	M	58
A	West	urban	combination	111	4	12	2		M	35
AB	East	rural	Volunteer	70	4	8	4		M	25
B	Midwest	suburban	Career	135	2	24	1	Y	M	47
B	Midwest	suburban	Volunteer	35	4	5	2		F	47
B	South	urban	Career	650	3	20	3		M	42
B	West	urban	Career	1200	2	36	3		F	59
C	East	urban	Career	300	3	39	3		M	58
C	Midwest	suburban	combination	50	3	31	3	Y	M	59
C	West	urban	Career	4900	2	37	2	Y	F	59
C	West	urban	Career	200	4	19	2	Y	M	43
CD	Midwest	rural	Volunteer	25	2	17	3		M	34
D	East	suburban	combination	38	4	22	3		M	45
D	East	urban	Career	150	4	23	3		F	49
E	South	suburban	volunteer	42	1	18	2		M	42
F	East	suburban/rural	combination	90	2	15	3	Y	M	40
F	South	urban	career	1,250	2	32	2	Y	M	55
F	West	rural	combination	70	3	18	3		M	40
none	East	suburban	combination	35	4	16	2		M	36
A ¹	East	rural	volunteer	70	4	8	4		M	25
A ¹	East	suburban	combination	35	4	16	2		M	36
A ¹	East	suburban	combination	38	4	22	3		M	45
A ¹	East	urban	career	20	1	16	2	Y	M	40
A ¹	East	urban	career	150	4	23	3		F	49
A ¹	Midwest	rural	volunteer	25	2	17	3		M	34
A ¹	Midwest	suburban	combination	454	4	9	2		M	37
A ¹	South	suburban	career	82	2	22	3		M	38
A ¹	West	urban	career	150	2	32	2	Y	M	56
A ¹	West	urban	career	200	1	36	2	Y	M	58
A ¹	West	urban	combination	111	4	12	2		M	35
A ¹ D ¹	Midwest	suburban	combination	32	4	17	3		M	41
D ¹	West	urban	career	458	2	22	3		M	50

Note: Rank is indicated by number, with level 1 corresponding to Department Chief, level 2 corresponding to Assistant, Deputy, or Division Chief, level 3 corresponding to Battalion Chief, and level 4 corresponding to the officer rank below Battalion Chief (name of rank varies by department); highest level of education is indicated by number with level 1 corresponding to doctoral level, level 2 corresponding to master's level, level 3 corresponding to baccalaureate level, and level 4 corresponding to associate's level.

For those Q-sorts with significant loading on Perspective A, all of the geographic regions (East, South, Midwest, West) are represented, both urban and suburban departments and career and combination departments are represented. Rural and volunteer departments are not

represented among the eight participants whose Q-sorts significantly loaded on Perspective A, but the participant whose Q-sort has a confounded significant loading on Perspectives A and B is from a rural volunteer department of 70 members. The range of department size for the participants with Q-sorts significantly loading on Perspective A is 20 members to 458 members. All but Battalion Chief-level ranks are represented; years of service ranges from 12 to 36. The highest level of education for those with significant loading on Perspective A includes both baccalaureate and master's level degrees (the Q-sort confounded on Perspectives A and B represents the associate's level degree as the highest level of education), and three of the participants with Q-sorts significantly loading on Perspective A have completed the Executive Fire Officer program while five have not. The age range for those significantly loading on Perspective A is 35 years to 58 years.

The four Q-sorts that significantly load on Perspective B represent the Midwest, South, and West regions, suburban and urban community types, and all three department types, volunteer, career, and combination, ranging from 35 members to 1200 members. All rank levels except Chief are represented with years of service ranging from five years to 36 years. Three levels of educational degrees are represented: doctoral, master's, and bachelor's, and one of participants with a Q-sort significantly loading on Perspective B has completed the EFO program. The age range for those significantly loading on Perspective B is 42 years to 59 years.

Four Q-sorts significantly load on Perspective C, representing the East, Midwest, and West regions, suburban and urban community types, and career and combination department types, ranging in size from 50 members to 4900 members. One Q-sort with confounded significant loading on Perspectives C and D represents a rural, volunteer department of 20 members. All but the Chief level of rank are represented, and years of service ranges from 19

years to 39 years. The participants have master's and baccalaureate degrees, and three of the four completed the EFO Program. The age range for those significantly loading on Perspective C is 43 years to 59 years.

Both Q-sorts significantly loading on Perspective D represent the East region, one urban, career department with 150 members, and one suburban, combination department with 38 members. Both represent level four in rank, but because other Q-sorts of participants with level four in rank significantly load on Perspectives A, B, C, and A¹ it is not a predictor for Perspective D. The years of service are similar too, with 23 years and 22 years. The participants with Q-sorts significantly loading on Perspective D both represent baccalaureate level of education, one is male and one is female, ages 45 and 49.

Only one participant's Q-sort significantly loads on Perspective E besides the negative, significant loading for the public administration theoretical Q-sort, representing a suburban, volunteer department of 42 members in the South region. There is a participant whose Q-sort was only 0.004 below significantly loading on Perspective E (0.442) who represents the East region and an urban, career department of 150 members. The participant whose Q-sort loads significantly on Perspective E is 42 years old, of level 1 rank with 18 years of service and a master's degree, and the participant whose Q-sort nearly significantly loads on Perspective E is 49 years old, of level 4 rank with 23 years of service and a baccalaureate degree.

Three Q-sorts significantly load on Perspective F, representing the East, South, and West regions, and three different community types – urban, rural, and suburban/rural. Two represent combination departments of 90 and 70 members, and one represents a career department with 1250 members. They represent levels 2 and 3 in rank and years of service range from 15 years to

32 years. Master's and baccalaureate levels of education are represented, and two have completed the EFO Program. The age range is 40 years to 55 years.

Perspective A¹ has 11 Q-sorts significantly loading on it. They represent all the regions, all the community and department types, and department sizes ranging from 25 members to 454 members.

Perspectives A, A¹, D, and E do not have Q-sorts significantly loading from participants belonging to large fire departments of over 1,000 members, whereas Perspectives B, C, and F do have Q-sorts significantly loading from participants belonging to such large departments. The range of department size in all of these is broad, however, so department size would not be a reliable predictor for any of these Perspectives.

Q-sort 8, representing a unique perspective on USFES organizational leadership is the only participant with a doctoral level degree and is among the 12 participants who completed the EFO Program. It is possible that education level is a predictor of this unique perspective.

Research Questions Summary

All of the above analysis and discussion of the results provides information relevant to the research questions, summarized in this section.

Research Question 1

What are the perspectives of officers in the USFES Departments regarding effective *Department-wide* leadership? There are ten different perspectives on effective organizational leadership revealed in the study. Each one has a unique combination of underlying leadership constructs within it. This is significant for the fire service in several ways. More than one perspective may be represented in a fire department, producing the potential for competing priorities among the department leaders and/or mismatches between the leadership approach

from department leaders and the expectations of department members and the community. The leadership perspective may also match or mismatch the dominant organizational culture. Research indicates that organizational effectiveness improves when the leadership style and the dominant culture of the organization are aligned (Cameron & Quinn, 2006; Cameron, Quinn, DeGraff, & Thakor, 2014). Leadership effectiveness and organizational effectiveness in the fire service has implications for safety in communities and safety for first-responders. The potential for competing priorities and mismatches and reduced leadership and organizational effectiveness can occur when the number of different perspectives is greater. However, it is also possible that diversity of thought and style regarding organizational leadership can have benefits. In other, non-fire-service contexts, different priorities, skills, behaviors, or styles are effective and important dependent upon the level of leadership (dyadic, team, organizational) and organizational context (administrative or operations, for example), and the ability to manage conflicting tensions, as opposed to developing cohesion, has been shown to be the most predictive of top-team performance (Day et al., 2012; Lencioni, 2006; Leonard, Wiita, & Milane, 2017; Yukl, 2010).

Research Question 2

How do these perspectives differ and how are they similar? The perspectives on effective organizational leadership converge and diverge along a number of leadership dimensions, including internal or external focus, present or future orientation, emphasis on collaboration and teamwork or individual growth and empowerment. Analysis of the differences and similarities provides further insight about how competing approaches and mismatches can emerge. Fire department leaders may consider how they can learn about perspectives on organizational leadership that exist in their departments. If multiple perspectives exist, understanding the

similarities and differences can provide opportunities to align and develop a shared understanding of effective leadership across the department. Such shared understanding impacts department member engagement and performance, since effective leadership may be about the resonance of leaders and members, alignment regarding priorities and goals, and fit across leaders, the organization, and contexts, but as noted above related to Research Question 1, diversity and tensions, when well-managed, can also be important (Canyon & Morrison, 2010; Day et al., 2012; Duncan et al., 2012; Goleman, Boyatzis, & McKee, 2001; Lencioni, 2006; Leonard et al., 2017; Yukl, 2010).

Research Question 3

How do these perspectives correspond to existing leadership constructs, concepts, theories, and models? A key finding related to the third research question is that none of the perspectives align cleanly with leadership theories. By making subjective thoughts about effective organizational leadership operant through Q-sorts, Q methodology is ideally suited to show in detail how individuals think about the topic, and demonstrate that perspectives on the topic are not purely theoretical through the inclusion of the theoretical Q-sorts. Where there is intersection and commonality of the perspectives with theory it may indicate that particular theories have informed practice, a key purpose for the development of leadership theories, their research and education (Northouse, 2013). For example, nine participants have completed the National Fire Academy's Executive Fire Officer Program, which emphasizes adaptive leadership theory, and the perspectives of three of those participants had a significant correlation with adaptive leadership theory. Six of those participants' perspectives did not align with adaptive leadership theory in spite of completing an extensive program emphasizing adaptive leadership theory. Similarly, three of the four participants who completed Master of Public Administration

degrees have perspectives that do not align with public administrative leadership theory. Perspectives E and F did not have a theoretical Q-sort that significantly loaded in the factor analysis, although other theories and models that were not included as theoretical sorts have some similarities with these perspectives.

Several possibilities may explain why the theoretical sorts do not strongly align with existing perspectives on organizational leadership in the American fire service, and why there are similarities, but there are not exact matches with the theories included in the study as theoretical Q-sorts nor with other theories and models. It is possible that the lack of strong alignment of the perspectives with theories is due to the centrality of self-reference and the development of an individual, personal approach to the exercise of leadership (Day et al., 2012; Wheatley, 2006). It is also possible that the theories are not intended to be pragmatic and useful, rather to demonstrate an important concept or component to leadership or the effects of a mediating variable to be predictive or explanatory (Albert, n.d.; Yukl, 2010).

Limitations and Delimitations of the Study

These points related to the USFES perspectives and how closely they may fit with leadership theories and models involves one of the limitations of this study. The Q-sample of statements was developed naturalistically within the USFES, which means it does not include statements that come directly from theory. There are some statements that do align with theory, such as “effective fire department leaders should be future-oriented” and “effective fire department leaders should understand the whole system and consider the ripple effects of every action,” but most Q-sample statements do not align neatly and clearly with statements defining theory and models. The explanation for this has three parts. First, USFES professionals are not likely to think exactly like theories because they have not been consistently exposed to any

prescribed theory or combination of theories in fire service education and professional development, and even if there were a prescribed theory or theories for fire service professionals, it is unlikely that the theoretical statements are the only ones that a qualitative study would reveal. Second, the Q-sample is purposefully selected to include only statements rich in meaning and open to interpretation. Third, existing theories and models may not reflect reality because of the biases of the scholar proposing the theory, the narrow focus of the theory development, or over-simplification (Yukl, 2010).

Another possible limitation relates to the selection of the P-sample. It is possible that additional perspectives exist because the initial, purposive nucleus of participants for the P-sample may have held too much in common across their viewpoints or because their networks of others to refer in the snowballing process were too similar. For example, the nucleus of participants from which the purposive snowballing proceeded did not include a non-sworn fire officer. Sworn fire service members swear an oath to protect and serve the community and have usually completed training in firefighting; non-sworn fire service members are those who serve in capacities that do not involve firefighting, such as administrative areas. The snowballing led to participation of only three fire service officers who are likely non-sworn personnel rather than sworn members of the department (the demographic questionnaire did not collect this information, so it is a conclusion drawn from the titles of the participants). While the three likely non-sworn participants have Q-sorts that load significantly on three different perspectives (B, C, and confounded significant loading on C and D), it is possible that additional purposive snowballing seeking the participation of more non-sworn personnel might have uncovered additional perspectives on effective organizational leadership in the fire service.

Additionally, there are delimitations it is important to note. The study involved the discovery of perspectives on organizational leadership in only American municipal fire departments. There may be different or more perspectives on organizational leadership in other types of fire departments, such as private, corporate, or military fire service departments, or wild fire protection agencies, and in other parts of the world. It is also beyond the scope of this study to correlate the different perspectives with levels of department performance. It did not include any data collection related to department performance, nor did it measure the effectiveness of leadership performance for the different perspectives. Such correlation would not be possible because of the small sample size for this study, and the nature of the sample related to discovery of the full variety of subjective perspectives on USFES organizational leadership. A study to examine the relationships between the perspectives on organizational leadership and organizational performance would also require a large sample of departments to be valid. In short, whether any perspective is associated with performance is not answerable from this study.

This study does not provide information regarding the prevalence of the varied organizational leadership perspectives. The methodology and sample size do not allow for any determination of the majority and minority perspectives across USFES organizations. To learn how the perspectives are manifested in the entire population would require a quantitative study with random selection from a large sample.

Recommendations for Further Study

This study is foundational to further rigorous investigations of organizational leadership in the American fire service. The National Fallen Fire Fighters Foundation 2007 white paper recommended attention to the stagnation in reducing fire deaths, injuries, damages, and loss and lack of research in the areas of organizational behavior, organizational culture, organizational

change, and organizational leadership that it is fundamental to improving the USFES safety culture and continue reducing firefighter and civilian deaths and injuries, and damages and loss due to fire. While the scope of this study does not allow for any recommendations regarding a leadership framework, model or theory as a foundation for fire service leadership education, for example, it does inform the next research that is necessary before such recommendations could be made. Quantitative studies can answer research questions related to determining the majority views on effective USFES organizational leadership and if any of the perspectives are correlated with performance measures.

As previously noted, it is significant that there is not a single perception of what constitutes effective organizational leadership and that there is alignment, but not perfect alignment with so many existing leadership theories and models. Identifying if any of the perspectives and the theories aligning with them are dominant and any underlying reasons for the dominance, as well as what explains the existence of minority perspectives can shed light on what theories and models USFES members are exposed to the most and how fire service professionals are learning leadership since it is not included in the NFPA job performance requirements for fire service officers and there is not an agreed upon leadership theoretical framework included in fire academies and fire service-related degree programs.

It is also significant that there is not a single perception of what constitutes effective organizational leadership because it is a possible explanation for the variation in department performance, if one or more perspectives on effective organizational leadership is/are associated with higher levels of performance, but that is not answerable from this study. Therefore, further quantitative research involving measures of organizational performance such as those associated with fire department accreditation, and identifying departments' organizational leadership

practices associated with each perspective would allow researchers to assess if any of the perspectives relate to higher levels of performance.

This study also implicates several theories or models as better matches for the perspectives revealed in the study than those already applied or promoted as relevant in the American fire service, and can inform the research design of future studies, ensuring that all the theories or models associated with the perspectives in this study are included. For example, public administrative leadership theory and boundary-spanning leadership, which have some similarity to Perspective C in the study, have not been studied in the USFES.

Since the perspectives revealed in the study do not align perfectly with theories, another recommendation is the analysis of the elements from the different theories and models included in the USFES perspectives to consider the development of one or more integrated theories that include *all* the elements of a particular perspective and may better reflect reality in fire departments where the leaders hold that perspective. Such development could lead to improved theoretical utility by enhancing the understanding of leadership complexity and increasing the ways that the theory can be applied and used and for leadership development of pre-supervisory personnel (Yukl, 2010; Day, 2014). As renowned psychologist Kurt Lewin noted, “There’s nothing so practical as good theory” (quoted in Schein, 1995), and it may be that existing theories are not so practical because they are not “good” in terms of reflecting the reality. When theory is practical, it improves understanding of underlying causes and explanatory processes, enhancing usage. This is the aim of investigations from the realist paradigm—to uncover the underlying causal mechanisms that explain what we can observe. Creating a fully descriptive or explanatory model of the phenomenon of effective organizational leadership in the American fire service that builds on this exploration of the perspectives in the industry would perhaps lead to

higher performance and more consistency in performance across the industry (Archer et al., 2016; Day et al., 2009; Yukl, 2010).

Beyond such studies on the correlation of leadership perspectives with performance, assessing the impact in organizations of change leadership specifically is an additional recommendation that would add to the depth of knowledge about leadership in USFES organizations and contribute to the achievement of the NFFF Life Safety Initiatives. The NFFF Life Safety Initiatives recommend a number of cultural or systems changes that cannot be imposed by NFPA standards, including the integration of risk management/safety management/behavior-based systems. Such change initiatives could be studied vis-à-vis organizational leadership to determine the most effective and efficient ways to integrate new cultural norms or systems importance to life safety in the fire service and in communities. Six of the ten perspectives revealed in this study include the change leadership construct, but each may have different leadership behaviors manifested, underlying mechanisms of achieving change, and different levels of effectiveness at implementing change.

All the perspectives on organizational leadership in the fire service agree that leadership development is necessary for department leaders, but disagreed on the type of leadership development that is best. While two approaches have been studied in the fire service, developmental experiences (Jones, 2007) and leadership training (Beaton et al., 2001), there have not been studies that compare approaches or that consider combinations or integration of leadership development approaches that may be most effective, including mentoring, developmental experiences, formal education, and in-service training experiences, and combinations of cognitive and reasoning capacities with skills development (Day et al., 2009).

This study did not indicate any demographic criteria as predictors for a particular leadership perspective, and additional quantitative studies are recommended that aim to determine if the leadership perspective or perspectives linked to performance is/are mediated by any of these criteria such as department type and size or community type.

Implications

There are no job performance requirements (JPRs) related to individual or organizational leadership currently included in the National Fire Protection Association extensive JPRs (NFPA, 2014b). The absence of any leadership competencies in the NFPA JPRs is notable given the investigations of poorly executed emergency managements such as the responses to hurricanes Katrina and Sandy, and that practitioners in the fire service and related industries acknowledge the importance of leadership and the relationship between leadership and performance. The shared characteristics, skills, abilities, and behaviors identified in the study can be the beginning of the development of job performance requirements (JPRs) for fire service officers that relate to leadership competencies for individuals or organizations.

The fact that multiple perspectives on organizational leadership exist and that they do not fit perfectly with theoretical models that have already been applied in studies of leadership in the American fire service can mean that there have been mismatches between theory and practice when a particular leadership theory or model is assumed *a priori* and studied in an organization where it may not be practiced. Researchers might have studied something that does not exist. Findings in such studies may be coincidental or spurious and unlinked to the theoretical framework applied. Going forward, using instruments to apply leadership theories *a priori* can be questioned as imperfect measures of the organizational leadership and a limitation since this

study has shown that the perspectives on organizational leadership in the industry do not clearly or fully match with theories.

With as many varied perspectives on organizational leadership that exist in the American fire service, it is possible that multiple perspectives exist within a single department. This implies the possibility of mismatches among department members in expectations, actions and responses that affect leadership effectiveness and organizational performance. Fire department leaders may consider the possibility of multiple perspectives within their departments and intentionally work to build shared expectations regarding leadership, or to effectively manage the tensions that different perspectives can create to take advantage of the benefits of different perspectives, including more effective problem-solving (Canyon & Morrison, 2010; Day et al., 2012; Duncan et al., 2012; Goleman et al., 2001; Lencioni, 2006; Leonard et al., 2017; Yukl, 2010).

Personal Reflections

This reflective section is intended to show how I have both shaped the study based upon personal experiences and been shaped by the process of undertaking this study. As a methodological requirement in a study with qualitative elements, acknowledging personal experiences and attitudes is important because attaining a more neutral posture is not possible without acknowledging the influence of personal frames of reference. Objectivity is operationally defined as treating data fairly, not the absence of opinions, beliefs, preferences, or experiences (Gould, 1996). In Chapter 3, I acknowledged experiences and attitudes that influenced my personal frame of reference and prepared me to treat the data fairly in this study. This section goes further to explore what the study has added to my frame of reference.

My pre-study frame of reference and awareness of questions surrounding leadership in the fire service added up to a strong case for me to look into researching fire service leadership in

an organizational leadership doctoral program. The information gaps about leadership in the American fire service were a source of dissatisfaction among my acquaintances in the fire service and for me, and sparked intellectual curiosity related to my chosen discipline. Both dissatisfaction and curiosity have provided intrinsic motivation to undertake and complete the study. Intellectual curiosity does not a dissertation make, of course, but it did lead me to continue asking questions and to realize that before I could accurately study if there is a best approach to organizational leadership in the American fire service, I would have to know what different approaches actually exist, and that there were probably more approaches than the two I already knew about.

My frame of reference now includes much more – from the trivial (my personal Q-sort was confounded on Factors A and B, somewhat aligned with a conglomeration of theories and models: adaptive leadership theory, the learning organization model, servant leadership, transformational leadership, and complexity leadership theories) – to the more significant, such as extensive knowledge of prior research in the fire service and related industries. I have also more closely examined my personal ontological view and acknowledged a realist paradigm that supported taking an exploratory approach and employing the hybrid, Q methodology. Completion of the study has affirmed this paradigm for me. In answering the foundational questions about the perspectives on organizational leadership that exist in the American fire service using Q methodology, I have reconciled my prior cognitive tension between the choice to pursue a study with a qualitative, interpretivist approach or a purely statistical approach. The statistical rigor embedded in the qualitative, interpretivist approach of Q methodology created insights related to existing perspectives on effective organizational leadership in the American

fire service and to what extent the perspectives include theoretical constructs -- insights that it would not have been possible to pursue with a purely qualitative or purely quantitative approach.

I am hopeful for future studies that come about from this foundational study. The impact future research on organizational leadership in the fire service can have on standards, policies, education, and performance in the fire service will be significant. Most importantly, continued research on organizational leadership in the fire service can make important contributions to the industry's Life Safety Initiatives to reduce injuries and deaths.

Conclusion

Chapter 5 presented the summary of findings and details regarding the similarities and differences among the perspectives on organizational leadership in the American fire service revealed in the study. It included a discussion of limitations and noted the delimitations of the study, recommendations for future study, and the implications of the study. The study isolated ten perspectives that have varying degrees of fit with existing theories, including (listed alphabetically) adaptive leadership theory, authentic leadership theory, boundary-spanning leadership theory, complexity leadership theory, crew resource management, leader-member exchange theory, learning organization model, Mintzberg's model of management, public administrative leadership theory, regulatory focus theory, servant leadership theory, transactional leadership, and transformational leadership. The fact that the perspectives as expressed by industry professionals through their Q-sorts and interviews do not perfectly fit with theory are an indication of the lack of knowledge and agreement in the industry on a theory or model that facilitates organizational performance, the imperfection or over-simplification of theories, the lack of required educational, training, or professional development related to leadership, and varied responses to the many complexities and contexts within a fire department. Fire service

professionals have used their own experiences, education, and judgment to develop perspectives on what constitutes effective organizational leadership.

References

- Ahrens, M. (2013). America burning 1973 – 2013: Work in progress. *National Fire Protection Association Journal*, (May/June). Retrieved from <http://www.nfpa.org/newsandpublications/nfpa-journal/2013/may-june-2013/features/work-in-progress>
- Ahrens, M. (2016). *Trends and patterns of U.S. fire loss*. National Fire Protection Association. Retrieved from <http://www.nfpa.org/news-and-research/fire-statistics-and-reports/fire-statistics/fires-in-the-us/overall-fire-problem/trends-and-patterns-of-us-fire-losses>
- Albert, M. (nd). Radical theory instructional. Retrieved from <https://zcomm.org/wp-content/uploads/zinstructionals/htdocs/RTInstruc/id6.htm>
- Andrews, R., Ashworth, R., & Meier, K. J. (2014). Representative bureaucracy and fire service performance. *International Public Management Journal*, 17(1), 1-24.
- Antonakis, J., Avolio, B. J., & Sivasubramaniam, N. (2003). Context and leadership: An examination of the nine-factor full-range leadership theory using the Multifactor Leadership Questionnaire. *The Leadership Quarterly*, 14(3), 261-295.
- Archer, M., Decoteau, C., Gorski, P., Little, D., Porpora, D., Rutzou, T., Smith, C., Steinmetz, G., & Vandenberghe, F. (2016, December 23). [web log post] What is critical realism? Retrieved from <http://www.asatheory.org/current-newsletter-online/what-is-critical-realism>
- Ash, J. (2014, November 20). Firefighters taking heat: Independent report sharply critical of city department. *Main Street Journal*. Retrieved from <http://msjnews.com/firefighters-taking-heat-p7016-91.htm>
- Auernhammer, T. (1994). Fresh heirs. *Fire Chief Magazine*, (September), 50-54.

- Avolio, B. J., & Gardner, W. L. (2005). Authentic leadership development: Getting to the root of positive forms of leadership. *The Leadership Quarterly*, 16(3), 315-338.
- Avolio, B. J., Walumbwa, F. O., & Weber, T. J. (2009). Leadership: Current theories, research, and future directions. *Annual Review of Psychology*, 60, 421-449.
- Baran, B. E., & Scott, C. W. (2010). Organizing ambiguity: A grounded theory of leadership and sensemaking within dangerous contexts. *Military Psychology*, 22(S1), S42-S69.
doi:10.1080/08995601003644262
- Barr, R. C., & Eversole, J. (Eds.). (2003). *The fire chief's handbook*. Tulsa, OK: PennWell Books.
- Beaton, R., Johnson, L. C., Infield, S., Ollis, T., & Bond, G. (2001). Outcomes of a leadership intervention for a metropolitan fire department. *Psychological Reports*, 88(3c), 1049-1066.
- Blanchard, K.H. (1985). *SLII: A situational approach to managing people*. Escondido, CA. Blanchard Training and Development.
- Boon-itt, S., & Pongpanarat, C. (2011). Measuring service supply chain management processes: The application of the Q-sort technique. *International Journal of Innovation, Management and Technology*, 2(3), 217.
- Bowen, L. (2016, January 7). Black firefighters ask feds to investigate hiring at Chicago Fire Department. *The Chicago Tribune*. Retrieved from <http://www.chicagotribune.com/news/local/breaking/ct-firefighters-claim-discrimination-met-0108-20160107-story.html#>

- Brame, K. (2001). Overview for the California public safety leadership and ethics program sponsored by the California community college Chancellor's office public safety education advisory committee. Unpublished manuscript.
- Brinker, K., Lumia, M., Markiewicz, K. V., Duncan, M. A., Dowell, C., Rey, A., & Funk, R. (2015). Assessment of emergency responders after a vinyl chloride release from a train derailment-New Jersey, 2012. *MMWR: Morbidity and Mortality Weekly Report*, 63, 1233-1237.
- Brown, C. S. (2014). The who of environmental ethics: Phenomenology and the moral self. *Ecopsychology, Phenomenology, and the Environment* (pp. 143-158), Springer.
- Brown, S. R. (1993). A primer on Q methodology. *Operant Subjectivity*, 16(3/4), 91-138.
- Brown, S. R. (1997). The history and principles of Q methodology in psychology and the social sciences. Read at the British Psychological Society Symposium on, A Quest for a Science of Subjectivity: The Lifework of William Stephenson, University of London.
- Brunacini, A. V. (2002). *Fire command*. Burlington, MA: Jones & Bartlett Learning.
- Bryman, A., Stephens, M., & aCampo, C. (1996). The importance of context: Qualitative research and the study of leadership. *The Leadership Quarterly*, 7(3), 353-370.
- Bucci, S. P., Inserra, D., Lesser, J., Mayer, M. A., Slattery, B., Spencer, J., & Tubb, K. (2013). After Hurricane Sandy: Time to learn and implement the lessons in preparedness, response, and resilience. *The Heritage Foundation Emergency Preparedness Working Group*, (144).
- Buckman, J. (2014, November). Can the fireground be research driven? *FireRescue1.com*, Retrieved from <http://www.firerescue1.com/fire-chief/articles/2020664-Can-the-fireground-be-research-driven/>

- Burgess, J. L., Duncan, M., Mallett, J., LaFleur, B., Littau, S., & Shiwaku, K. (2014). International comparison of fire department injuries. *Fire Technology*, 50(5), 1043-1059.
- Cameron, K., & Quinn, R. (2006). *Diagnosing and changing organizational culture*. San Francisco, CA: Jossey-Bass Publications.
- Cameron, K. S., Quinn, R. E., DeGraff, J., & Thakor, A. V. (2014). *Competing values leadership*. Edward Elgar Publishing.
- Canyon, D. V., & Morrison, P. (2010). Do the emergency management and emergency medicine professions require a new leadership theory? *Journal of Rural and Tropical Public Health*, 9, 51-52.
- Carter, H. R. (2005). *Member retention in the volunteer fire service: An analysis of the impact of leadership* (Ph.D.). Retrieved from ProQuest Dissertations & Theses Full Text. (305349073.)
- Carter, H. (2007). Approaches to leadership: The application of theory to the development of a fire service-specific leadership style. *International Fire Service Journal of Leadership and Management*, 1(1), 28 -37.
- Center for Public Safety Excellence. (2014). About professional credentialing. <http://publicsafetyexcellence.org/professional-credentialing/about-credentialing-cpc.aspx>
- Charmaz, K., & Henwood, K. (2008). Grounded theory. In: C. Willig and W. Stainton-Rogers (Ed.s.) *Handbook of qualitative research in psychology*. London: Sage, 240–260.
- Clark, B. A. (2015). *I can't save you but I'll die trying: The American fire culture*. Nashville: Premium Press.
- Clark, W. (1976). Wingspread II conference on fire service administration, education and research. *Fire Files Digital Library*. Retrieved from <https://fire.omeka.net/items/show/8>

- Coleman, R. J., Revere, M. A., & Craig, F. M. (2010). *The compleat [sic] fire officer: Officer self-assessment & professional development*. Chantilly, VA: Center for Public Safety Excellence, Inc.
- Comfort, L. K., & Okada, A. (2013). Emergent leadership in extreme events: A knowledge commons for sustainable communities. *International Review of Public Administration*, 18(1), 61-77.
- Comfort, L. K., & Resodihardjo, S. (2013). Leadership in complex adaptive systems. *International Review of Public Administration*, 18(1), 1-5.
- Cochran, K. (2013, January). Servant leadership theory in practice. In *The New Chief Leadership Symposium*, conducted at the meeting of the International Association of Fire Chiefs, Chicago, Illinois.
- Compton, D. (2010). *Progressive leadership principles, concepts, and tools: The fire officer's guide to excellence in leadership*. Stillwater, OK: Fire Protection Publications.
- Covey, S. R. (1989). *The 7 habits of highly successful people*. New York: Fireside.
- Crosby, B. C., & Bryson, J. M. (2014). Public integrative leadership. In D. Day (Ed.), *The Oxford handbook of leadership and organizations* (pp. 57-72). New York, NY: Oxford University Press.
- Cross, R. M. (2005). Exploring attitudes: The case for Q methodology. *Health Education Research*, 20(2), 206-213.
- Crowe, J., Allen, J. A., & Bowes, B. (2014). Multi-crew responses to a structure fire: Challenges of multiteam systems in a tragic fire response context. In *Pushing the Boundaries: Multiteam Systems in Research and Practice* (pp. 205-219). Emerald Group Publishing Limited.

- Day, D. V. (Ed.). (2014). *The Oxford handbook of leadership and organizations*. Oxford, U.K. Oxford University Press.
- Day, D. V., Harrison, M. M., & Halpin, S. M. (2012). *An integrative approach to leader development: Connecting adult development, identity, and expertise*. New York, NY: Routledge.
- DeJoy, D. M., Smith, T. D., & Dyal, M. A. (2017). Safety climate and firefighting: Focus group results. *Journal of Safety Research*, 62, 107-116. doi.org/10.1016/j.jsr.2017.06.011
- Ditch, R. L. (2012). *An analysis of factors influencing the achievement of higher education by chief fire officers* (Doctoral dissertation). Retrieved from ProQuest Dissertations & Theses. (1039271297.)
- Dixon, B. C. (2014). *Leadership under extreme conditions: A qualitative study of volunteer firefighter leader's [sic] experiences* (Doctoral dissertation). Retrieved from ProQuest Dissertations & Theses. (3615111.)
- Donner, J. C. (2001). Using Q-sorts in participatory processes: An introduction to the methodology. *Social Development Papers*, 36, 24-49.
- Duncan, R. C., Matthew, D. A., & Scarpa, S. J. (2012). Fire service influence on homeland security regionalism and councils of government (COGs). *International Fire Service Journal of Leadership and Management*, 6, 27 – 31.
- Durso, F. (2015, January 15). Homes are burning so darn fast: Fire chiefs make convincing argument for home fire sprinklers. [Web log post]. Retrieved from <http://nfpatoday.blog.nfpa.org/2015/01/homes-are-burning-so-darn-fast-fire-chiefs-make-convincing-argument-for-home-fire->

sprinklers.html?utm_source=feedburner&utm_medium=feed&utm_campaign=Feed%3A+nfpablog+%28NFPA+Today+BLOG%29

- Eigel, K. M., & Kuhnert, K. W. (2005). Authentic development: Leadership development level and executive effectiveness. *Monographs in Leadership and Management*, 3, 357-385.
- Ellis, G., Barry, J., & Robinson, C. (2007). Many ways to say 'no', different ways to say 'yes': Applying Q-methodology to understand public acceptance of wind farm proposals. *Journal of Environmental Planning and Management*, 50(4), 517-551.
- Ent, M., Baumeister, R., Vonasch, A. (2012). Power, leadership, and self-regulation. *Social and Personality Psychology Compass*, 6(8), 619-630. doi:10.1111/j.1751-9004.2012.00446
- Ernst, C., & Yip, J. (2009). Boundary spanning leadership: Tactics for bridging social boundaries in organization. In T. Pittinsky (Ed.) *Crossing the divide: Intergroup leadership in a world of difference*. Boston, MA: Harvard Business School Press.
- Esteve, M., Boyne, G., Sierra, V., & Ysa, T. (2013). Organizational collaboration in the public sector: Do chief executives make a difference? *Journal of Public Administration Research and Theory*, 23(4), 927-952.
- Eyre, A. (2014). The making of a hero: An exploration of heroism in disasters and implications for the emergency services. *International Fire Service Journal of Leadership and Management*, 8, 7-16.
- Faturechi, R. (2016). *The fire sprinkler war, state by state*. Propublica. Retrieved from <https://www.propublica.org/article/the-fire-sprinkler-war-state-by-state>
- Fiedler, F. E. (1995). Cognitive resources and leadership performance. *Applied Psychology*, 44(1), 5-28.

- Fleming, C. J., McCartha, E. B. & Steelman, T. A. (2015). Conflict and collaboration in wildfire management: The role of mission alignment. *Public Administration Review*.
doi:10.1111/puar.12353
- Focht, W. (2002). Assessment and management of policy conflict in the Illinois River watershed in Oklahoma: An application of Q methodology,” *International Journal of Public Administration* 25:1311-1349.
- Franco, Z., Blau, K., and Zimbardo, P. 2011. Heroism: A conceptual analysis and differentiation between action and altruism. *Review of General Psychology*, 15(2), 99 – 113.
- Gasaway, R. B. (2008). Fireground command decision making: Understanding the barriers challenging commander situation awareness (Ph.D.). Retrieved from ProQuest Dissertations & Theses Full Text. (230704335.)
- Gay, M. (2015, October 6). FDNY starts diversity training for all firefighters. *The Wall Street Journal*. Retrieved from <http://www.wsj.com/articles/fdny-starts-diversity-training-for-all-firefighters-1444179604>
- Gist, R., Taylor, V. H., & Raak, S. (2011). Suicide surveillance, prevention, and intervention measures for the US Fire Service: Findings and recommendations for the Suicide and Depression Summit. National Fallen Firefighters Foundation.
- Goleman, D., Boyatzis, R., & McKee, A. (December 2001). Primal leadership. In *On managing yourself* (p. 169-188, 2011). Boston: Harvard Business Review Press.
- Gorman, C. A., Meriac, J. P., Overstreet, B. L., Apodaca, S., McIntyre, A. L., Park, P., & Godbey, J. N. (2012). A meta-analysis of the regulatory focus nomological network: Work-related antecedents and consequences. *Journal of Vocational Behavior*, 80(1), 160-172.

- Gould S. J. (1996). *The mismeasure of man: Revised and expanded edition*. New York: W. W. Norton.
- Griffin, M., & Talati, Z. (2014). Safety leadership. In D. Day (Ed.), *The Oxford handbook of leadership and organizations* (pp. 638-656). New York, NY: Oxford University Press.
- Griffith, J. C., Roberts, D. L., & Wakeham, R. T. (2015). A meta-analysis of crew resource management/incident command systems implementation studies in the fire and emergency services. In *A3iR Conference Proceedings, Jan 2015*.
- Guzzi, A. (2011, July). The company officer: Cultivating reliable team members. *Fire Engineering* 170(11). Retrieved from <http://www.fireengineering.com/articles/print/volume-164/issue-7/features/the-company-officer-cultivating-reliable-team-members.html>
- Hall, J. (2013). *U.S. experience with sprinklers*. National Fire Protection Association. Retrieved from <http://www.nfpa.org/news-and-research/fire-statistics-and-reports/fire-statistics/fire-safety-equipment/us-experience-with-sprinklers>
- Hannah, S. T., Avolio, B. J., & Walumbwa, F. O. (2011). Relationships between authentic leadership, moral courage, and ethical and pro-social behaviors. *Business Ethics Quarterly*, 21(4), 555-578.
- Hannah, S., & Parry, K. (2014). Leadership in extreme contexts. In D. Day (Ed.), *The Oxford handbook of leadership and organizations* (pp. 613-637). New York, NY: Oxford University Press.
- Hannah, S. T., Uhl-Bien, M., Avolio, B. J., & Cavarretta, F. L. (2009). A framework for examining leadership in extreme contexts. *The Leadership Quarterly*, 20(6), 897-919.

- Hayes, B. (2014, December 1). How the fire service assassinates its leaders [Web log post].
Retrieved from <http://www.firechief.com/2014/12/01/how-the-fire-service-assassinates-its-leaders/>
- Heifetz, R. A. (2010). Adaptive leadership. In G. R. Hickman (Ed.), *Leading organizations: Perspectives for a new era*, 2nd ed., (pp. 505-509). Thousand Oaks, CA: Sage.
- Hickman, G. R. (Ed.). (2010). *Leading organizations: Perspectives for a new era*, 2nd ed. Thousand Oaks, CA: Sage.
- Hoffman, E. L., & Lord, R. G. (2013). A taxonomy of event-level dimensions: Implications for understanding leadership processes, behavior, and performance. *The Leadership Quarterly*, 24(4), 558-571.
- Holton, B., & Roden, E. (2014, September 24). Firefighting coast to coast [Web log post].
Retrieved from <http://www.fireengineering.com/articles/2014/09/humpday-hangout-2.html>
- Hong, O., Phelps, S., Feld, J., & Vogel, S. (2012). Occupational injuries, duty status, and factors associated with injuries among firefighters. *Workplace Health & Safety*, 60(12), 517-523.
- Hughes, R., Ginnett, R., & Curphy, G. (2010). Contingency theories of leadership. In G. Hickman (Ed.), *Leading organizations: Perspectives for a new era*, 2nd ed. (pp. 101 – 121). Thousand Oaks, CA: Sage.
- Hunter, S. T., Bedell-Avers, K. E., & Mumford, M. D. (2009). Impact of situational framing and complexity on charismatic, ideological and pragmatic leaders: Investigation using a computer simulation. *The Leadership Quarterly*, 20(3), 383-404.

- Hunter, S., Cushenberry, L., Thoroughgood, C., Johnson, J., & Ligon, G. (2011). First and ten leadership: A historiometric investigation of the CIP leadership model. *The Leadership Quarterly*, 22(1), 70 – 91.
- International Fire Service Training Association. (2011). *Fire and emergency services orientation and terminology*, 5th ed. Stillwater, OK: Fire Protection Publications.
- International Fire Service Training Association. (2013). *Essentials of firefighting*, 6th ed. Stillwater, OK: Fire Protection Publications.
- International Fire Service Training Association, Fire Service Reputation Management Development Group. (2010). *Fire Service Reputation Management White Paper*. Retrieved from <http://firerescue1-praetorian.netdna-ssl.com/whitepaper2010highres-Final-for-publication-3-8-2010.pdf>
- Jahnke, S. A., Poston, W. S. C., & Haddock, C. K. (2014). Perceptions of alcohol use among US firefighters. *Journal of Substance Abuse Alcoholism*, 2, 1012.
- Jahnke, S. A., Poston, W. S., Jitnarin, N., & Haddock, C. K. (2012). Health concerns of the US fire service: Perspectives from the firehouse. *American Journal of Health Promotion*, 27(2), 111-118.
- Jensen, J., & Waugh, W. L. (2014). The United States' experience with the incident command system: What we think we know and what we need to know more about. *Journal of Contingencies and Crisis Management*, 22: 5–17. doi: 10.1111/1468-5973.12034
- Jones, W. (2007). The role of developmental experience in the career development of fire chiefs in the United States. Retrieved from ProQuest Dissertations & Theses (3299808.)
- Kastros, A. (2013, April). *The American fire service leadership pandemic*. Paper presented at the meeting of Fire Engineering Training International Conference, Indianapolis, Indiana.

- Kime, C. H. (1999). *Leadership in organizations: A paradigmatic shift in thinking*. Retrieved from ProQuest Dissertations & Theses Global. (304523075.)
- Kime, C. H. (2001). *Organizational leadership: Fire services in the United States*. New York, NY: JAI.
- King, J. (2011). Silent trumpets: When leadership fails. *Fire Engineering*, 164(7), 37-42.
- Klein, G., Calderwood, R., & Clinton-Cirocco, A. (2010). Rapid decision making on the fire ground: The original study plus a postscript. *Journal of Cognitive Engineering and Decision Making*, 4(3), 186-209.
- Kline, R. (2011). A planned approach to firefighter safety. *Firehouse Magazine* (February 24). Retrieved from <http://www.firehouse.com/article/10463596/a-planned-approach-to-firefighter-safety>
- Kouzes, J., & Posner, B. (2002). *The leadership challenge* (3rd ed.). San Francisco, CA: Jossey Bass.
- Kouzes, J., & Posner, B. (2010). *The leadership practices inventory* (4th ed.). New York, NY: Wiley.
- Kuehl, K. S., Elliot, D. L., Goldberg, L., Moe, E. L., Perrier, E., & Smith, J. (2013). Economic benefit of the PHLAME wellness programme on firefighter injury. *Occupational Medicine*, 63(3), 203-209.
- Kuehl, H., Mabry, L., Elliot, D. L., Kuehl, K. S., & Favorite, K. C. (2013). Factors in adoption of a fire department wellness program: Champ-and-chief model. *Journal of Occupational and Environmental Medicine*, 55(4), 424-429. doi:10.1097/JOM.0b013e31827dba3f

- Kupietz, K. (2010). To change the world or to rule it: Applying transformational leadership theory to the American fire service. *International Fire Service Journal of Leadership and Management*, 4(2), 23-29.
- Lencioni, P. (2006). *The five dysfunctions of a team*. John Wiley & Sons.
- Leonard, O., Wiita, N., & Milane, C. (2017). The best senior teams thrive on disagreement. *Harvard Business Review*. Retrieved from <https://hbr.org/2017/09/the-best-senior-teams-thrive-on-disagreement>
- LeSage, P., Dyar, J. T., & Evans, B. (2010). *Crew resource management: Principles and practice*. Jones & Bartlett Publishers.
- Lowe, W. A., & Barnes, B. (2000). An examination of the relationship between leadership practices and organizational commitment in the fire service. (Unpublished doctoral dissertation). Nova Southeastern University.
- Markley, R. (2016, January 8). Fire chiefs and wisely chosen battles [Weblog post]. Retrieved from <http://www.firechief.com/2016/01/08/fire-chiefs-and-wisely-chosen-battles/>
- McChesney, F. (2002, June 24). Smoke and errors [Web log post]. Library of Economics and Liberty. Retrieved from <http://www.econlib.org/library/Columns/Mcchesneyfire.html>
- McChesney, F. (2015, September 4). Fewer fires, so why are there more firefighters? *Washington Post*. Retrieved from https://www.washingtonpost.com/opinions/2015/09/04/05316abe-517c-11e5-933e-7d06c647a395_story.html
- McGuire, M., Agranoff, R., & Silvia, C. (2011). Putting the “public” back into collaborative public management. Paper presented at the meeting of *Public Management Research Conference*, Syracuse, NY.

- McGuire, M., & Silvia, C. (2010). The effect of problem severity, managerial and organizational capacity, and agency structure on intergovernmental collaboration: Evidence from local emergency management. *Public Administration Review*, 70(2), 279-288.
- McKeown, B., & Thomas, D.B. (2013). *Q methodology*, 2nd ed. Los Angeles, CA: Sage.
- Militello, M., & Benham, M. (2010). Sorting out collective leadership: How Q- methodology can be used to evaluate leadership development. *The Leadership Quarterly* (21), 620-632.
- Militello, M., & Janson, C. (2014). The urban school reform opera: The obstructions to transforming school counseling practices. *Education and Urban Society*, 46(7), 743-772.
- Mintzberg, H. (1998). Covert leadership: Notes on managing professionals. *Harvard Business Review*, 7(6), 140-148.
- Mintzberg, H. (2009). *Managing*. Berrett-Koehler Publishers.
- Mirvis, P. H., Hurley, S. T., & MacArthur, A. (2014). Transforming executives into corporate diplomats: The power of global pro bono service. *Organizational Dynamics*, 43(3), 235-245.
- Moeller, B. J. (2007). Answering “big questions” in the fire service. *International Fire Service Journal of Leadership and Management*, 1(2), 11-16.
- Moschella, J. M. (2008). *Setting academic fires: Towards a best practice graduate curriculum for fire studies*. Retrieved from ProQuest Dissertations & Theses Full Text. (250925119.)
- Moynihan, D. P., Pandey, S. K., & Wright, B. E. (2012). Setting the table: How transformational leadership fosters performance information use. *Journal of Public Administration Research and Theory*, 22(1), 143-164.
- National Commission on Fire Prevention and Control. (1989, 1973). *America burning*. Washington, D.C.: Author.

National Fallen Firefighters Foundation. (2007). Firefighter life safety initiative 1 white paper.

Emmitsburg, MD: NFFF. Retrieved from

<http://www.lifesafetyinitiatives.com/1/Initiative1.pdf>

National Fire Protection Association. (2014a). The U.S. fire problem, updated September 2014.

Retrieved from [http://www.nfpa.org/research/reports-and-statistics/fires-in-the-](http://www.nfpa.org/research/reports-and-statistics/fires-in-the-us/overall-fire-problem/overview-of-the-us-fire-problem)

[us/overall-fire-problem/overview-of-the-us-fire-problem](http://www.nfpa.org/research/reports-and-statistics/fires-in-the-us/overall-fire-problem/overview-of-the-us-fire-problem)

National Fire Protection Association. (2014b). *NFPA 1021: Standard for fire officer professional*

qualifications. Retrieved from [http://www.nfpa.org/codes-and-standards/document-](http://www.nfpa.org/codes-and-standards/document-information-pages?mode=code&code=1021)

[information-pages?mode=code&code=1021](http://www.nfpa.org/codes-and-standards/document-information-pages?mode=code&code=1021)

Neyfakh, L. (2013, September 8). Plenty of firefighters, but where are the fires? *The Boston*

Globe. Retrieved from <http://www.bostonglobe.com>

Northouse, P. (2013). *Leadership theory and practice* (6th ed.). Thousand Oaks, CA: Sage.

Odom, S. R. F. (2011). *An examination of the preference for leadership style among different*

rank generational cohorts. (Unpublished dissertation). Texas A&M University, TX.

Onieal, D. G. (1990). A handbook to train members of fire departments in the state of New

Jersey to use the microcomputer-based national fire incident reporting system. Retrieved

from ProQuest Dissertations & Theses Full Text. (303870359.)

Onieal, D. (2003a, August). Professional status: The future of fire service training and education.

Part One: Introduction. *Fire House Magazine*, 79-82.

Onieal, D. (2003b, September). Professional status: The future of fire service training and

education. Part Two: Training and education. *Fire House Magazine*, 92-93.

Onieal, D. (2003c, October). Professional status: The future of fire service training and

education. Part Three: A model for training and education. *Fire House Magazine*, 80-84.

- Onieal, D. (2003d, November). Professional status: The future of fire service training and education. Part Four: Combining the systems: Independent assessment of skills/reciprocity. *Fire House Magazine*, 90-92.
- Onieal, D. (2003e, December). Professional status: The future of fire service training and education. Part Five: The future: Where we go from here. *Fire House Magazine*, 102-104.
- Onieal, D. (2007a). A message from Denis Onieal. *International Fire Service Journal of Leadership and Management*, 1(1), 5-6.
- Onieal, D. (2007b). Potential research topics. *International Fire Service Journal of Leadership and Management*, 1(1), 6-7.
- Osborn, R. N., Uhl-Bien, M., & Milosevic, I. (2014). The context and leadership. In D. Day (Ed.), *The Oxford handbook of leadership and organizations* (pp. 589-612). New York, NY: Oxford University Press.
- Peters, T. J., & Waterman, R. H. (1982). *In search of excellence: Lessons from America's best-run companies*. New York, NY: Harper & Row.
- Peterson, K., Witt, M., Morton, K., Olmsted, M., Amandus, H. E., Proudfoot, S.L., & Wassell, J. T. (2008). *Fire fighter fatality investigation and prevention program evaluation*. Morgantown, WV: National Institute for Occupational Safety and Health.
- Pillai, R., & Williams, E. (2004). Transformational leadership, self-efficacy, group cohesiveness, commitment, and performance. *Journal of Organizational Change Management*, 17(2), 144-159.

- Porter, L. W., Steers, R. M., Mowday, R. T., & Boulian, P. V. (1974). Organizational commitment, job satisfaction, and turnover among psychiatric technicians. *Journal of applied psychology*, 59(5), 603.
- Ramlo, S. (2016). Mixed method lessons learned from 80 years of Q methodology. *Journal of Mixed Methods Research*, 10(1), 28-45.
- Ramlo, S., & Newman, I. (2011). Q methodology and its position in the mixed methods continuum. *Operant Subjectivity: The International Journal of Q Methodology*, 34(3), 173-192.
- Rao, V., & Woolcock, M. (2003). Integrating qualitative and quantitative approaches in program evaluation. In F. Bourguignon, *The impact of economic policies on poverty and income distribution: Evaluation techniques and tools* (pp. 165-190). World Bank Publications.
- Revere, M. (2014, June). *Preventing professional drift: Best practices in training and education*. Symposium conducted at the meeting of the United States Fire Administration's Fire and Emergency Services in Higher Education Program and Training Resources and Data Exchange Program, Emmitsburg, MD.
- Rielage, R. (2015, December 7). How to get fire/EMS noticed during terrorist response [Weblog post]. Retrieved from <http://www.firechief.com/2015/12/07/how-to-get-fireems-noticed-in-terrorism-response/>
- Rocha, V., Willon, P., St. John, P., & Karlamangla, S. (2016, December 6). Investigation into Oakland fire intensifies. *Los Angeles Times*. Retrieved from <http://www.latimes.com>.
- Rowold, J., & Borgmann, L. (2013). Are leadership constructs really independent? *Leadership & Organization Development Journal*, 34(1), 20-43.
- Rubin, D. (2012). *Rube's rules for leadership*. Half Cooked Octopus.

- Rubin, D. (2013). *D.C. Fire*. Tulsa, OK: Pen Well.
- Rubin, D. (2014, February 20). Hiring plan: How not to hire idiots, thugs and misfits [Web log post]. Retrieved from <http://www.firechief.com/2014/02/20/hiring-plan-hire-idiots-thugs-misfits/>
- Russell, E. J. (2014). *The desire to serve: Servant leadership for fire and emergency services*. Westfield, IN: Greenleaf Center for Servant Leadership.
- Salka, J. (2005). *First in, last out: Leadership lessons from the New York Fire Department*. New York, NY: Penguin.
- Schmeltz, M. T., González, S. K., Fuentes, L., Kwan, A., Ortega-Williams, A., & Cowan, L. P. (2013). Lessons from Hurricane sandy: A community response in Brooklyn, New York. *Journal of Urban Health*, 90(5), 799-809. doi: 10.1007/s11524-013-9832-9
- Schmolck, P. (2014). *PQMethod 2.35*. Retrieved from <http://schmolck.userweb.mwn.de/qmethod/downpqwin.htm>
- Senge, P. M. (1994). *The fifth discipline: The art and practice of the learning organization* (1st ed.). New York, NY: Currency Doubleday.
- Shinebourne, P. (2009). Using Q method in qualitative research. *International Journal of Qualitative Methods* 8(1), 93-97.
- Sjöberg, M., Wallenius, C., & Larsson, G. (2006). Leadership in complex, stressful rescue operations: A qualitative study. *Disaster Prevention and Management: An International Journal*, 15(4), 576-584.
- Smeby, Jr., L.C. (2006). *Fire and emergency services administration: Management and leadership practices*. Sudbury, MA: Jones & Bartlett.

- Smith, J. A., Flowers, P., & Larkin, M. (2009). *Interpretative phenomenological analysis: Theory, method and research*. London: Sage.
- Smith, T. D., Eldridge, F., & Dejoy, D. M. (2016). Safety-specific transformational and passive leadership influences on firefighter safety climate perceptions and safety behavior outcomes. *Safety Science*, 86, 92-97. doi:10.1016/j.ssci.2016.02.019
- Standridge, S. (2012). Using action research methodology for developing an archetype fire officer development program. *International Fire Service Journal of Leadership and Management*, 6(1), 43-51.
- Tabarrok, A. (2013, September 7). Firefighter hysteresis [Web log post]. *Marginal Revolution*. Retrieved from <http://marginalrevolution.com/marginalrevolution/2013/09/firefighter-hysteresis.html>.
- The Johnson Foundation. (1966). Wingspread Conference on fire service administration, education and research, *Fire Files Digital Library*. Retrieved from <https://fire.omeka.net/items/show/6>
- Thiel, A. (2013). Fire news in focus: Firefighting physics, current research will change when we go in. Retrieved from <https://www.firerecruit.com/articles/1401197-Firefighting-physics-Current-research-will-change-when-we-go-in>
- Trottier, T., Van Wart, M., & Wang, X. (2008). Examining the nature and significance of leadership in government organizations. *Public Administration Review*, 68(2), 319-333.
- United States Fire Administration. (2014a). *Creating a competency-based professional development system*. Retrieved from http://www.usfa.fema.gov/training/prodev/national_professional_development.html

United States Fire Administration. (2014b). *National fire department census quick facts*.

Retrieved from <http://apps.usfa.fema.gov/census/summary.cfm#a>

United States Fire Administration (2014c). *U.S. fire statistics*. Retrieved from

<http://www.usfa.fema.gov/data/statistics/>

United States House Select Bipartisan Committee to Investigate the Preparation for and

Response to Hurricane Katrina. 2006. *A Failure of initiative*. Washington, DC: U.S.

Government Printing Office.

United States National Commission for the Protection of Human Subjects of Biomedical and

Behavioral Research. (1978). *The Belmont report: Ethical principles and guidelines for*

the protection of human subjects of research. Washington, D.C.: U.S. Government

Printing Office.

Uhl-Bien, M., & Marion, R. (2009). Complexity leadership in bureaucratic forms of organizing.

The Leadership Quarterly, 20(4), 631-650. doi:10.1016/j.leaqua.2009.04.007

Uhl-Bien, M., Marion, R., & McKelvey, B. (2007). Complexity leadership theory: Shifting

leadership from the industrial age to the knowledge era. *The Leadership Quarterly*, 18(4),

298-318.

Uhl-Bien, M., & Ospina, S. (Ed.s) (2012). *Advancing relational leadership research: A dialogue*

among perspectives. IAP.

Van der Wal, Z., De Graaf, G., & Lasthuizen, K. (2008). What's valued most? Similarities and

differences between the organizational values of the public and private sector. *Public*

Administration, 86(2), 465-482.

Van Exel, J., & de Graaf, G. (2005). *Q methodology: A sneak preview*. Retrieved from

<http://www.Qmethodology.net/PDF/Q-Methodology>,

- Van Wart, M. (2003). Public-sector leadership theory: An assessment. *Public Administration Review*, 63(2), 214-228.
- Van Wart, M. (2013). Administrative leadership theory: A reassessment after 10 years. *Public Administration*, 91(3), 521-543.
- Waite, M. R. (2007). *Fire service leadership: Theories and practices*. Burlington, MA: Jones & Bartlett Learning.
- Watts, S., & Stenner, P. (2012). *Doing Q methodological research: Theory, method and interpretation*. Thousand Oaks, CA: Sage.
- Waugh, W. L., & Streib, G. (2006). Collaboration and leadership for effective emergency management. *Public Administration Review*, 66(s1), 131-140.
- Webler, T., Danielson, S., & Tuler, S. (2009). *Using Q method to reveal social perspectives in environmental research*. Greenfield MA: Social and Environmental Research Institute. Downloaded from <http://www.seri-us.org/sites/default/files/Qprimer.pdf>
- Wheatley, M. J. (2006). *Leadership and the new science: Discovering order in a chaotic world* (Ed. 3). Berrett-Koehler.
- Willing, L. (2015, December 7). 10 things fire chiefs must do [Weblog post]. Retrieved from <http://www.firechief.com/2015/12/07/10-things-fire-chiefs-must-do-2/>
- Willing, L. (2015, June 9). Fire chief spending reveals dept. priorities: How a fire department allocates its money says more about its priorities than does any mission statement; how you spend is who you are [Weblog post]. Retrieved from <https://www.firerescue1.com/fire-chief/articles/2184354-Fire-chief-spending-reveals-dept-priorities/>

- Woods, C. E. (2011). Using Q methodology to explore leadership: The role of the school business manager. *International Journal of Leadership in Education*, 14(3), 317-335.
- Woods, D. D., & Branlat, M. (2011). Basic patterns in how adaptive systems fail. *Resilience Engineering in Practice*, 127-144.
- Yukl, G. (2010). *Leadership in organizations*, 7th ed. Upper Saddle River, NJ: Prentice Hall.

Appendix A: Types of Leadership Conceptual Frameworks and Results of Studies of USFES

Leadership within Each Framework Type

The information presented in Appendix A provides an overview of leadership research in USFES. It includes brief summaries of 28 USFES studies published between 1985 and 2014 that relate to leaders and leadership. The studies summarized apply many conceptual frameworks or have results and conclusions involving a variety of conceptual frameworks. Within this body of research, there is also diversity in the level of analysis (individual or departmental) and the methodology. Information presented here is arranged according to the type of conceptual framework: *leader-centric*, *follower-centric*, *team/dyadic*, or *other*. These are the same categories David Day established for organizing the chapters of the comprehensive *Oxford Handbook of Leadership and Organizations* (2014). Table 37 summarizes 11 studies applying leader-centric conceptual frameworks; Table 38 summarizes six studies applying follower-centric conceptual frameworks; Table 39 summarizes seven studies applying team or dyadic conceptual frameworks; and Table 40 summarizes four studies applying conceptual frameworks that do not fit any of the aforementioned categories. When the study has a conceptual framework named in Day's (2014) *Oxford Handbook*, it is in the same category; when the study has a conceptual framework not included in Day's *Oxford Handbook*, I categorized the framework.

Table 37

Leader-centric Studies of USFES Leadership

Author(s), Year	Study topic, type, hypothesis, participants, etc.	Results or conclusions and recommendations
Carter, 2005	Volunteer retention in a New Jersey county, participants were Chief Officers, mixed methods with MLQ and phenomenology	Transformational chief officers are attentive to retention and turnover; themes that negatively impact volunteer retention include leadership, economics, interpersonal relations, organizational and structural issues, personal matters of volunteers
Carter, 2007	Identifying leadership theory for the fire service, a theoretical study	mix of charismatic, trait, transformational deemed most appropriate
Clark, 2015 (originally 1985)	Do leaders in fire departments deemed excellent practice the same eight management principles identified in Peters and Waterman's 1982 book, <i>In Search of Excellence</i>	5 of the 8 principles are present: bias for action, stay close to the customer, autonomy and entrepreneurship, productivity through people, and hands-on/value-driven; 1 was not present: stick to the knitting; two were not discoverable through interviews, requiring on-site observation
Dixon, 2014	Fire ground leadership, phenomenological study of 12 fire suppression leaders, single volunteer department (Delaware)	5 common themes important to fire ground leadership: safety, leading by example, ownership/responsibility, striving for excellence, learning from experience
Gasaway, 2008	Fire ground decision making, barriers to situational awareness, multi-departments in a single metro area (Minnesota), mixed methods of card sort and interviews	3 most cited types of barriers: staffing, communications, physical/mental health and stress
Greco, 2014	Case Study of a Fire department/EMS merger, assessing impact of adopting a common book – Covey's <i>Seven Habits of Highly Successful People</i> – and concepts to facilitate merger success and improve performance as assessed by Insurance Standards Office; conceptual foundation was competing values framework (Cameron & Quinn, 2006)	4 areas of growth and improvement identified: passive-aggressive, divisive spirit replaced with sense of renewal and unification; improved communication through changed attitudes, vocabulary, and behavior; structure established to provide better services; triangulated validated interview data with improved ISO rating and legal approval for updated Standard Operating Guidelines
Jahnke, Haddock, & Poston, 2014	Leadership behaviors that promote firefighter health, national qualitative study (focus groups in 34 departments)	Themes identified: setting expectations and tone, leading by example, empowerment, allocation of resources, policy formation and enforcement

(continued)

Author(s), Year	Study topic, type, hypothesis, participants, etc.	Results or conclusions and recommendations
Klein, Calderwood, & Clinton-Cirocco, 2010	Fire ground decision making, qualitative analysis of 26 fire ground commanders' decisions	study developed recognition-primed decision model, distinct from other models which include calculation, options, and analysis
Kuehl, H., Mabry, Elliot, Kuehl, K., & Favorite, 2013	Qualitative study to identify factors in <i>adoption</i> of department wellness program (12 departments adopting, 24 department non-adopting)	Leadership of a health champion and the chief important, but no leadership theory in study
Lowe & Barnes, 2000	Hypothesis that leadership practices improve organizational commitment and performance in fire departments; single department; 73 participants completed the LPI and the Organizational Commitment Questionnaire; published article did not specify statistical methods or significance levels	Higher reported LPI levels for direct supervisors correlated positively with higher self-reported commitment level; no additional analysis to determine if some factors (e.g., age, tenure, education) were more significantly correlated than LPI levels
Lunde, 2013	Testing hypothesis that moral cognition levels correlate positively with self-reported ethical leadership character traits, Defining Issues Test (DIT2) instrument	Statistically insignificant results
Odom, 2011	Generational preferences for commanders' styles in an urban fire department; fictional descriptions of laissez-faire chief, transactional chief, and transformational chief	All generations and ranks preferred transformational style

Table 38

Follower-Centric Studies of USFES Leadership

Author(s), Year	Study topic, type, hypothesis, participants, etc.	Results or conclusions and recommendations
De Jesus Paris (2003)	Quantitative study of how employee satisfaction is affected by the interaction of cultural differences, cultural assimilation, and perceived/preferred leadership style; single fire department (Honolulu); used Hofstede's values survey module scale (culture dimensions), and two parts of Schaupp's Employee Satisfaction scale to measure job satisfaction and preferred/perceived leadership styles scale (likert-type, authoritarian – participative range to identify styles "telling," "selling," "consulting," "joining"); confirm or disconfirm Schaupp (1978) low correlation between leadership style and job satisfaction (other factors more significant such as employee expectations, culture)	Results indicated that the six cultural groups had assimilated, with little closeness to the "birth country" values survey module scale. Confirmed the Schaupp study showing little association between leadership style and employee satisfaction
Ditch, 2012	Factors contributing to higher education achievement among fire chiefs in Virginia, qualitative study of 20 chiefs in 10 departments	Top influences on achievement were internal motivation to gain personal gratification, desire to learn, succession planning
Gist, Taylor & Raak, 2011	Suicide meta-analysis	Little research exists concerning suicide in the USFES, specifically; call for research on applicability of existing theory/interventions.
Jones, 2007	Fire officer development, qualitative study of 14 fire service aspiring leaders (interviews)	Confirmed the Center for Creative Leadership's developmental experience model of positive change in leadership and management behaviors derived from challenging on-the-job experiences; strongest changes occurred from experiences with the first fire chief, and hardship; recommends more development of political skills
Rowold, 2010	Study of the relationship between leadership behaviors and performance and whether or not heterogeneity of age, gender, and culture has moderating effect; n=283 members of German fire departments	Team heterogeneity moderated the relationship between 3 leadership behaviors (transformational, laissez-faire, and consideration) and performance; transformational leadership and consideration work best for teams with gender heterogeneity
Woods & Branlat, 2011	Theoretical example of fire ground firefighting operations to illustrate how adaptive systems fail	Fire ground is a specific context to illustrate the basic patterns of system failure: decompensation (exhausted capacity for adaptation); working at cross purposes (local adaptation contradicts global solutions); "getting stuck" in outdated behaviors

Table 39

Team and Dyadic Studies of USFES Leadership

Author(s), Year	Study topic, type, hypothesis, participants, etc.	Results or conclusions and recommendations
Beaton, Johnson, Infield, Ollis, & Bond, 2001	Leader influence on follower job satisfaction, following training based on contingency theory and stress management training; quasi-experimental with 2 fire departments – 1 with leadership training intervention for supervisors and 1 without; training intervention included only 8 hours of “Leader Match” leadership training	At 3- and 9-month intervals, department with intervention showed improvements for firefighters in perception of ability to attain career goals and in managing stress; however, performance measures of on-the-job injuries, absenteeism, and apparatus incidents had no change (either contingency theory is not relevant to fire service, or 8 hours training is not sufficient)
Duncan, Matthew & Scarpa, 2012	Stated purpose to examine the influence of fire departments on Homeland Security Councils of Government, but published report identified elements for effective Homeland Security – Councils of Government (HSCG)	Elements for effective HSCG’s: Structure (communications, processes), culture of teamwork, leadership (defined as “an influence relationship among colleagues . . . a dynamic owned by no one, yet shared among and between collaborators who intend changes reflective of their mutual interests and values”)
Griffin, 2013	Organizational learning post-line-of-duty-death incident, single-department case study; mixed methods with Dimensions of Learning Organizations Questionnaire, interviews	Individual learning, team/group learning, and organizational learning occurred through focus on training
Kuehl, Elliot, Goldberg, Moe, Perrier, & Smith, 2013	ROI study on team-implemented wellness program (quasi-experimental, 4 Oregon fire departments)	Fire department workers compensation claims and medical costs were reduced after implementation of the PHLAME workplace health promotion program. This is a low cost, team-based, peer-led, wellness program that may provide a feasible, cost-effective means to reduce firefighter injury and illness rates
Pillai & Williams, 2004	Quantitative study of transformational leadership correlation to group cohesiveness, self-efficacy, organizational commitment, and perception of performance; single fire department	Positive correlations, statistically significant for direct effects of transformational leadership on cohesiveness, self-efficacy, organizational commitment, and perception of performance; indication that transformational leadership model may be valid for USFES
Smith, 2014	Qualitative study of factors affecting volunteer retention in 3 suburban fire departments (NE); social exchange and reciprocity theory; Herzberg’s two-factor theory of job satisfaction	relationships, department environment important (Herzberg’s hygiene factors)
Standridge, 2012	Using an action-research model in a Colorado fire department to research and propose a fire officer development program involving out-of-state exchanges	Model provided team-oriented departmental process to efficiently develop detailed proposal

Table 40

Other Conceptual Frameworks Applied in Studies of USFES Organizations

Author(s), Year	Study topic, type, hypothesis, participants, etc.	Results or conclusions and recommendations
Hong, Chin, Phelps, Feld, & Vogel, 2012	Factors associated with firefighter injuries (N=435 in 3 states)	Age is a significant factor correlating with on- and off-duty firefighter injuries; the organization's commitment to health and safety a significant factor (but leadership theory/model/style not studied)
Kime, 1999, 2001	Theoretical and empirical study. Development of an organizational leadership model and scale for measuring organizational leadership capacity, using the USFES industry as a case study; 336 USFES departments completed questionnaires; uses the complex adaptive organizations conceptual framework since many public organizations are most accurately described as complex systems operating in complex environments. Defined organizational leadership capacity: the capacity of the organization to collectively respond to endogenous and exogenous stimuli. Operationalized the concept as the following dimensions: tolerance for questioning and challenging assumptions, extent to which an organization values its employees, perception of how the organization responds to failures and errors, perception of the decision-making environment, nature of communication, philosophy and practices for handling change.	Measured the Org. Leadership Capacity score for 242 fire departments and learned the following: as fire department size increases, org. leadership capacity decreases, no statistically significant differences in org. leadership capacity for union and non-union departments, the tenure of management is not statistically significant as a factor affecting Org leadership capacity, the number of kinds of tasks/services a department performs is negatively correlated with org leadership capacity (explains 14% of variance in scores), same for ration of employees to managers
Moeller, 2007	Meta-analysis of research supporting NFPA Standard 1710 (staffing policy recommendation to maximize response-time measures of department effectiveness)	Analysis of 4 studies showed that 3 of the studies did not collect original data; the 4 th had methodological errors related to the study designers who were also study participants; overall conclusion of the meta-analysis is that there is a paucity of rigorous research and limited information to support and justify NFPA 1710

(continued)

Author(s), Year	Study topic, type, hypothesis, participants, etc.	Results or conclusions and recommendations
Pessemier, 2012	Quantitative study applying organizational culture construct to assess factors that are important to USFES organizational safety culture, validated in three municipal fire departments; used a UK survey on fire service safety systems, an oil industry survey on safety climate, and developed a questionnaire to assess safety behaviors among firefighters, and also collected job function, age and experience information	Safety systems (policies, management, evaluation) and safety behaviors predict organizational safety climate; 0-5 years of service and over 20 years of service score higher on individual perception of safety systems, safety behaviors and organizational safety climate; presents practical matrix model for determining safety behavior and safety systems variables for interventions

Appendix B: E-mail to and Response from the Acting Superintendent of the National Fire
Academy

Dr. Kirby Kiefer
National Fire Academy
Acting Superintendent

Hello Dr. Kiefer:

I am writing to you in your capacity as the Acting Superintendent the National Fire Academy to request permission and learn the correct protocol to contact the National Fire Academy Board of Visitors and invite their participation in a research study. This study explores leadership in United States fire and emergency services (USFES) municipal departments. I am conducting this research to fulfill part of The Chicago School of Professional Psychology's requirements for the Doctor of Philosophy, Organizational Leadership degree. Dr. Kathleen Kleinfeldt is overseeing this study.

With your affirmative response, I would invite the Board of Visitors members to participate in telephone interviews I will conduct in order to develop a complete concourse of subjective perspectives on effective, department-wide leadership in USFES municipal departments. The Board of Visitors would be an ideal, initial sample for interviews because of the nation-wide representation and the variety of stakeholder perspectives they represent in USFES.

The text of a letter and the informed consent document that I would send to them is attached for your review. I would prefer to contact them via e-mail if you are able to provide their e-mail addresses.

Thank you for your consideration!
Sincerely,

Monika S. Byrd, M.A.

From: Kiefer, Kirby <Kirby.Kiefer@fema.dhs.gov>
Reply all
Fri 6/17
Monika Byrd - Student;
Newlin, Ellen <Ellen.Newlin@fema.dhs.gov>;
Brame, Kevin <kevin.brame@fema.dhs.gov>

Dear Ms. Byrd,

Congratulations on your academic journey! I am pleased to see that your dissertation has progressed so far.

Members of our Board of Visitors are private citizens; it would be inappropriate for me to either endorse your study by contacting them on your behalf or to discourage you from contacting them. That said, my guess is that they will be happy to help your study. Since their contact information is a matter of public record, it may help your study to check the web page:

<https://www.usfa.fema.gov/training/nfa/about/bov.html>

Please keep me informed of your progress (and Good Luck!)

Sincerely,
Kirby E. Kiefer, Ed.D.
Acting Superintendent
National Fire Academy
(301) 447-1117

Appendix C: Concourse Development Participant Communication and Informed Consent Form

Concourse Development Participant Email

Hello,

My name is Monika Byrd, and I am a doctoral student at The Chicago School of Professional Psychology. I am contacting you to invite you to participate in a research study. This study explores leadership in United States fire and emergency services (USFES) departments. I am conducting this research to fulfill part of The Chicago School's requirements for the Doctor of Philosophy, Organizational Leadership degree. Dr. Kathleen Kleinfeldt is overseeing this study.

Participation involves a brief demographic questionnaire via e-mail and a telephone interview. In the interview, I will ask questions about your subjective perspective regarding effective department-wide leadership in USFES departments. I will also ask about other subjective perspectives of which you are aware that are different in one or more ways from your own perspective, and if you know one or two individuals with another perspective than your own who would participate in an interview to whom you could forward this study recruitment e-mail.

The interview is expected to last 20 minutes, and should not exceed an hour. Following the telephone interview, and forwarding this e-mail to one or two additional people (if applicable), nothing more will be asked of you related to this phase of the study. You may be contacted to participate in an additional phase of the study, and I will send a separate request to participate in it.

Participation in this interview phase of the study and the additional phase is voluntary, and declining to participate will not affect any positions you hold. If you do decide to participate, and then change your mind, you are still free to withdraw from the study without penalty at any time. You also have the right to skip any questions that you feel uncomfortable answering.

The risks that go along with this study are no greater than risks encountered when engaged in professional dialogue in your industry. If known to others, the responses you give could affect relationships, a promotional process, or credentialing for you. To minimize the potential of such harm, I will follow strict procedures to maintain and protect your confidentiality. You will be assigned a participant code number which will prevent personal identifying information from being known by anyone other than the investigator, supervisor, and transcriber. The transcriber will be required to sign a confidentiality agreement to protect personal identifying information. All information provided during the study will remain confidential and will only be reported with no identifying information. There is no compensation, nor are there any other direct benefits to participation, but the data you provide will be used to increase knowledge about leadership in USFES municipal departments.

To re-iterate, all of the information you provide is confidential. Only the primary researcher, Monika Byrd, the project supervisor, Dr. Kathleen Kleinfeldt, and a transcriber will have access to the data collected. Your interview will be recorded and securely stored in a

password protected computer account and will not be released. The interview recording will be transcribed without personal identifying information, and following the transcription, the recorded interview audio file will be deleted. The transcriber will complete a confidentiality agreement. Back-up files of your Informed Consent form and interview transcription will be stored on a password-protected digital storage drive kept in a locked safe box. The transcribed responses will be used anonymously to contribute to the development of a complete concourse about the knowledge, skills, abilities, traits, and behaviors that contribute to effective department-wide leadership in USFES municipal departments. Five years following the study, the Informed Consent form and interview transcription and will be destroyed.

If you are interested in participating, further details about the study are in the attached Informed Consent form for your review. Please review it and if you have any questions, feel free to contact the researcher, Monika Byrd, by replying to this email at mr6646@ego.thechicagoschool.edu. Or by telephone at . You can also direct any questions to Dr. Kathleen Kleinfeldt, the project Chair, at kkleinfeldt@thechicagoschool.edu. If you have any questions regarding your rights as a research participant, you can contact the Institutional Review Board (IRB) by phone at (312) 467-2343 or by email at IRB@thechicagoschool.edu.

You can accept this request to participate by returning the signed Informed Consent form to me by e-mail to mr6646@ego.thechicagoschool.edu or mail to

Monika Byrd | [REDACTED]

After returning the Informed Consent form, you can click on the link below to a Doodle® scheduling web page. After clicking on this link, you will be taken to a web page with date and time options from which you may select one or more times that are convenient for you to participate in the telephone interview. While the time options provided on the web page are 30 minutes in length, the interviews are not anticipated to exceed 20 minutes. When you have selected a date and time, I will receive an e-mail from Doodle® and I will follow-up with you to confirm and to provide a telephone number for the interview.

Doodle® scheduling web page:
<http://doodle.com/poll/n2tp7m8pag73iips>

Thank you for your time and consideration!

Monika S. Byrd, M.A.

Concourse Interviewee's Informed Consent



Investigator: Monika Byrd

Study Title: The Perceptions of Effective Department-wide Leadership in American Fire and Emergency Services

I am a student at The Chicago School of Professional Psychology. This study is being conducted as a part of my dissertation requirement for the Ph. D. in Organizational Leadership.

I am asking you to participate in a research study involving an interview about organizational, department-wide leadership in the United States Fire and Emergency Services. Please take your time to read the information below and feel free to ask any questions before signing this document.

Purpose: You are being asked to participate in a research study which involves an exploration of organizational leadership in American fire and emergency services municipal departments. The purpose of this research study is to identify different perspectives within American fire and emergency services departments regarding effective *department-wide* leadership. Second, the data will be analyzed to determine the extent to which there is one or more perspectives that are shared across types or sizes of departments. Third, the data will be analyzed to determine whether or not any of the perspectives are similar to existing models, theories, or approaches to organizational leadership in leadership research.

Procedures: If you agree to participate in this study, you will be asked to complete a demographic questionnaire via e-mail, and to participate in an interview about your perspective on department-wide, organizational fire and emergency services leadership expected to be twenty minutes and which will not be more than one hour. You will also be asked about other subjective perspectives of which you are aware that are different in one or more ways from your own perspective, and if you know one or two individuals with another perspective than your own who would participate in an interview and if you are willing to forward the recruitment e-mail to them. Interviews will be recorded and transcribed with procedures to ensure confidentiality. Details about maintaining and protecting confidentiality are in the “Confidentiality” section of this document. The purpose and benefit of voice recording is to ensure the accuracy of the data collection and to minimize disruption or lengthen the time with note taking during the interview. You will be contacted via e-mail with the option to participate in a second phase of the research. The details about the second phase of the research will be fully described in a separate Informed Consent form.

Risks to Participation: There are minimal risks associated with participation in this study; however, the following have the potential to occur:

- You may feel concern or have questions related to department leadership, and you may have feelings related to department-wide leadership that persist beyond the period of the study, and you may recall situations or events that were unpleasant. To minimize risk, the

investigator will assure you that you always have the choice to withdraw and end the interview at any time. The likelihood of these incidents occurring is minimal however, because the interview process does not require that you delve deeply into any personal experiences. Your responses will remain confidential.

- If known to others, the responses you give could affect a promotional process, credentialing, or relationships at work for you. To minimize the potential of such harm, the investigator will maintain and protect your confidentiality. Refer to the “Confidentiality” section below for additional details on maintaining and protecting confidentiality.
- You may not feel comfortable completing the interview. To minimize risk the investigator will allow you to select the time and location for the interview, and ask at the beginning of the interview if you are in a location where you feel free and safe to share your perspective. If you do not feel that you are in a safe environment, the investigator will suggest that the sorting process and the interview be rescheduled when a safe and comfortable location can be established. You may withdraw from the interview at any time.
- Participation in the interviews involves no more than one hour. You may feel that this is too long or not long enough. To minimize risk that you desire more or less time, the investigator will have some flexibility to conclude the interview in less than hour or continue for more than an hour.

Throughout the study, the investigator will seek to minimize each of the potential risks as described.

Benefits to Participants: There is no compensation for participation, nor are there other direct benefits for participation. However, we hope the information learned from this study may benefit the United States fire and emergency services industry and communities with contributions to our understanding of effective department-wide leadership. Participation may provide you with an opportunity to reflect upon leadership in American fire and emergency services, and the reflection may benefit your personal exercise of leadership and contribute to the leadership capacity of your organization.

Alternatives to Participation: Participation in this study is voluntary. You may withdraw from study participation at any time without any penalty.

Confidentiality: Participation requires completion of a demographic questionnaire that collects information about you for the purpose of this research. This includes

- Geographic location and type of community (urban, suburban, rural)
- Department size
- Type of fire department (career, volunteer, combination)
- Title, Rank
- Years of service
- Highest level of education completed
- Type and number of professional development opportunities (not fire service or medical training, certifications, or continuing education and development)
- Gender
- Age

There will be measures to ensure the participants that their participation and the information provided during the study will remain confidential. You will be assigned a participant code number which will prevent personal identifying information from being known by anyone other than the investigator, supervisor, and transcriber. The transcriber will be required to sign a confidentiality agreement to protect your personal identifying information. All information provided during the study will remain confidential and will only be reported with no identifying information. Signed Informed Consent forms and data collected will be stored in a password-protected computer account. Interview audio files will be deleted after transcription. Following transcription, data will be directly accessed and reviewed by only the researcher and the study's supervisor, Dr. Kathleen Kleinfeldt. Back-up files of the data will be stored on a storage drive that is also password-protected and kept in a locked safe box. E-mails and other written correspondence will be deleted or destroyed when the study is concluded. Signed Informed Consent forms and data collected will be secured for a period of five years following the completion of the study and then deleted completely and confidentially.

The research records may be reviewed by federal agencies whose responsibility is to protect human subjects participating in research, including the Office of Human Research Protections (OHRP) and by representatives from The Chicago School of Professional Psychology Institutional Review Board, a committee that oversees research.

Questions/Concerns: If you have questions related to the procedures described in this document please contact Monika Byrd at mr6646@ego.thechicagoschool.edu or by phone at , or the supervisor of the research, Dr. Kathleen Kleinfeldt at kkleinfeldt@thechicagoschool.edu.

If you have questions concerning your rights in this research study, you may contact the Institutional Review Board (IRB), which is concerned with the protection of subjects in research project. You may reach the IRB office Monday-Friday by calling 312.467.2343 or writing: Institutional Review Board, The Chicago School of Professional Psychology, 325 N. Wells, Chicago, Illinois, 60654.

Consent to Participate in a Research Interview:

Interviewee:

I have read the above information and have received satisfactory answers to my questions. I understand the research project and the procedures involved have been explained to me. I agree to participate in an interview for this study. My participation is voluntary and I do not have to sign this form if I do not want to be part of this research project. I will receive a copy of this consent form for my records.

Name of Interviewee (print)

Signature of Interviewee

Date: _____

Monika Byrd

Name of the Person Obtaining Consent (print)

Signature of the Person Obtaining Consent

Date: _____

Appendix D: E-mail (4/29/16) Requesting a review of the Q-sample Statements, and Mr. Kevin Brame's Response

Hello Kevin –

I am writing to ask for your review of a sample of statements regarding United States Fire and Emergency Services (USFES) department-wide leadership that I will develop following a series of interviews with USFES leaders and published materials by USFES leaders such as books on leadership and blog posts regarding leadership. This review of statements is *not* participation in my study as a participant providing data or in any other way involving the data collection; rather it is part of the preparation for data collection.

The sample of statements that I will send will not include every single statement expressed in the interviews and publications – through categorizing all of the statements, I will reduce the large, comprehensive sample of statements to 40, 50, or 60 (maximum) *representative* statements of the entire content of the interviews and published materials.

If you are able to complete the review, I am interested in just two things regarding each statement:

1. is the statement readable and comprehensible?
2. is the statement open to interpretation?

The answer to both questions should be “yes.” If it is not, just let me know the statement number that needs revision and if it is because of 1., 2., or both. It is expected that you will agree with or like some statements, disagree with some, and think some are inconsequential – but that is NOT what you need to tell me!

It would be ideal if two people could look over the statements, so if you can ask a colleague involved with the Executive Fire Officer Program to go over the statements with you, it may go faster for you, and I will have the assurance of two reviews of the statements.

Because of your experience as a USFES leader and your involvement in the professional development of many other USFES leaders, you and another person from the EFOP are ideally suited to review the statements and help identify the statements that could be improved before I use them with the actual participants for data collection in my dissertation study on USFES department-wide leadership.

Let me know what questions you have, and of course I understand if you will not be able to help – just let me know!

Thank you!
Monika

From Kevin Brame <kbrame54@gmail.com>
5/1/2016

To Monika Byrd – Student <mrb6646@ego.thechicagoschool.edu>

Hi Monika - please consider this email as my official notification of my willingness to review and/or provide input to your dissertation research proposal.

Kevin S. Brame

From Kevin Brame, <kevin.brame@fema.dhs.gov>

3/1/17

To Monika Byrd – Student <mrb6646@ego.thechicagoschool.edu>

Hi Monika – A total of 5 faculty members at the NFA reviewed your list of statements. All 5 concurred that all of the statements are readable and understandable and that all of them are open to interpretation. It was interesting to see your list!

Best regards,

Kevin

Appendix E: Q-Sort Participant Email Letter and Informed Consent Form

Q-Sort Participant Email

Hello,

My name is Monika Byrd, and I am a doctoral student at The Chicago School of Professional Psychology. I am contacting you to invite you to participate in a research study. This study explores leadership in United States fire and emergency services (USFES) municipal departments. I am conducting this research to fulfill part of The Chicago School's requirements for the Doctor of Philosophy, Organizational Leadership degree. Dr. Kathleen Kleinfeldt is overseeing this study. This is phase 2 of the research, and if you have already participated in phase 1, thank you, and I hope you will also consider participating in phase 2.

Participation involves completing a demographic questionnaire via e-mail (if you did not participate in phase 1 and complete the demographic questionnaire already), and then completing a specialized survey and a post-survey interview designed to take no more than an hour in total. There is no right or wrong way to complete the survey; you will complete it based upon your personal, subjective perspective. The survey involves your subjective perspective regarding effective department-wide leadership in USFES municipal departments. Following the survey and interview, you may be asked to forward this e-mail to someone whom you know whose perspective differs from your own who would be willing to participate by completing the demographic questionnaire, survey and post-survey interview. From my analysis of the data, I will be describing and labeling the perspectives that exist on USFES department-wide leadership, and I may contact you to provide feedback about the descriptions and labels. Nothing additional will be asked of you.

Participation in this study is voluntary, and declining to participate will not affect any positions you hold. If you do decide to participate, and then change your mind, you are still free to withdraw from the study without penalty. You also have the right to skip any questions that you feel uncomfortable answering.

The risks that go along with this study are no greater than risks encountered when engaged in professional dialogue in your industry. If known to others, the responses you give could affect relationships, a promotional process, or credentialing for you. To minimize the potential of such harm, I will follow strict procedures to maintain and protect your confidentiality. You will be assigned a participant code number which will prevent personal identifying information from being known by anyone other than the investigator, supervisor, and transcriber. The transcriber will be required to sign a confidentiality agreement to protect personal identifying information. All information provided during the study will remain confidential and will only be reported with no identifying information. There is no compensation, nor are there any other direct benefits to participation, but the data you provide will be used to increase knowledge about leadership in USFES municipal departments.

To re-iterate, all of the information you provide is confidential. Only the primary researcher, Monika Byrd, the project supervisor, Dr. Kathleen Kleinfeldt, and a transcriber will have access to the data. Your Informed Consent form, survey responses and post-survey

interview will be recorded and securely stored on a password protected computer and will not be released. The interview recording will be transcribed without personal identifying information, and following the transcription, the recorded interview audio file will be deleted. The transcriber will complete a confidentiality agreement. Back-up files will be stored on a password-protected digital storage drive kept in a locked safe box. The transcribed responses will be used anonymously to increase knowledge about the knowledge, skills, abilities, traits, and behaviors that contribute to effective department-wide leadership in USFES municipal departments. Five years following the study, the interview transcription will be destroyed.

If you are interested in participating, you can click on the link to a Doodle® scheduling web page at the bottom of this email. After clicking on this link, you will be taken to a web page with date and time options from which you may select that is convenient for you to complete the survey and telephone interview. When you have selected a date and time, I will receive an e-mail from Doodle® and I will follow-up with you to confirm and to request your address where I can send the survey materials and instructions.

If you have any questions, feel free to contact the researcher, Monika Byrd, by email at mrb6646@ego.thechicagoschool.edu. You can also direct any questions to Dr. Kathleen Kleinfeldt, the project Chair, at kkleinfeldt@thechicagoschool.edu. If you have any questions regarding your rights as a research participant, you can contact the Institutional Review Board (IRB) by phone at (312) 467-2343 or by email at IRB@thechicagoschool.edu.

Doodle® scheduling web page:
[link to be added when Doodle Poll created]

Thank you for your time and consideration!

Monika S. Byrd, M.A.

Q-Sorter's Informed Consent



Investigator: Monika Byrd

Study Title: The Perceptions of Effective Department-wide Leadership in American Fire and Emergency Services

I am a student at The Chicago School of Professional Psychology. This study is being conducted as a part of my dissertation requirement for the Ph. D. in Organizational Leadership.

I am asking you to participate in a research study. Please take your time to read the information below and feel free to ask any questions before signing this document.

Purpose: You are being asked to participate in a research study which involves an exploration of organizational leadership in United States fire and emergency services municipal departments. The first purpose of this research study is to identify different perspectives within United States fire and emergency services Departments regarding effective Department-wide leadership. Second, the data will be analyzed to determine the extent to which there is one or more perspectives that are shared across types or sizes of departments. Third, the data will be analyzed to determine whether or not any of the perspectives are similar to existing models, theories, or approaches to organizational leadership in leadership research.

Procedures: If you agree to participate in this study, you will be asked to complete a brief demographic questionnaire via e-mail, or if you already completed it as a participant in an earlier phase of the research you may confirm your responses to the the demographic questionnaire. The data collection involves sorting 40 – 50 statements according to your opinion of their importance to organizational, department-wide, fire and emergency services leadership, using materials that will be sent to you in the mail, and respond to a post-sorting interview. The sorting will take place at a time convenient for you between the dates of _____, 2016 and _____, 2016, after you receive the set of statement cards and instructions in the mail. You will sort the statement cards after you have connected with me via Zoom online video conference so that you may ask questions at any time and for the post-sort interview. If an internet connection and web camera or mobile device with a camera are not available to you, we will connect via telephone. Sorting the statement cards should take no more than 30 to 40 minutes. At the conclusion of the sorting process, there will be an interview via Zoom online video call or telephone in which I will ask you questions about why you ordered the statements in the way that you did and about any experiences that have shaped your opinions. The post-sort interview will take approximately 20 minutes. At the conclusion of the post-sort interview, you may be asked to forward the recruitment e-mail to an additional person whose perspective may differ from yours to participate in the research. Prior to the sorting and the interview, you will be asked to confirm that the sorting process and the interview are being conducted in a safe environment in which you feel comfortable to complete the sort and to speak freely. You will be asked to grant permission for the interview to be voice recorded. The purpose and benefit of voice recording is to ensure the accuracy of the data collection and to minimize disruption or lengthen the time with

note taking during the interview. Your sort and your responses to the interview will remain confidential. Details about maintaining and protecting confidentiality are in the “Confidentiality” section of this document. The researcher will analyze the data and develop descriptions and labels for the perspectives that exist regarding effective department-wide leadership, and you may be contacted via e-mail to provide feedback about a description and label to validate that they are representative of your perspective. If you are contacted to provide feedback about a description and label, it is expected that it will take approximately 20 minutes to review and provide feedback either by e-mail response or by telephone, based upon your preference.

Risks to Participation: There are minimal risks associated with participation in this study; however, the following have the potential to occur:

- You may feel concern or have questions related to your opinions about Department-wide leadership expressed in the study that persist beyond the period of the study, or the sorting process and interview may lead you to recall situations or events that were unpleasant and that you would not feel comfortable mentioning under normal circumstances. To minimize risk, the investigator will assure you that you may speak freely without concern about judgment from the investigator and that you always have the choice to respond to a question or not, or to stop your responses at any time. During the interview, when such discomfort is expressed or witnessed by non-verbal communication, the investigator will avoid further questioning or probing that may contribute to distress. In extreme cases of distress or discomfort you may also be encouraged to seek out additional coaching or counseling support (e.g., perhaps from your organization’s HR department) as necessary. The likelihood of these incidents occurring is minimal however, because the questions are about the opinions formed related to department-wide leadership and will not delve deeply into the personal experiences. Your sorting order and interview responses will remain confidential.
- If known to others, the opinions expressed through participation in the study could affect a promotional process or credentialing or relationships at work. To minimize the potential of such harm, the investigator will maintain and protect your confidentiality. Refer to the “Confidentiality” section below for additional details on maintaining and protecting confidentiality.
- Whether you complete the sorting process online or with a set of physical cards with the sorting statements on them, you may not feel comfortable completing the process. To minimize risk the investigator will allow you to select the location where you complete the sorting process. Prior to the sorting process and to the interview, the investigator will ask you to confirm that you are in a safe environment. If you do not feel that you are in a safe environment, the investigator will suggest that the sorting process and the interview be rescheduled when a safe and comfortable location can be established. You may withdraw from participation at any time.
- The sorting process and interview are planned to take place during a one-hour time period. You may feel that an hour is not enough time to complete the sorting process and to discuss your opinions and experiences related to your sorting order. To minimize risk the investigator will remain flexible to meet your needs by providing additional time if necessary.
- A transcriber will be involved in this study for the purpose of transferring the audio recorded data obtained in your interview into written format. The transcriber will have access to the research information that means that there is the potential for a breach of confidentiality. To minimize the risk of such a breach the transcriber will be asked to sign a confidentiality

agreement, confirming that he or she agrees to conditions that protect your confidential participation in this study. For additional information regarding potential breaches and protection of confidentiality, refer to the “Confidentiality” section below.

Throughout the study, the investigator will seek to minimize each of the potential risks as described.

Benefits to Participants: There is no compensation for participation, nor are there other direct benefits for participation. However, we hope the information learned from this study may benefit the United States fire and emergency services industry and communities with contributions to our understanding of effective department-wide leadership. Participation may provide you with an opportunity to reflect upon leadership in American fire and emergency services, and the reflection may benefit your personal exercise of leadership and contribute to the leadership capacity of your organization.

Alternatives to Participation: Participation in this study is voluntary. You may withdraw from study participation at any time without any penalty.

Confidentiality: Participation requires completion of a demographic questionnaire that collects information about you for the purpose of this research. This includes

- Geographic location and type of community (urban, suburban, rural)
- Department size
- Type of fire department (career, volunteer, combination)
- Title, Rank
- Years of service
- Highest level of education completed
- Type and number of professional development opportunities (not fire service or medical training, certifications, or continuing education and development)
- Gender
- Age

There will be measures to ensure the participants that their participation and the information provided during the study will remain confidential. You will be assigned a participant code number which will prevent personal identifying information from being known by anyone other than the investigator, supervisor, and transcriber. The transcriber will be required to sign a confidentiality agreement to protect your personal identifying information. All information provided during the study will remain confidential and will only be reported with no identifying information. Signed Informed Consent forms and data collected will be stored in a password-protected computer account. Interview audio files will be deleted after transcription. Following transcription, data will be directly accessed and reviewed by only the researcher and the study’s supervisor, Dr. Kathleen Kleinfeldt. Back-up files of the data will be stored on a storage drive that is also password-protected and kept in a locked safe box. E-mails and other written correspondence will be deleted or destroyed when the study is concluded. Signed Informed Consent forms and data collected will be secured for a period of five years following the completion of the study and then deleted completely and confidentially.

The research records may be reviewed by federal agencies whose responsibility is to protect human subjects participating in research, including the Office of Human Research

Protections (OHRP) and by representatives from The Chicago School of Professional Psychology Institutional Review Board, a committee that oversees research.

Questions/Concerns: If you have questions related to the procedures described in this document please contact Monika Byrd at mr6646@ego.thechicagoschool.edu or by phone at , or the supervisor of the research, Dr. Kathleen Kleinfeldt at kkleinfeldt@thechicagoschool.edu.

If you have questions concerning your rights in this research study you may contact the Institutional Review Board (IRB), which is concerned with the protection of subjects in research project. You may reach the IRB office Monday-Friday by calling 312.467.2343 or writing: Institutional Review Board, The Chicago School of Professional Psychology, 325 N. Wells, Chicago, Illinois, 60654.

Consent to Participate in Research

Participant:

I have read the above information and have received satisfactory answers to my questions. I understand the research project and the procedures involved have been explained to me. I agree to participate in this study. My participation is voluntary and I do not have to sign this form if I do not want to be part of this research project. I will receive a copy of this consent form for my records.

Name of Participant (print)

Signature of Participant

Date: _____

Preferred Mailing Address to receive the participation materials:

Monika Byrd

Name of the Person Obtaining Consent (print)

Signature of the Person Obtaining Consent

Date: _____

Appendix F: Q-Sort Participant Demographic Questionnaire

Participant I.D. Code (completed by the investigator) _____

Geographic location (East, South, Midwest, Northwest, Southwest)

Type of community (urban, suburban, rural)

Type of fire department (career, volunteer, combination) _____

Department size (number of employees and/or volunteers) _____

Title, Rank _____

Years of service _____

Describe your level of education and the professional development opportunities you have had that are not considered technical or required training:

Gender _____

Age _____

telephone at (Refer to the e-mail confirmation you would have received following scheduling at the Doodle scheduling site; contact me if you need it resent). Plan to read the instructions prior to the sorting time you scheduled. At the time scheduled for the sorting and post-sorting interview, please connect to the Zoom online video conference with the link provided to you via e-mail.

There are four parts to complete your participation in the study:

1. Informed Consent Review: you previously read and signed the Informed Consent form. Please review it and note any questions that you have.
2. Complete the demographic questionnaire (only if you did not already complete it as a participant in an earlier phase of the research).
3. Review this complete document of Q-sort Instructions and sort the statement cards at the agreed upon time according to the instructions contained in this document and with the researcher as a resource for completing the process via Zoom video call (preferred) or telephone.
4. Respond to the post-sort interview questions.

We will complete these parts in numerical order. Parts 1, 2, and 4 are self-explanatory. The following section contains more detailed instructions for Part 3. Please plan to read or re-read the Part 3 instructions close to the scheduled time for your sorting process. Q-methodology researchers have wide experience providing instructions similar to these so participants know more about the process and what to expect (Watts & Stenner, 2012).

Part 3 – Q-Sorting

Step 1:

Locate an area that will allow you the space to set out the sorting form board and the sorting cards and that you can position your computer and webcam or mobile device to provide the researcher video of your sorting process.

Step 2:

Connect to the Zoom video call with the link e-mailed to you (preferred) or telephone, . Once you are connected, do not hesitate to ask the researcher any questions about the Q-sorting process. The researcher will be a resource about Q-sorting throughout the process described in the following steps.

Step 3:

Carefully read and reflect on the question called the *Condition of Instruction* at the top of the sorting form board. Each of the Q-sorting cards contains a statement that provides an answer to the question posed in the *Condition of Instruction*. You will be placing each card in position on the sorting form board based upon how strongly you feel about the statement as important/unimportant as an answer to the question posed in the *Condition of Instruction*. The more you feel a statement is important, the further to the right side of the sorting form board you will place it. The more you feel a statement is unimportant, the further to the left side of the sorting form board you will place it. You will notice that the sorting form board limits the number of statement cards that you can indicate are most important and that you can indicate are most unimportant. This is an important part of Q methodology for capturing your viewpoint on

the subject. Following the 10 steps of the Q-sorting process, there are some Helpful Sorting Hints that can provide further assistance to you for sorting.

Step 4:

Take the stack of sorting cards. The small numbers in the lower right hand corner have been randomly assigned to more succinctly identify each statement in the data entry process, and they have no other meaning. Read each statement and make an initial, provisional sorting of each into one of five categories. To the right, place those statements **MOST LIKE** your point of view. To the left, place those statements **MOST UNLIKE** your point of view. Next, place those statements that are **SOMEWHAT LIKE** your point of view in a pile in front of you, but closer to the middle. Place those statements that are **SOMEWHAT UNLIKE** your point of view near the “unlike you” pile already identified, on toward the middle. For statements that you feel ambivalent about, have no opinion about, or are confused about the meaning, place them in a pile in the middle of the other four piles (we will refer to it as the **NO OPINION** pile). There are no limits to the number you place in each pile in this initial, provisional sorting step, just be faithful to your viewpoint, reactions, and feelings.

Step 5:

Move the **MOST UNLIKE** and **SOMEWHAT UNLIKE** piles and the **NO OPINION** pile further to the left side so the space in front of you is clear, and then spread out the statement cards in your **MOST LIKE** and **SOMEWHAT LIKE** piles to review them all together. You will now determine where to place each statement in the **MOST LIKE** pile on the sorting form board. The highest ranking positions are at the far right of the sorting form board, and that is where you will place the statements that you consider the **MOST LIKE YOUR POINT OF VIEW** important. In the column on the extreme right, you can place the two statements that you consider the most significantly like your point of view. In the next column second from the right, there are places for three statements that you also consider very much like your point of view. Continue placing your **SOMEWHAT LIKE YOUR POINT OF VIEW** statement cards on the sorting form board until they are all placed on the right side of the form board. Use as many spaces on the form board as you need to place all of the **LIKE YOUR POINT OF VIEW** and **SOMEWHAT LIKE YOUR POINT OF VIEW** statement cards

Step 6:

Continue sorting the statements by using the same process with the piles of cards you initially sorted as **MOST UNLIKE YOUR POINT OF VIEW** and **SOMEWHAT UNLIKE YOUR POINT OF VIEW**, except you will now use the far left columns of the form board.

Spread out the piles in front of you to review them all at once. When you have identified two that you consider the **MOST UNLIKE**, you will place them on the sorting form board in the column on the extreme left. Place the next most **UNLIKE** statements in the column second from the left, and so on. Keep going until you have placed all of the statements from your **UNLIKE** and **SOMEWHAT UNLIKE** piles of cards. When you have finished Step 6, there will be cards placed on the right side of the form board and on the left side of the form board, and a number of blank spaces in the middle of the form board that corresponds to the number of statements in your **NO OPINION** pile of cards. Complete Step 7 before placing the **NO OPINION** cards on the form board.

Step 7:

Using a pencil, on the blank Q-Sort Recording Sheet, fill in the small numbers on each statement card in the corresponding location for the statement on the form board. Draw a dark line to indicate the border line of the placement of statements initially sorted as MOST LIKE / SOMEWHAT LIKE (on the right side of the form board and distribution template sheet), and the border line of the statements initially sorted into the MOST UNLIKE/SOMEWHAT UNLIKE piles (on the left side of the form board and distribution template sheet), as illustrated below in Figure 13. Please note that this is an example only, and the number of cards and the placement of the bold lines will be different on your Q-Sort Recording Sheet!

Condition of Instruction: What is your view of effective Department-wide leadership in USFES Departments?										
Most unlike my view			Indifference, Ambivalence, or Ambiguity						Most like my view	
A	B	C	D	E	F	G	H	I	J	K
47	29	36				10	40	39	01	17
23	33	35				46	02	19	04	28
	24	27				03	26	25	08	
		20					41	45		
							42			

Figure 13. Illustration of the Q-Sort Recording Sheet with bold lines drawn by the sorter to indicate the statements that were in the MOST LIKE and SOMEWHAT LIKE sorting pile (on the right) and the MOST UNLIKE and SOMEWHAT UNLIKE sorting pile (on the left)

Step 8:

Spread out the last pile of cards in front of you to view them all at once – those initially sorted into the NO OPINION pile. It may seem difficult to place these statements on the form board because you probably do not hold a strong opinion about them in either direction. You will notice, though, that there are more places in the columns in the middle of the form board, meaning that there are fewer hard decisions to make. Place those statements that you think are more like your view in the spaces remaining that are more toward the right, and those that are less like your view in the spaces remaining that are more toward the left.

Step 9: Finish filling in the numbers of the NO OPINION statement cards on the Q-Sort Recording Sheet

Step 10:

Well-done! You have finished sorting and there is a complete Q-sort in front of you. Take a few minutes to review the placements of all the statements and feel free to make any changes you want to make. Once you are happy and satisfied with the placement of all the statements, finalize in the blank distribution template using the tiny number on each statement card in the blank space on the template corresponding to the space where you placed the card.

Step 11:

If you have a mobile device that you can use to take a digital photograph of the filled-in distribution template sheet of paper, snap a photo and e-mail it to the researcher at mr6646@ego.thechicagoschool.edu.

Helpful Sorting Hints

1. It can be difficult to decide immediately which two statements should be ranked as most MOST LIKE my point of view especially if you have a large number of items that you initially sorted into the MOST LIKE pile. One possible strategy is to read each statement carefully and when you have finished reading it, slide each one that generates stronger feelings of importance to the right and those that you feel less strongly about toward the left. This will physically spread the statements and create a sense of how they should be distributed within the MOST LIKE my point of view initial sort category.
2. Try not to get “hung up” on the ranking of any particular statement. For example, if you have three statements that you really want to rank as most important and need to select one that will have to be ranked in the next most important column, do not spend many minutes to decide which one you will not place in the most important column. The sorting process is for learning a general sense of your perspective, and the researcher will have that general sense no matter which one you place in the second column from the right rather than the first.
3. Do not worry if your MOST LIKE my point of view statements cross over into the columns headed by zero or negative numbers. It does not mean that you consider a statement most like or somewhat like your point of view unimportant if it is in a column with a negative number heading it. It just means that those statements are less like your point of view than those in columns with higher numbers. The rankings in Q-sorting are all relative to each other.

Appendix H: Research Transcriber Confidentiality Agreement



Investigator: Monika Byrd

Study Title: The Perceptions of Effective Department-wide Leadership in American Fire and Emergency Services

Purpose: The research study which involves an exploration of organizational leadership in U.S. fire and emergency services municipal departments. The purpose of this research study is to identify different perspectives within U.S. fire and emergency services regarding effective department-wide leadership. Second, the data will be analyzed to determine the extent to which there is one or more perspectives that are shared across types or sizes of departments. Third, the data will be analyzed to determine whether or not any of the perspectives are similar to existing models, theories, or approaches to organizational leadership in leadership research.

In order to be hired for the purpose of transcribing research interviews, your explicit consent not to reveal any of the contents of the tapes, notes or to reveal the identities of the participants must be obtained. Please read each of the statements below and sign to confirm your agreement to the conditions.

I, (name of the research transcriber) agree to:

1. Keep all research information I will be asked to record/and or transcribe confidential by not discussing or sharing the data in any form or format (e.g., consent forms, interview tapes, sound files or interview notes) with anyone other than the primary investigator.

2. Keep all research information in any form or format secure while it is in my possession.

3. Return all research information in any form or format to the primary investigator when I have completed the transcription tasks.

4. After consulting with the primary investigator, erase or destroy all research information in any form or format regarding this project that is not returnable to the primary investigator (e.g., interview and other relevant files stored on the computer hard drive)

Research Transcriber:

 (print name) (signature) (date)

Principal Investigator

 (print name) (signature) (date)

Questions/Concerns: If you have questions related to the procedures described in this document please contact Monika Byrd at mr6646@ego.thechicagoschool.edu or by phone at .

Appendix I: The Concourse and Q-Sample

The Concourse

The complete concourse on organizational leadership in the USFES follows, listed by the broad themes the researcher initially identified: a.) statements that recurred frequently in the interviews and trade literature, indicating that all sources agree they are important, b.)

Administration, c.) Change, d.) Command, e.) Monitoring, f.) Service, and g.) Teambuilding.

a.) Statements and variations of these that recurred frequently, that all sources agree are important

1. believe that on-the-job training is more important than formal education
2. believe that the only time to be authoritative is if there is a safety risk -- on or off the scene
3. fire department leaders emulate and embody the core values of the department
4. department leaders expect rigorous, tough, and regular training
5. department leaders maintain unity in the organization
6. have to operate in All would agree the realms that impact departments - legal, fiscal, political, strategic, technical
7. fire department leaders focus on crisis management
8. department leaders communicate effectively
9. department leaders communicate, communicate, communicate
10. department leaders minimize interference of their communications
11. department leaders department leaders vary their types of communications
12. department leaders study leadership
13. department leaders get out from behind their desks
14. department leaders support a respectful environment
15. department leaders are good at personal and professional writing as well as technical writing
16. department leaders are good public speakers
17. department leaders are self-aware
18. department leaders have exceptional organizational skills
19. department leaders know that actions speak loudly and they must "walk the talk"
20. department leaders make communication critical
21. department leaders admit mistakes and address them
22. department leaders have to be a lot more than good firefighters
23. department leaders separate what is right for the department from what is right for them as individuals
24. department leaders have consistent expectations
25. seek to comply and contribute to the 16 National Fire Service Life Safety Initiatives
26. have knowledge of the civil service and law
27. "lead by walking around"
28. are efficient and do not "have meetings about meetings"

29. stay on point, are not easily sidetracked
30. do not hesitate to un-do their own bad decisions and mistakes
31. make people comfortable by being clear and consistent
32. stay consistent on the priorities and issues until resolved
33. fire department leaders serve with integrity
34. department leaders walk the walk
35. department leaders always tell the truth
36. department leaders are calm under pressure
37. department leaders do not get emotional
38. department leaders are role models
39. department leaders do not whine
40. department leaders value professionalism
41. department leaders are consistently fair
42. consider strong, positive labor/management relations the top priority
43. do not "fake it till they make it" (not self-serving)
44. "have two buckets -- one with water and one with gas; which do they prefer to use?"
45. convey that it is an honor and a privilege to serve the department
46. are honest about what they cannot do well and what they need from others
47. have tenacity for the long haul
48. never "clone" another leader
49. expect loyalty
50. have studied supervision, management, and leadership whether or not there are training and professional development opportunities within the department
51. fire department leaders maintain good public relations
52. fire department leaders focus on public relations
53. department leaders are "on" All would agree the time, whether on the job or off
54. department leaders don't tell, they show
55. department leaders maintain the bridge between the leadership and members
56. department leaders encourage members who desire to achieve
57. are politically astute
58. are never too busy keeping their job rather than busy doing their job
59. have presence that is apparent throughout the organization so leadership is not an abstraction
60. interface well with other agencies, state-level fire officials
61. set priorities according to the public interest
62. keep clear lines of authority and uphold the chain of command
63. keep people focused on the public mission
64. admit they don't know everything
65. recognize that they don't know everything

b.) Administration Theme

66. fire department leaders identify professional development needs throughout the department
67. fire department leaders engage in succession planning
68. department leaders open up professional development opportunities for department members

69. department leaders focus less on firefighting skills and more on skills of finance, budgets, political and administrative issues
70. department leaders are relentless about follow up
71. department leaders take hiring very seriously
72. department leaders focus on risk-reduction
73. department leaders have knowledge of both national and local issues relevant to the fire department
74. focus on expanding or protecting resources
75. coordinate work across functional areas of the Department
76. frequently examine bureaucratic processes in detail to avoid excessive bureaucracy
77. department leaders have an expert understanding of dialogue processes and problem-solving
78. department leaders take on controversial, complex issues
79. pursue grants successfully to stay current and progressive
80. hold "listening meetings"
81. department leaders have deep awareness of the community
82. department leaders understand the metrics that are meaningful to external stakeholders
83. department leaders focus less on firefighting skills and training and more on skills of finance, budgets, and political and administrative issues
84. fire department leaders close the loop for issues so everyone understands what has happened
85. department leaders are analytical thinkers
86. department leaders focus on community risk reduction through research, education, proactive lobbying
87. department leaders have a little knowledge about every support area of the department
88. department leaders do risk/benefit analysis for All would agree decisions
89. department leaders are effective delegators, to balance their workload and focus on strategic change
90. department leaders have a broad focus and "eyes wide open" about the culture across different contexts
91. department leaders have exceptional cognitive abilities to see and analyze multiple perspectives and All would agree sides of the issues
92. department leaders must consider and weigh the statistics, the laws, the philosophical, and the human aspects of the issues
93. department leaders stress that public safety is truly impacted when the trucks are inside the station
94. department leaders get out of the way
95. department leaders look at the big picture and don't hesitate to lobby for resources from anywhere that can contribute to effectiveness
96. department leaders have skills and education related to public policy and public administration
97. are smart about how to influence different individuals
98. department leaders need to have knowledge and at least a passing knowledge or a little bit of experience or dabbled in just about every aspect of the fire department so they can understand.
99. department leaders need to have good knowledge of the community demographics

100. department leaders should influence the budget process
101. department leaders do their homework about the parts of the system the department is in
-- budget, labor, community demographics
102. department leaders have a college degree that emphasizes transferable skills like
writing, public speaking
103. department leaders understand the whole system that the department exists in and
considers the ripple effects of every action
104. department leaders stay current technically and regarding hr
105. department leaders provide the professional development opportunities for many people
to be ready to move up in the organization
106. rely on many in the department to provide input and expertise
107. avoid the temptation to focus only on areas where they have expertise or interest or are
popular
108. know that they are guiding a system, with the whole of the process being only as strong
as the weakest part
109. are attentive and alert at All would agree times to the big picture influences on the
department
110. act as the conduits that connect organizational support to the delivery of the services
111. champion public education efforts
112. know that human resource management is critical for the organization
113. good at presenting arguments to "sell" the department
114. have knowledge of what strategic planning, hr, IT, Finance, psychology contribute to
the department's effectiveness
115. understand supporting functions even if they have an operations background
116. department leaders facilitate the transfer of knowledge and experience between
labor/management, young/old, line/staff, operations/administration
117. have immense patience, because developing and improving a department is like
"watching the grass grow"
118. have a broad, general education so that technical expertise does not become like
blinders that prevent them from seeing and using the skills and abilities of others across
the entire department
119. department leaders are transparent,
120. department leaders are not politic
121. department leaders remain calm, respectful, and objective
122. can tolerate ambiguity
123. appreciate the growing supporting, non-sworn personnel necessary for an effective fire
department
124. embrace the complexity of things and never try to simplify things or gloss over it
125. department leaders develop relationships and coalitions with other public entities to
work together for the public good
126. seek alternate viewpoints, positive and negative
127. department leaders know that someone is always watching
128. department leaders have a laser focus on serving the public
129. department leaders ask and analyze what the needs and wants of the community are
130. department leaders address and connect with the community

131. department leaders reach out to All would agree groups in a community because of the importance of diversity and inclusion
132. give extra attention to community involvement
133. balance inward and outward-facing actions and behaviors
134. department leaders build strong labor/management relationships
135. department leaders build the bridge between labor and management
136. department leaders build bridges linking groups in a community so it can be resilient
137. are observant to see who "is on the fence" about leadership, policies, changes, etc., and creates relationships to persuade
138. department leaders attend public meetings even when they haven't been asked or aren't required to make a report
139. department leaders manage expectations of stakeholders
140. department leaders command respect from and build trust with "the bosses" (emergency director, mayor, city manager)
141. department leaders have to engage with local and state legislators
142. department leaders are connected to the world, not insulated
143. department leaders work within political realities
144. department leaders have strategic, cross-divisional teams for All would agree planning with both labor and management representatives
145. department leaders communicate effectively and frequently with external stakeholders
146. department leaders understand that the fire department isn't the only department providing services in the community and develops relationships and partnerships to better understand the services other agencies are providing
147. ensure the needs and desires of external stakeholders are met
148. bridge boundaries
149. department leaders need to know the government leaders they work with
150. department leaders make sure that administrative staff will support their decisions
151. department leaders make connections throughout the organization
152. compromise on single issues to preserve long-term relationships for future issues
153. actively explore ways to positively partner and participate with other public entities
154. uphold accountability to the public more than to fellow department members
155. make subject-matter-experts available for questions so not every level of leadership has to know everything

c.) Change Theme

156. department leaders plan for future changes beyond replacing apparatus and building new stations
157. department leaders expect department members to continue their education
158. fire department leaders must have a strategic plan that guides the department
159. department leaders equip employees with the training and tools needed to be creative in the problem solving process
160. department leaders are contemporary and up-to-date
161. department leaders are responsive to demographic and political shifts
162. department leaders need training and knowledge in strategic planning
163. break barriers to change

164. guard against the need to always agree; that is not a good way to guide the decision-making process
165. avoid "group think"
166. talk about what they are learning (and unlearning) to model the way for the importance of ongoing learning
167. encourage diversity of thought
168. pay attention to current and possible future trends
169. share lots of information and educate about the reasons for change or new things
170. fire department leaders communicate a well-organized strategic plan
171. fire department leaders must over-communicate the plan
172. fire department leaders form teams for planning and problem-solving
173. department leaders listen twice as much as they speak
174. department leaders are discerning about strategy
175. department leaders take a long-term view
176. department leaders expect department members to think creatively and solve problems
177. department leaders help members overcome being set in their ways
178. department leaders take risks with positions that are right for the department and community in spite of opposition
179. department leaders take a long-range view
180. department leaders create a safe zone for questioning processes and decisions where new ideas can emerge
181. department leaders have had exposure to different ways of thinking and seek out opportunities for further exposure
182. department leaders are persuasive to convince members that change has to happen
183. department leaders build buy-in for change
184. know how to "unstick" people stuck in the past
185. go beyond "getting along" to improve the services (external) and support for the members (internal)
186. department leaders are future-oriented
187. department leaders have had college/university, on-campus experiences, not just online education, that makes them worldly and appreciative of many perspectives
188. are introspective for the purpose of improving
189. start any change with self-assessment by asking is the change that needs to happen an issue with me?
190. department leaders are visionary
191. department leaders flex according to situations
192. department leaders reflect a lot about how to be a better department
193. department leaders reflect a lot about what changes to expect in the future
194. department leaders are brave about advancing new ideas and creating change
195. department leaders are never satisfied with the status quo, no matter how good it is
196. department leaders reflect a lot about what they still have to learn
197. department leaders have tough skin to take resistance and criticism about change
198. are forward-thinking
199. are not set in their ways or rigid
200. take on responsibility for contributing to the community's infrastructure to prevent fire and life safety emergencies

- 201. fire department leaders are skilled at overcoming objections
- 202. department leaders create freedom for free thinkers
- 203. department leaders welcome free thinking and new ideas
- 204. department leaders are pro-active in addition to being response-oriented
- 205. department leaders decentralize authority
- 206. department leaders embrace opportunities to do more functions that contribute to the public safety

d.) Command Theme

- 207. department leaders set up organizing principles that guide everything that happens in the department
- 208. department leaders admit mistakes and move on
- 209. department leaders act like a corporate CEO
- 210. department leaders give clear orders
- 211. department leaders do the tough stuff first
- 212. department leaders prevent and control rumors
- 213. department leaders should not rely on charisma
- 214. keep their personalities out of their leadership
- 215. tow the clear, hard line
- 216. try not to run a "team of equals"
- 217. rely on the chain of command to enforce policies and procedures
- 218. run a "tight ship," enforcing all policies and procedures with diligence
- 219. department leaders are loyal to the organization
- 220. fire department leaders have stalwart demeanor
- 221. fire department leaders do what they say, when they say they will do it, in the way that they say they will do it
- 222. department leaders are respected for being decisive; people generAll would agreeey would rather the Chief makes a bad decision than no decision
- 223. department leaders are bosses, not buddies
- 224. department leaders are consistent authoritarians
- 225. do not make exceptions or grant special permissions outside of policies
- 226. look out for those who show promise of advancement and understand that not everyone "should get a trophy" and be eligible for promotion
- 227. are wary of too much camaraderie and "buddy/buddy" behavior
- 228. department leaders have to focus on the big picture
- 229. department leaders are attentive to the big picture, have the 30,000 foot view
- 230. department leaders are visible
- 231. department leaders show up at incidents
- 232. department leaders are visible
- 233. "hold their cards close" until the support is there
- 234. ensure that everyone is in-tune with the leadership
- 235. provide clarity about the change to take place, about the performance expectation, about the timeline, and about consequences (if any piece is missing, the change won't be fulfilled)

e.) Monitoring Theme - This theme reflects an emphasis on responsibility (self and others') and accountability (self and others')

- 236. fire department leaders discipline members of the department consistently and fairly
- 237. department leaders monitor performance for consistency
- 238. department leaders expect thorough incident review for every single incident
- 239. keep SOPs up-to-date and create more for new situations, "you can never have too many SOPs"
- 240. department leaders are strong disciplinarians
- 241. department leaders are consistent disciplinarians
- 242. department leaders uphold accountability
- 243. department leaders understand the "hot stove" rule of discipline (punishment fits the offense)
- 244. develop metrics that are meaningful to different stakeholders
- 245. departments leaders ask for anonymous feedback
- 246. department leaders know that self-discipline is the best discipline
- 247. department leaders assess and evaluate processes and performance
- 248. department leaders perfect processes and performance
- 249. department leaders set consistent expectations
- 250. should run the department like a business
- 251. actively and regularly critique All would agree practices and processes
- 252. strictly uphold medical and physical standards for All would agree firefighters based on the duties they are to perform
- 253. have thick skin because of the criticism they inevitably face
- 254. fire department leaders have impeccable appearance
- 255. department leaders do not accept excuses
- 256. department leaders own the problem
- 257. department leaders don't pass the buck
- 258. department leaders recognize that they are responsible for everything, so must be up to speed on everything
- 259. department leaders are vigilant stewards of the public funds
- 260. department leaders enforce all the rules all the time

f.) Service Theme

- 261. department leaders are relentless about feedback
- 262. department leaders raise the bar
- 263. department leaders paint the picture with a clear "why"
- 264. strive to deeply understand issues and what the department needs
- 265. department leaders explain "the why" about the fire service as customer service
- 266. department leaders help department members humanize the tasks so they are more than "task masters"
- 267. know what true mentorship is to ensure that others are prepared for leadership
- 268. provide frequent positive reinforcement and recognition to prevent members from operating beneath expectations
- 269. department leaders get input
- 270. department leaders communicate bad news right away
- 271. department leaders are present for the whole department

- 272. department leaders focus on the "people side" not the administrative side
- 273. department leaders are visible, regardless of the size of the department
- 274. department leaders high in the hierarchy need interpersonal skills more than technical knowledge
- 275. department leaders are responsible for morale
- 276. fire department leaders mentor; build others up
- 277. department leaders give individualized attention to department employees
- 278. department leaders adjust communications for levels of experience and generational differences
- 279. department leaders are effective delegators, to develop future leaders
- 280. department leaders know how to have difficult conversations
- 281. department leaders help All would agree members develop a wider, more informed, more inclusive perspective
- 282. department leaders seek to be fair, not necessarily consistent
- 283. department leaders understand good followership
- 284. department leaders step aside to let others lead
- 285. "ask for the heart before they ask for a hand"
- 286. share lots of information for the greater good
- 287. are skilled at interpreting culture and non-verbal communications in the organization
- 288. support every aspect of wellness for members
- 289. department leaders believe in equity
- 290. are open to correction
- 291. department leaders believe the fire service is customer service
- 292. department leaders want to better the community
- 293. have faith in the members of the department and All would agreeow them to do their jobs
- 294. are patient to earn trust, loyalty
- 295. department leaders encourage the heart for department members
- 296. fire department leaders are human relations oriented
- 297. fire department leaders serve with humility
- 298. department leaders remember where they came from
- 299. department leaders always put others first
- 300. department leaders follow the golden rule (do unto others as you would have done unto you)
- 301. department leaders reflect a lot about how to be a better leader
- 302. department leaders have genuine empathy
- 303. department leaders are humble
- 304. department leaders are committed to diversity and inclusivity initiatives
- 305. department leaders have to get and sustain faith and trust in officers beneath them and let them do their jobs without micromanaging
- 306. department leaders are genuinely empathetic
- 307. department leaders believe that the fire service is more than organizations that provide emergency response, the fire service should also participate in helping the community recover from emergencies
- 308. department leaders make the personal choice to have a positive attitude
- 309. department leaders know that "they're not All would agree that"

- 310. department leaders believe everyone brings something to the table
- 311. have great humility
- 312. let go of ego
- 313. department leaders explain things often and thoroughly to involve people in the vision
- 314. give people permission and freedom to learn and share lessons learned
- 315. department leaders treat the community as a living organism that requires nurturing
- 316. department leaders work to make the community feel safe at All would agree times (not just during fires)
- 317. department leaders engage every member in leadership
- 318. department leaders walk around the neighborhoods and make casual, unofficial visits to businesses
- 319. department leaders truly care about the perspectives of members of the department, the community, and the political leadership of the community
- 320. department leaders give attention to career development for All would agree members
- 321. place a priority on being perceived as concerned participants who are interested in issues relating to the well-being of the people of the community
- 322. balance their attention to internal operations, internal support functions, and the external environment
- 323. fire department leaders share responsibility
- 324. department leaders walk around the jurisdiction, visit businesses, notice what is going on, and are approachable
- 325. department leaders ensure that members of the departments develop a wider, more informed, more inclusive perspective
- 326. department leaders are foremost concerned with community relations
- 327. department leaders propose and host community events
- 328. department leaders step aside sometimes and let other people have opportunities to exercise leadership
- 329. department leaders show up at fire stations
- 330. department leaders give away power
- 331. department leaders show up at community events
- 332. department leaders are customer service advocates
- 333. department leaders take care of the department members (because then they can take care of the public effectively)
- 334. department leaders serve the public *and* the department members
- 335. department leaders serve everybody else but themselves
- 336. department leaders lead in the community by volunteering for organizations that need leadership
- 337. department leaders express encouragement
- 338. department leaders have a broad understanding of who the fire department's stakeholders are
- 339. department leaders are concerned with quality of life in the community
- 340. department leaders have relationships with individual members -- they know what motivates each and what their needs are
- 341. department leaders have a village mentality -- the opposite of a hierarchical approach
- 342. department leaders are aware of the fire service as family for its members
- 343. department leaders are invested in concerns of members and seek solutions

- 344. department leaders help people feel that they are part of something bigger
- 345. department leaders assess their impact on others constantly
- 346. department leaders seek a lot of input
- 347. department leaders let people know what's going on, even when it's bad news
- 348. Allow time for rehab in overall workplans for all department members
- 349. know that "the white shirt puts you in "no-man's land" unless you succeed at relating to the real-life experiences of members
- 350. give members a sense of belonging to something bigger and beyond self

g.) Teambuilding Theme

- 351. department leaders must have a cognitive, intentional, and thoughtful approach to working with people
- 352. must network effectively and know "the players" in the community, the state, and the industry
- 353. department leaders get out of the way and give power to teams
- 354. department leaders support supervisors
- 355. department leaders share information openly
- 356. department leaders use people skills more than technical skills
- 357. department leaders are focused on the environment and mood of the department
- 358. fire department leaders keep the lines of communication open
- 359. fire department leaders empower teams to meet objectives
- 360. department leaders recognize how All would agree personnel support the mission
- 361. department leaders are good at collaborating
- 362. department leaders network frequently with other public safety agency leaders
- 363. department leaders use informal opportunities to open communications
- 364. department leaders share a sense of direction
- 365. department leaders develop cross-division interactions and project teams
- 366. department leaders should know the state of labor relations
- 367. department leaders understand the skill levels and commitment levels of employees to create the best teams
- 368. department leaders know that for people to learn, they have to give them a chance to correct issues on their own
- 369. department leaders know how to combine the right mix of people for great teamwork
- 370. are skilled at facilitating organizational dialogue
- 371. department leaders have a collaborative, inclusive style
- 372. are collaborative
- 373. department leaders are open to hearing from many people inside and outside of the organization
- 374. department leaders support having fun with the community to create social and political capital
- 375. fire department leaders share authority
- 376. department leaders are democratic
- 377. department leaders foster positive attitudes
- 378. department leaders ensure people feel safe in coming forward with ideas, suggestions, and input
- 379. department leaders are foremost concerned with workplace culture

- 380. fire department leaders decentralize authority
- 381. department leaders are very political in the community
- 382. department leaders make time for face time
- 383. department leaders rely on others to keep them up to speed
- 384. department leaders allow others to make decisions if they have more information
- 385. department leaders need to focus on the politics in the community
- 386. department leaders understand that fire departments are above All would agree social organizations
- 387. department leaders can lead more effectively if the department is open and connected to the community
- 388. department leaders take time with the people who got you to the dance

The Q-Sample

Tables 41, 42, and 43 indicate the themes and sub-themes represented by each Q-sample statement. Q-sample statements are self-referent and naturalistic from interviews and trade publications in the USFES, and are therefore often rich in meaning, integrating more than one concept. This richness of meaning is demonstrated by these tables when a statement has more than one X in its row. Table 43 includes a column to indicate the total number of sub-themes integrated in each Q-sample statement from all three tables.

Table 41

Q-sample statements and the sub-themes related to the broad themes of Administration and Change

Statement	Administration Theme					Change Theme				
	support	Accommodate	boundary-spanning	Network	Diplomatic	inclusive	open	visionary	adaptive	Creative
effective fire department leaders should actively explore ways to partner with other public entities			X	X						
effective fire department leaders should create freedom for free thinkers										X
effective fire department leaders should embrace opportunities to do more functions that contribute to			X						X	
effective fire department leaders need to be decisive, sometimes without complete information										
effective fire department leaders should pursue grants to stay current and progressive			X	X					X	

(continued)

Statement	Administration Theme					Change Theme				
	support	Accommodate	boundary-spanning	Network	Diplomatic	inclusive	open	visionary	adaptive	Creative
effective fire department leaders should treat human resource management as critical for the organization	X									
effective fire department leaders need to be vigilant stewards										
effective fire department leaders should have empathy										
effective fire department leaders should not accept excuses										
effective fire department leaders form cross-division teams to address issues and challenges			X							X
effective fire department leaders seek informal opportunities to interact				X						
effective fire department leaders own the department's problems									X	
effective fire department leaders give extra attention to community involvement			X							
effective fire department leaders need to develop metrics that are meaningful for different stakeholders	X	X		X						
effective fire department leaders should be future-oriented								X	X	
effective fire department leaders should act like corporate CEOs										
effective fire department leaders should hold their cards close until the support is there					X					
effective fire department leaders need to be smart about how to influence different individuals		X			X					X
effective fire department leaders should let go of ego										
effective fire department leaders should never be satisfied with the status quo								X	X	
effective fire department leaders should do the tough stuff first										
effective fire department leaders have to keep their personalities out of their leadership				X	X					
effective fire department leaders should value objectivity							X			
effective fire department leaders should give away power										
effective fire department leaders should share lots of information for the greater good	X									
effective fire department leaders need to facilitate organizational dialogue			X			X				
effective fire department leaders should get things done collaboratively						X				
effective fire department leaders have to actively and regularly critique all practices and processes										
effective fire department leaders should be wary of too much "buddy/buddy" behavior						X				
effective fire department leaders should guard against the need to always agree		X			X		X			
effective fire department leaders need to tow the clear, hard line										
effective fire department leaders should maintain a stalwart demeanor					X					
effective fire department leaders' most important work is risk reduction										
effective fire department leaders should facilitate career development for all members	X									
effective fire department leaders should believe everyone brings something to the table						X				
effective fire department leaders should decentralize authority										

(continued)

Statement	Administration Theme					Change Theme				
	support	Accommodate	boundary-spanning	Network	Diplomatic	inclusive	open	visionary	adaptive	Creative
effective fire department leaders need to pick the spots where to bend for the sake of morale										
effective fire department leaders need to be strong disciplinarians										
effective fire department leaders should know that good SOPs run the department										
effective fire department leaders need to create a safe zone for members to question processes and decisions	X						X		X	
effective fire department leaders should operate at the 30,000' level										
effective fire department leaders should understand the whole system and consider the ripple effects of										
effective fire department leaders should seek to be politically astute				X	X					
effective fire department leaders should measure success through the betterment of department	X									
effective fire department leaders value equity						X				
effective fire department leaders should give members time to correct issues on their own										
effective fire department leaders should be democratic	X									
effective fire department leaders need to break barriers to change			X					X		
Total statements representing the leadership quality/knowledge/behavior	7	3	7	6	6	5	3	3	6	3

Table 42

Q-sample statements and the sub-themes related to the broad themes of Command and Monitoring

Statement	Command Theme				Monitoring Theme			
	Discernment, selectivity	Contingency thinking	Strong	decisiveness	accountable	big picture	fair/just	systems/complexity
effective fire department leaders should actively explore ways to partner with other public entities		X						
effective fire department leaders should create freedom for free thinkers								
effective fire department leaders should embrace opportunities to do more functions that contribute		X						X
effective fire department leaders need to be decisive, sometimes without complete information	X		X	X				
effective fire department leaders should pursue grants to stay current and progressive		X						
effective fire department leaders should treat human resource management is critical for the						X		X
effective fire department leaders need to be vigilant stewards					X	X		
effective fire department leaders should have empathy								

(continued)

Statement	Command Theme				Monitoring Theme			
	Discernment, selectivity	Contingency thinking	Strong	decisiveness	accountable	big picture	fair/just	systems/complexity
effective fire department leaders should not accept excuses							X	
effective fire department leaders form cross-division teams to address issues and challenges								
effective fire department leaders seek informal opportunities to interact								
effective fire department leaders own the department's problems					X			
effective fire department leaders give extra attention to community involvement						X		X
effective fire department leaders need to develop metrics that are meaningful for different								
effective fire department leaders should be future-oriented								
effective fire department leaders should act like corporate CEOs			X		X	X		
effective fire department leaders should hold their cards close until the support is there	X							
effective fire department leaders need to be smart about how to influence different individuals								
effective fire department leaders should let go of ego	X							
effective fire department leaders should never be satisfied with the status quo					X			
effective fire department leaders should do the tough stuff first	X		X		X			
effective fire department leaders have to keep their personalities out of their leadership								
effective fire department leaders should value objectivity							X	
effective fire department leaders should give away power								
effective fire department leaders should share lots of information for the greater good								
effective fire department leaders need to facilitate organizational dialogue								X
effective fire department leaders should get things done collaboratively								
effective fire department leaders have to actively and regularly critique all practices					X			
effective fire department leaders should be wary of too much "buddy/buddy" behavior							X	
effective fire department leaders should guard against the need to always agree	X							
effective fire department leaders need to tow the clear, hard line			X	X	X			
effective fire department leaders should maintain a stalwart demeanor			X					
effective fire department leaders' most important work is risk reduction		X						X

(continued)

Statement	Command Theme				Monitoring Theme			
	Discernment, selectivity	Contingency thinking	Strong	decisiveness	accountable	big picture	fair/just	systems/complexity
effective fire department leaders should facilitate career development for all							X	
effective fire department leaders should believe everyone brings something to the								
effective fire department leaders should decentralize authority								
effective fire department leaders need to pick the spots where to bend for the sake of	X							
effective fire department leaders need to be strong disciplinarians			X	X			X	
effective fire department leaders should know that good SOPs run the department				X			X	
effective fire department leaders need to create a safe zone for members to question								
effective fire department leaders should operate at the 30,000' level						X		X
effective fire department leaders should understand the whole system and consider		X						X
effective fire department leaders should seek to be politically astute	X							
effective fire department leaders should measure success through the betterment of								
effective fire department leaders value equity							X	
effective fire department leaders should give members time to correct issues on their								
effective fire department leaders should be democratic							X	
effective fire department leaders need to break barriers to change								
Total statements representing the leadership quality/knowledge/behavior	7	5	6	4	7	5	8	7

Table 43

Q-sample statements and the sub-themes related to the broad themes of Service and Teambuilding, including the total number of sub-themes across all of the broad themes in Tables 41, 42, and 43

Statement	Service Theme					Teambuilding Theme					Total sub-themes
	Reflec- tive	nurture	greater good	empathy	humble	Trans- parency	Team- builder	Facilit- ating	High expect- ations	build trust	
effective fire department leaders should actively explore ways to partner with other			X				X				5
effective fire department leaders should create freedom for free thinkers						X					2
effective fire department leaders should embrace opportunities to do more functions			X								5
effective fire department leaders need to be decisive, sometimes without complete	X										4
effective fire department leaders should pursue grants to stay current and											4
effective fire department leaders should treat human resource management is critical										X	4
effective fire department leaders need to be vigilant stewards			X						X		4
effective fire department leaders should have empathy		X		X							2
effective fire department leaders should not accept excuses									X		2
effective fire department leaders form cross-division teams to address issues and							X	X			4
effective fire department leaders seek informal opportunities to interact					X		X			X	4
effective fire department leaders own the department's problems											2
effective fire department leaders give extra attention to community involvement			X								4
effective fire department leaders need to develop metrics that are meaningful for											3
effective fire department leaders should be future-oriented											2
effective fire department leaders should act like corporate CEOs											3
effective fire department leaders should hold their cards close until the support is											2
effective fire department leaders need to be smart about how to influence different											3
effective fire department leaders should let go of ego					X						2
effective fire department leaders should never be satisfied with the status quo	X								X		5
effective fire department leaders should do the tough stuff first											3
effective fire department leaders have to keep their personalities out of their											2

(continued)

Statement	Service Theme					Teambuilding Theme					Total sub-themes
	Reflec- tive	nurture	greater good	empath y	humble	Trans- parency	Team- builder	Facilit- ating	High expect- ations	build trust	
effective fire department leaders should value objectivity	X									X	4
effective fire department leaders should give away power					X			X			2
effective fire department leaders should share lots of information for the greater			X			X		X		X	5
effective fire department leaders need to facilitate organizational dialogue								X			4
effective fire department leaders should get things done collaboratively							X	X			3
effective fire department leaders have to actively and regularly critique all practices	X					X					3
effective fire department leaders should be wary of too much "buddy/buddy" behavior											2
effective fire department leaders should guard against the need to always agree											4
effective fire department leaders need to tow the clear, hard line											3
effective fire department leaders should maintain a stalwart demeanor											2
effective fire department leaders' most important work is risk reduction			X						X		4
effective fire department leaders should facilitate career development for all		X		X							4
effective fire department leaders should believe everyone brings something to the		X					X		X		4
effective fire department leaders should decentralize authority					X		X				2
effective fire department leaders need to pick the spots where to bend for the sake of		X									2
effective fire department leaders need to be strong disciplinarians									X		4
effective fire department leaders should know that good SOPs run the department						X					3
effective fire department leaders need to create a safe zone for members to question								X		X	5
effective fire department leaders should operate at the 30,000' level											2
effective fire department leaders should understand the whole system and consider											2
effective fire department leaders should seek to be politically astute											3
effective fire department leaders should measure success through the betterment of	X								X		3
effective fire department leaders value equity		X	X	X							5
effective fire department leaders should give members time to correct issues on their own							X			X	2
effective fire department leaders should be democratic						X				X	4

(continued)

	Service Theme					Teambuilding Theme					Total sub-themes
Statement	Reflec- tive	nurture	greater good	Empath y	Humble	Trans- parency	Team- builder	Facilit- ating	High expect- ations	build trust	
effective fire department leaders need to break barriers to change											2
Total statements representing the leadership quality/knowledge/behavior	5	5	7	3	4	5	7	6	7	7	

Appendix J: Q-Sort Intercorrelation Matrix

Table 44

Intercorrelation Matrix Between Sorts, Sorts 1 – 16

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	1.00	0.37	0.20	0.29	0.34	0.27	0.39	0.42	0.06	0.07	0.33	0.47	0.55	0.57	0.15	0.45
2	0.37	1.00	0.22	0.55	0.43	0.16	0.53	0.46	0.30	0.04	0.28	0.55	0.37	0.52	0.14	0.62
3	0.20	0.22	1.00	0.36	0.45	0.35	0.28	0.42	0.28	0.15	0.27	0.31	0.37	0.48	0.11	0.39
4	0.29	0.55	0.36	1.00	0.46	0.22	0.62	0.36	0.18	0.25	0.23	0.57	0.47	0.54	0.31	0.42
5	0.34	0.43	0.45	0.46	1.00	0.37	0.41	0.36	0.14	0.37	0.55	0.46	0.44	0.54	0.28	0.51
6	0.27	0.16	0.35	0.22	0.37	1.00	0.24	0.53	-0.18	0.30	0.16	0.37	0.38	0.44	0.33	0.32
7	0.39	0.53	0.28	0.62	0.41	0.24	1.00	0.37	0.18	0.22	0.12	0.53	0.44	0.57	0.36	0.52
8	0.42	0.46	0.42	0.36	0.36	0.53	0.37	1.00	-0.19	0.12	0.32	0.60	0.47	0.52	0.15	0.62
9	0.06	0.30	0.28	0.18	0.14	-0.18	0.18	-0.19	1.00	0.00	0.06	-0.02	0.04	0.24	0.12	-0.02
10	0.07	0.04	0.15	0.25	0.37	0.30	0.22	0.12	0.00	1.00	0.22	0.28	0.21	0.21	0.32	-0.02
11	0.33	0.28	0.27	0.23	0.55	0.16	0.12	0.32	0.06	0.22	1.00	0.29	0.45	0.37	0.31	0.22
12	0.47	0.55	0.31	0.57	0.46	0.37	0.53	0.60	-0.02	0.28	0.29	1.00	0.46	0.57	0.29	0.61
13	0.55	0.37	0.37	0.47	0.44	0.38	0.44	0.47	0.04	0.21	0.45	0.46	1.00	0.72	0.34	0.43
14	0.57	0.52	0.48	0.54	0.54	0.44	0.57	0.52	0.24	0.21	0.37	0.57	0.72	1.00	0.34	0.47
15	0.15	0.14	0.11	0.31	0.28	0.33	0.36	0.15	0.12	0.32	0.31	0.29	0.74	0.34	1.00	0.01
16	0.45	0.62	0.39	0.42	0.51	0.32	0.52	0.62	-0.02	-0.02	0.22	0.61	0.43	0.47	0.01	1.00
17	0.44	0.13	0.13	0.03	0.16	-0.03	0.30	0.04	0.27	-0.13	0.14	0.09	0.16	0.27	0.07	0.11
18	0.54	0.43	0.36	0.29	0.47	0.40	0.43	0.29	0.18	0.07	0.27	0.43	0.39	0.52	0.31	0.45
19	0.30	0.32	0.08	0.39	0.48	0.44	0.43	0.37	0.01	0.23	0.42	0.47	0.51	0.48	0.43	0.32
20	0.31	0.50	0.23	0.21	0.44	0.31	0.26	0.43	0.12	0.16	0.43	0.44	0.50	0.51	0.32	0.50
21	0.37	0.20	0.17	0.21	0.26	0.34	0.24	0.28	-0.18	0.32	0.18	0.35	0.27	0.32	0.02	0.28
22	0.42	0.47	0.35	0.32	0.43	0.27	0.39	0.37	0.12	0.11	0.44	0.57	0.55	0.63	0.36	0.45
23	0.36	0.37	0.24	0.28	0.45	0.41	0.35	0.50	-0.11	0.36	0.26	0.48	0.55	0.46	0.14	0.43
24	0.44	0.47	0.32	0.49	0.38	0.14	0.49	0.49	-0.02	0.22	0.33	0.55	0.61	0.52	0.27	0.61
25	0.44	0.33	0.30	0.43	0.47	0.26	0.50	0.39	0.21	0.06	0.31	0.36	0.57	0.65	0.30	0.42
26	0.43	0.39	0.25	0.43	0.26	0.33	0.67	0.52	0.02	0.10	0.10	0.63	0.33	0.39	0.29	0.70
27	0.49	0.59	0.48	0.66	0.49	0.27	0.47	0.59	0.26	0.08	0.37	0.54	0.57	0.57	0.06	0.59
28	0.28	0.07	0.13	0.24	0.26	0.19	0.10	0.36	-0.20	0.18	0.29	0.23	0.34	0.28	-0.13	0.23
29	0.43	0.30	0.34	0.28	0.42	0.30	0.39	0.44	0.00	-0.08	0.35	0.49	0.45	0.39	0.16	0.61
30	0.04	0.15	0.15	0.20	0.00	-0.01	0.17	0.12	0.28	0.18	-0.20	0.09	0.07	0.06	0.01	0.20
31	-0.12	-0.41	-0.13	-0.17	-0.25	-0.17	-0.15	-0.50	0.26	0.28	-0.21	-0.40	-0.12	-0.11	0.09	-0.61
32	0.62	0.39	0.28	0.17	0.40	0.36	0.27	0.64	-0.04	-0.09	0.37	0.49	0.55	0.49	-0.05	0.67

Note. Q-Sorts 1 – 25 are P-Sample sorts; Q-Sorts 26 – 32 are theoretical sorts; PQMethod 2.35

Table 45

Intercorrelation Matrix Between Sorts, Sorts 17 – 32

Sort	17	18	19	20	21	22	23	24	25	26 Serv	27 Adapt	28 Pubad min	29 Transf orm	30 CRM)	31 Transact	32 Complex
1	0.44	0.54	0.30	0.31	0.37	0.42	0.36	0.44	0.44	0.43	0.49	0.28	0.43	0.04	0.12	0.62
2	0.13	0.43	0.32	0.50	0.20	0.47	0.37	0.47	0.33	0.39	0.59	0.07	0.30	0.15	-0.41	0.39
3	0.13	0.36	0.08	0.23	0.17	0.35	0.24	0.32	0.30	0.25	0.48	0.13	0.34	0.15	-0.13	0.28
4	0.03	0.29	0.39	0.21	0.21	0.32	0.28	0.49	0.43	0.43	0.66	0.24	0.28	0.20	-0.17	0.17
5	0.16	0.47	0.48	0.44	0.26	0.43	0.45	0.38	0.47	0.26	0.49	0.26	0.42	0.00	-0.25	0.40
6	-0.03	0.40	0.44	0.31	0.34	0.27	0.41	0.14	0.26	0.33	0.27	0.19	0.30	-0.01	-0.17	0.36
7	0.30	0.43	0.43	0.26	0.24	0.39	0.35	0.49	0.50	0.67	0.47	0.10	0.39	0.17	-0.15	0.27
8	0.04	0.29	0.37	0.43	0.28	0.37	0.50	0.49	0.39	0.52	0.59	0.36	0.44	0.12	-0.50	0.64
9	0.27	0.18	0.01	0.12	-0.18	0.12	-0.11	-0.02	0.21	0.02	0.26	-0.20	0.00	0.28	0.26	-0.04
10	-0.13	0.07	0.23	0.16	0.32	0.11	0.36	0.22	0.06	0.10	0.08	0.18	-0.08	0.18	0.28	-0.09
11	0.14	0.27	0.42	0.43	0.18	0.44	0.26	0.33	0.31	0.10	0.37	0.29	0.35	-0.20	-0.21	0.37
12	0.09	0.43	0.47	0.44	0.35	0.57	0.48	0.55	0.36	0.63	0.54	0.23	0.49	0.09	-0.40	0.49
13	0.16	0.39	0.51	0.50	0.27	0.55	0.55	0.61	0.57	0.33	0.57	0.34	0.45	0.07	-0.12	0.55
14	0.27	0.52	0.48	0.51	0.32	0.63	0.46	0.52	0.65	0.39	0.57	0.28	0.39	0.06	-0.11	0.49
15	0.07	0.31	0.43	0.32	0.02	0.36	0.14	0.27	0.30	0.29	0.06	-0.13	0.16	0.01	0.09	-0.05
16	0.11	0.45	0.32	0.50	0.28	0.45	0.43	0.61	0.42	0.70	0.59	0.23	0.61	0.20	-0.61	0.67
17	1.00	0.29	0.13	-0.10	0.00	0.21	0.09	0.06	0.10	0.24	0.11	-0.11	0.25	-0.11	0.11	0.17
18	0.29	1.00	0.36	0.43	0.37	0.54	0.57	0.23	0.35	0.44	0.37	0.05	0.32	0.14	-0.15	0.43
19	0.13	0.36	1.00	0.53	0.42	0.47	0.41	0.47	0.45	0.44	0.35	0.19	0.44	-0.21	-0.25	0.32
20	-0.10	0.43	0.53	1.00	0.37	0.63	0.45	0.52	0.55	0.31	0.45	0.23	0.38	0.10	-0.29	0.45
21	0.00	0.37	0.42	0.37	1.00	0.30	0.47	0.27	0.33	0.19	0.26	0.23	0.26	0.03	-0.17	0.28
22	0.21	0.54	0.47	0.63	0.30	1.00	0.52	0.46	0.53	0.29	0.43	0.15	0.37	-0.08	-0.16	0.40
23	0.09	0.57	0.41	0.45	0.47	0.52	1.00	0.47	0.41	0.37	0.41	0.44	0.34	0.22	-0.07	0.44
24	0.06	0.23	0.47	0.52	0.27	0.46	0.47	1.00	0.40	0.52	0.45	0.29	0.46	0.19	-0.30	0.45
25	0.10	0.35	0.45	0.55	0.33	0.53	0.41	0.40	1.00	0.35	0.53	0.26	0.37	0.08	-0.18	0.42
26	0.24	0.44	0.44	0.31	0.19	0.29	0.37	0.52	0.35	1.00	0.33	0.11	0.64	0.21	-0.42	0.44
27	0.11	0.37	0.35	0.45	0.26	0.43	0.41	0.45	0.53	0.33	1.00	0.52	0.45	0.23	-0.28	0.62
28	-0.11	0.05	0.19	0.23	0.23	0.15	0.44	0.29	0.26	0.11	0.52	1.00	0.33	0.08	0.00	0.46
29	0.25	0.32	0.44	0.38	0.26	0.37	0.34	0.46	0.37	0.64	0.45	0.33	1.00	0.03	-0.55	0.66
30	-0.11	0.14	-0.21	0.10	0.03	-0.08	0.22	0.19	0.08	0.21	0.23	0.08	0.03	1.00	0.03	0.25
31	0.11	-0.15	-0.25	-0.29	-0.17	-0.16	-0.07	-0.30	-0.18	-0.42	-0.28	0.00	-0.55	0.03	1.00	-0.54
32	0.17	0.43	0.32	0.45	0.28	0.40	0.44	0.35	0.42	0.44	0.62	0.46	0.66	0.25	-0.54	1.00

Note. Q-Sorts 1 – 25 are P-Sample sorts; Q-Sorts 26 – 32 are theoretical sorts; PQMethod 2.35